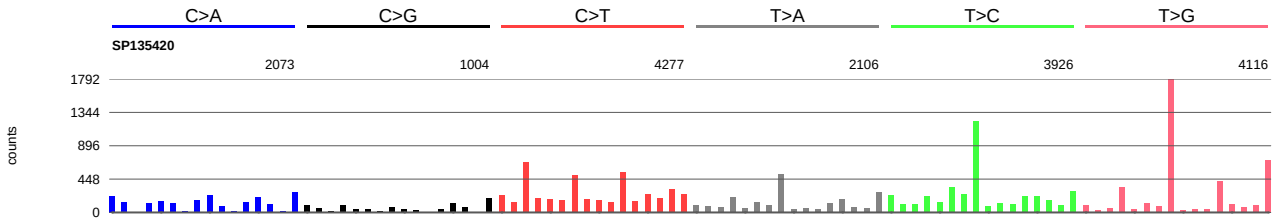
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SP135420
2073
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4277
2106
3926
4116
counts
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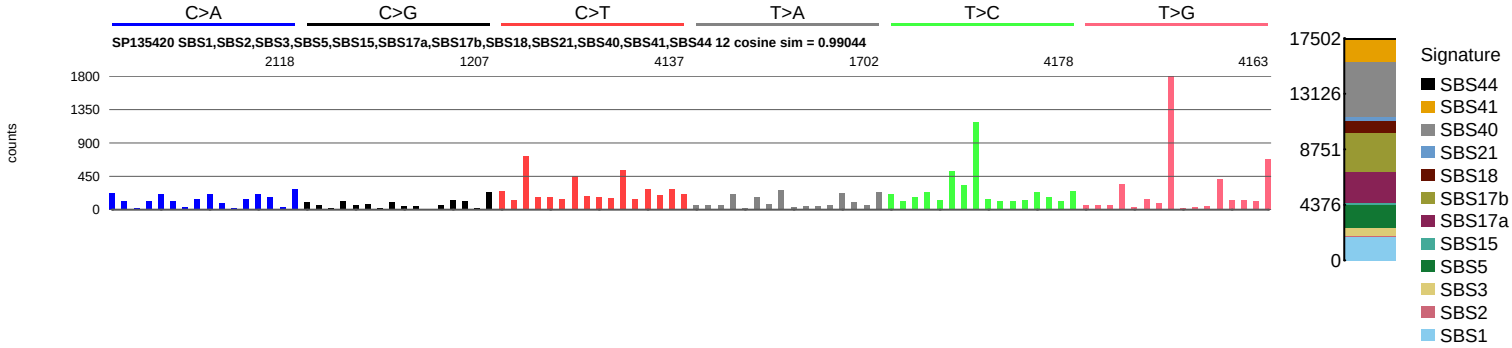
1344
896
448
0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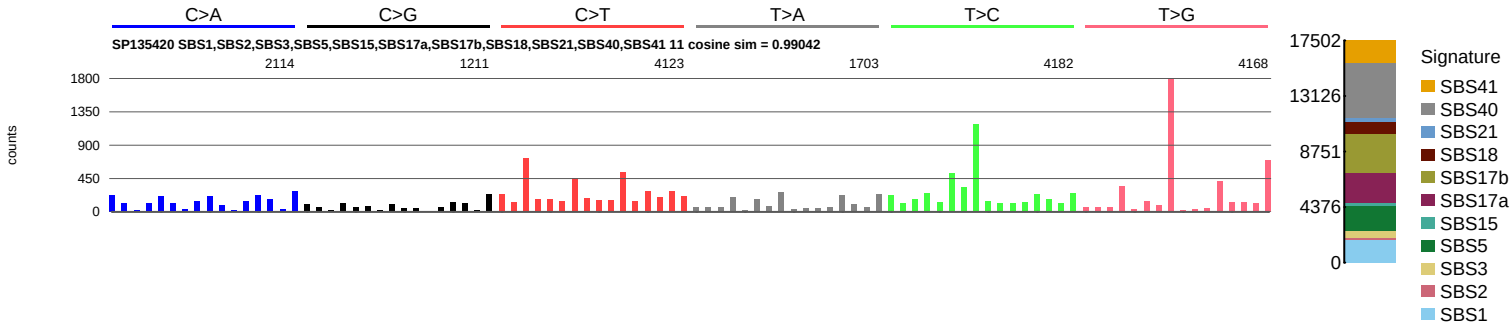
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99044
2118
1207
4137
1702
4178
4163
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99042
2114
1211
4123
1703
4182
4168
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
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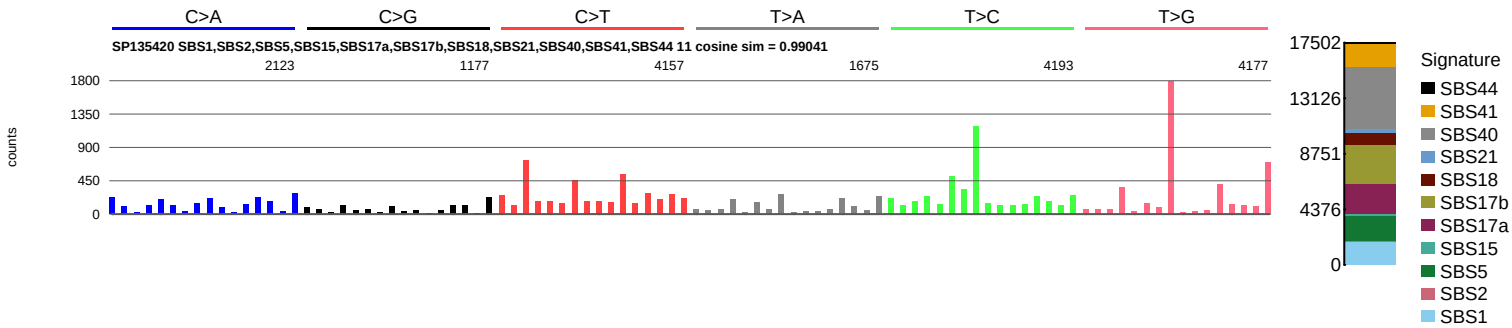
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99041
2123
1177
4157
1675
4193
4177
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
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SBS21
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SBS17b
SBS17a
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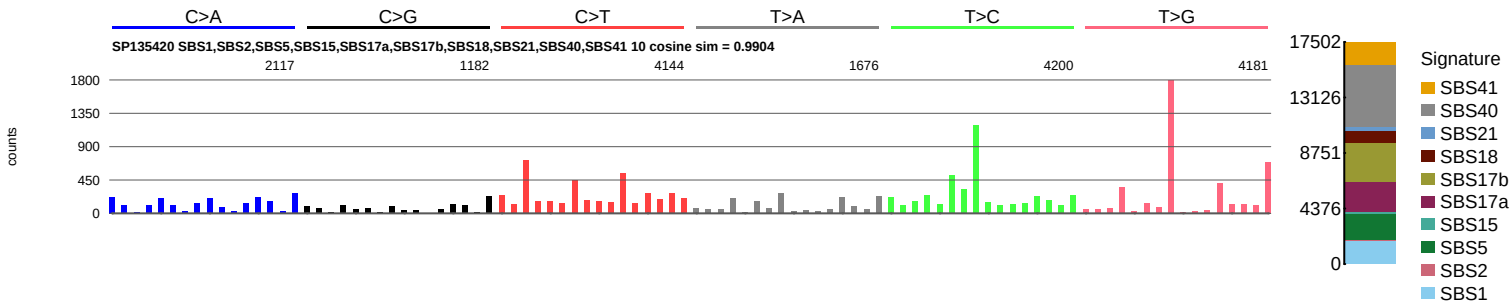
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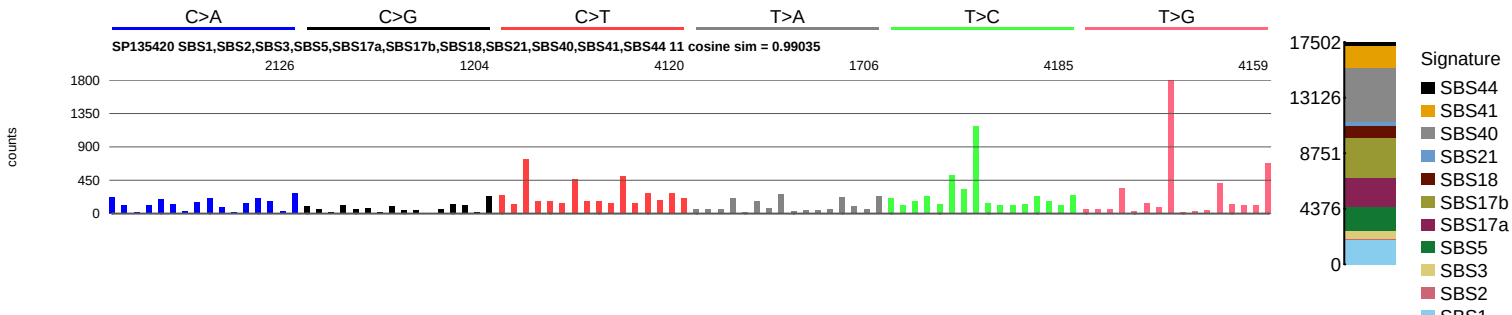
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9904
2117
1182
4144
1676
4200
4181
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99035
2126
1204
4120
1706
4185
4159
counts
1800
1350
900
450
0
17502
13126
8751
4376
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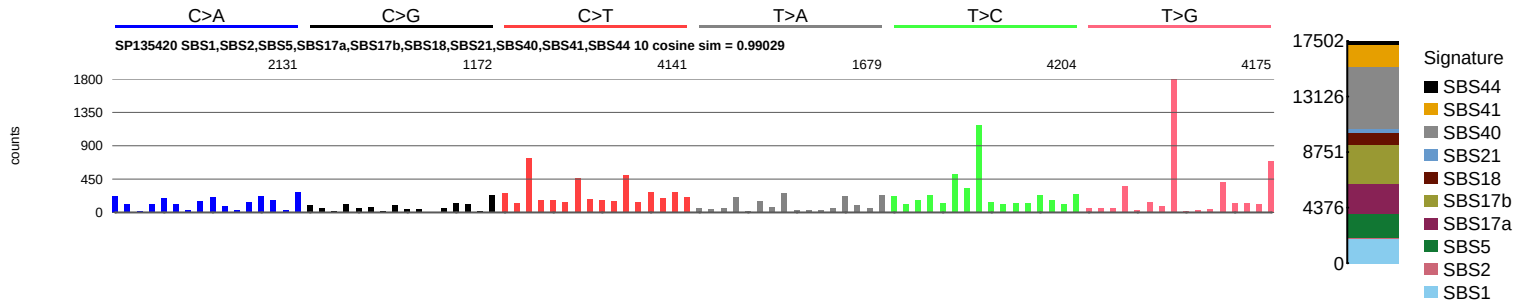
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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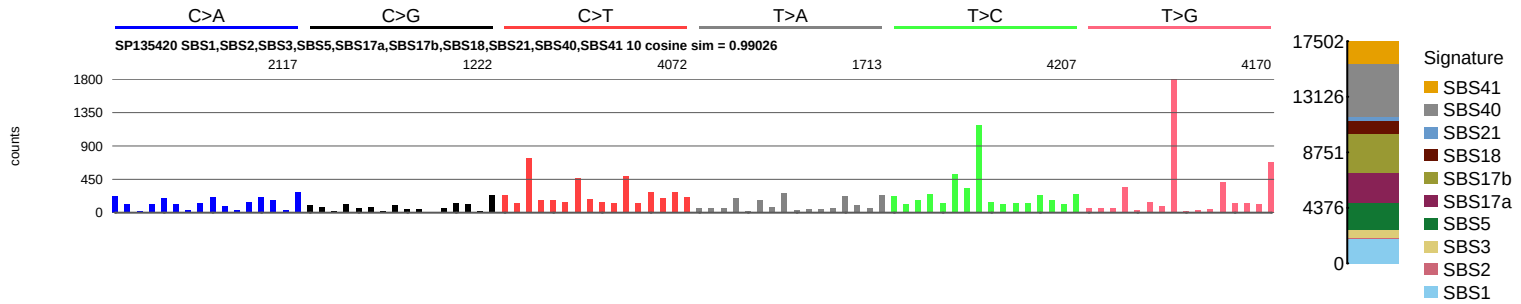
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99029
2131
1172
4141
1679
4204
4175
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99026
2117
1222
4072
1713
4207
4170
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
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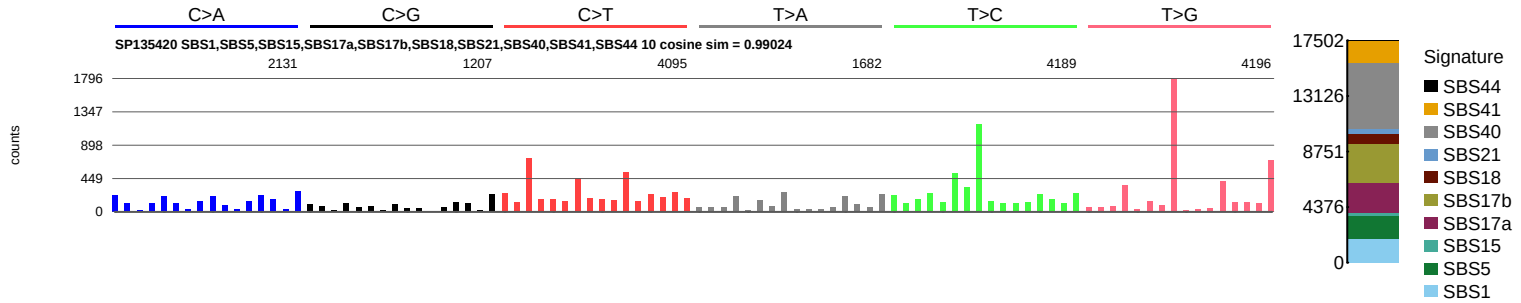
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99024
2131
1207
4095
1682
4189
4196
counts
1796
1347
898
449
0
17502
13126
8751
4376
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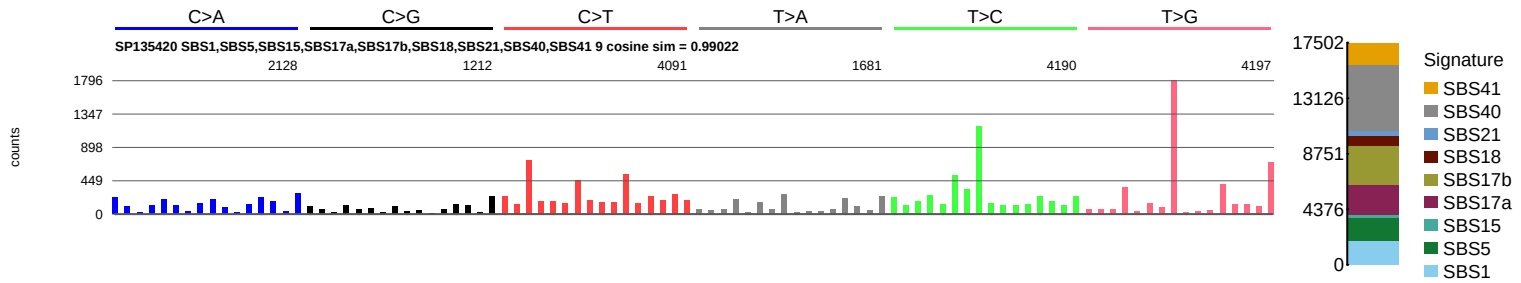
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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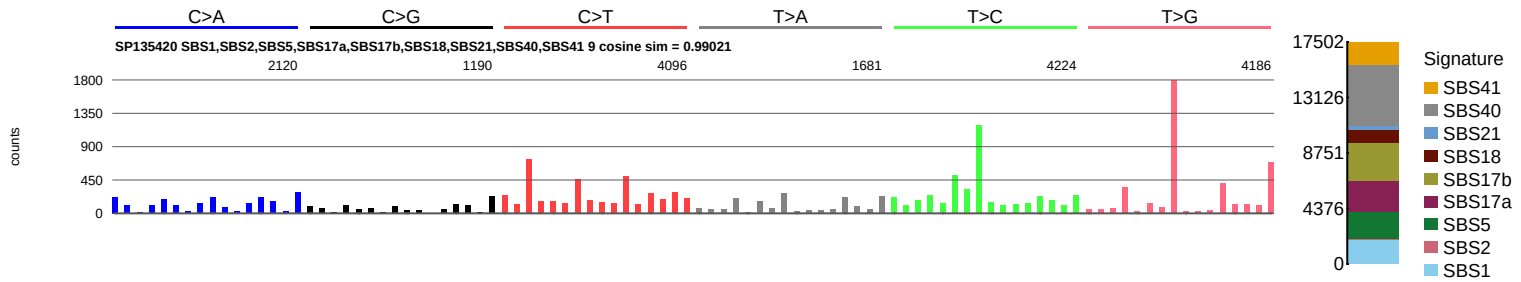
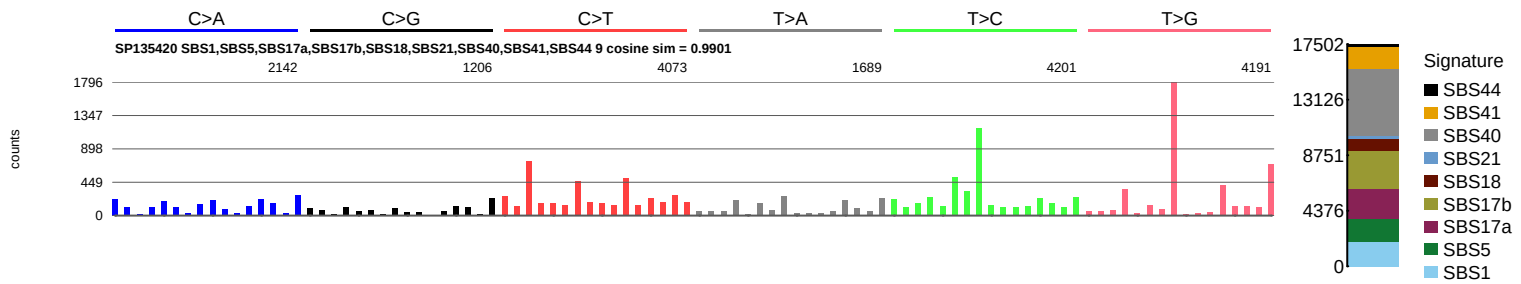
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99022
2128
1212
4091
1681
4190
4197
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99021
2120
1190
4096
1681
4224
4186
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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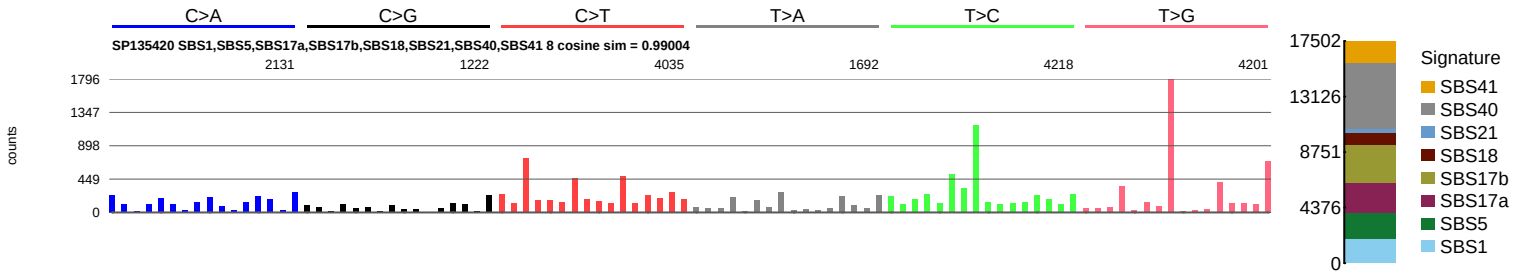
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9901
2142
1206
4073
1689
4201
4191
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99004
2131
1222
4035
1692
4218
4201
counts
1796
1347
898
449
0
17502
13126
8751
4376
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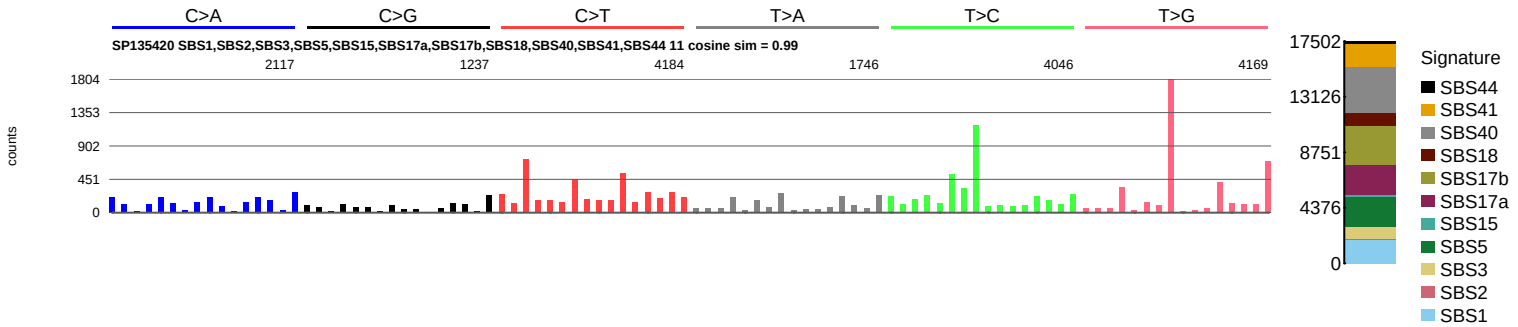
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
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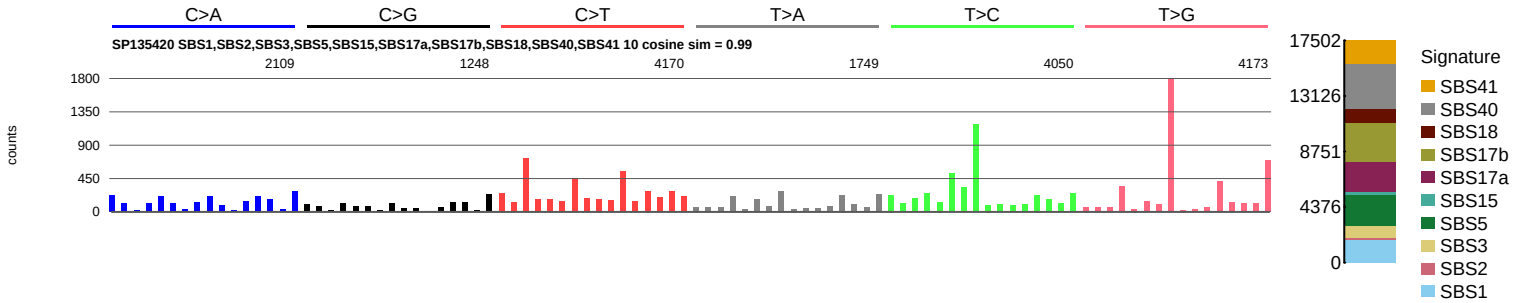
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99
2117
1237
4184
1746
4046
4169
counts
1804
1353
902
451
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99
2109
1248
4170
1749
4050
4173
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
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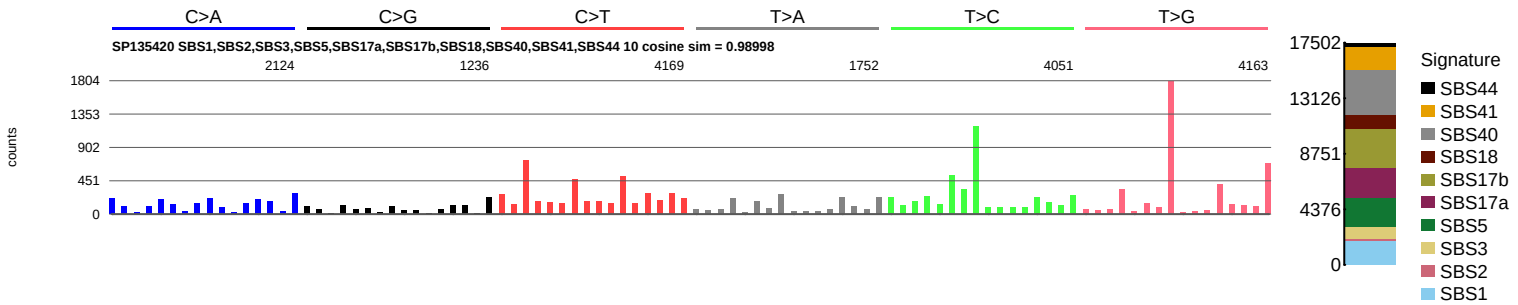
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98998
2124
1236
4169
1752
4051
4163
counts
1804
1353
902
451
0
17502
13126
8751
4376
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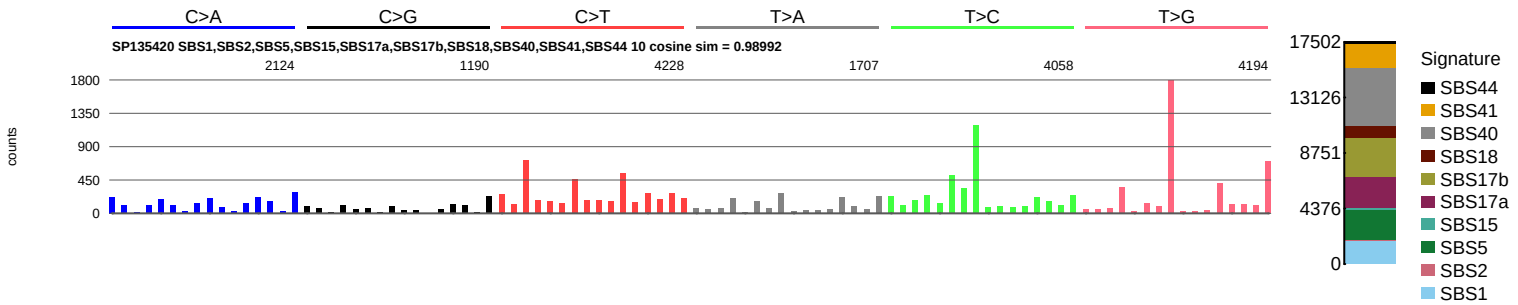
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98992
2124
1190
4228
1707
4058
4194
counts
1800
1350
900
450
0
17502
13126
8751
4376
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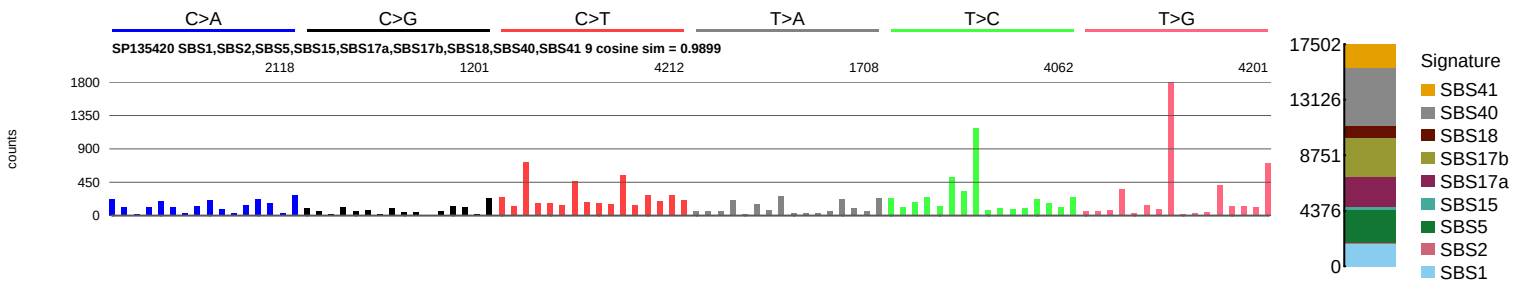
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>T
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T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9899
2118
1201
4212
1708
4062
4201
counts
1800
1350
900
450
0
17502
13126
8751
4376
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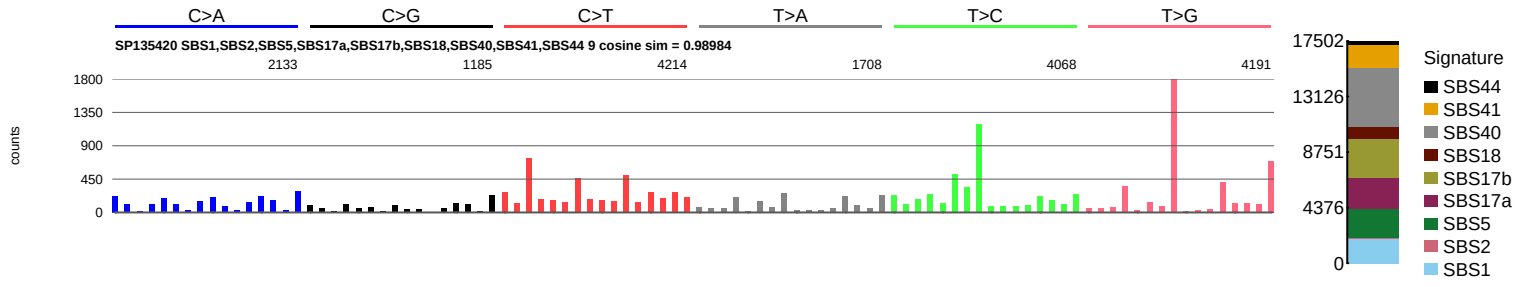
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98984
2133
1185
4214
1708
4068
4191
counts
1800
1350
900
450
0
17502
13126
8751
4376
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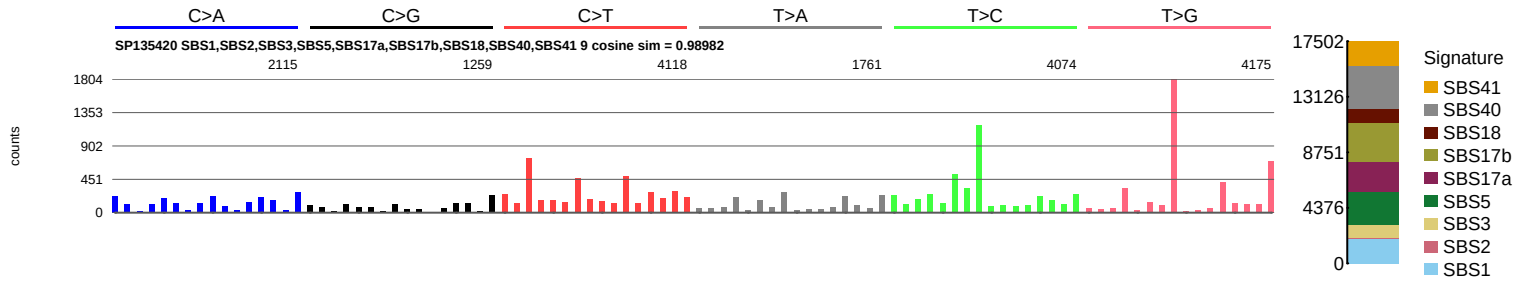
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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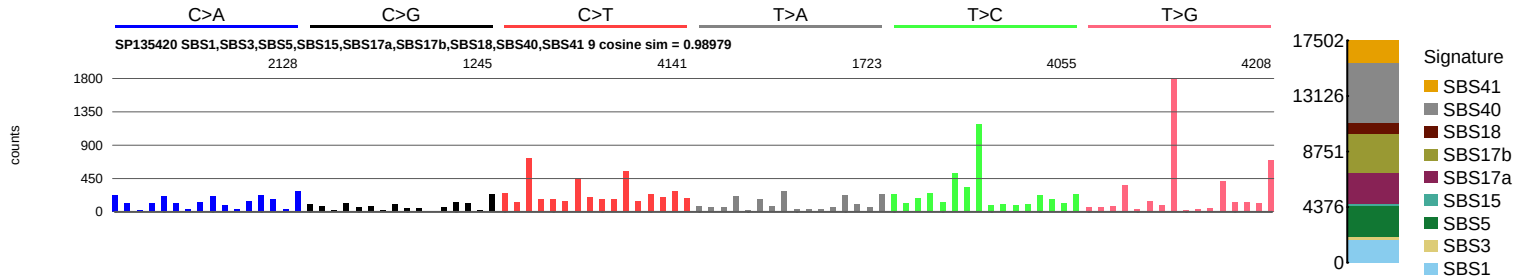
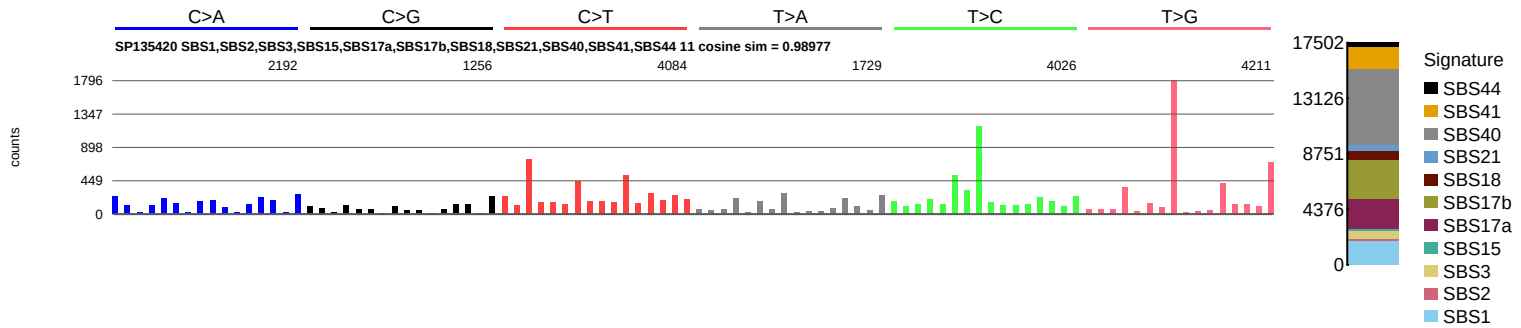
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98982
2115
1259
4118
1761
4074
4175
counts
1804
1353
902
451
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98979
2128
1245
4141
1723
4055
4208
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98977
2192
1256
4084
1729
4026
4211
counts
1796
1347
898
449
0
17502
13126
8751
4376
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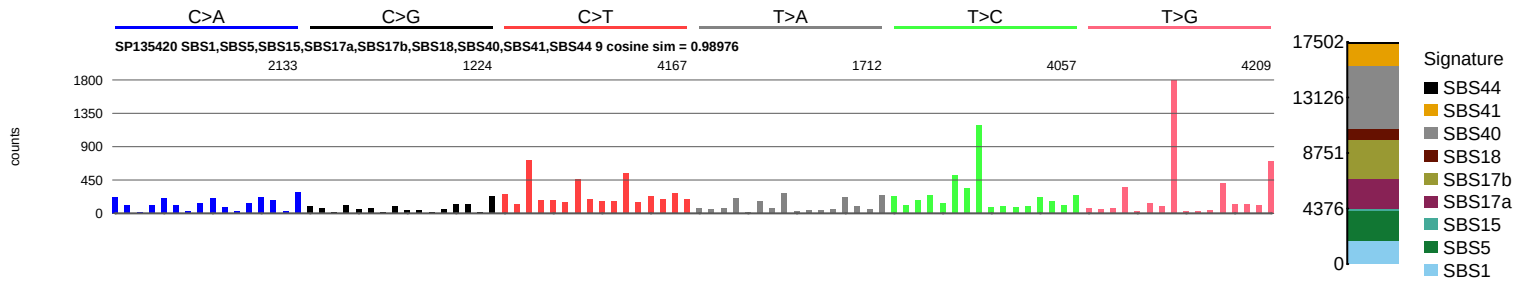
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98976
2133
1224
4167
1712
4057
4209
counts
1800
1350
900
450
0
17502
13126
8751
4376
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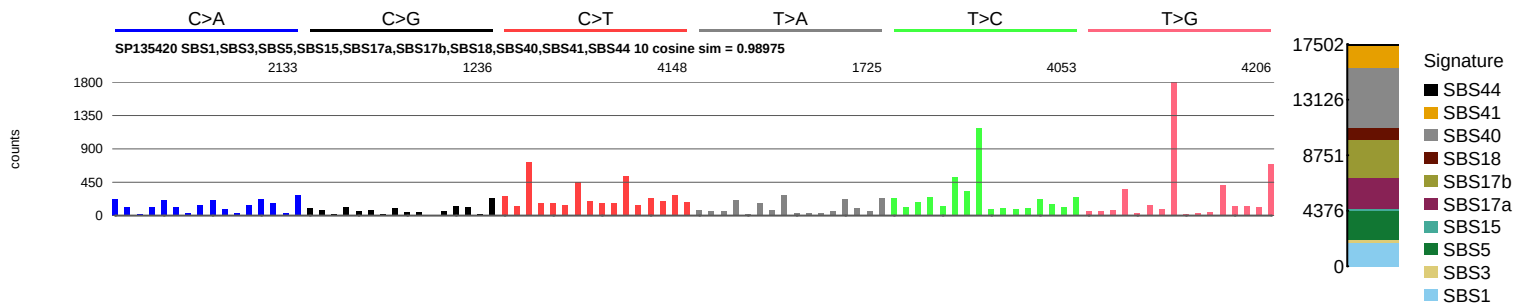
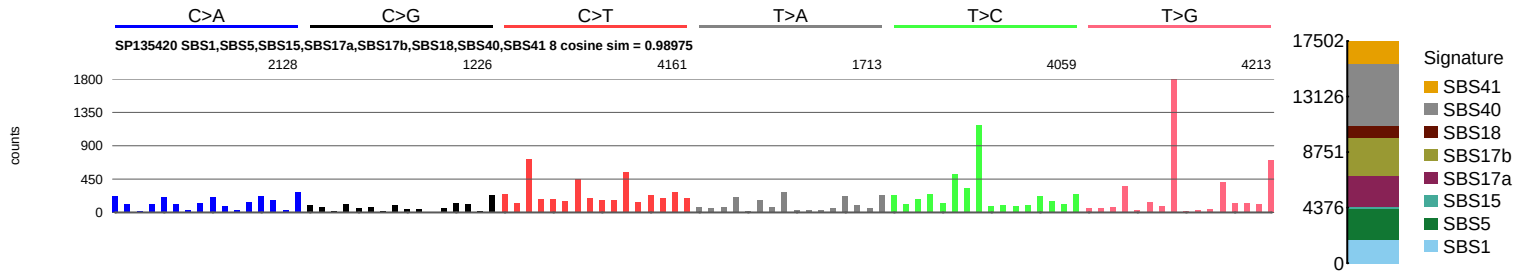
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98975
2133
1236
4148
1725
4053
4206
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98975
2128
1226
4161
1713
4059
4213
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
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SBS17a
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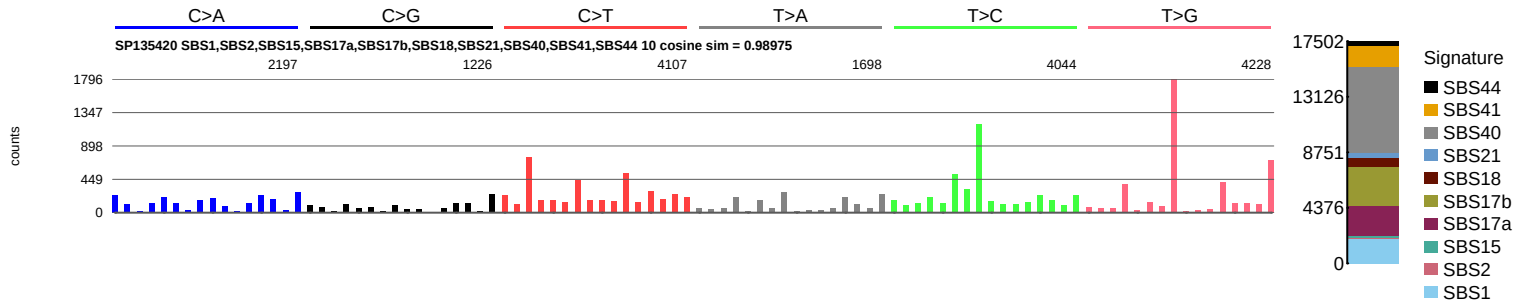
SBS1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SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98975
2197
1226
4107
1698
4044
4228
counts
1796
1347
898
449
0
17502
13126
8751
4376
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SBS18
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SBS17a
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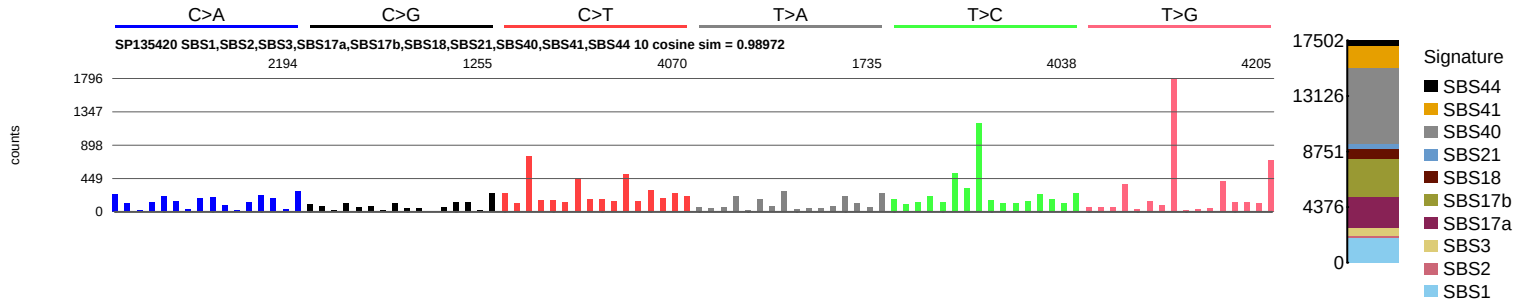
SBS2
SBS1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SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98972
2194
1255
4070
1735
4038
4205
counts
1796
1347
898
449
0
17502
13126
8751
4376
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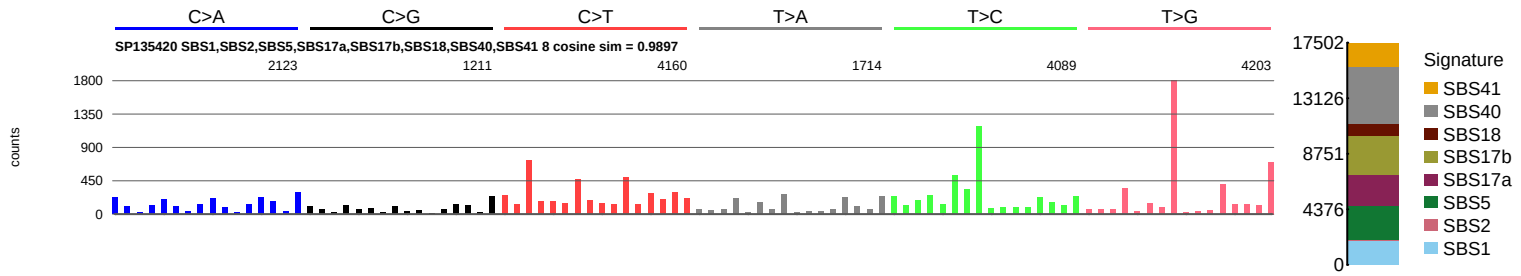
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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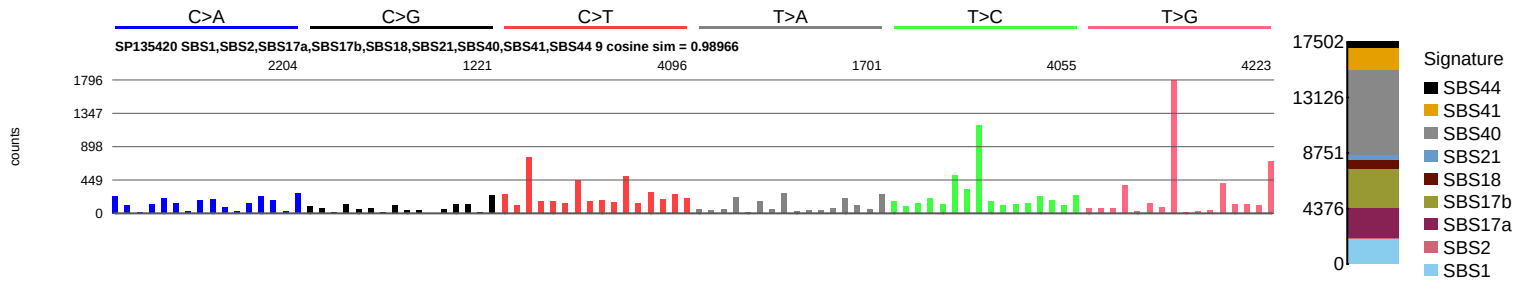
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9897
2123
1211
4160
1714
4089
4203
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98966
2204
1221
4096
1701
4055
4223
counts
1796
1347
898
449
0
17502
13126
8751
4376
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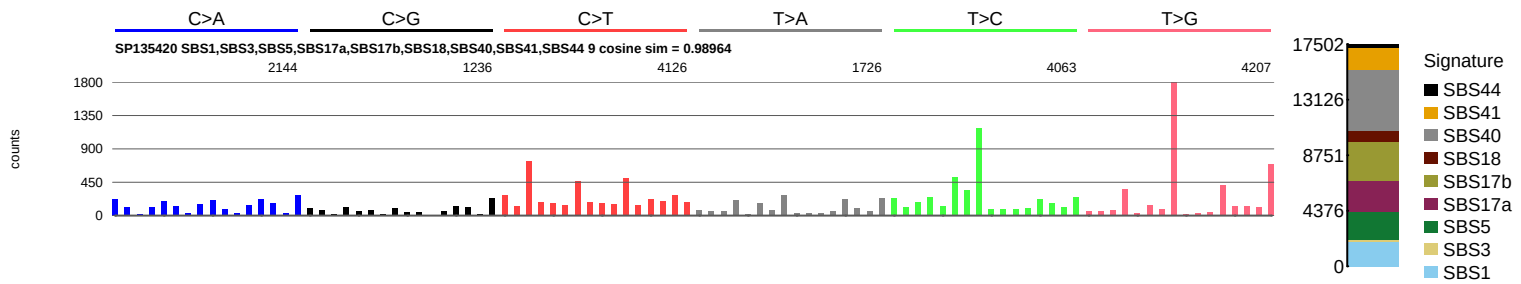
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98964
2144
1236
4126
1726
4063
4207
counts
1800
1350
900
450
0
17502
13126
8751
4376
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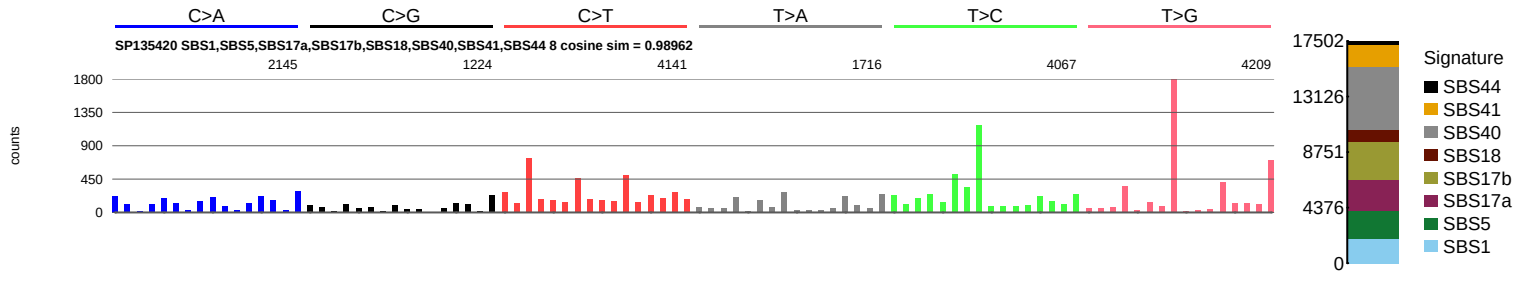
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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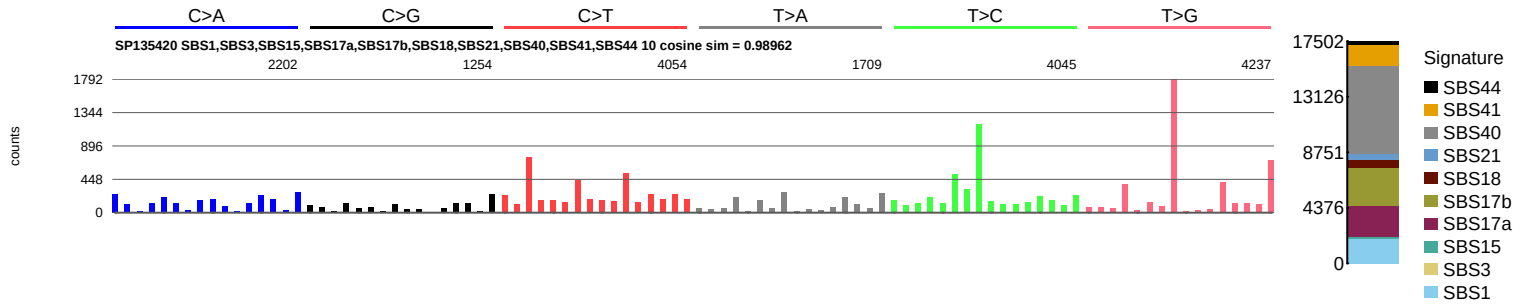
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98962
2145
1224
4141
1716
4067
4209
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
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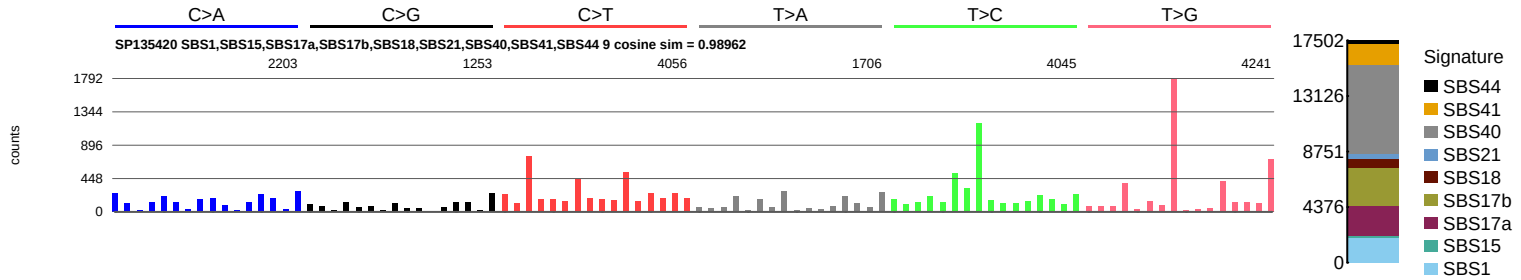
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98962
2202
1254
4054
1709
4045
4237
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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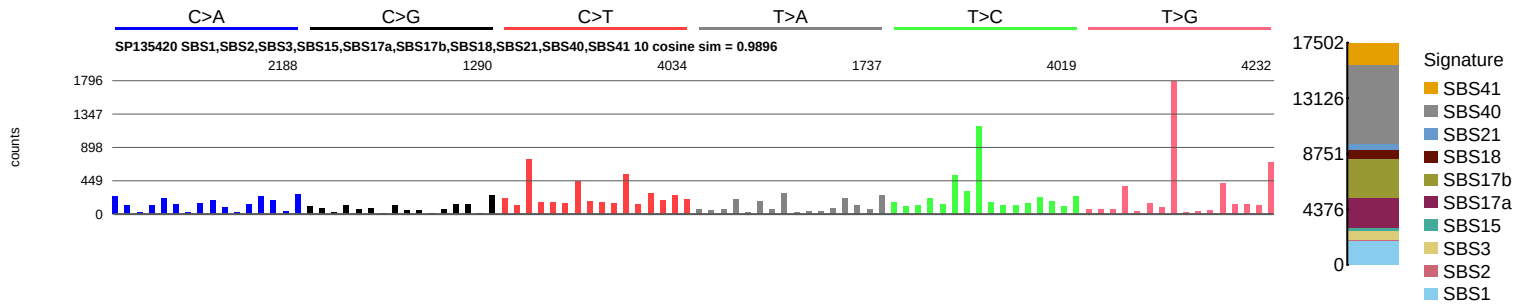
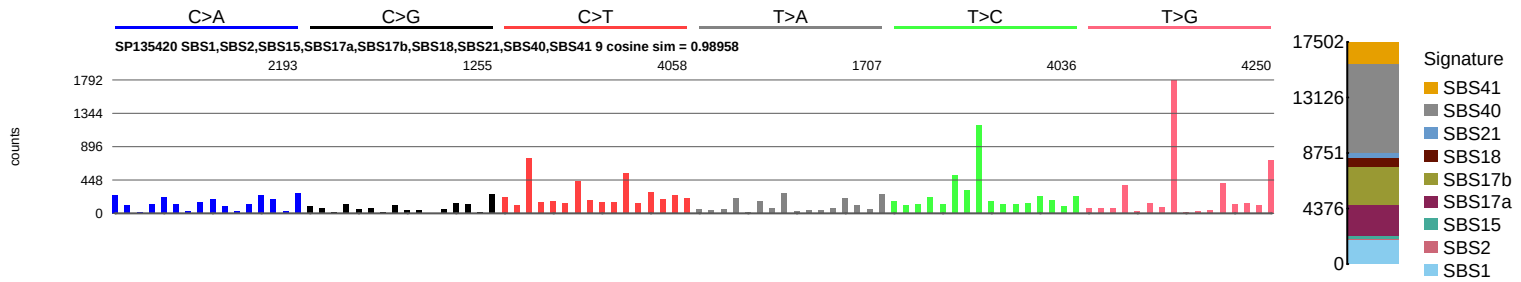
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98962
2203
1253
4056
1706
4045
4241
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9896
2188
1290
4034
1737
4019
4232
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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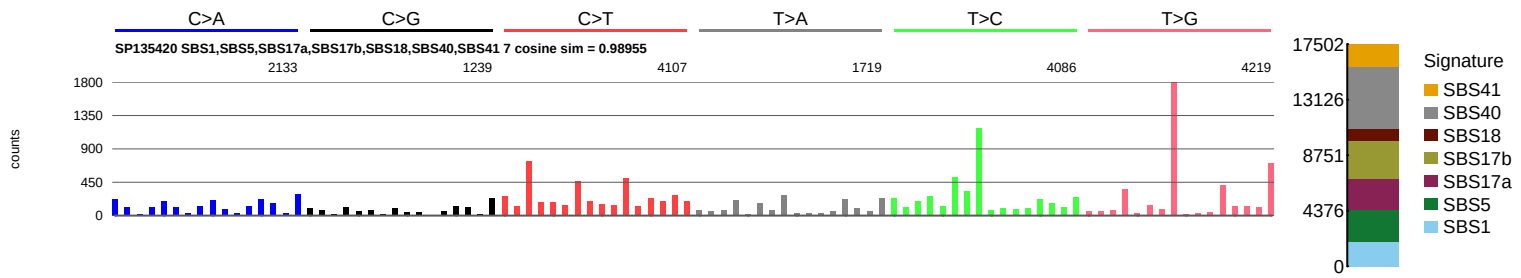
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98958
2193
1255
4058
1707
4036
4250
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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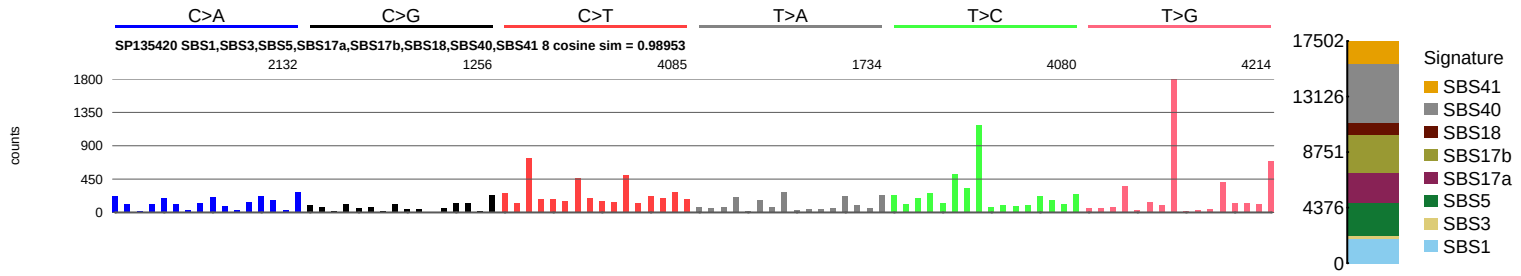
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98955
2133
1239
4107
1719
4086
4219
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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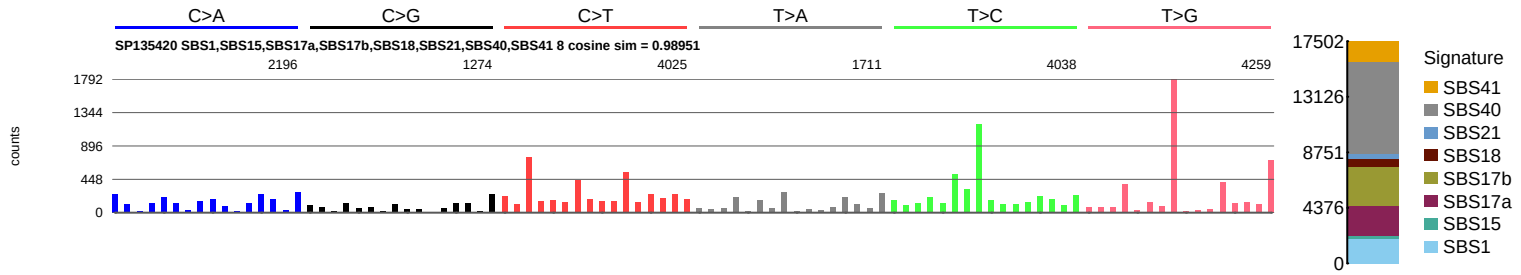
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98953
2132
1256
4085
1734
4080
4214
counts
1800
1350
900
450
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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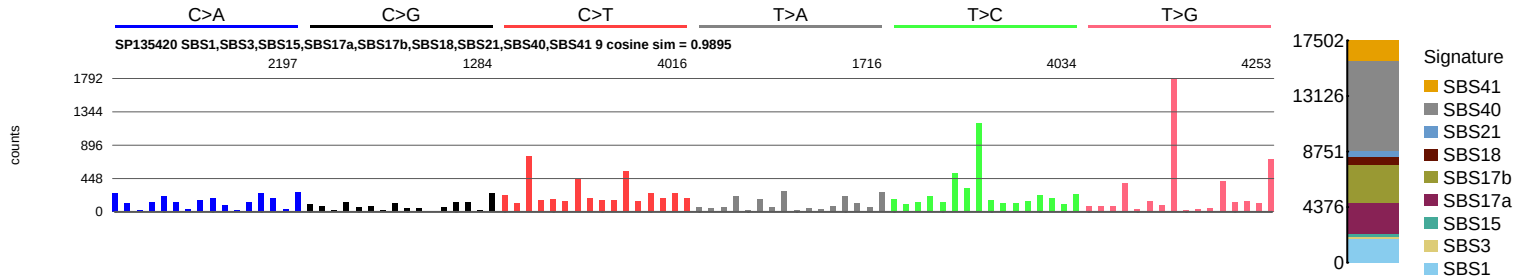
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98951
2196
1274
4025
1711
4038
4259
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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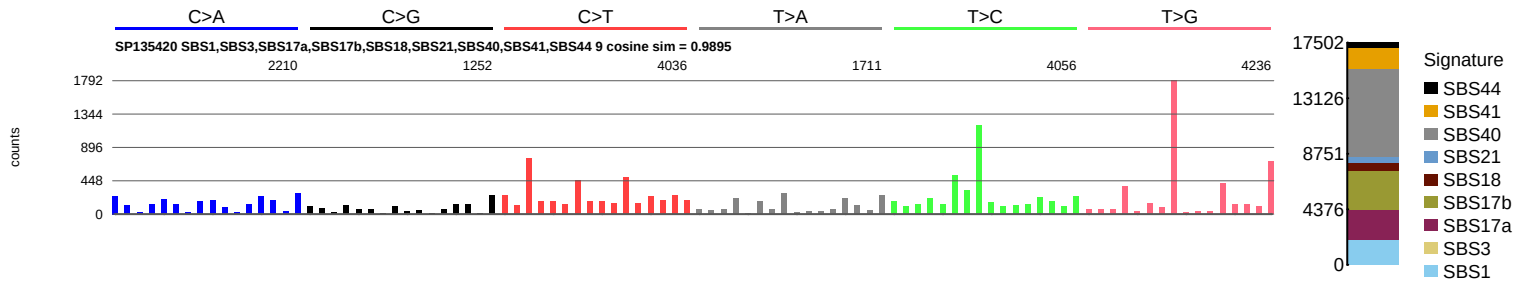
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9895
2197
1284
4016
1716
4034
4253
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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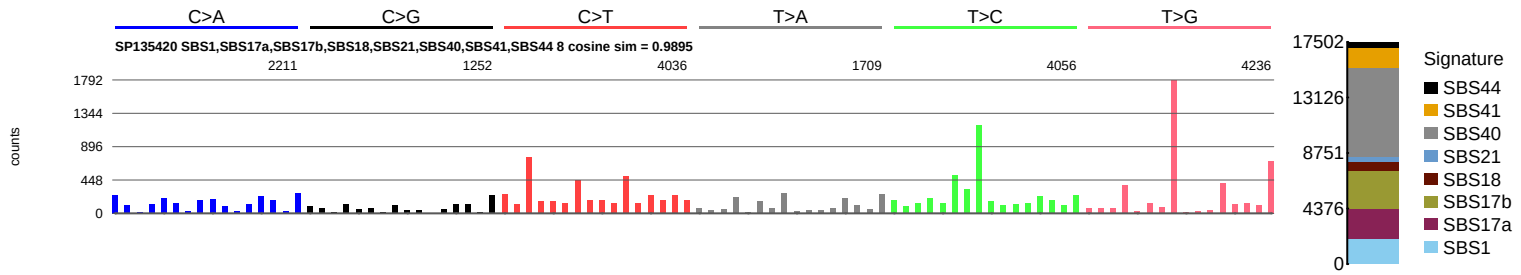
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9895
2210
1252
4036
1711
4056
4236
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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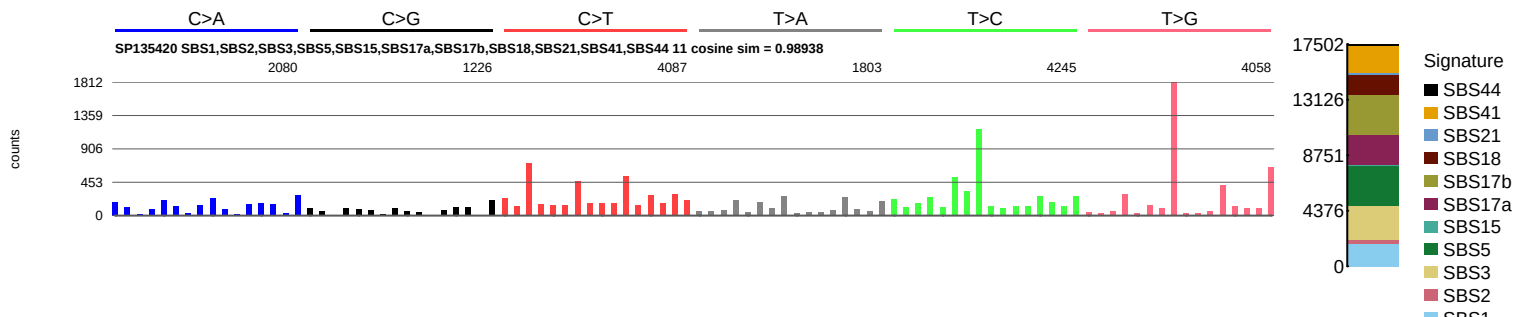
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9895
2211
1252
4036
1709
4056
4236
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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C>T
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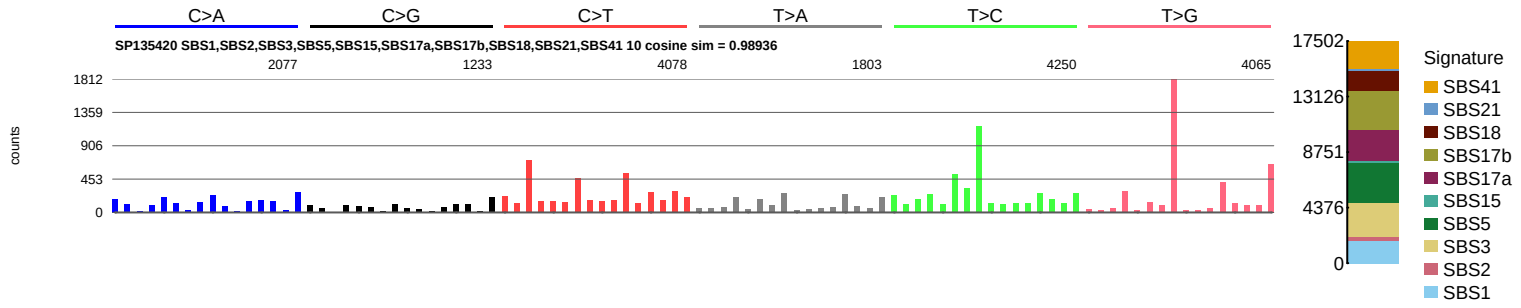
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98938
2080
1226
4087
1803
4245
4058
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98936
2077
1233
4078
1803
4250
4065
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
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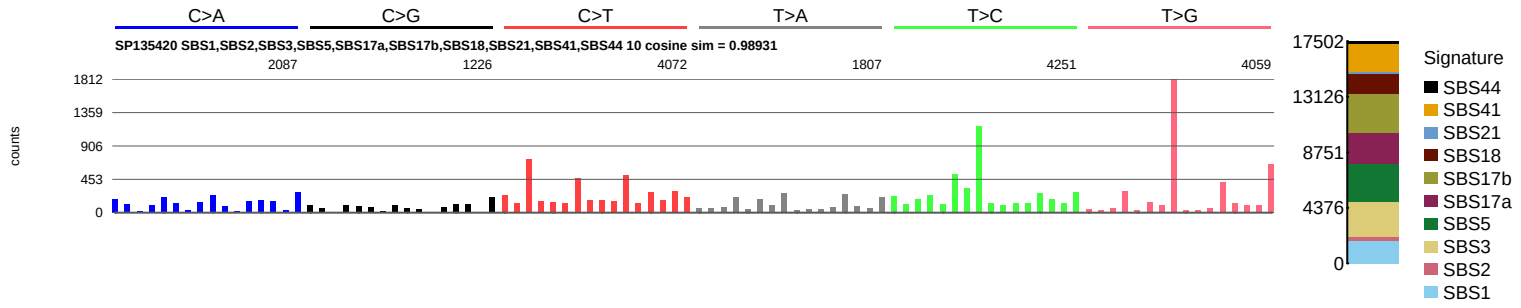
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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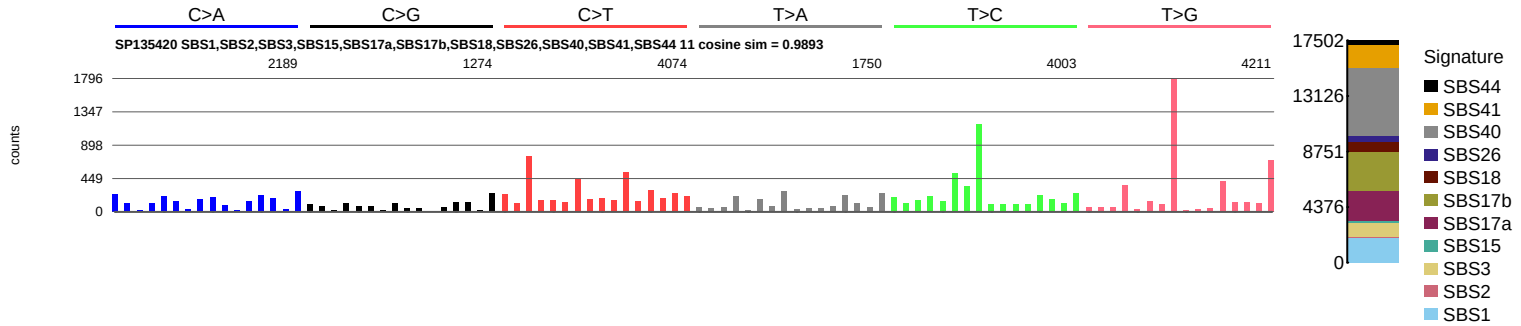
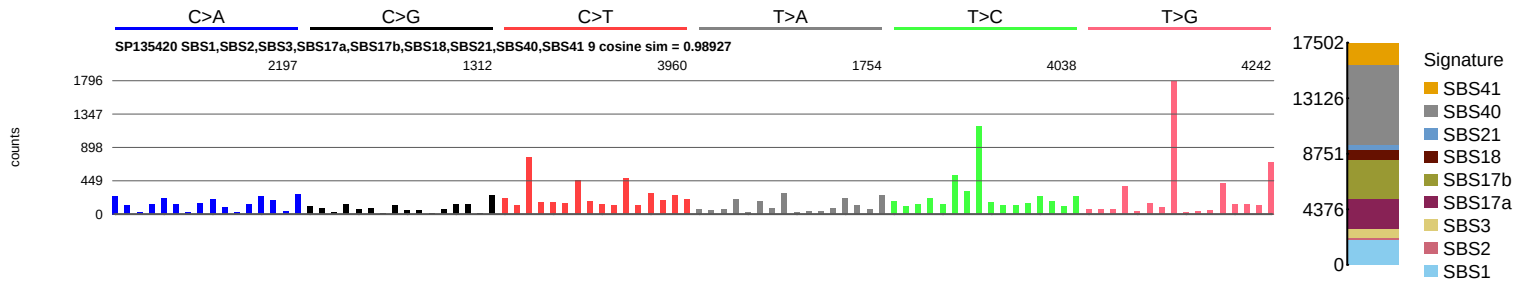
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98931
2087
1226
4072
1807
4251
4059
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9893
2189
1274
4074
1750
4003
4211
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98927
2197
1312
3960
1754
4038
4242
counts
1796
1347
898
449
0
17502
13126
8751
4376
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SBS41
SBS40
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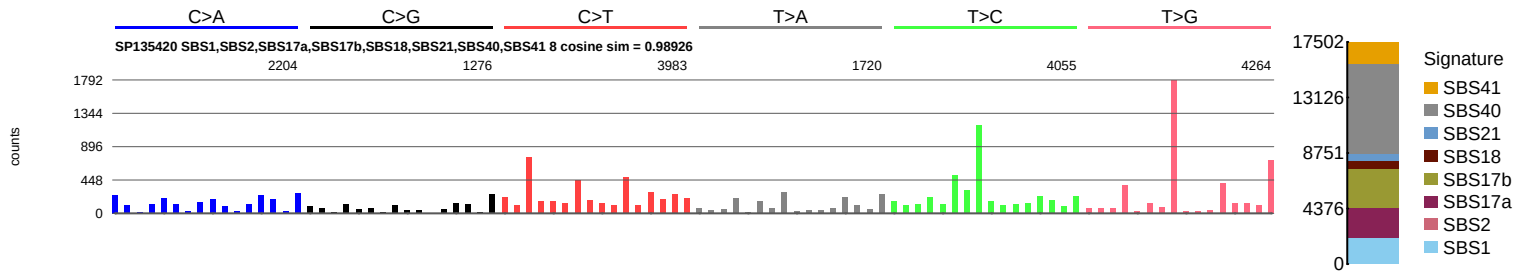
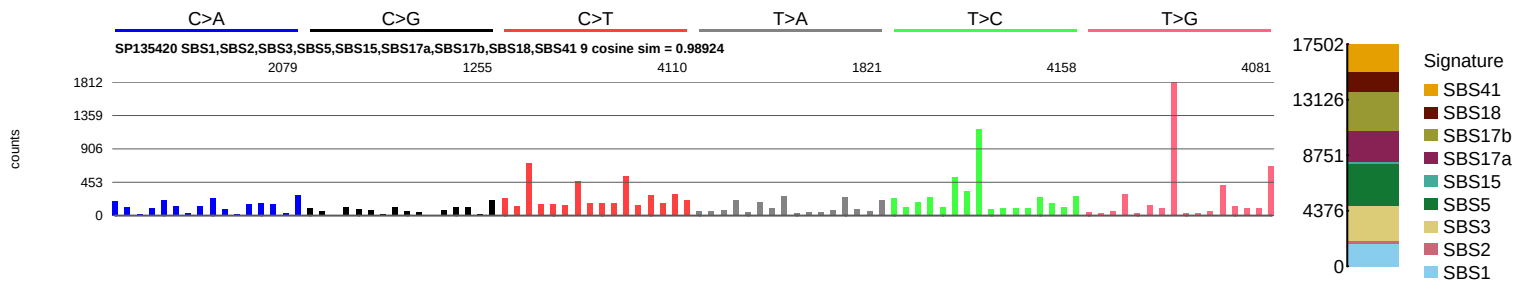
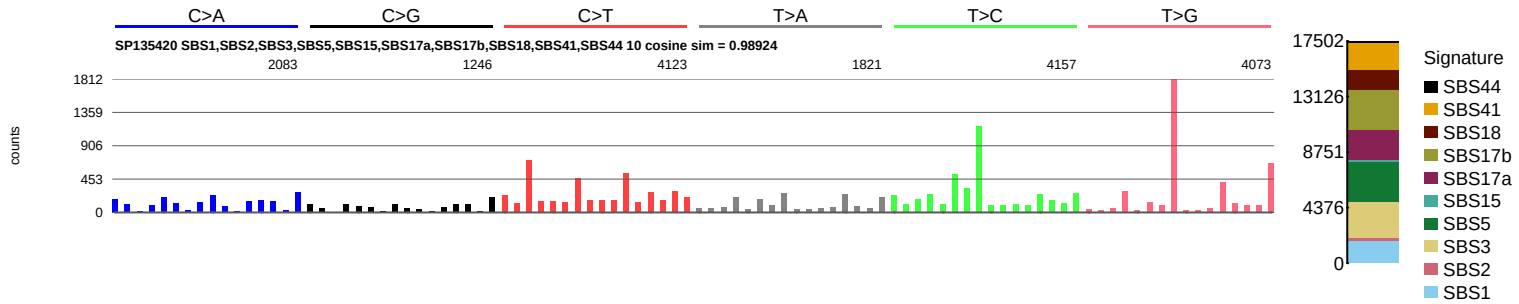
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98926
2204
1276
3983
1720
4055
4264
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98924
2079
1255
4110
1821
4158
4081
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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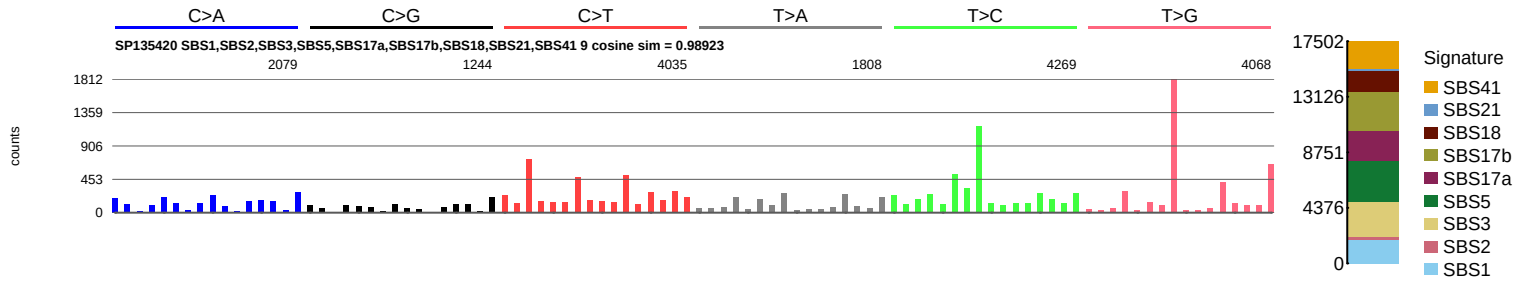
2083
1246
4123
1821
4157
4073
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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2079
1244
4035
1808
4269
4068
counts
1812
1359
906
453
0
17502
13126
8751
4376
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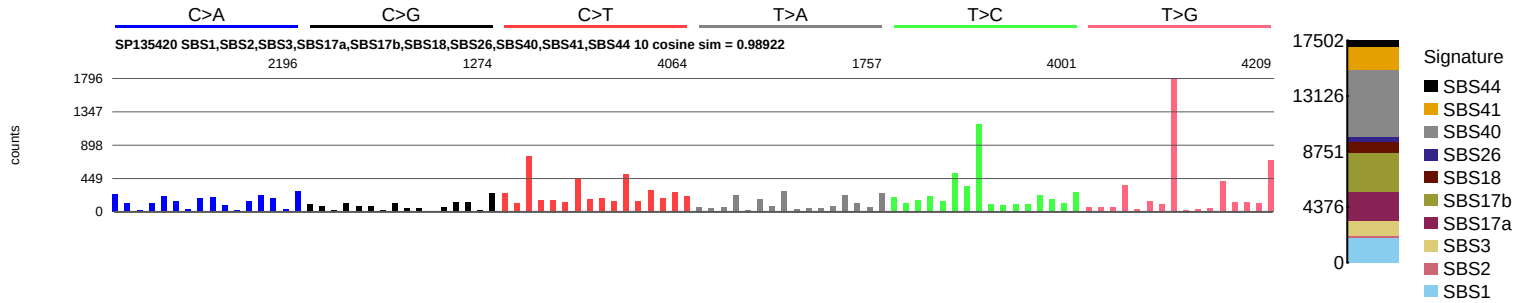
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98922
2196
1274
4064
1757
4001
4209
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
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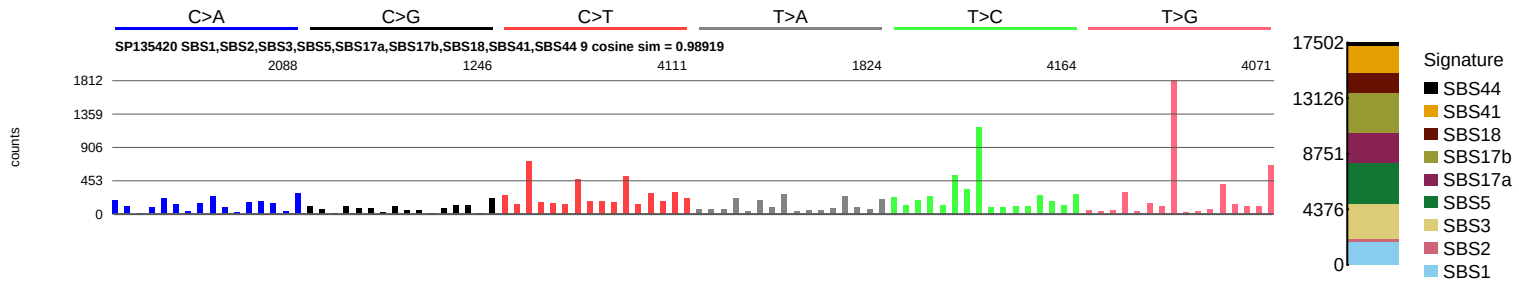
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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T>C
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SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98919
2088
1246
4111
1824
4164
4071
counts
1812
1359
906
453
0
17502
13126
8751
4376
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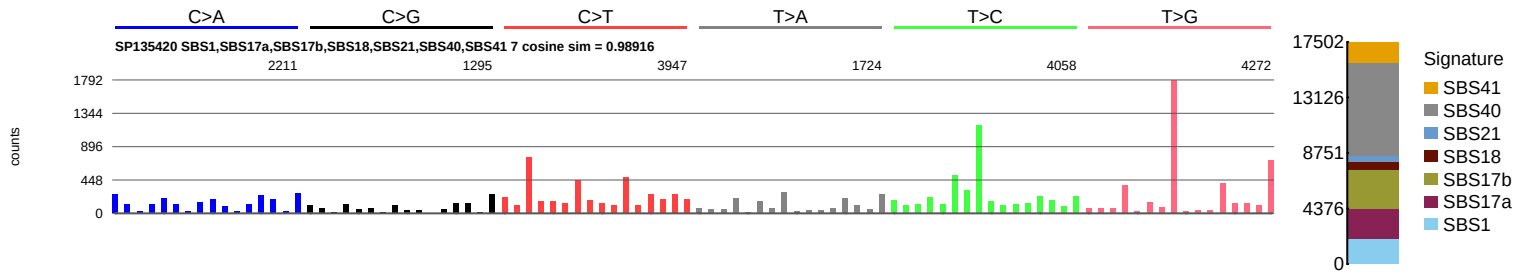
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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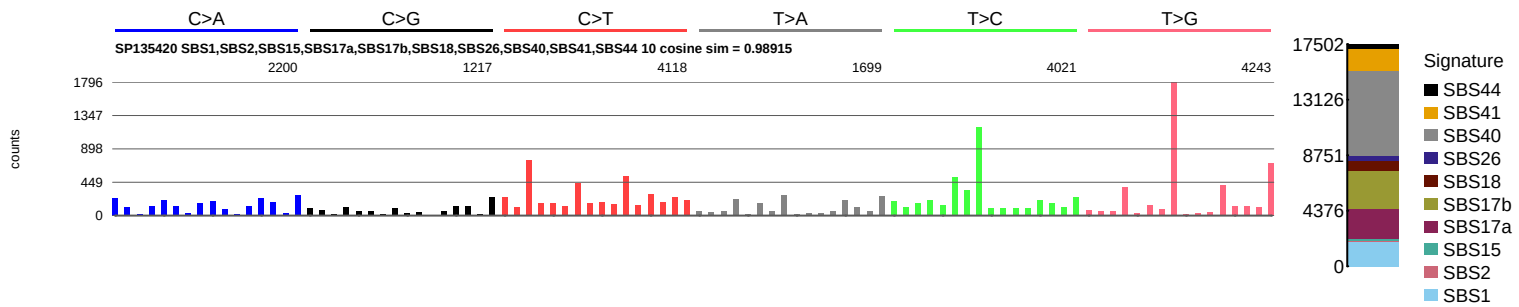
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98916
2211
1295
3947
1724
4058
4272
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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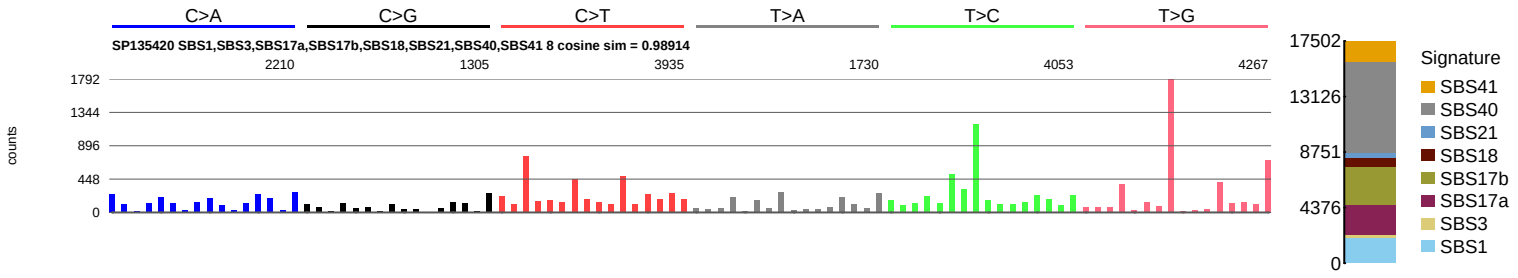
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98915
2200
1217
4118
1699
4021
4243
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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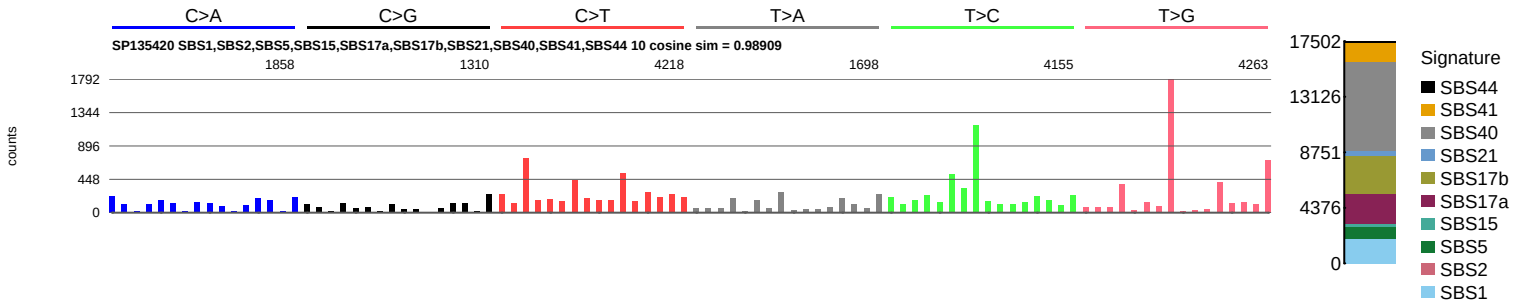
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98914
2210
1305
3935
1730
4053
4267
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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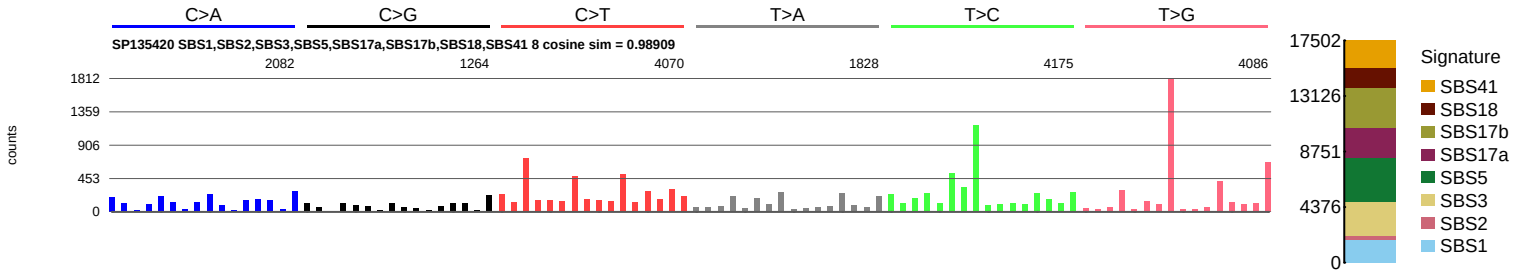
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98909
1858
1310
4218
1698
4155
4263
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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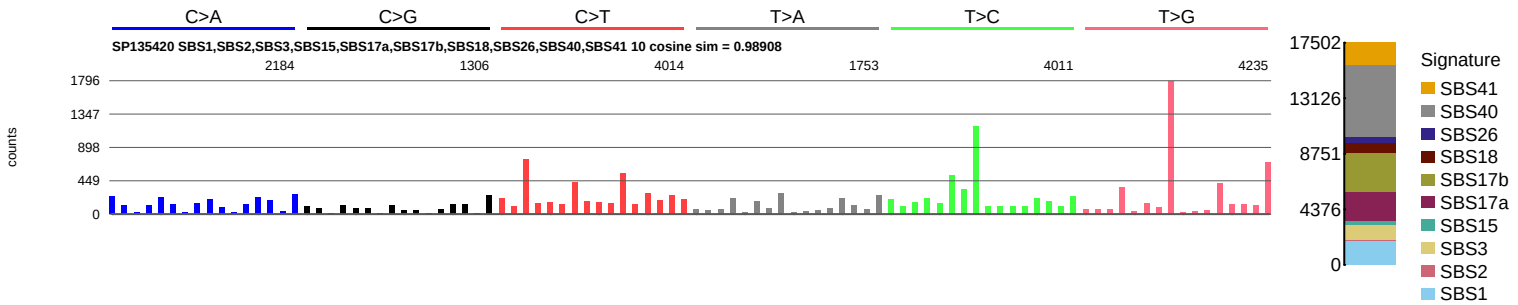
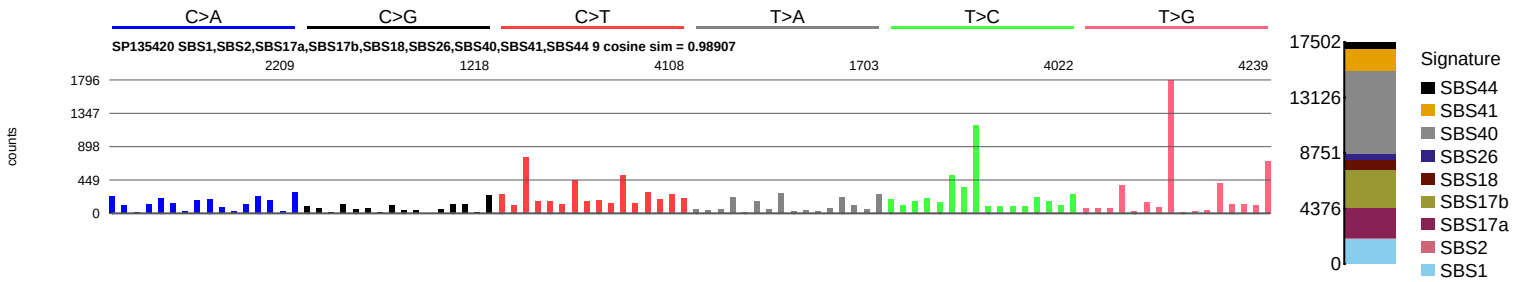
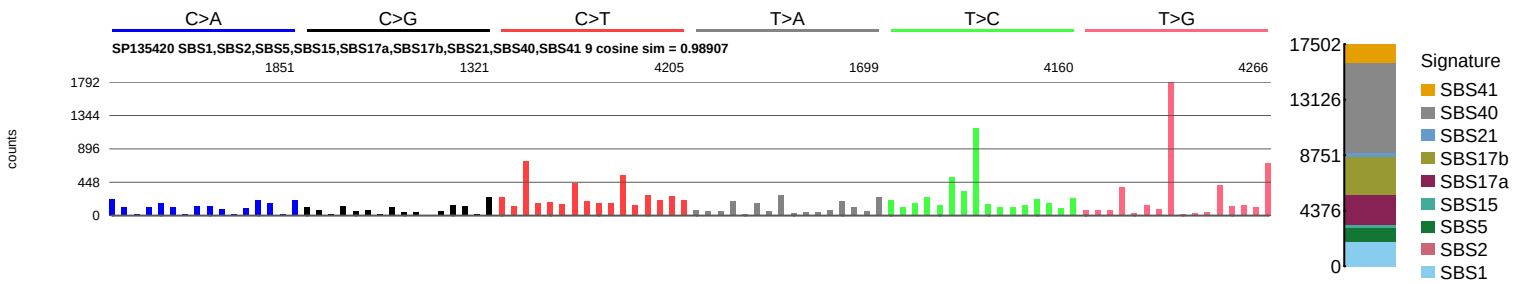
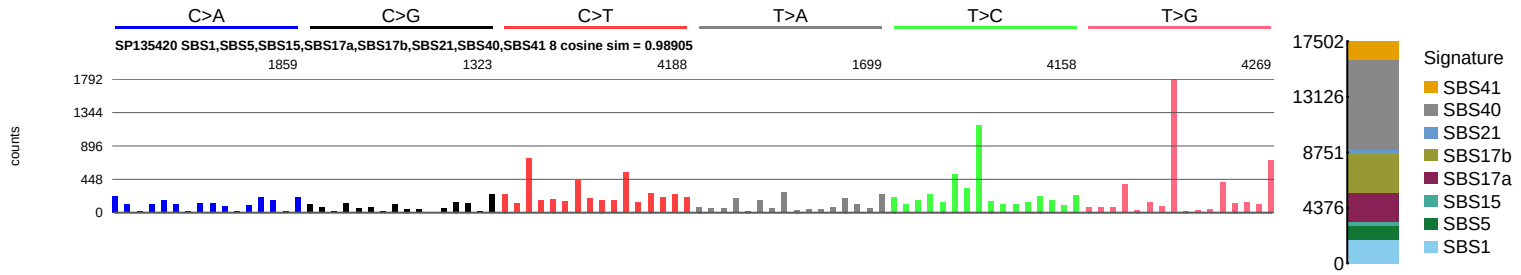
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98909
2082
1264
4070
1828
4175
4086
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98908
2184
1306
4014
1753
4011
4235
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98907
2209
1218
4108
1703
4022
4239
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98907
1851
1321
4205
1699
4160
4266
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98906
1866
1319
4196
1698
4155
4266
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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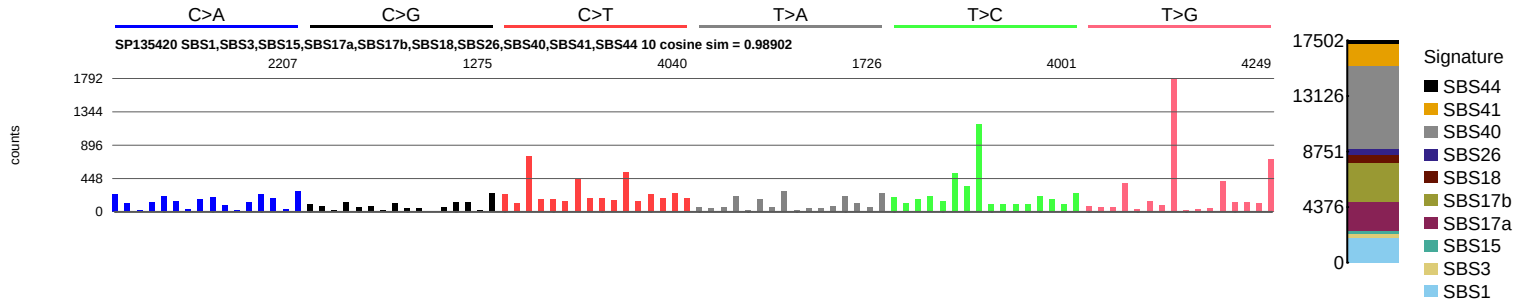
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98905
1859
1323
4188
1699
4158
4269
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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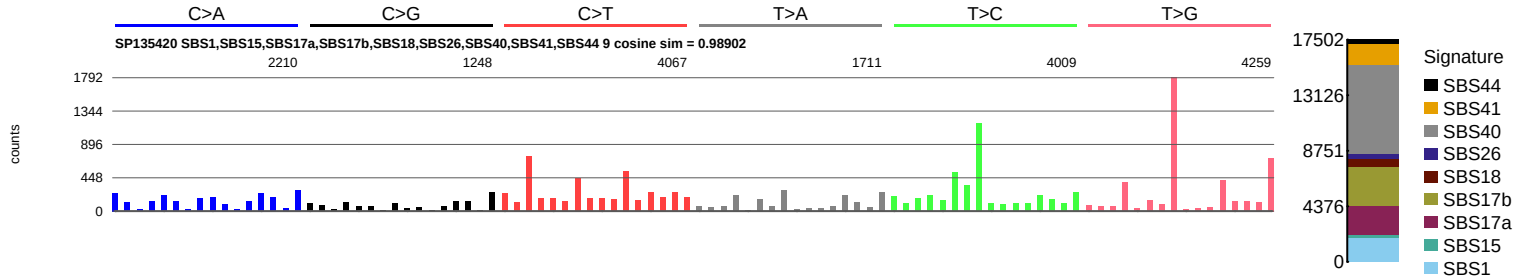
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98902
2207
1275
4040
1726
4001
4249
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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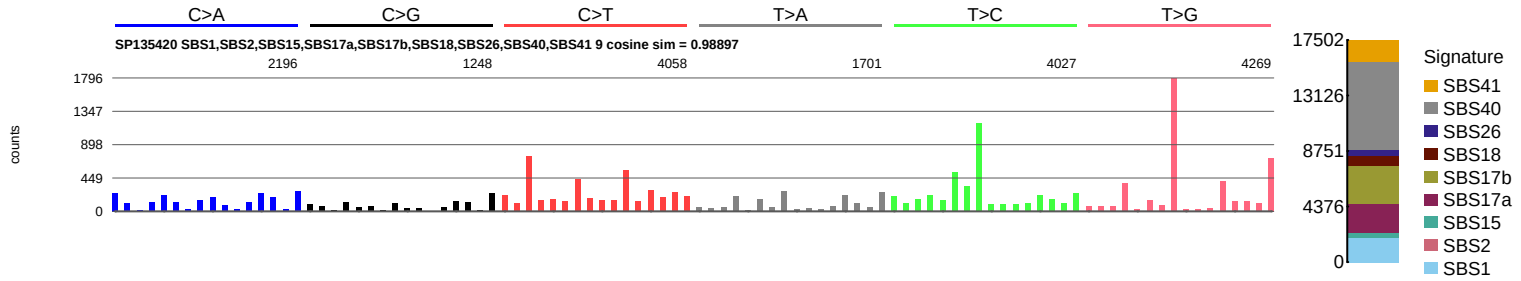
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98902
2210
1248
4067
1711
4009
4259
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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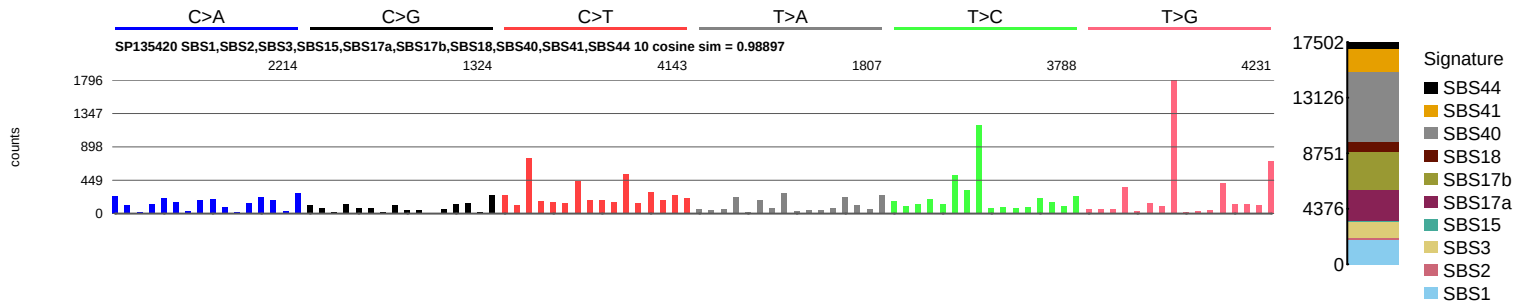
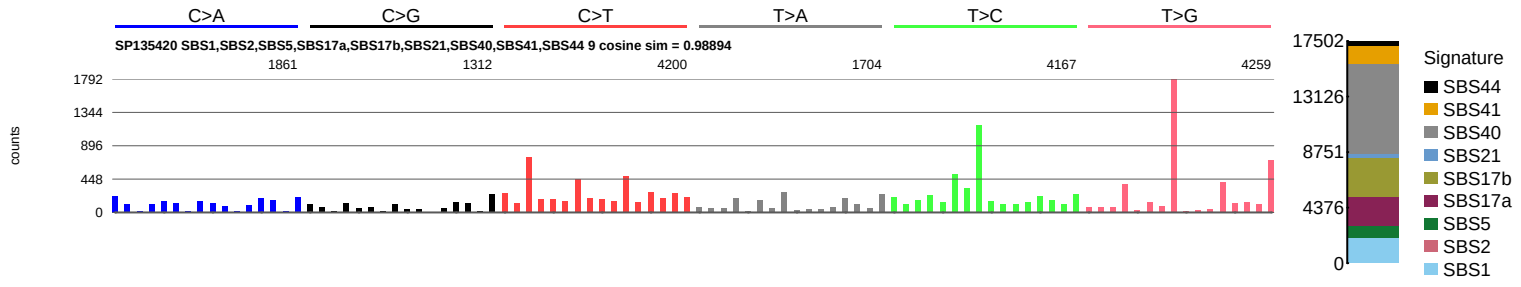
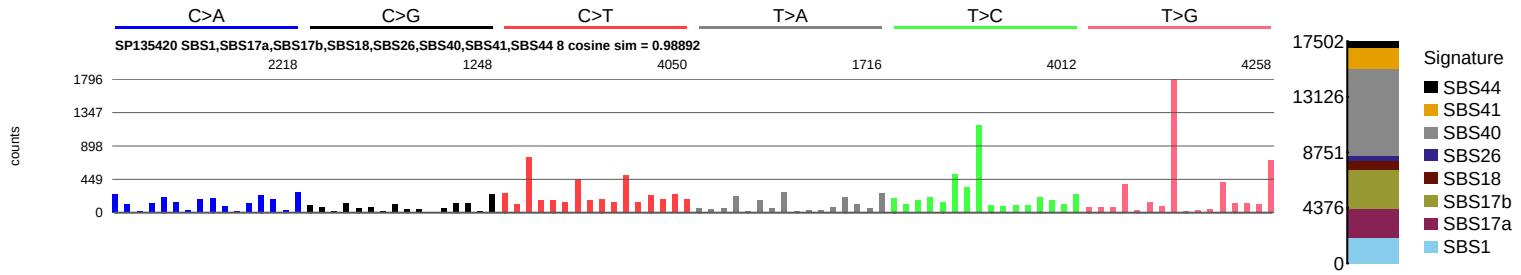
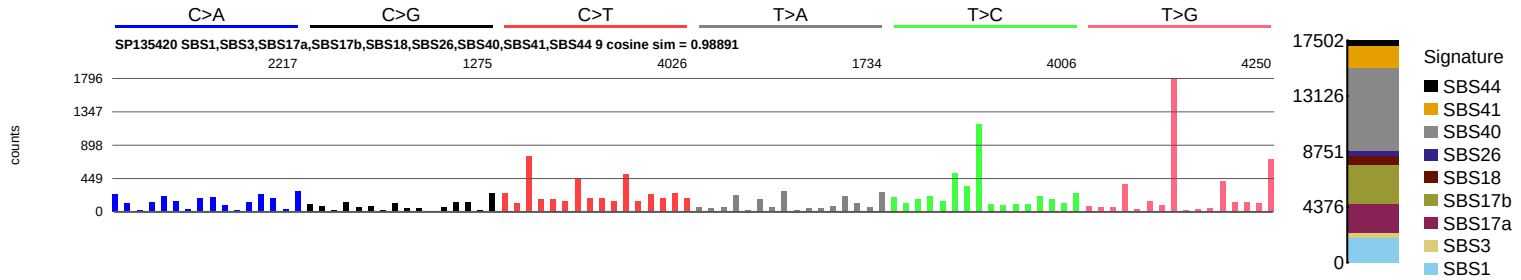
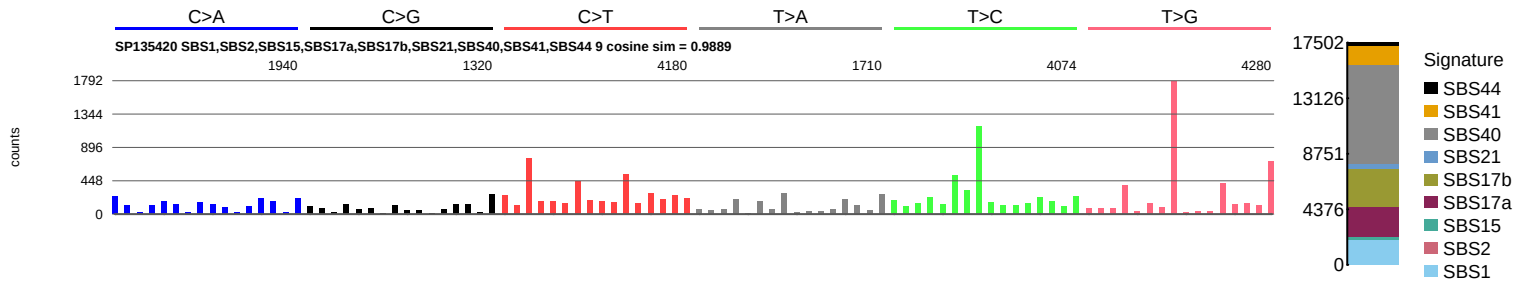
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98897
2196
1248
4058
1701
4027
4269
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

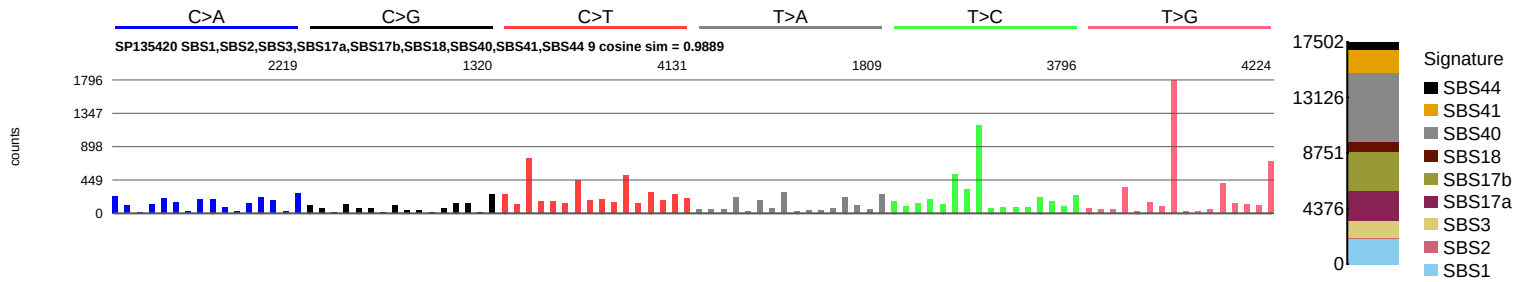

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98897
2214
1324
4143
1807
3788
4231
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

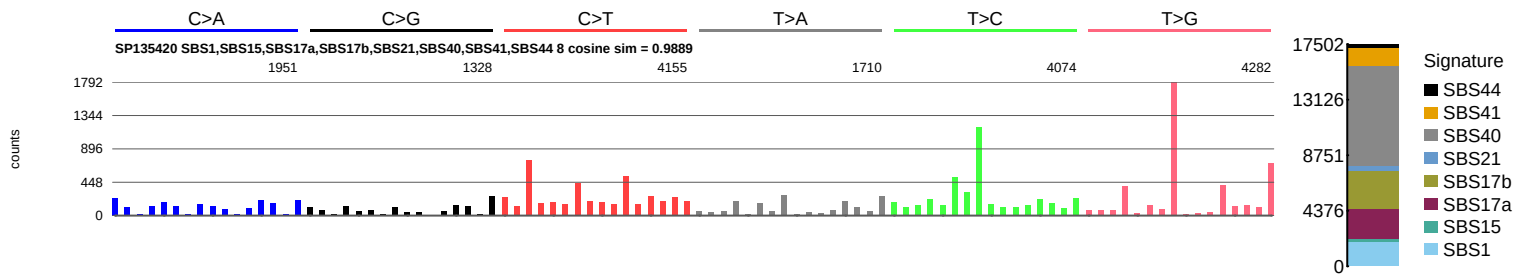

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98894
1861
1312
4200
1704
4167
4259
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

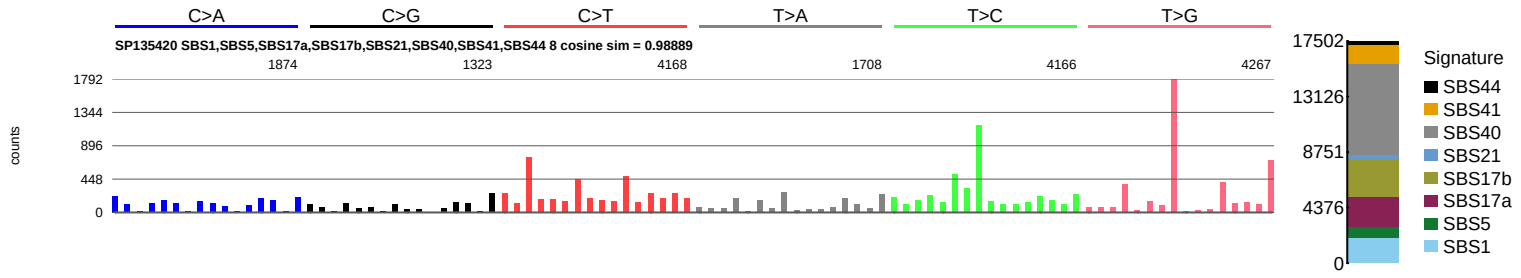

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98892
2218
1248
4050
1716
4012
4258
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

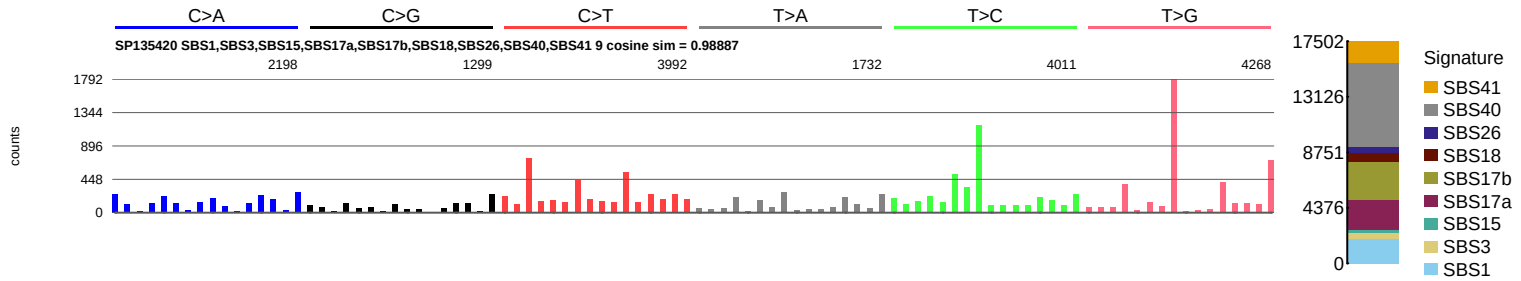

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98891
2217
1275
4026
1734
4006
4250
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9889
1940
1320
4180
1710
4074
4280
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9889
2219
1320
4131
1809
3796
4224
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9889
1951
1328
4155
1710
4074
4282
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98889
1874
1323
4168
1708
4166
4267
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98887
2198
1299
3992
1732
4011
4268
counts
1792
1344
896
448
0
17502
13126
8751
4376
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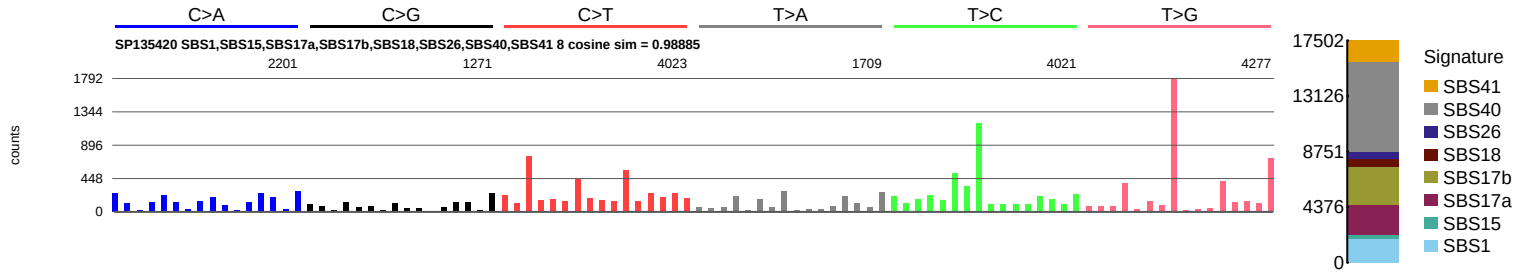
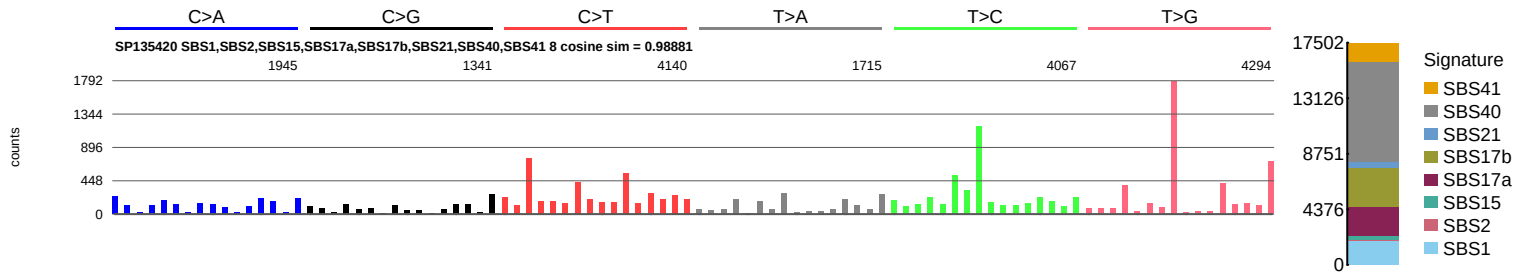
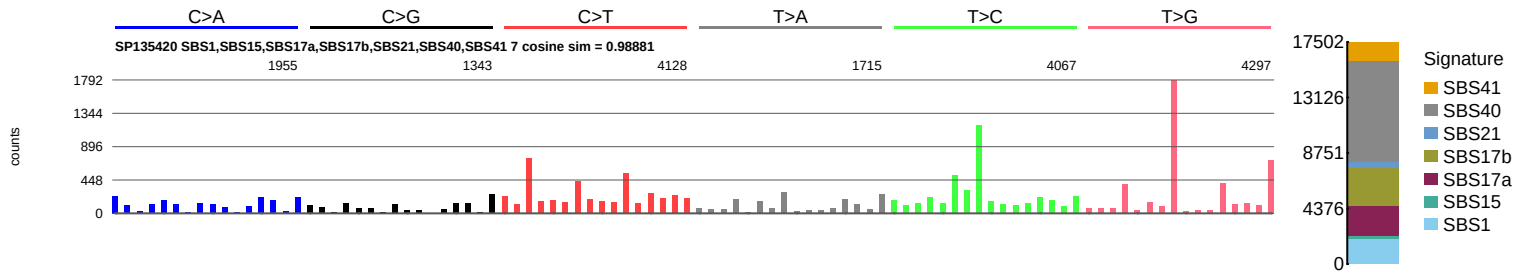
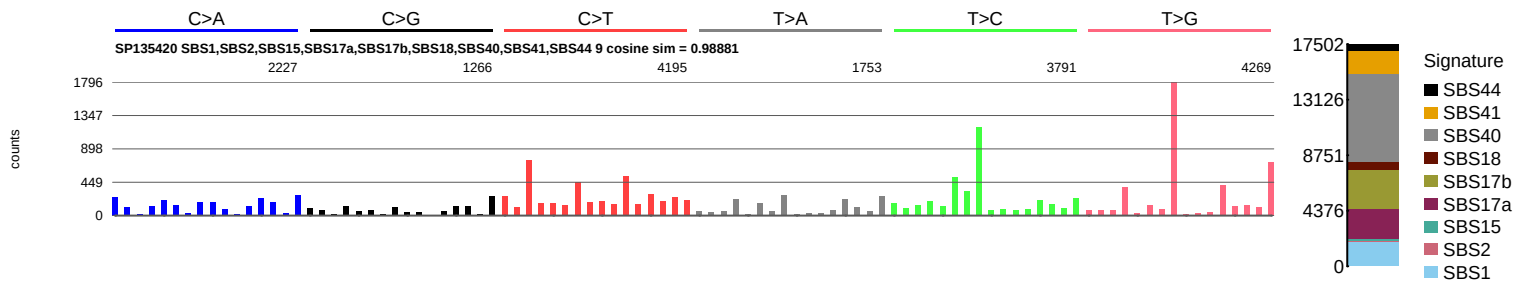
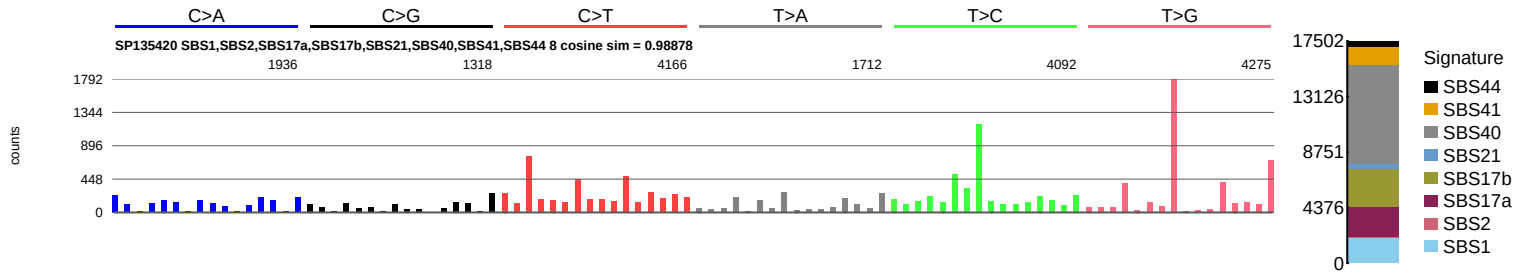
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

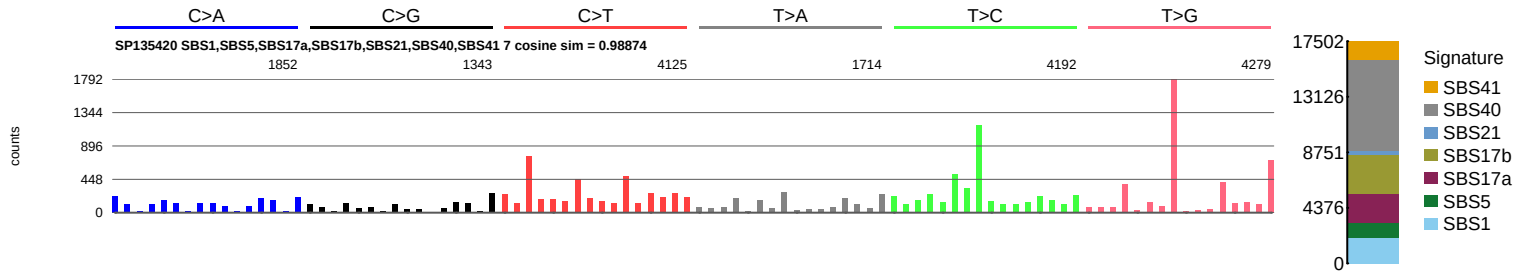

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98885
2201
1271
4023
1709
4021
4277
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

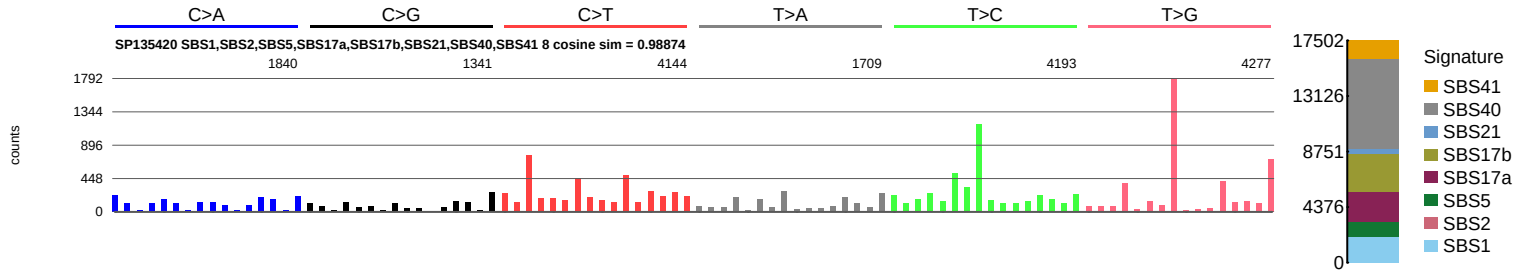

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98881
1945
1341
4140
1715
4067
4294
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

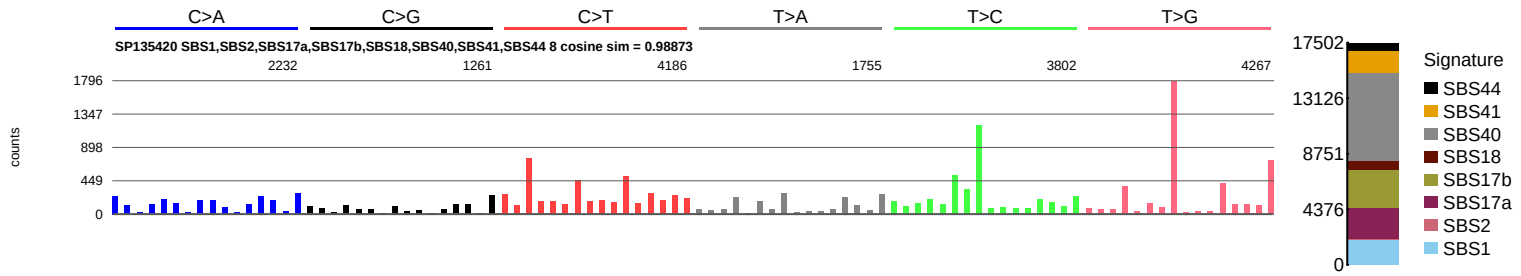

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98881
1955
1343
4128
1715
4067
4297
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

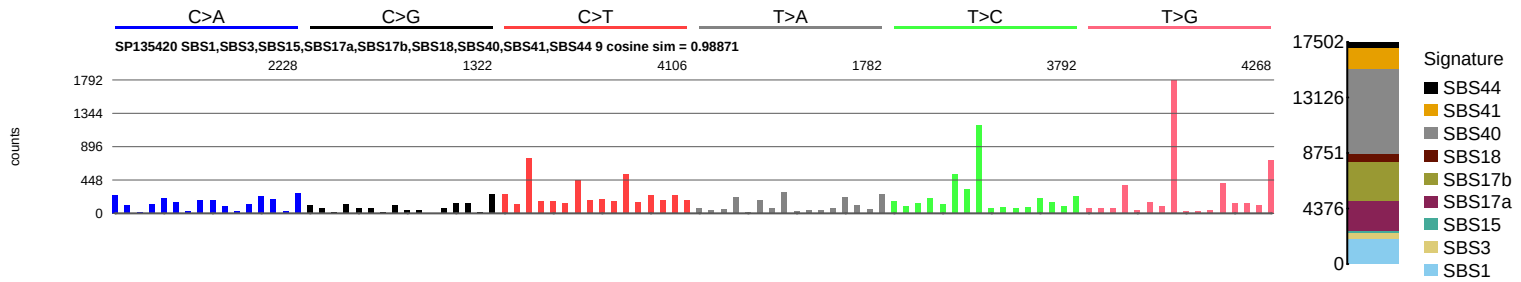

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98881
2227
1266
4195
1753
3791
4269
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

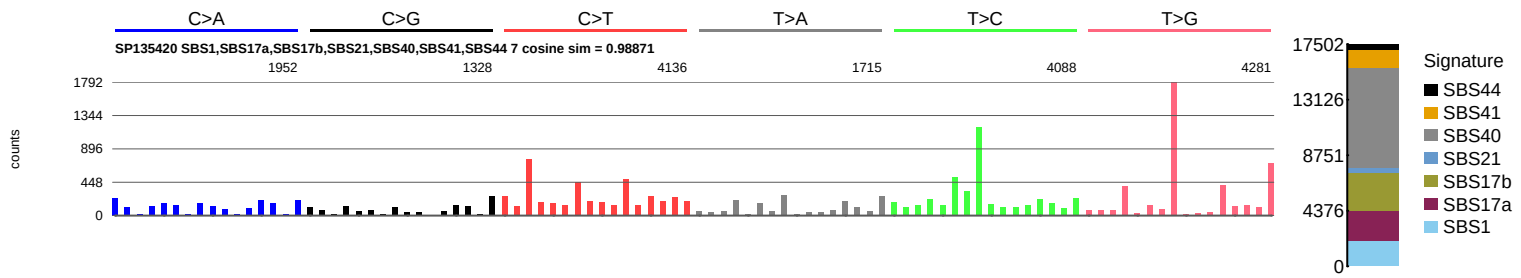

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98878
1936
1318
4166
1712
4092
4275
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

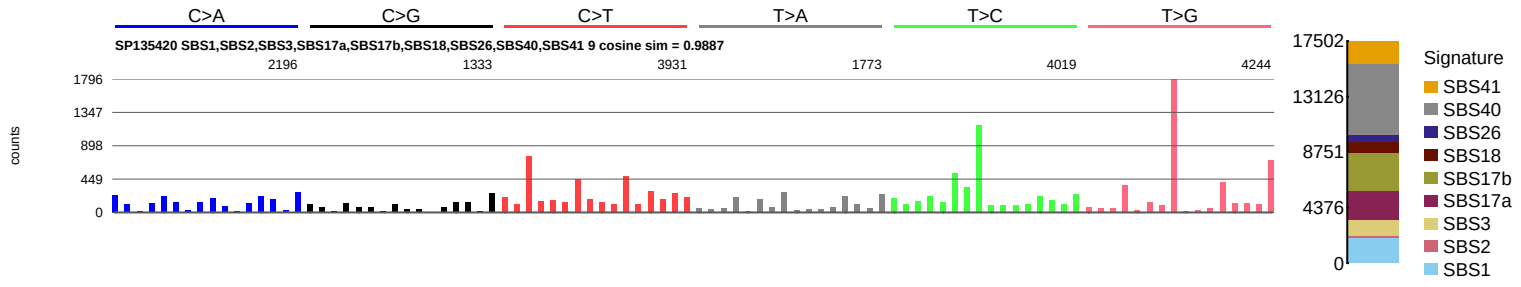

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98874
1852
1343
4125
1714
4192
4279
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

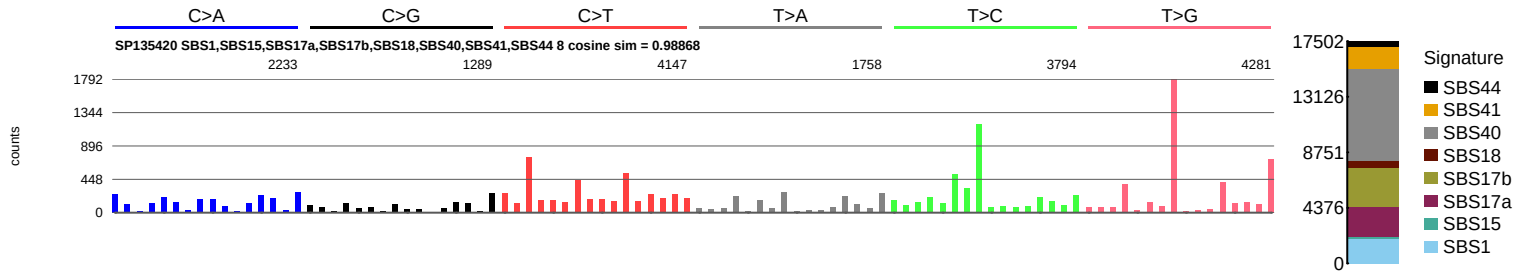

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98874
1840
1341
4144
1709
4193
4277
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

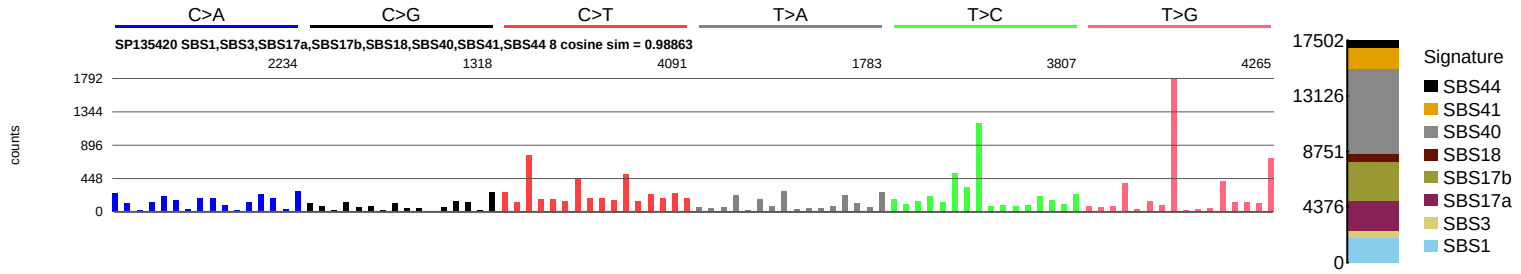

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98873
2232
1261
4186
1755
3802
4267
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

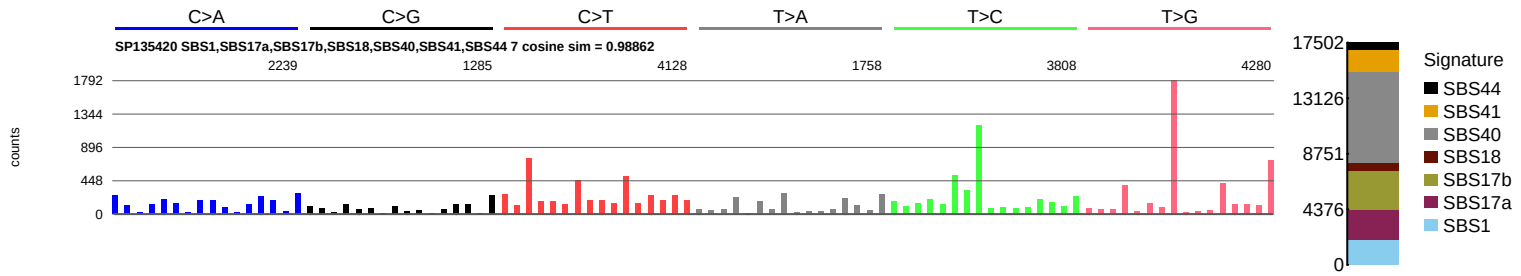

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98871
2228
1322
4106
1782
3792
4268
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

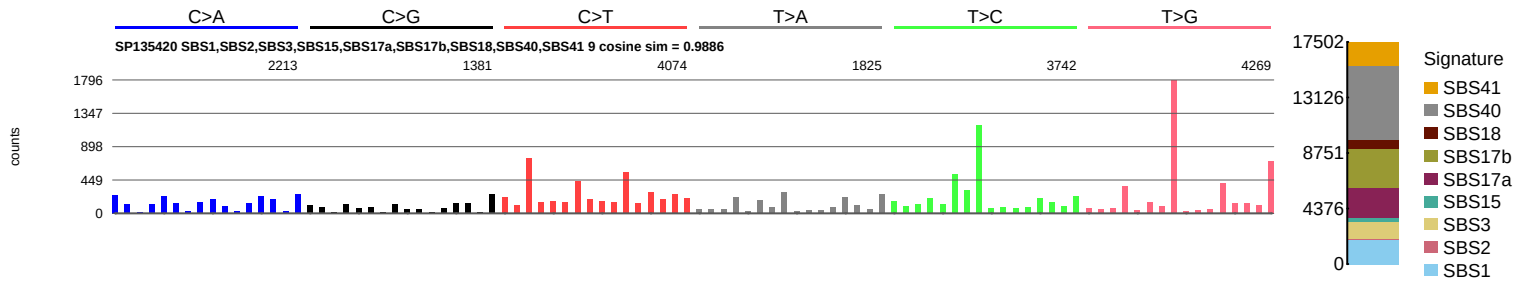

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98871
1952
1328
4136
1715
4088
4281
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

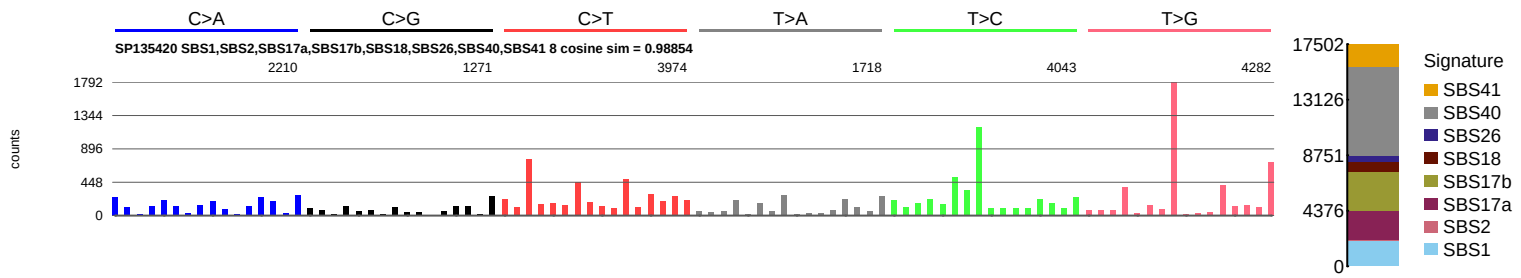

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9887
2196
1333
3931
1773
4019
4244
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

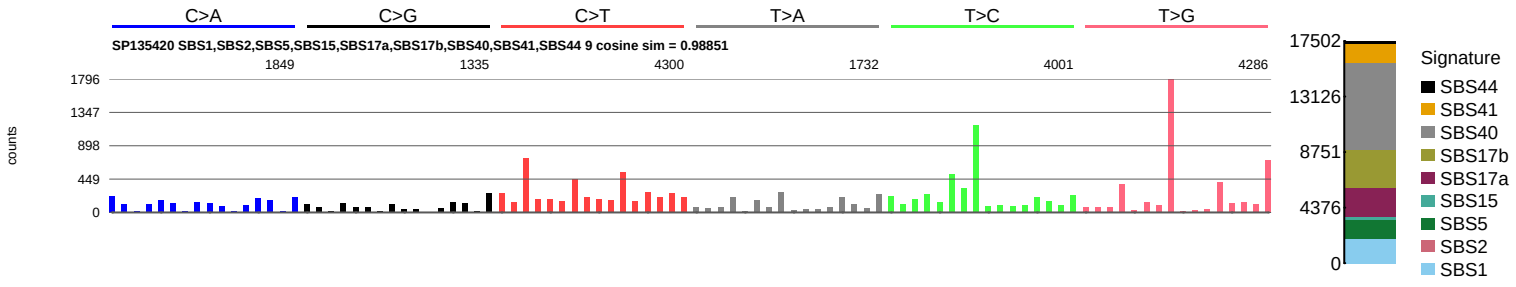

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98868
2233
1289
4147
1758
3794
4281
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

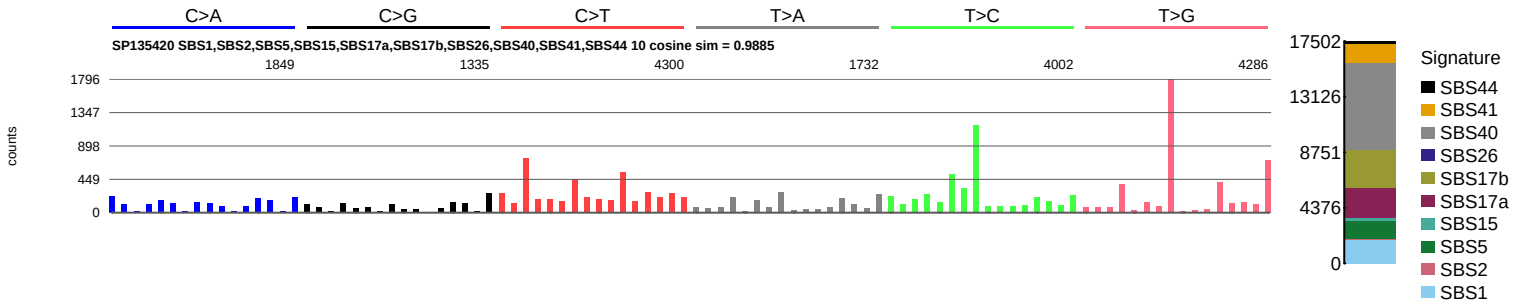

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98863
2234
1318
4091
1783
3807
4265
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1

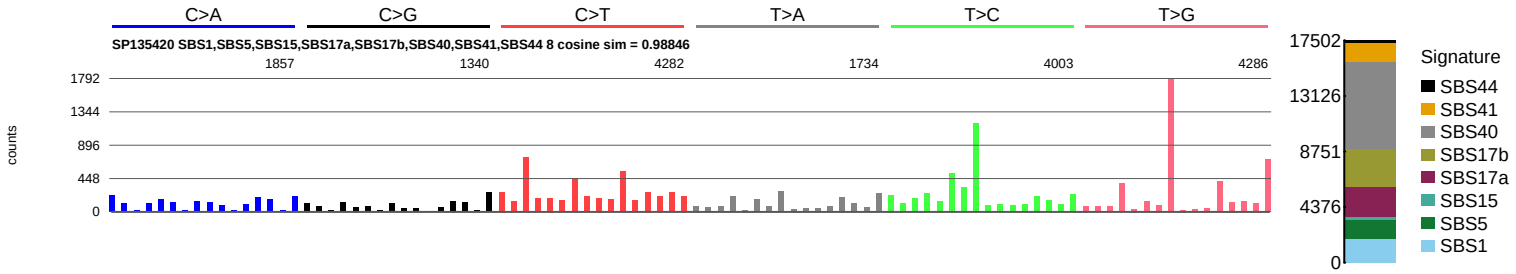

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98862
2239
1285
4128
1758
3808
4280
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1

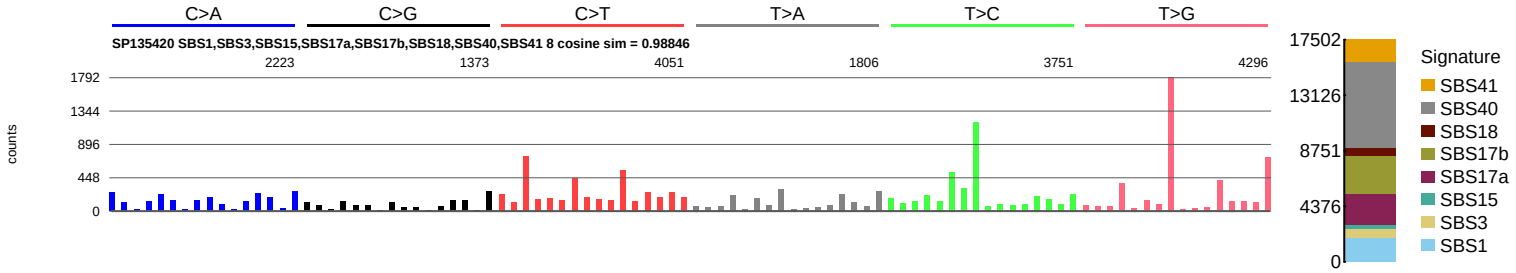

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9886
2213
1381
4074
1825
3742
4269
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

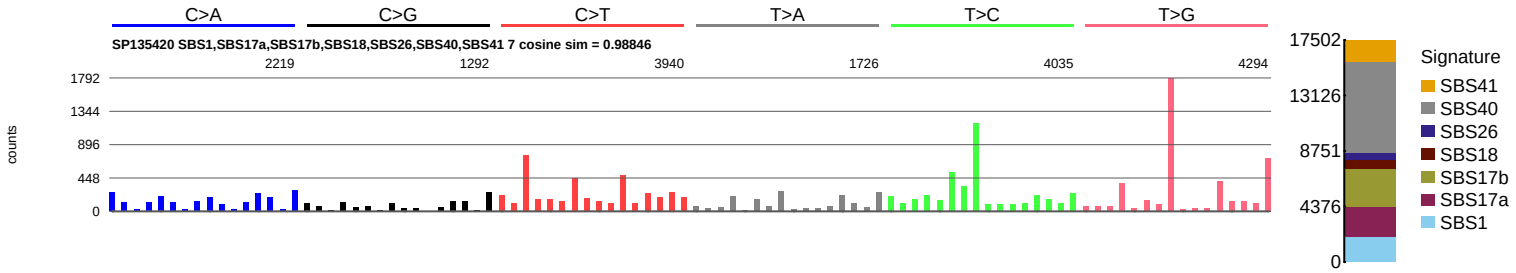

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98854
2210
1271
3974
1718
4043
4282
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

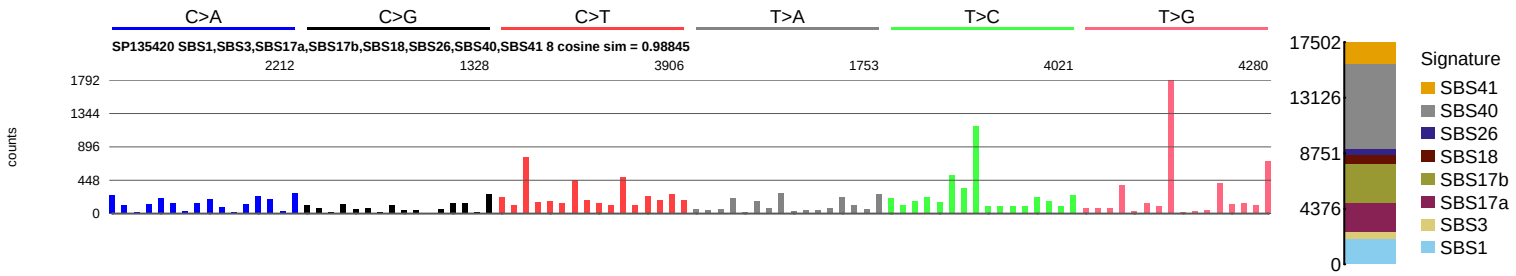

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98851
1849
1335
4300
1732
4001
4286
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

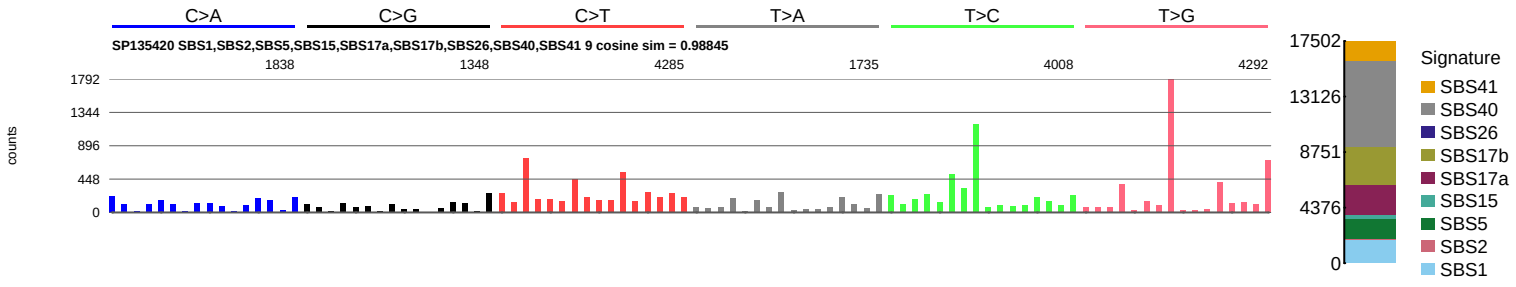

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9885
1849
1335
4300
1732
4002
4286
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

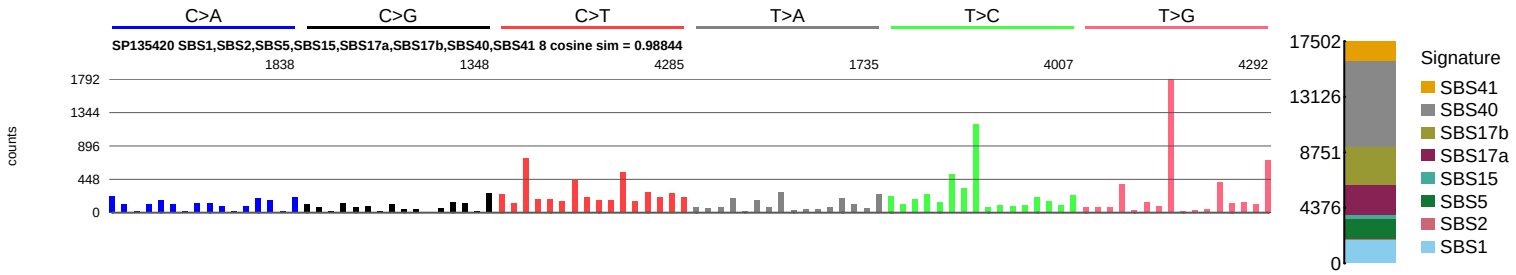

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98846
1857
1340
4282
1734
4003
4286
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98846
2223
1373
4051
1806
3751
4296
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98846
2219
1292
3940
1726
4035
4294
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98845
2212
1328
3906
1753
4021
4280
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98845
1838
1348
4285
1735
4008
4292
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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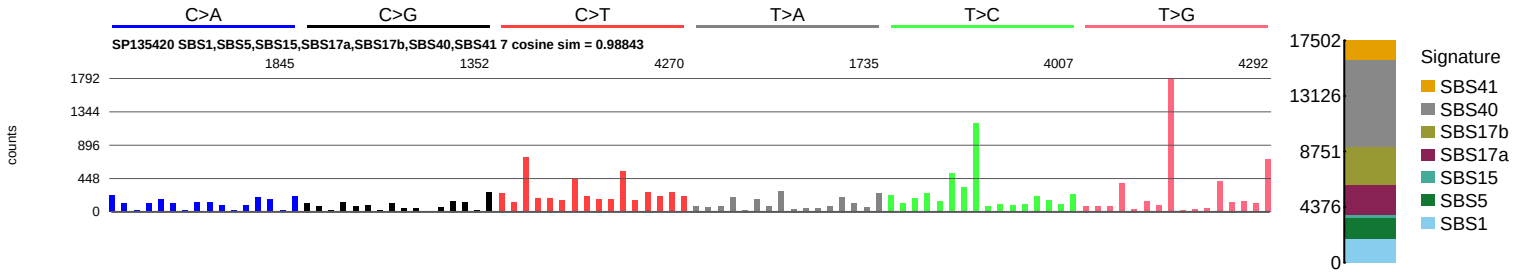
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98844
1838
1348
4285
1735
4007
4292
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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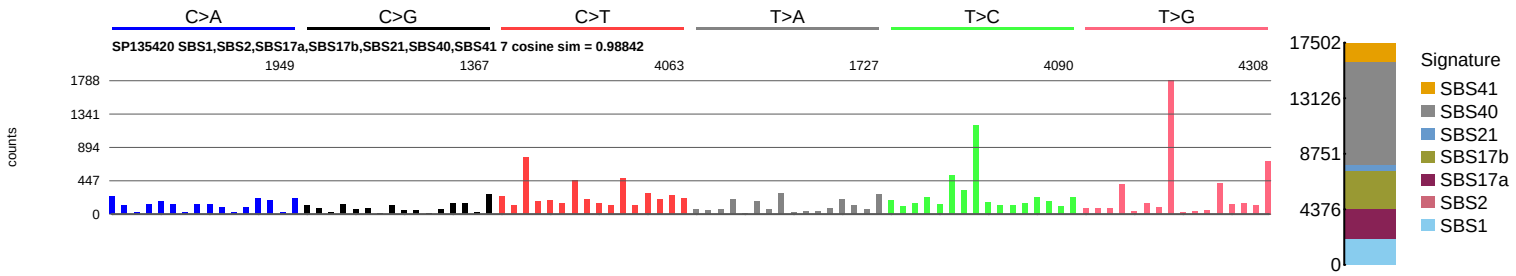
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98843
1845
1352
4270
1735
4007
4292
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1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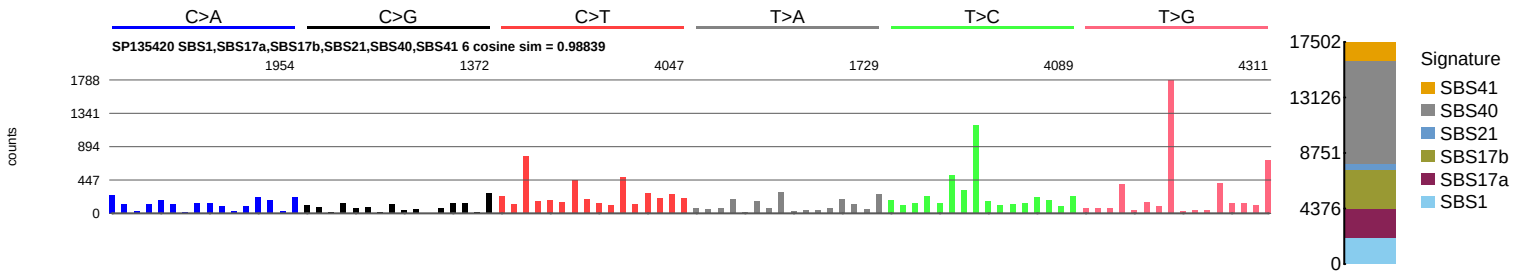
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98842
1949
1367
4063
1727
4090
4308
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1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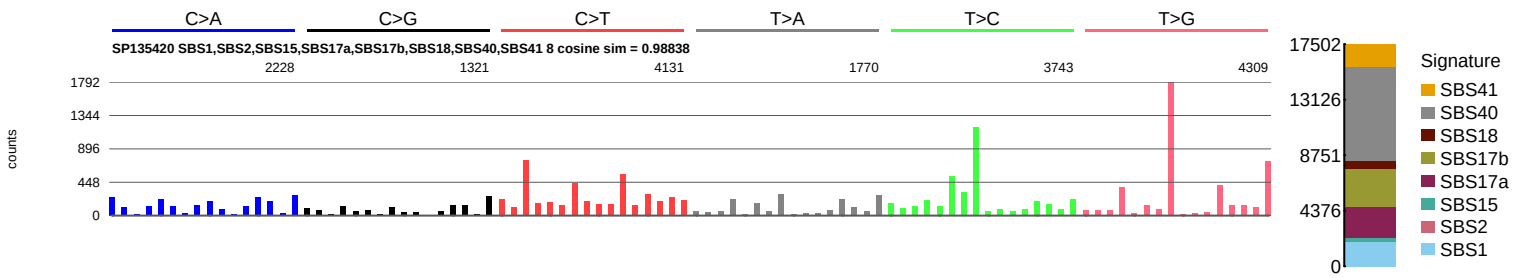
SP135420 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98839
1954
1372
4047
1729
4089
4311
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1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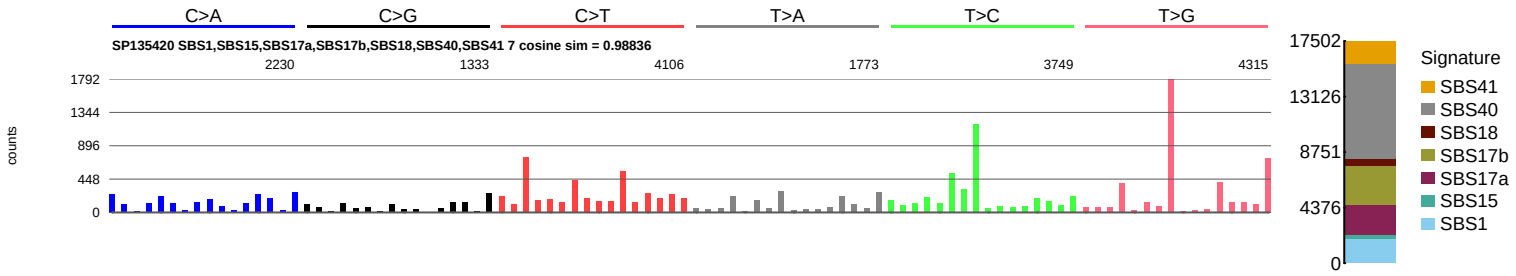
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98838
2228
1321
4131
1770
3743
4309
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
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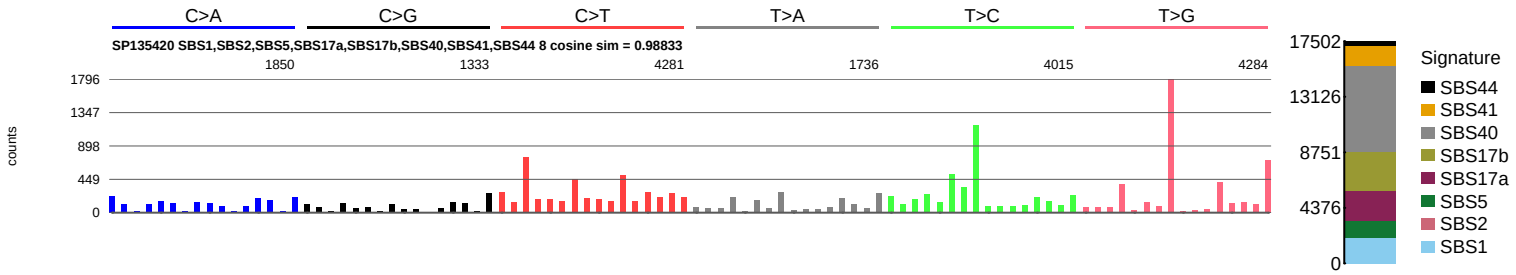
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98836
2230
1333
4106
1773
3749
4315
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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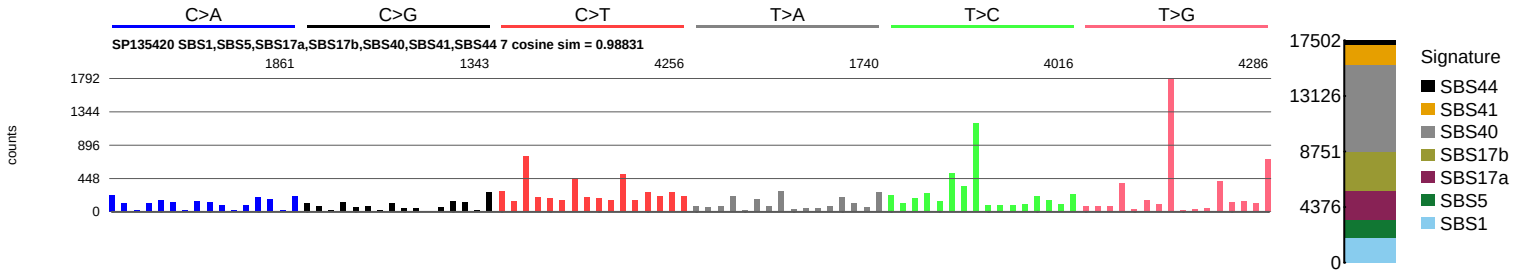
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98833
1850
1333
4281
1736
4015
4284
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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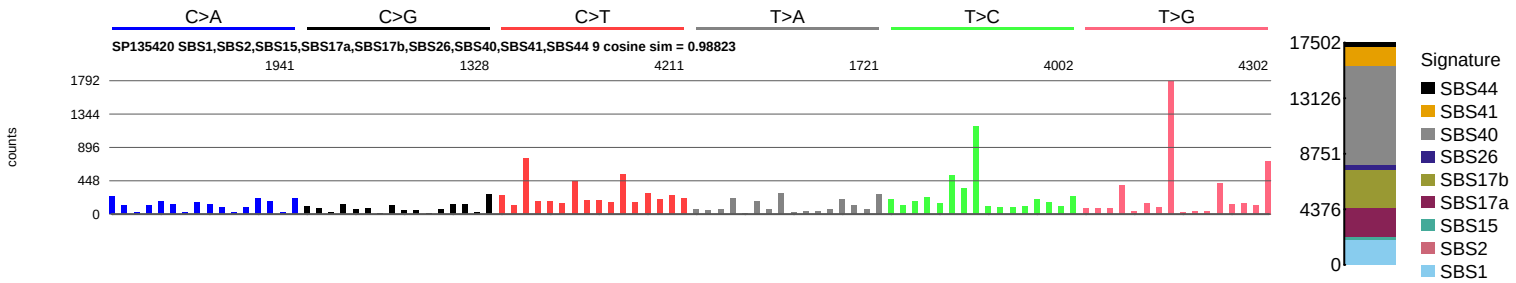
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98831
1861
1343
4256
1740
4016
4286
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1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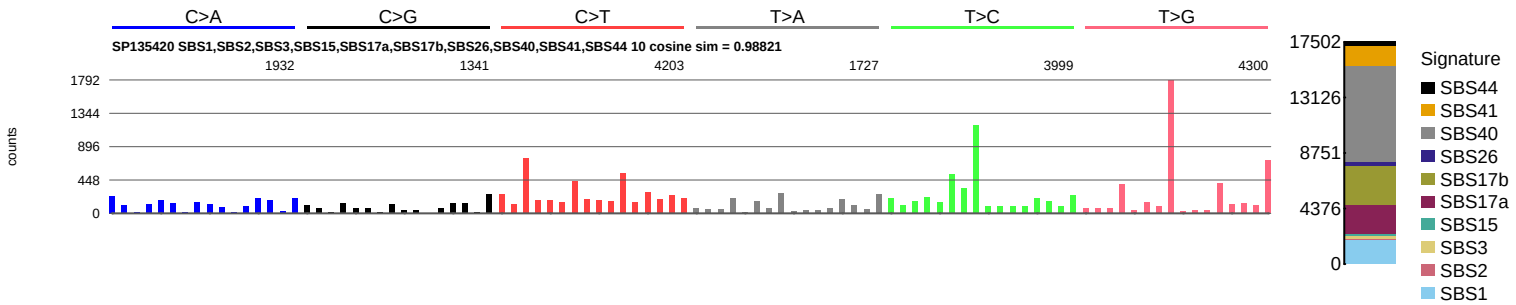
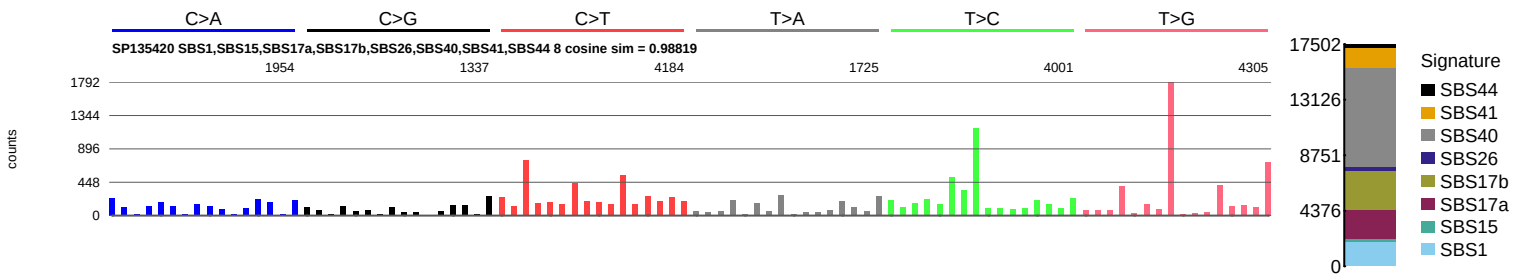
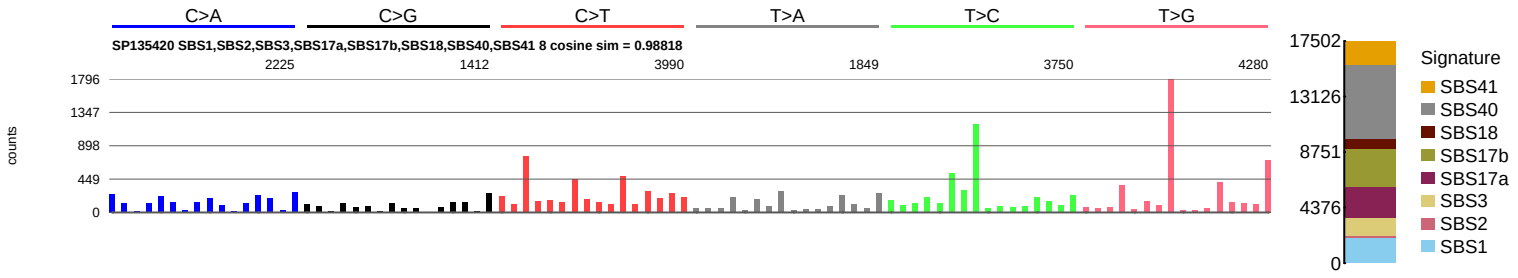
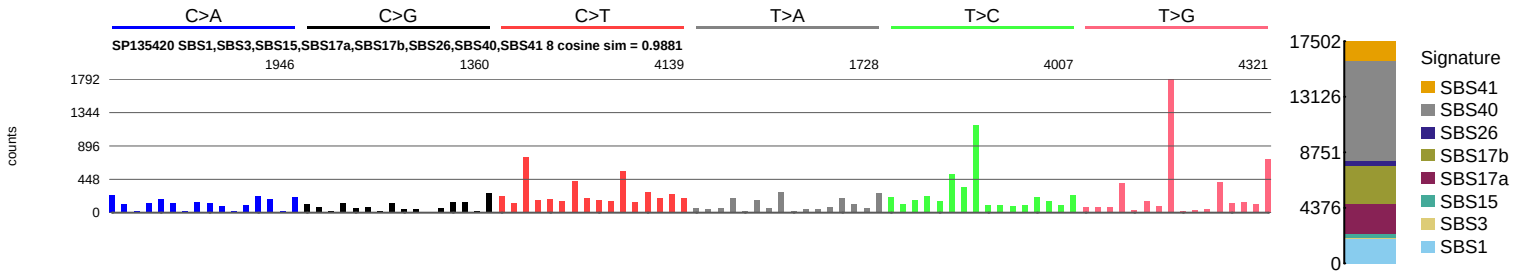
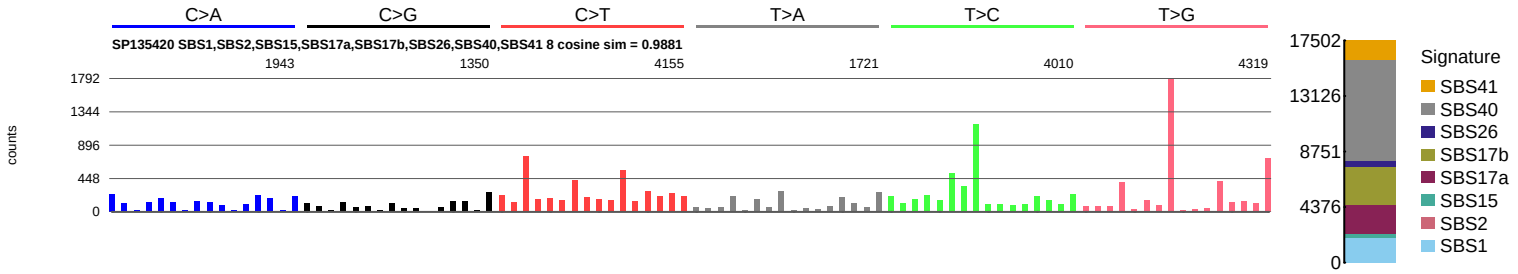
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98823
1941
1328
4211
1721
4002
4302
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

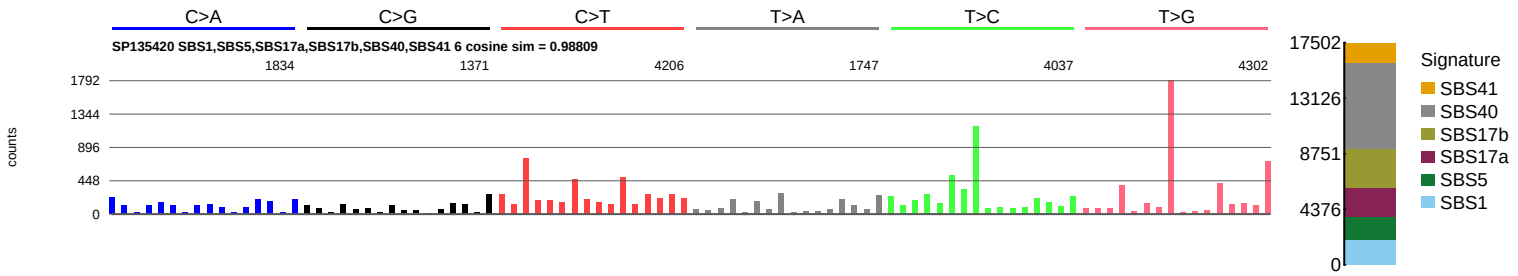

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98821
1932
1341
4203
1727
3999
4300
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

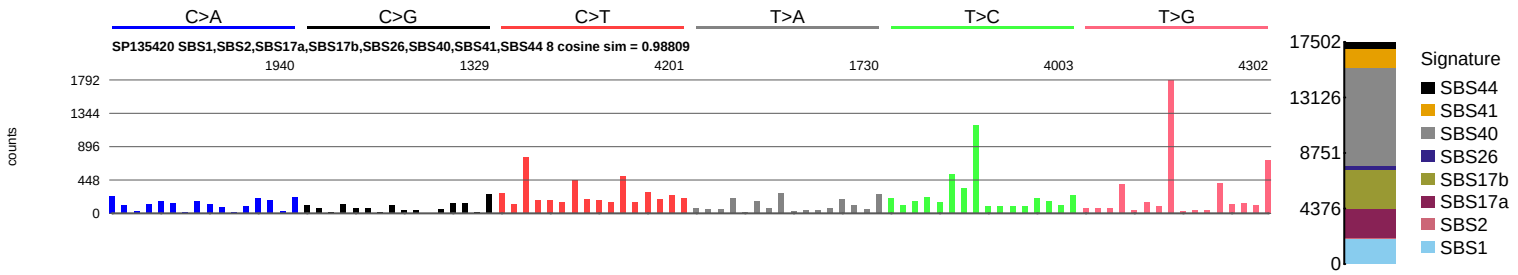

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98819
1954
1337
4184
1725
4001
4305
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98818
2225
1412
3990
1849
3750
4280
counts
1796
1347
898
449
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.9881
1946
1360
4139
1728
4007
4321
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.9881
1943
1350
4155
1721
4010
4319
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98809
1834
1371
4206
1747
4037
4302
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1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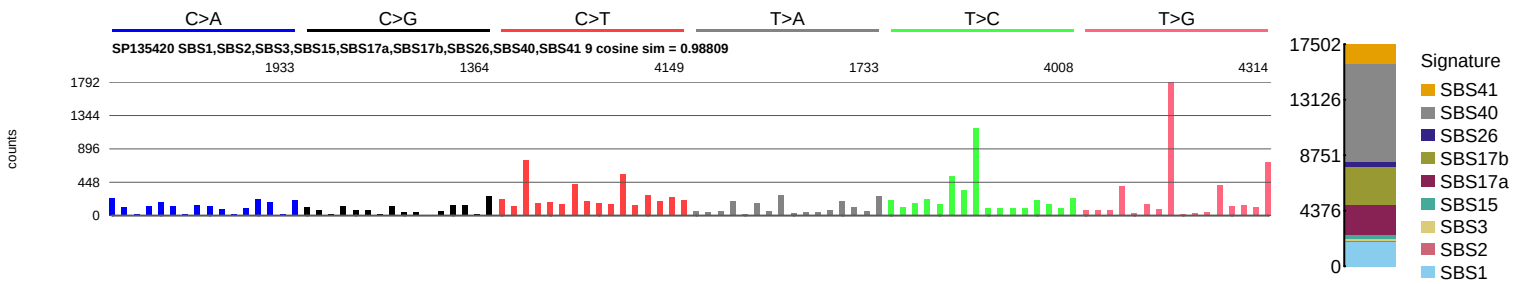
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98809
1940
1329
4201
1730
4003
4302
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98809
1933
1364
4149
1733
4008
4314
counts
1792
1344
896
448
0
17502
13126
8751
4376
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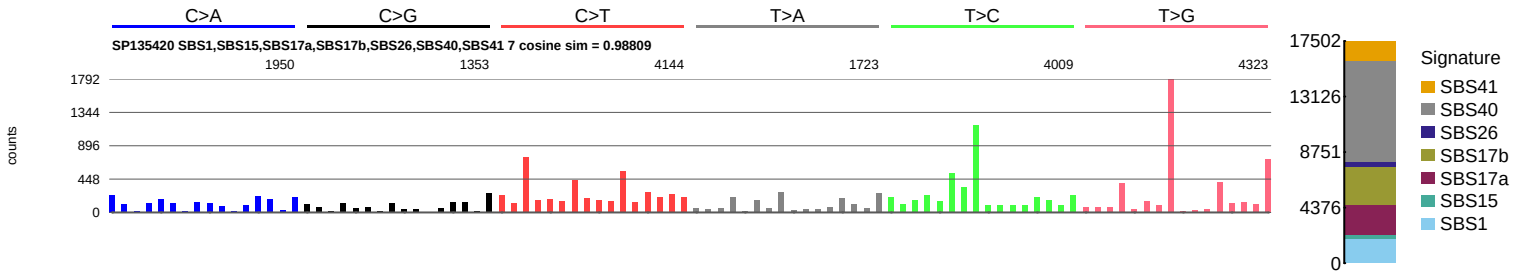
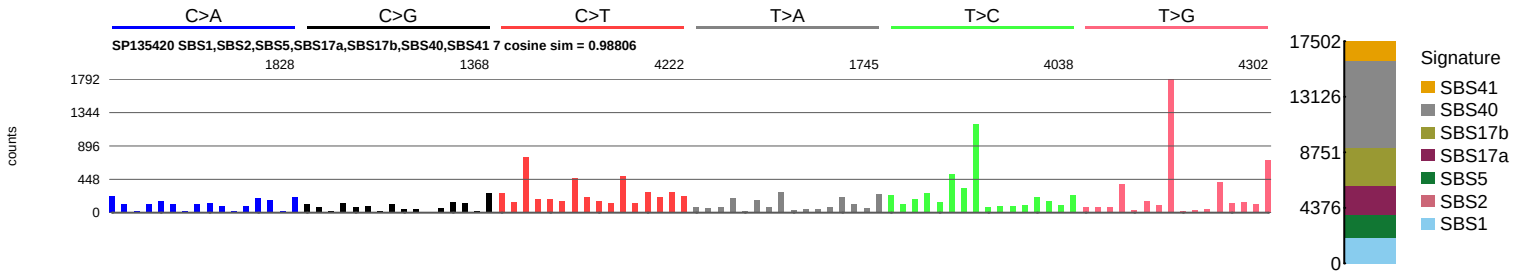
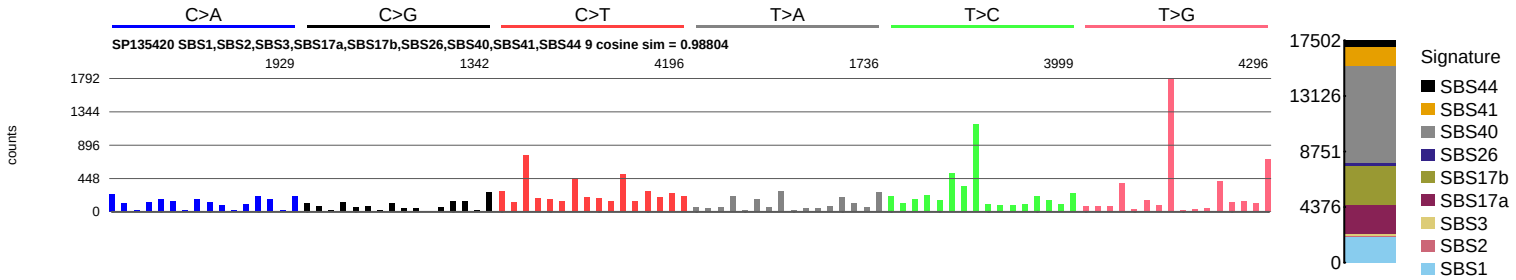
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98809
1950
1353
4144
1723
4009
4323
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98806
1828
1368
4222
1745
4038
4302
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98804
1929
1342
4196
1736
3999
4296
counts
1792
1344
896
448
0
17502
13126
8751
4376
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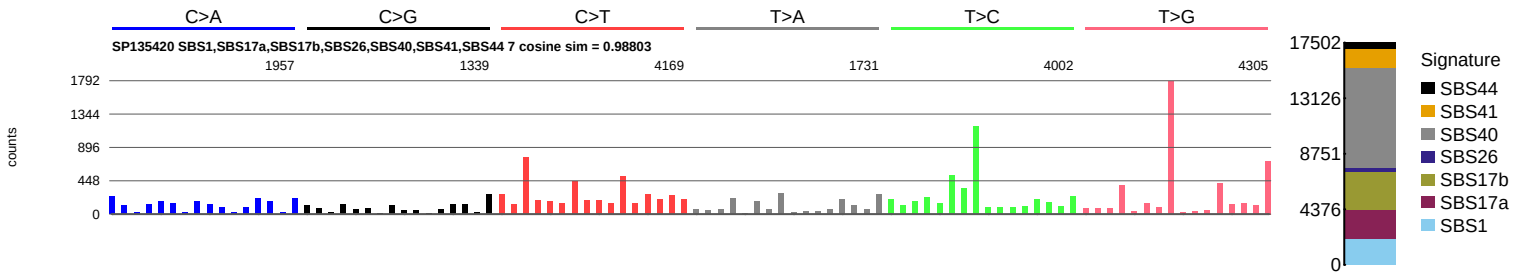
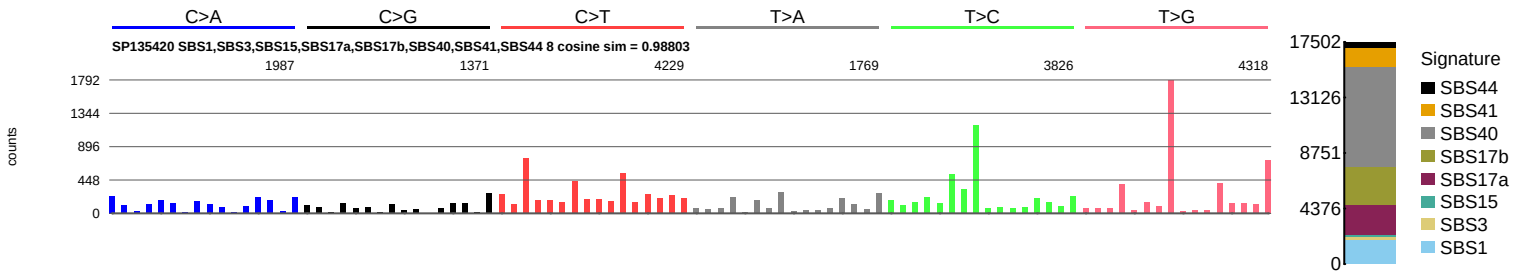
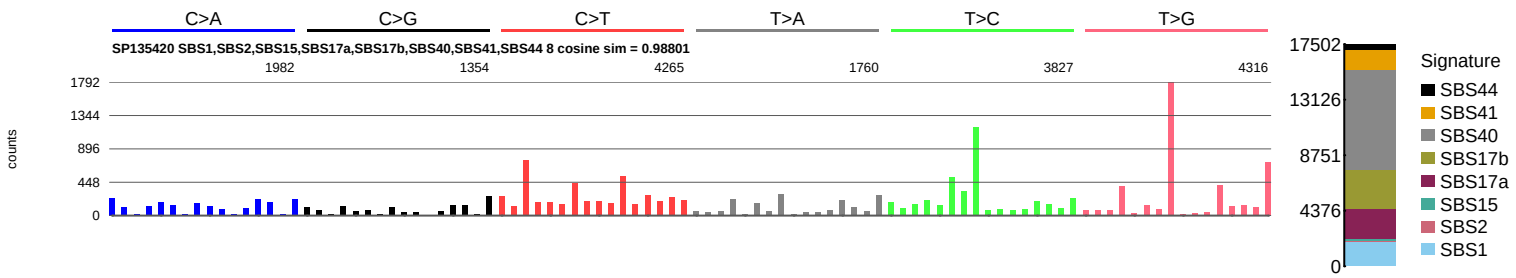
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98803
1957
1339
4169
1731
4002
4305
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98803
1987
1371
4229
1769
3826
4318
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98801
1982
1354
4265
1760
3827
4316
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
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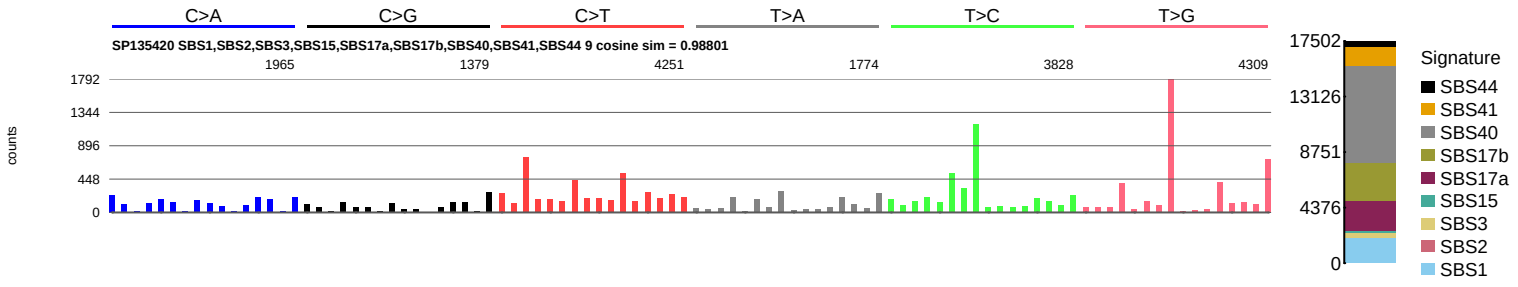
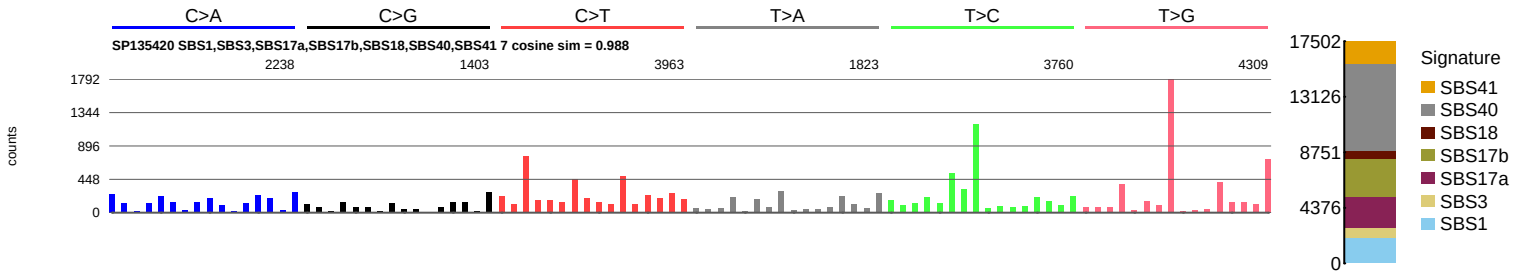
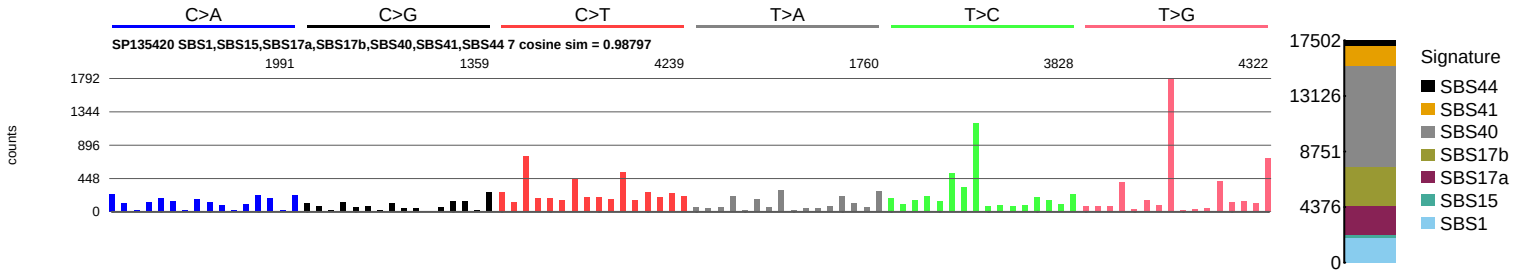
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98801
1965
1379
4251
1774
3828
4309
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.988
2238
1403
3963
1823
3760
4309
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1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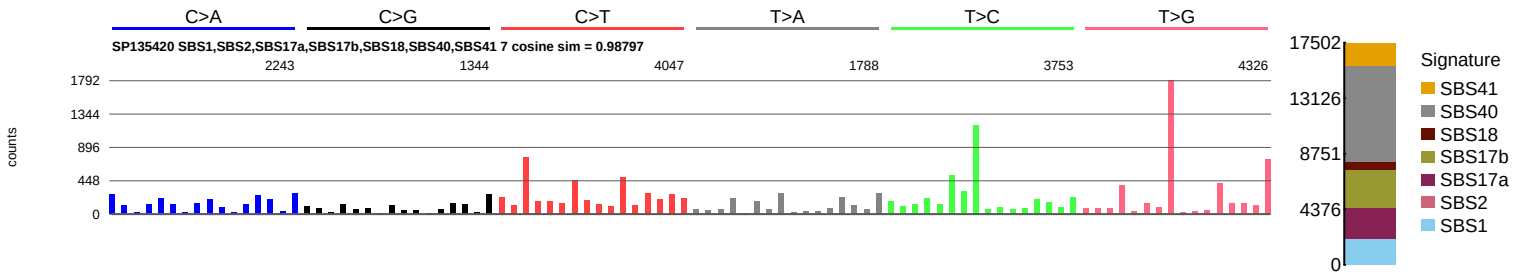
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98797
1991
1359
4239
1760
3828
4322
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1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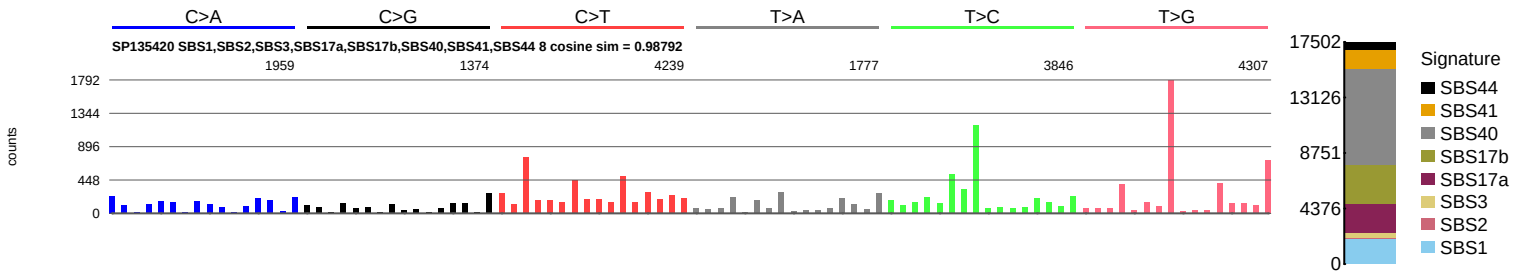
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98797
2243
1344
4047
1788
3753
4326
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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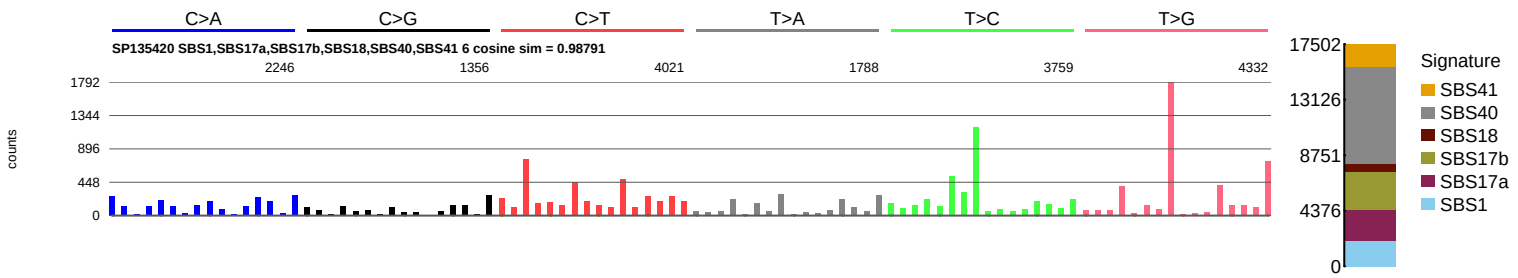
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98792
1959
1374
4239
1777
3846
4307
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1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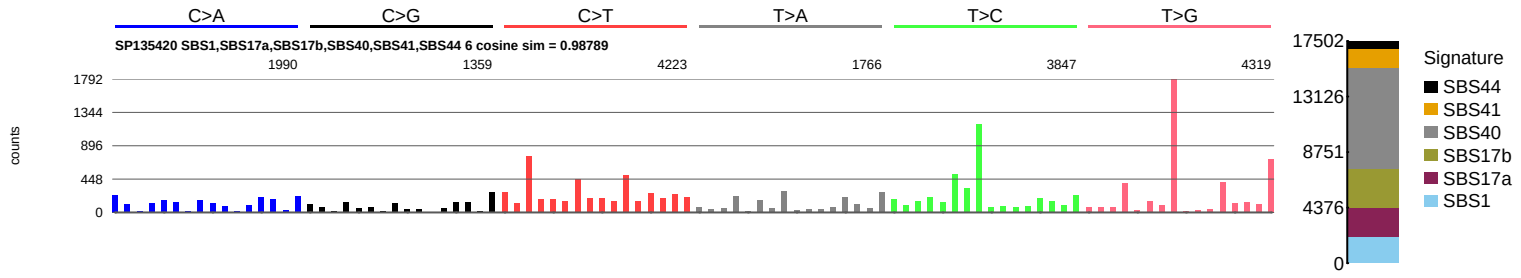
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98791
2246
1356
4021
1788
3759
4332
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1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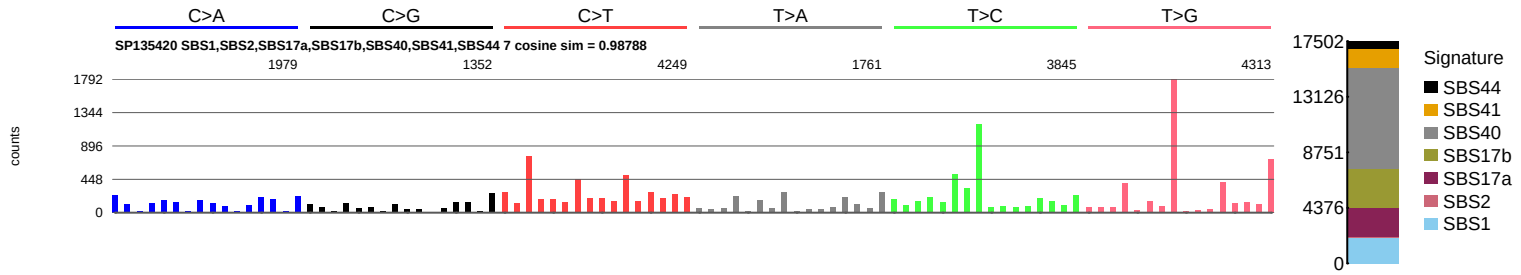
SP135420 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98789
1990
1359
4223
1766
3847
4319
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
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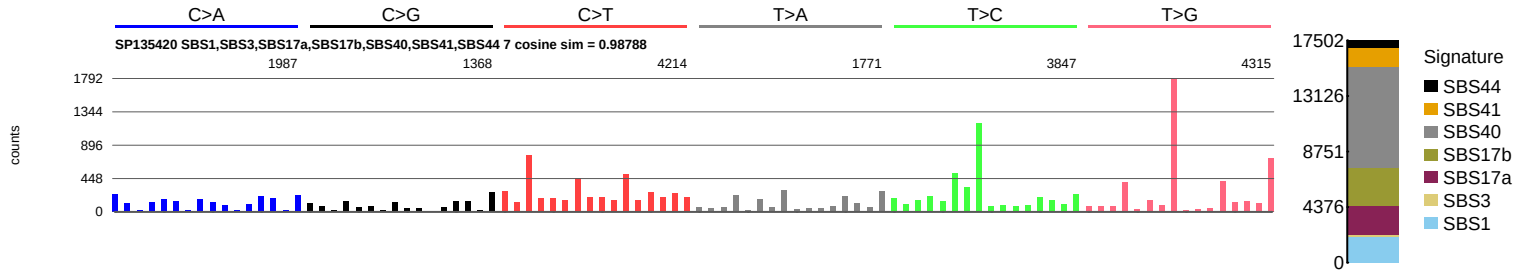
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98788
1979
1352
4249
1761
3845
4313
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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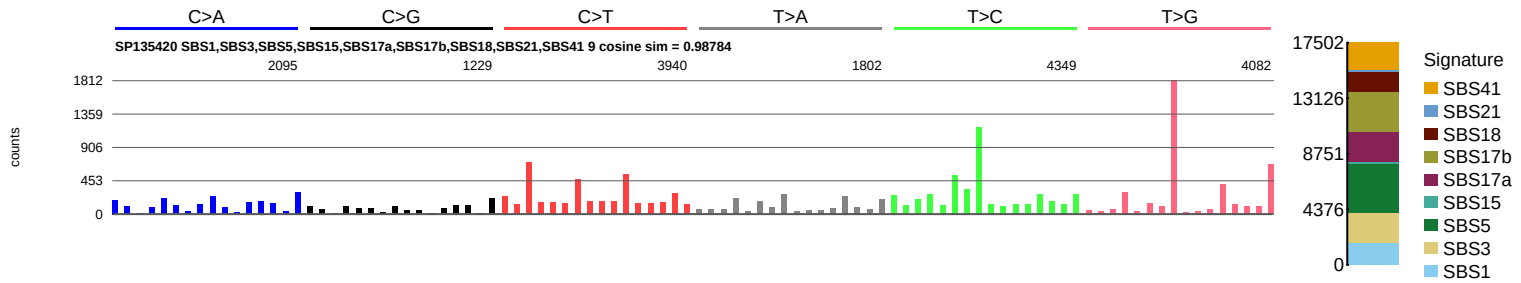
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98788
1987
1368
4214
1771
3847
4315
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS1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T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98784
2095
1229
3940
1802
4349
4082
counts
1812
1359
906
453
0
17502
13126
8751
4376
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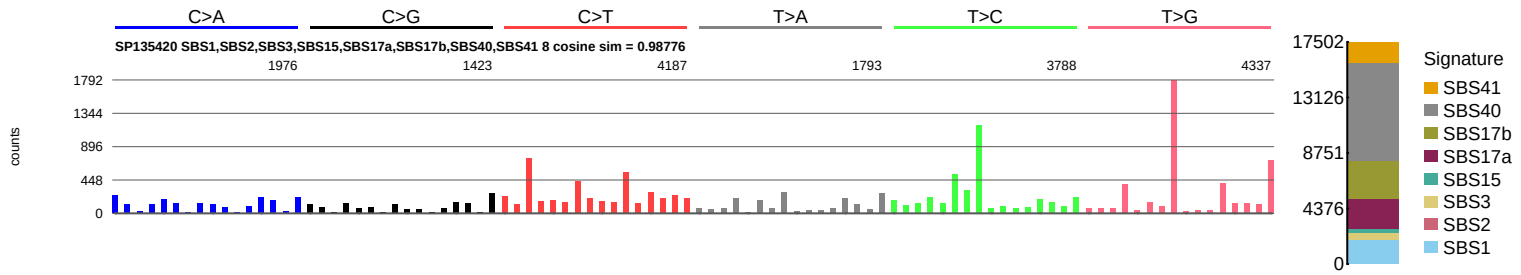
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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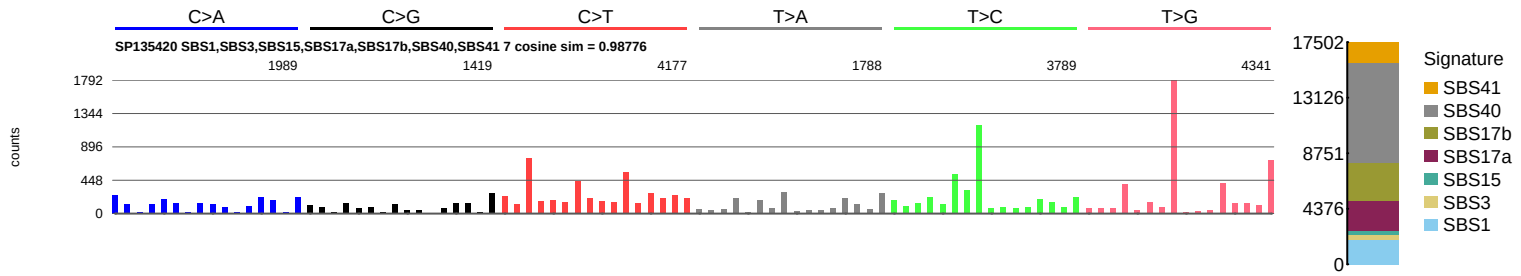
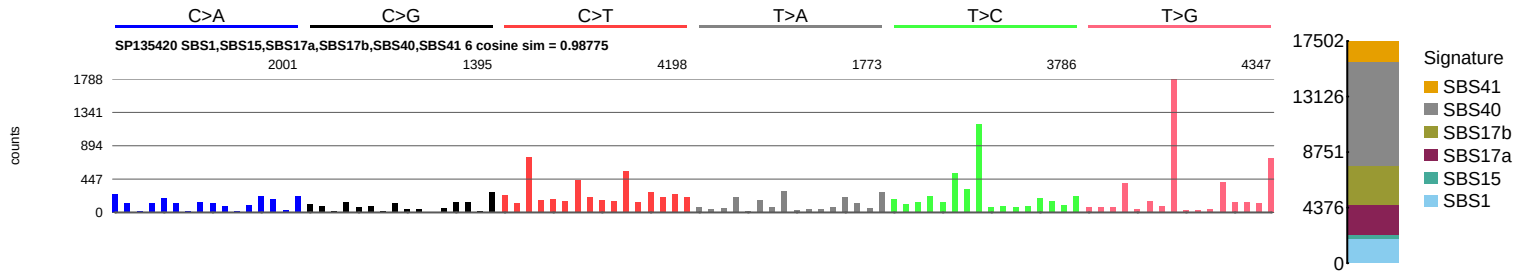
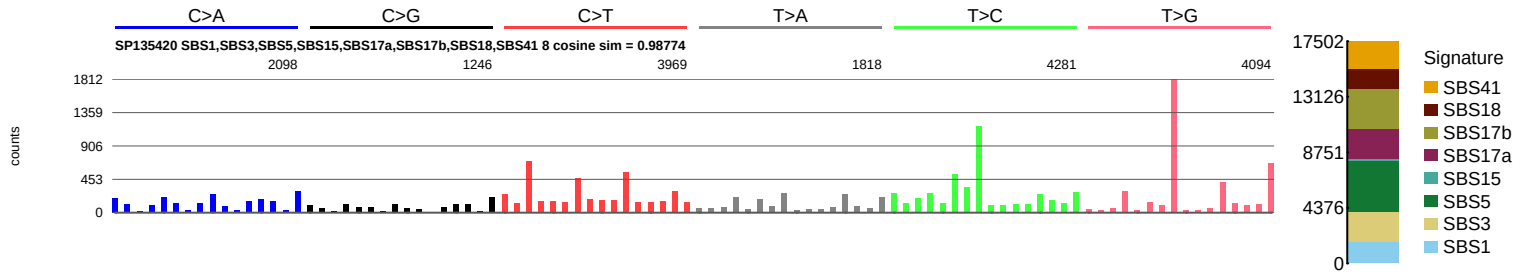
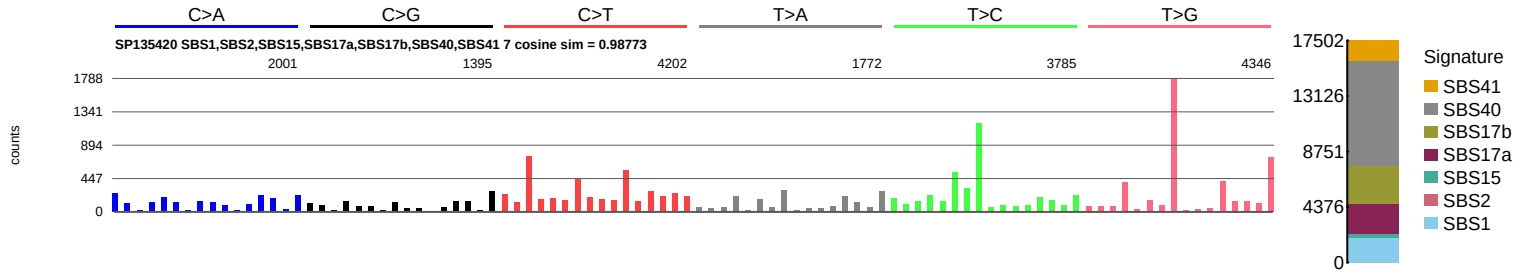
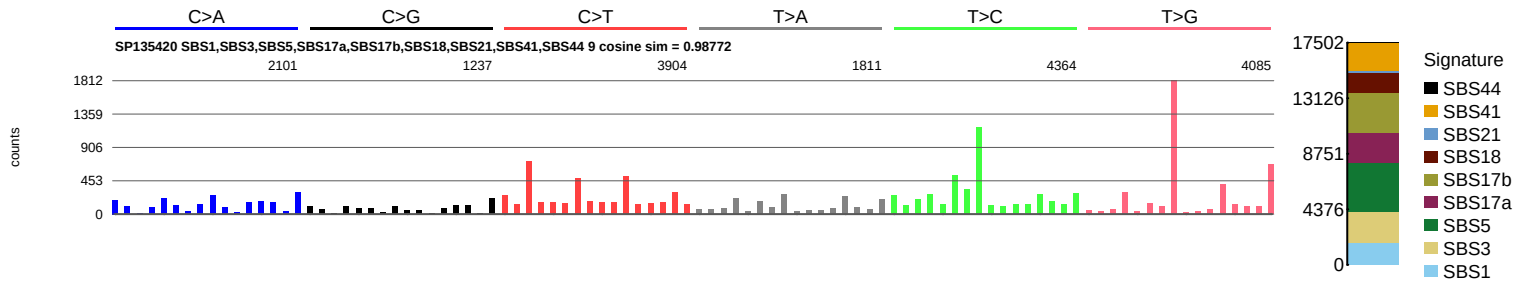
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98776
1976
1423
4187
1793
3788
4337
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

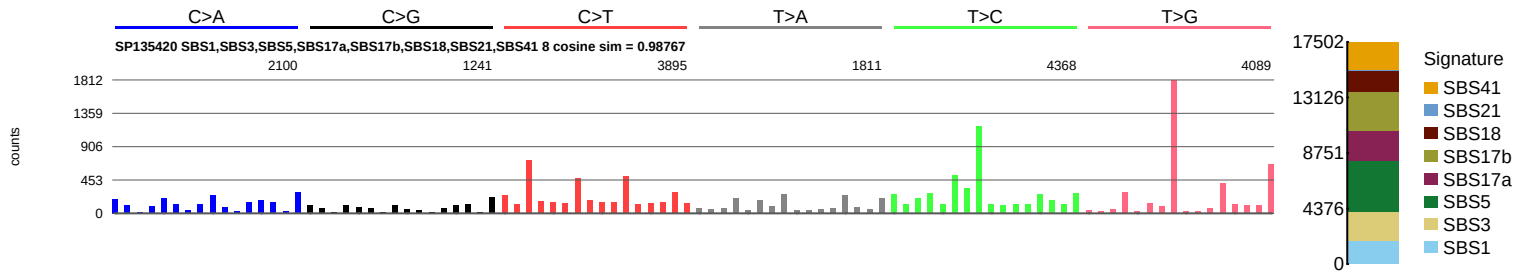

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98776
1989
1419
4177
1788
3789
4341
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98775
2001
1395
4198
1773
3786
4347
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98774
2098
1246
3969
1818
4281
4094
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98773
2001
1395
4202
1772
3785
4346
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98772
2101
1237
3904
1811
4364
4085
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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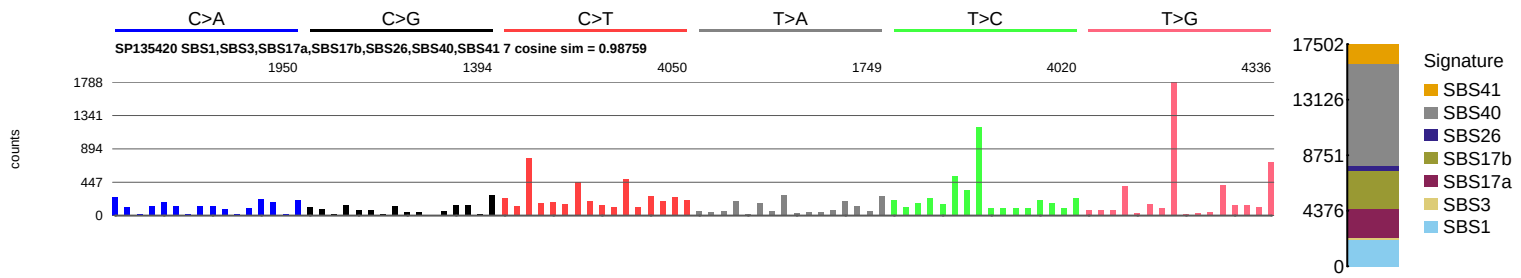
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98767
2100
1241
3895
1811
4368
4089
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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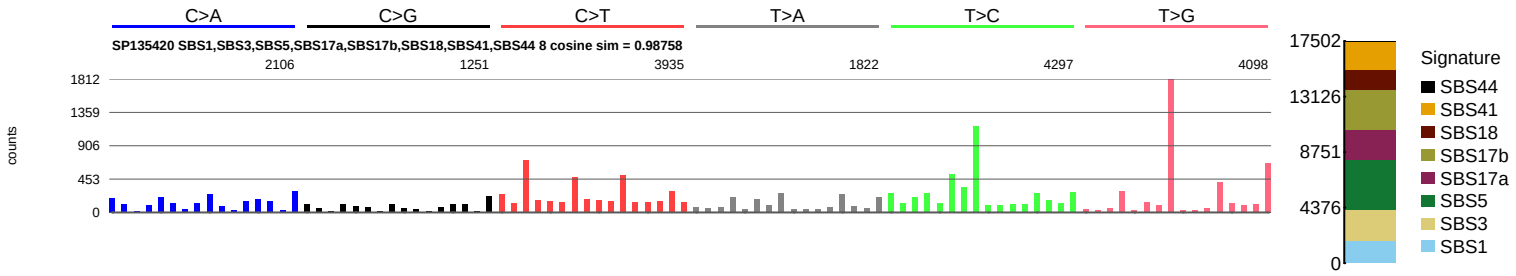
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98759
1950
1394
4050
1749
4020
4336
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1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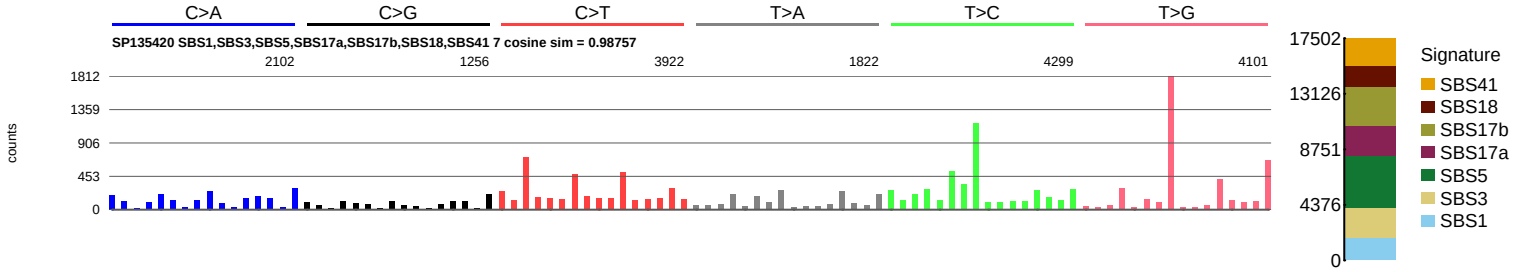
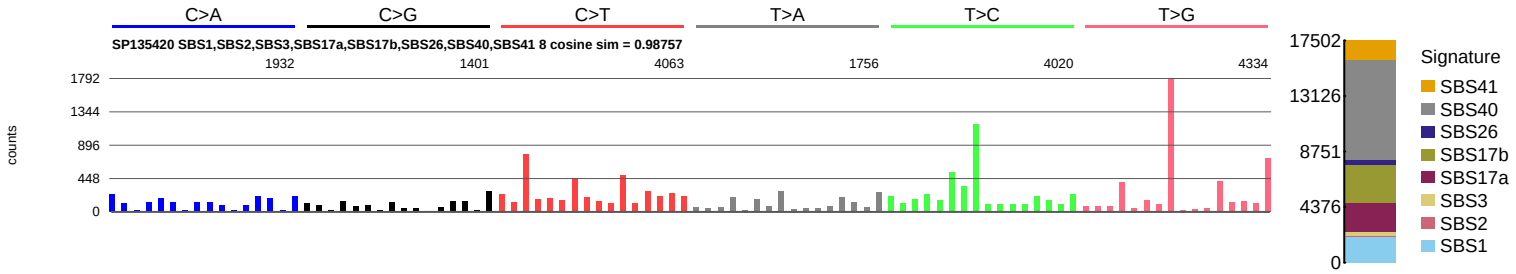
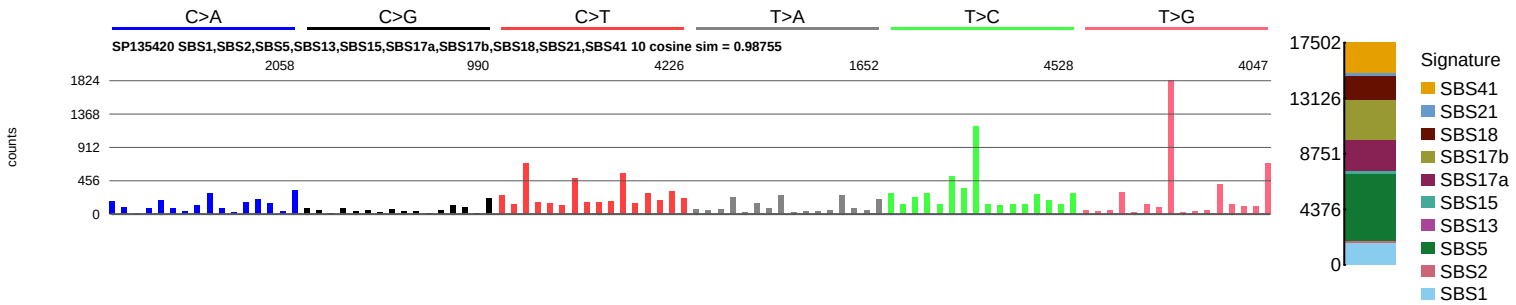
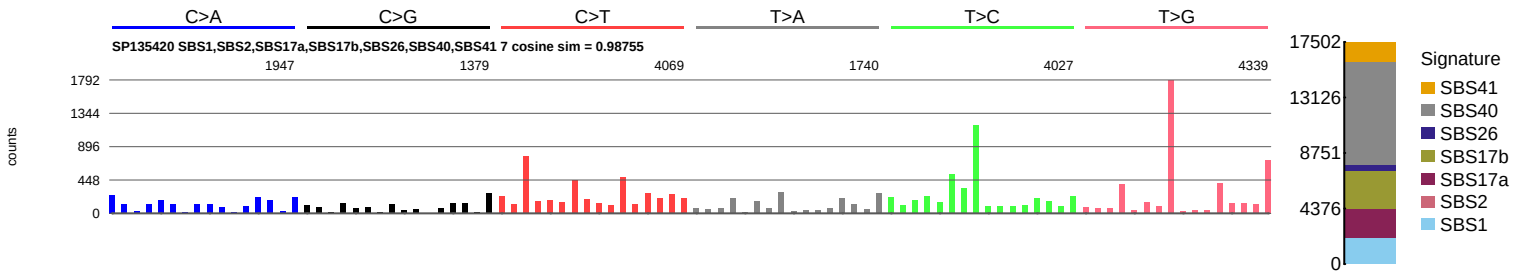
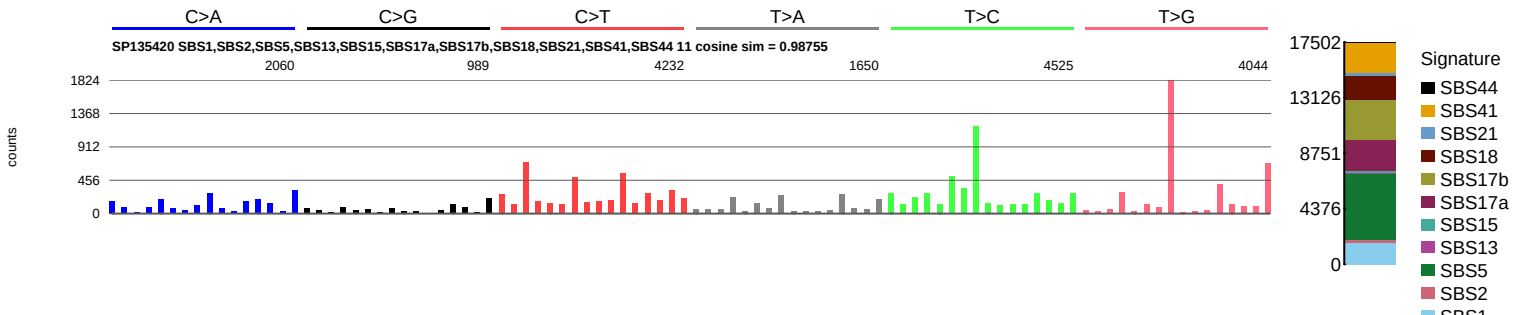
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98758
2106
1251
3935
1822
4297
4098
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

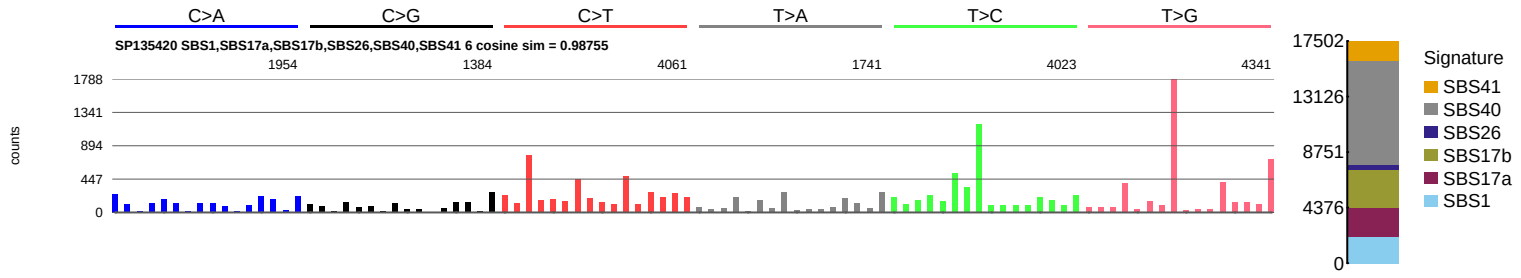

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98757
2102
1256
3922
1822
4299
4101
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

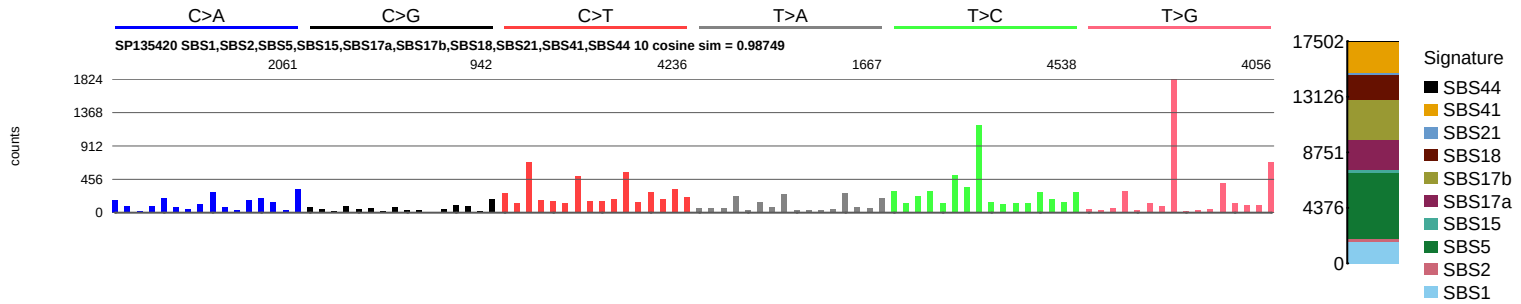

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98757
1932
1401
4063
1756
4020
4334
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

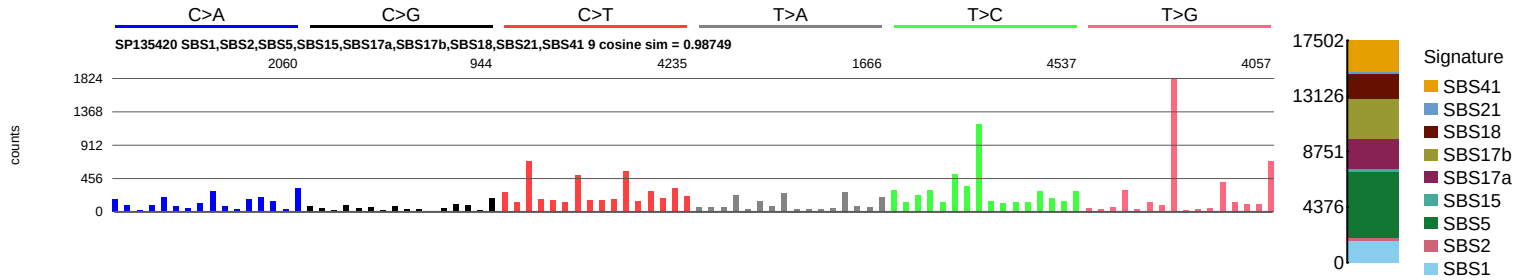

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98755
2058
990
4226
1652
4528
4047
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98755
1947
1379
4069
1740
4027
4339
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98755
2060
989
4232
1650
4525
4044
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98755
1954
1384
4061
1741
4023
4341
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98749
2061
942
4236
1667
4538
4056
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98749
2060
944
4235
1666
4537
4057
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
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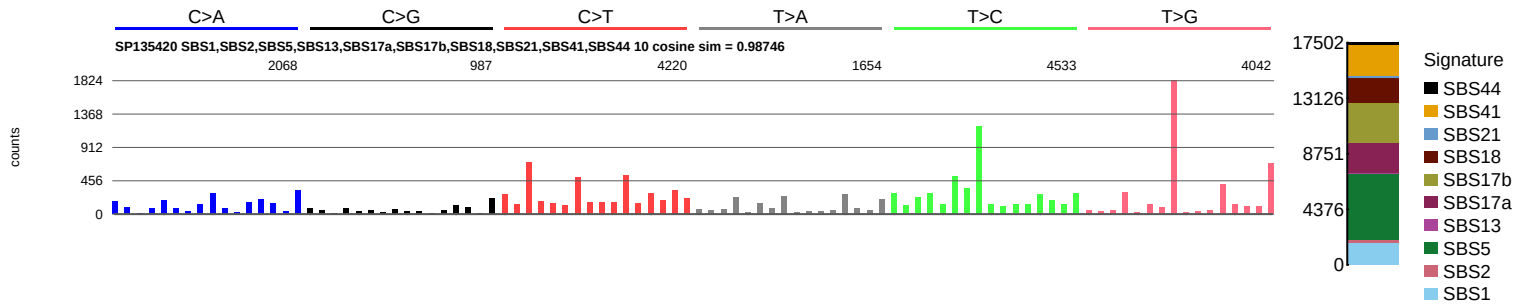
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SP135420 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98746
2068
987
4220
1654
4533
4042
counts
1824
1368
912
456
0
17502
13126
8751
4376
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Signature
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SBS21
SBS18
SBS17b
SBS17a
SBS13
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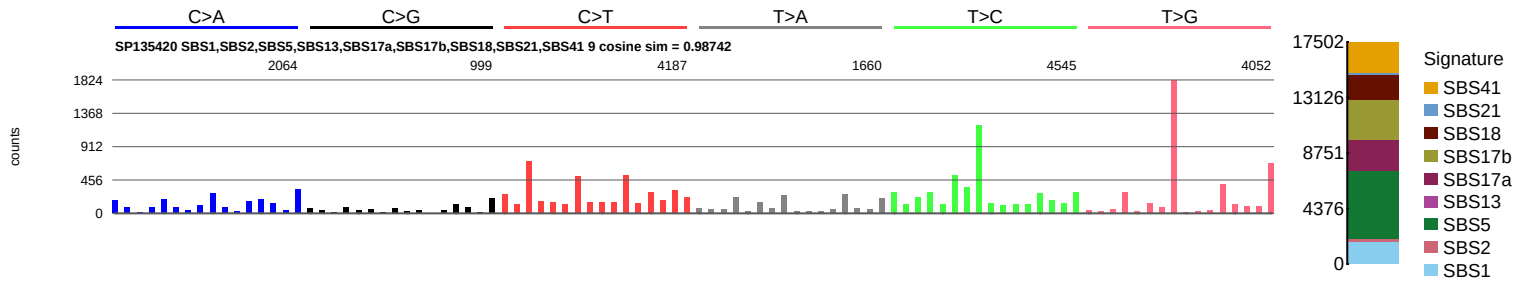
SBS1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2064
999
4187
1660
4545
4052
counts
1824
1368
912
456
0
17502
13126
8751
4376
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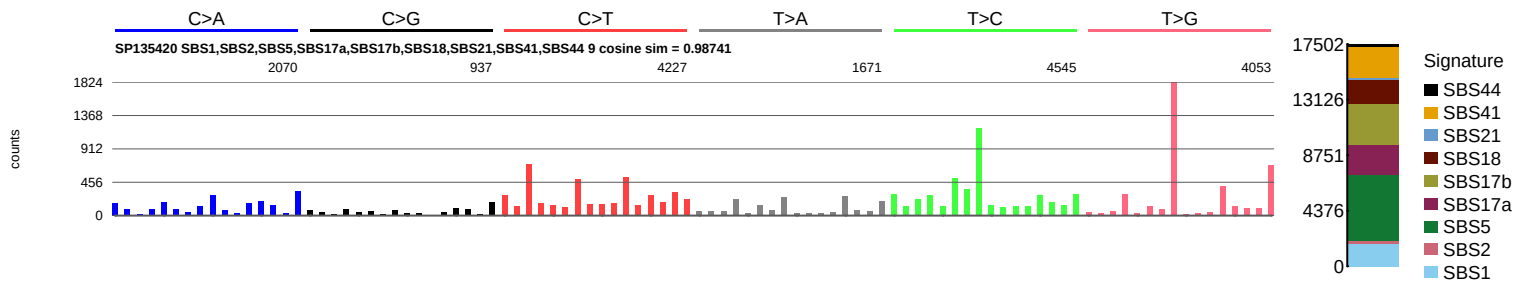
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98741
2070
937
4227
1671
4545
4053
counts
1824
1368
912
456
0
17502
13126
8751
4376
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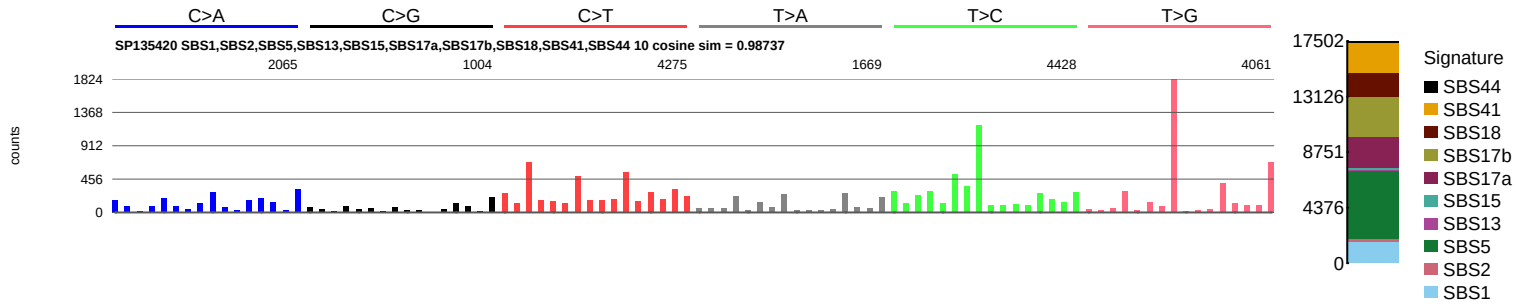
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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SP135420 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98737
2065
1004
4275
1669
4428
4061
counts
1824
1368
912
456
0
17502
13126
8751
4376
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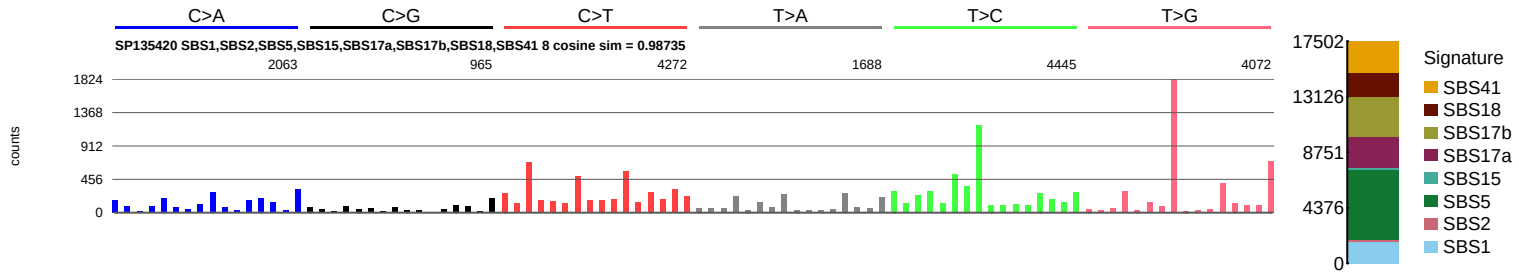
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98735
2063
965
4272
1688
4445
4072
counts
1824
1368
912
456
0
17502
13126
8751
4376
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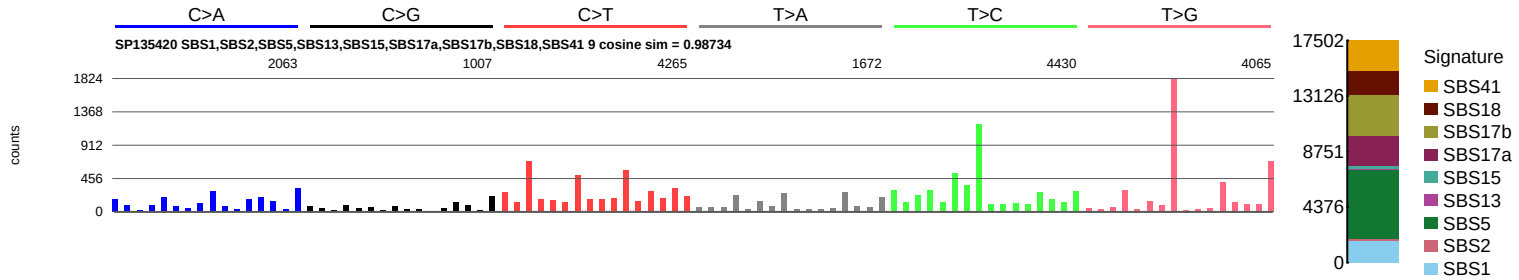
SBS2
SBS1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SP135420 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98734
2063
1007
4265
1672
4430
4065
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
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SBS41
SBS18
SBS17b
SBS17a
SBS15
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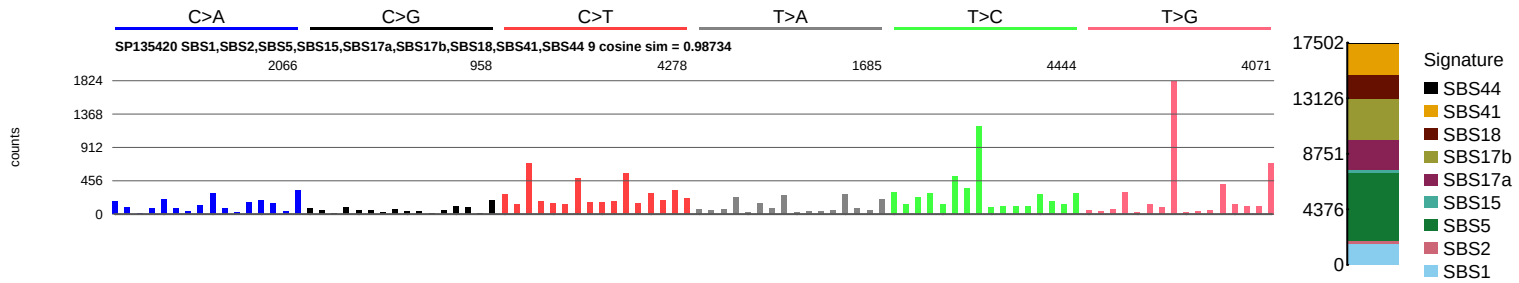
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98734
2066
958
4278
1685
4444
4071
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
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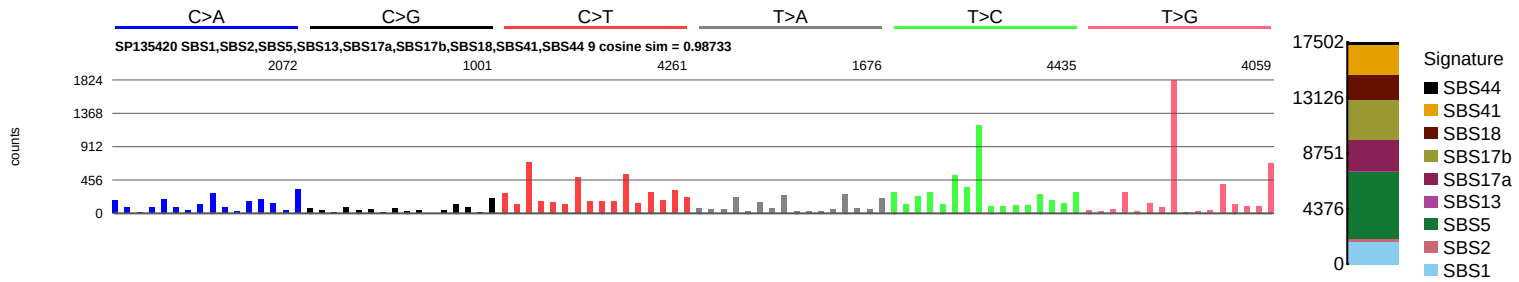
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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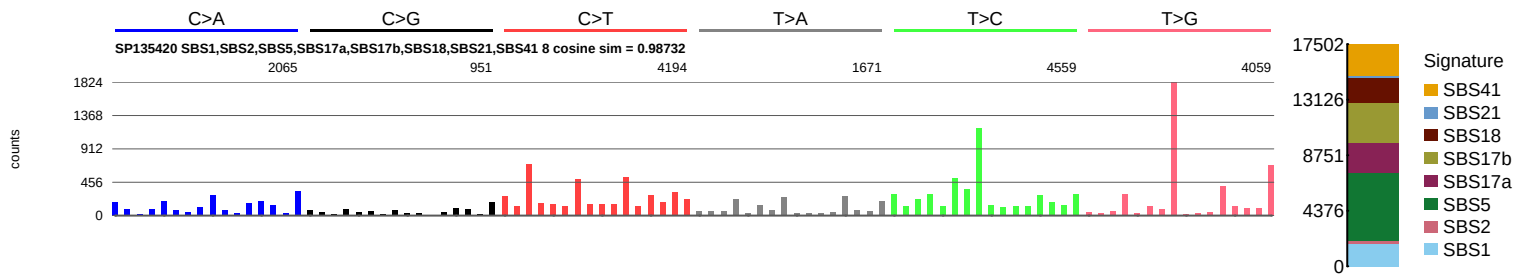
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98733
2072
1001
4261
1676
4435
4059
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
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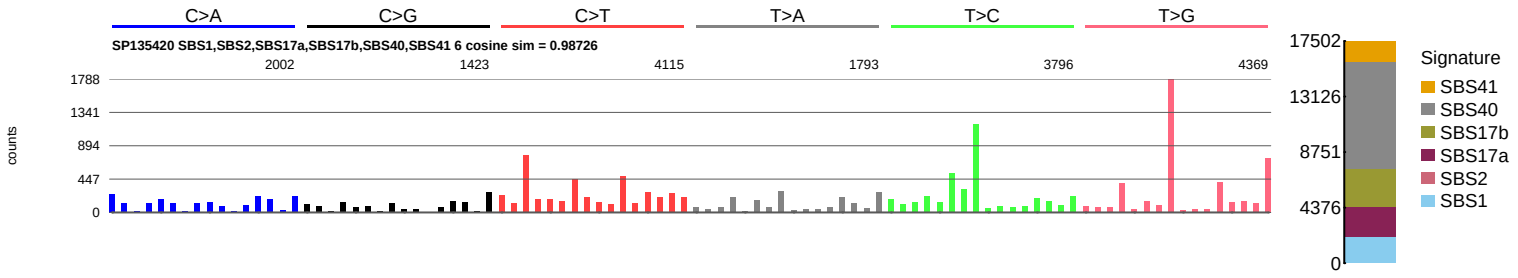
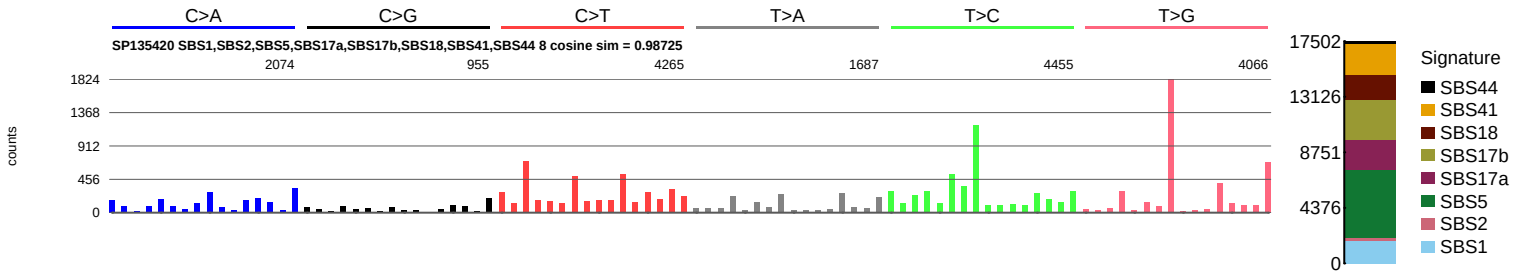
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98732
2065
951
4194
1671
4559
4059
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98726
2002
1423
4115
1793
3796
4369
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1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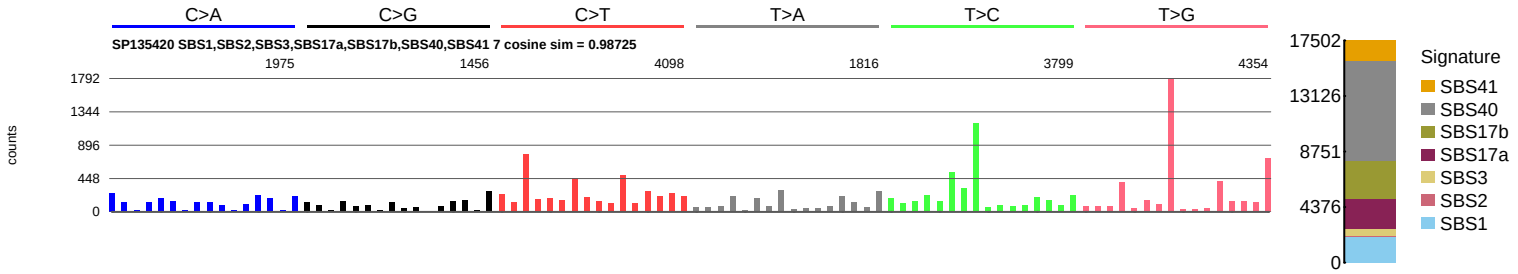
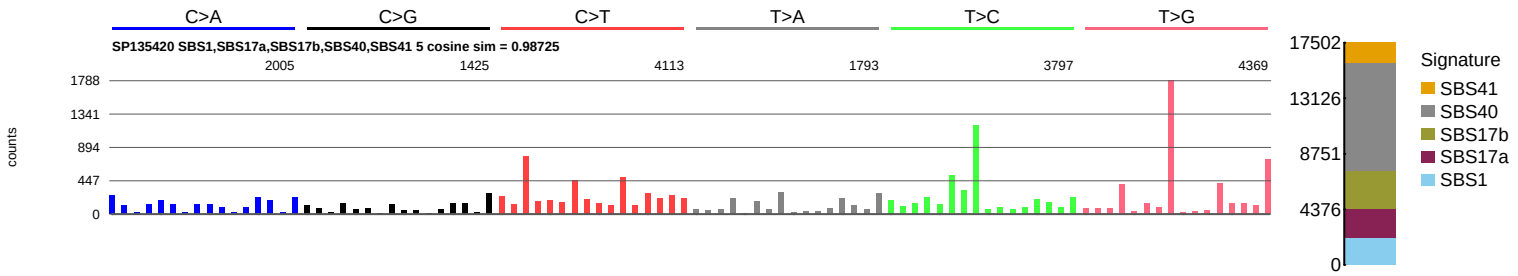
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98725
2074
955
4265
1687
4455
4066
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98725
1975
1456
4098
1816
3799
4354
counts
1792
1344
896
448
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1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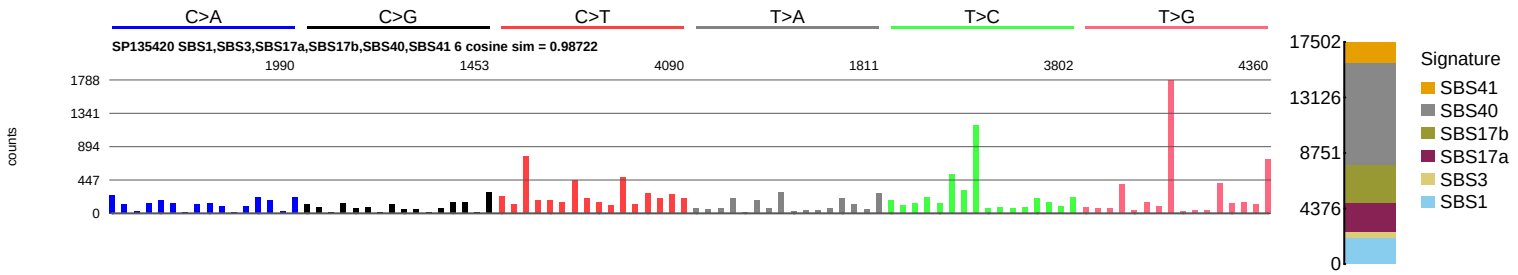
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98725
2005
1425
4113
1793
3797
4369
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1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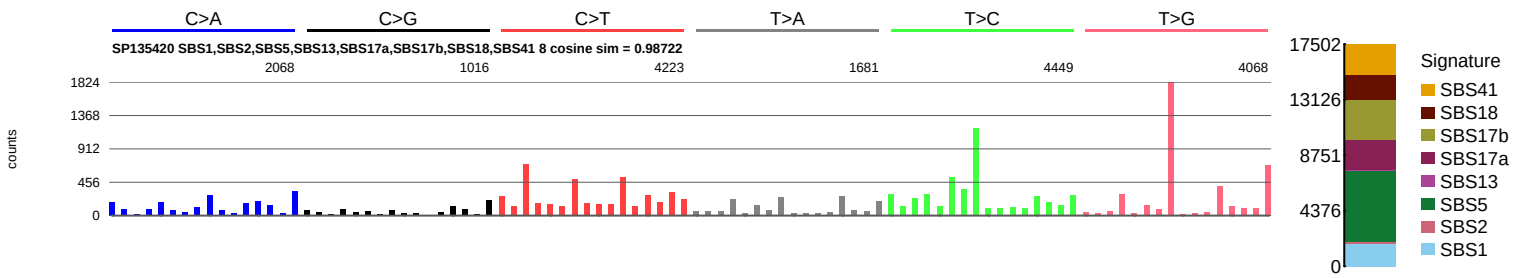
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98722
1990
1453
4090
1811
3802
4360
counts
1788
1341
894
447
0
17502
13126
8751
4376
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS1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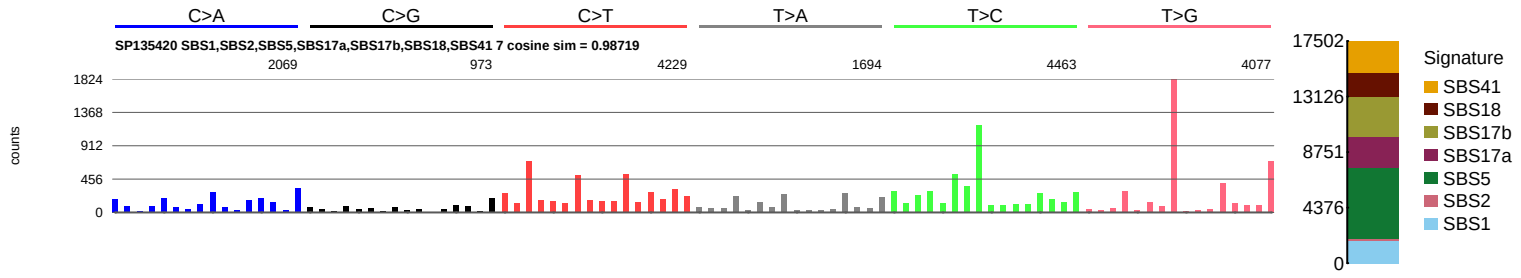
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98722
2068
1016
4223
1681
4449
4068
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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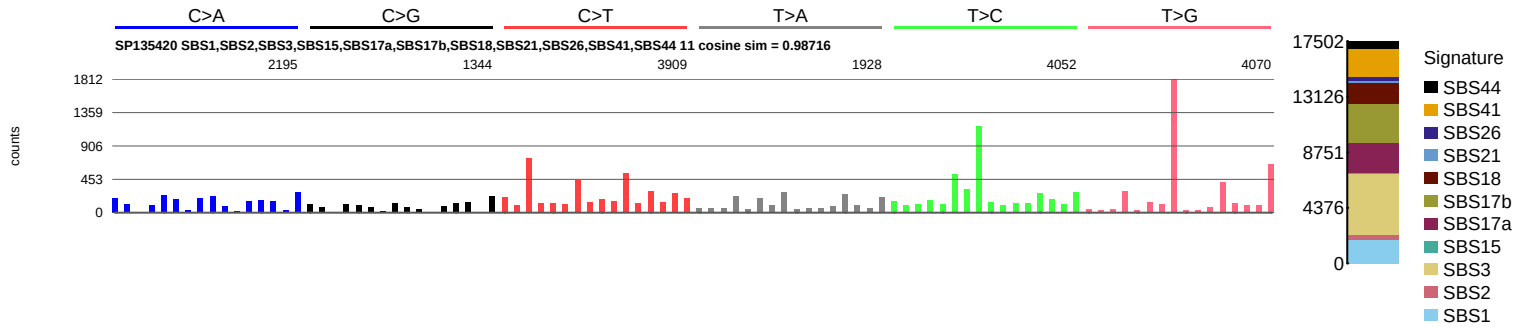
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98719
2069
973
4229
1694
4463
4077
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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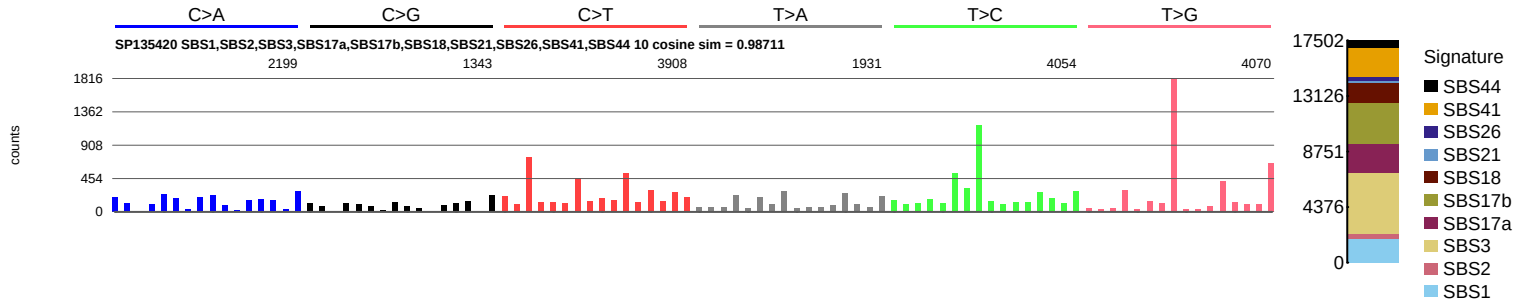
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98716
2195
1344
3909
1928
4052
4070
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98711
2199
1343
3908
1931
4054
4070
counts
1816
1362
908
454
0
17502
13126
8751
4376
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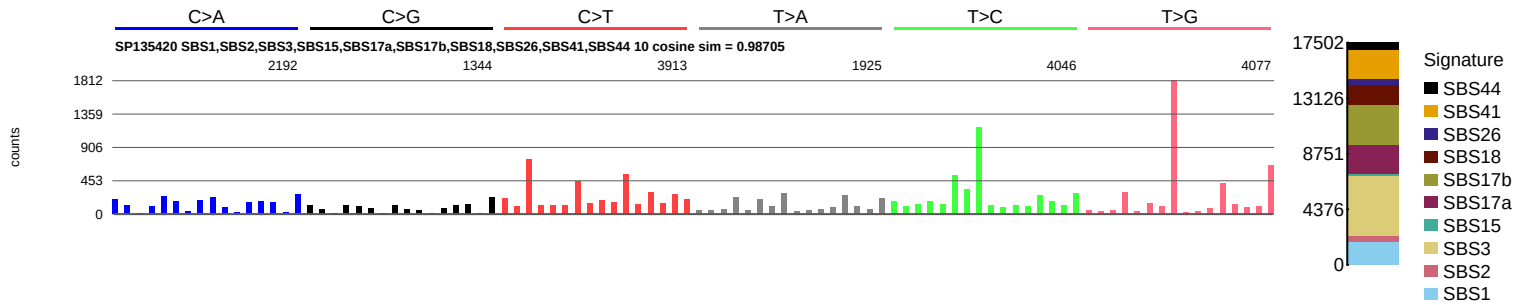
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
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T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98705
2192
1344
3913
1925
4046
4077
counts
1812
1359
906
453
0
17502
13126
8751
4376
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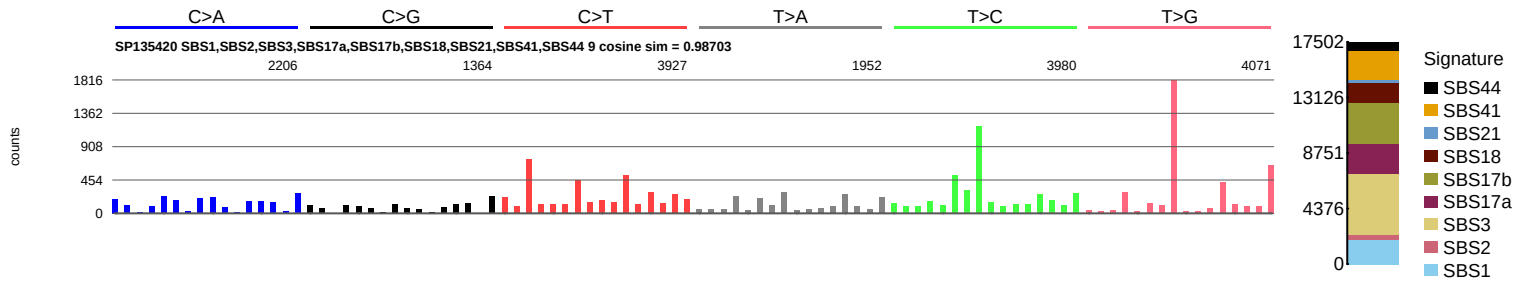
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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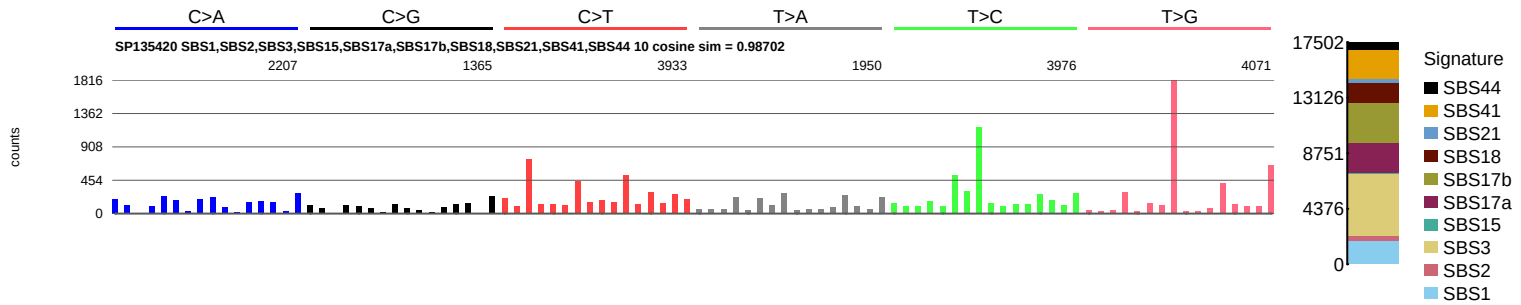
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98703
2206
1364
3927
1952
3980
4071
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
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SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98702
2207
1365
3933
1950
3976
4071
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
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SBS41
SBS21
SBS18
SBS17b
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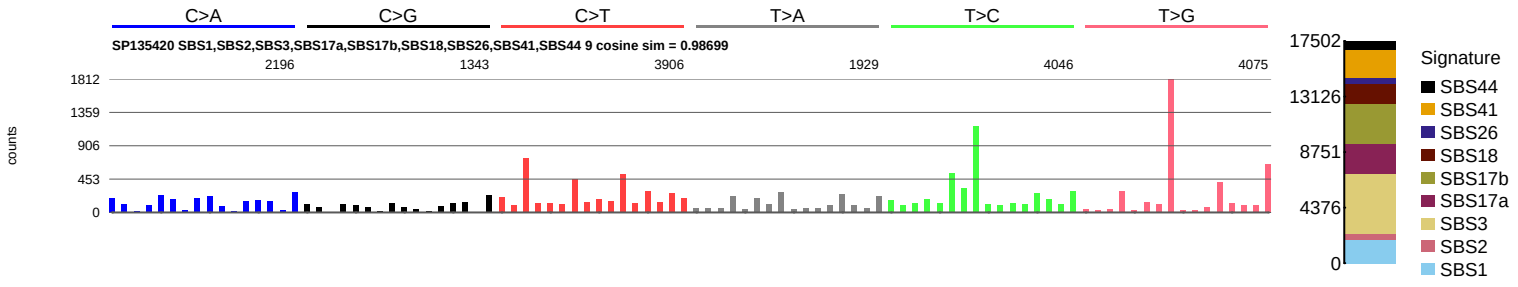
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98699
2196
1343
3906
1929
4046
4075
counts
1812
1359
906
453
0
17502
13126
8751
4376
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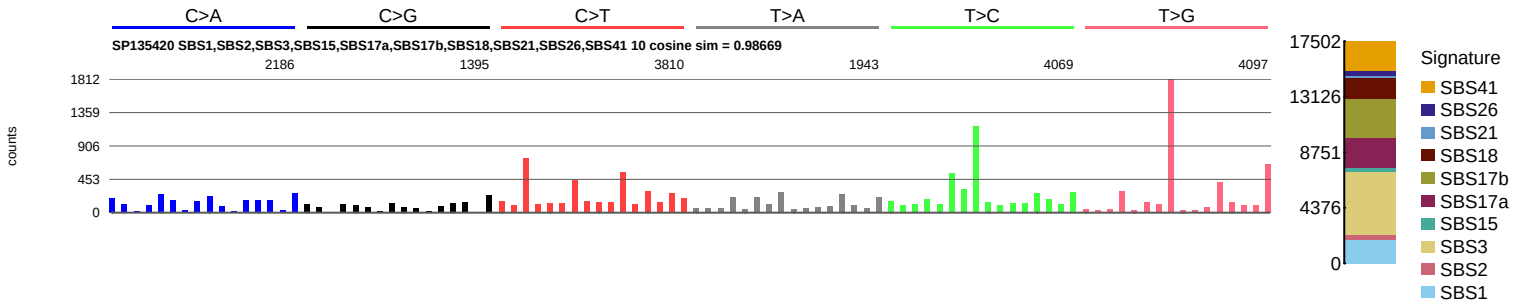
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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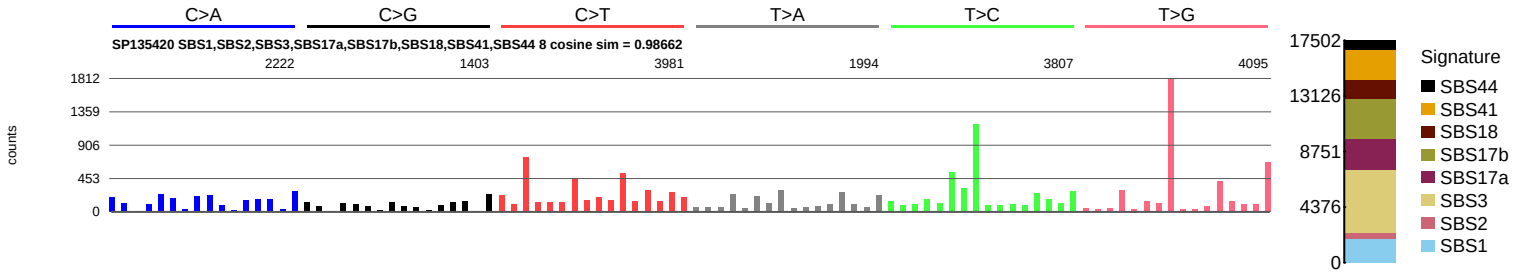
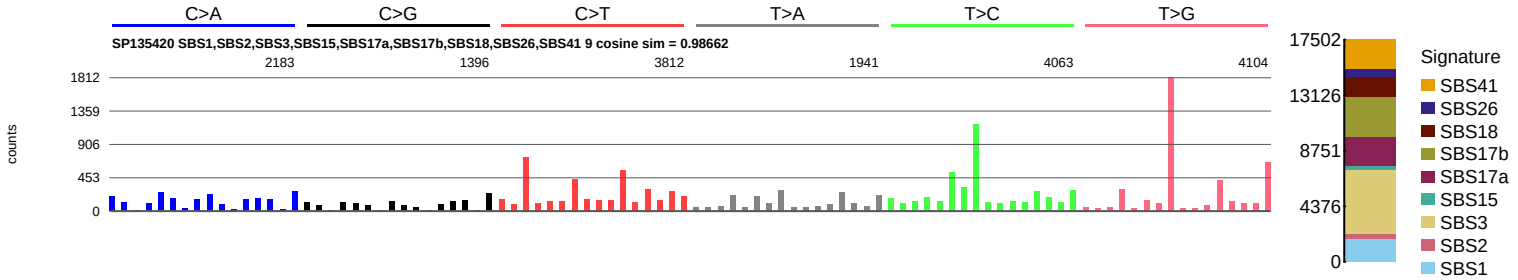
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98669
2186
1395
3810
1943
4069
4097
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98662
2222
1403
3981
1994
3807
4095
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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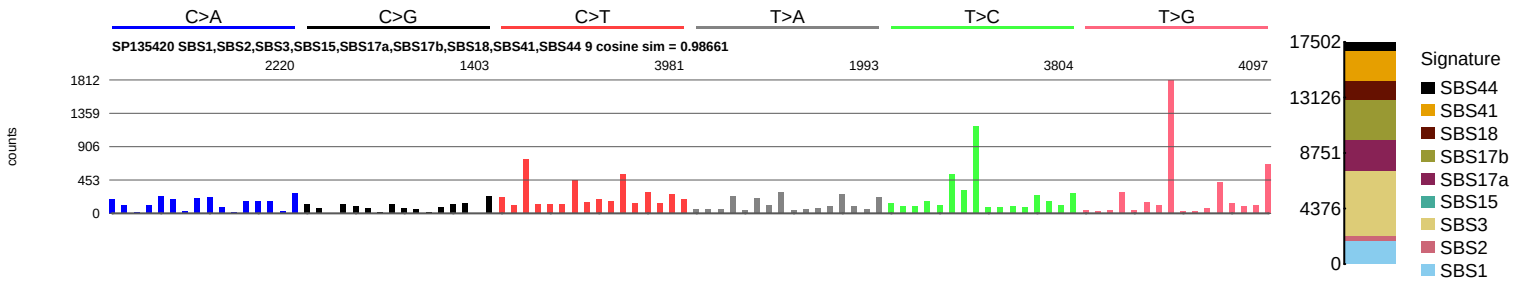
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98662
2183
1396
3812
1941
4063
4104
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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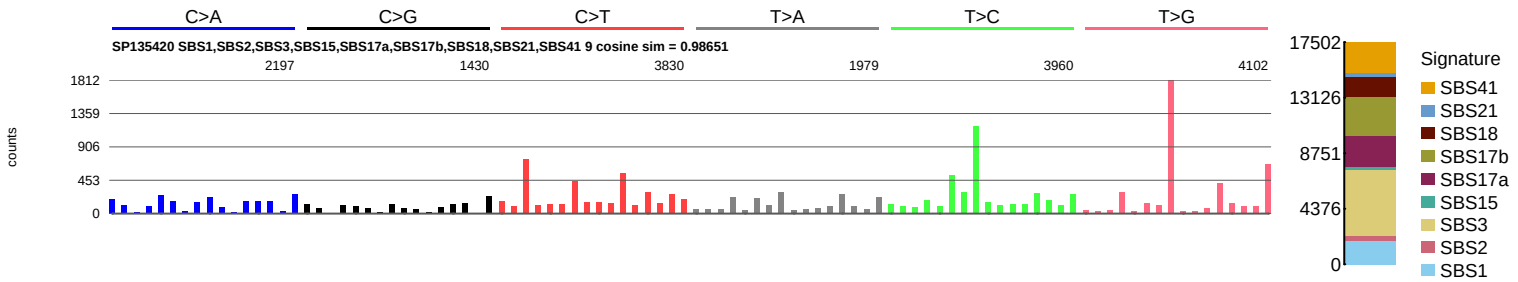
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98661
2220
1403
3981
1993
3804
4097
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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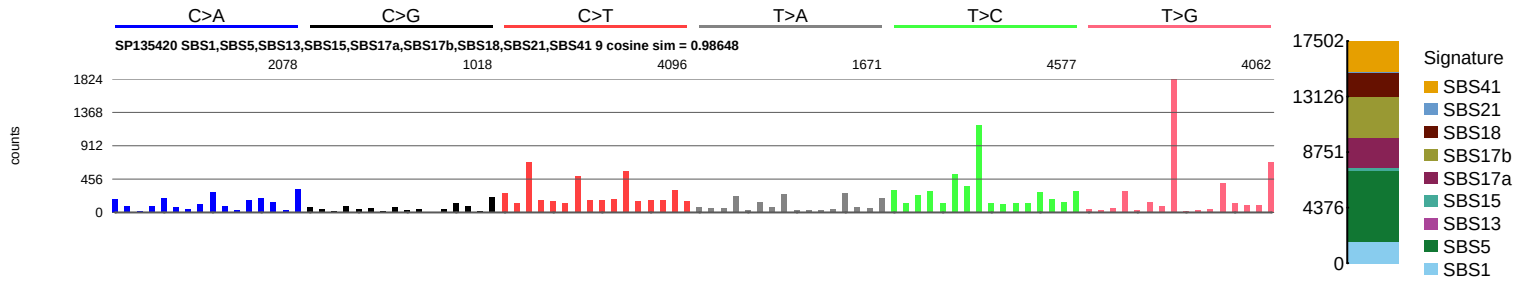
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98651
2197
1430
3830
1979
3960
4102
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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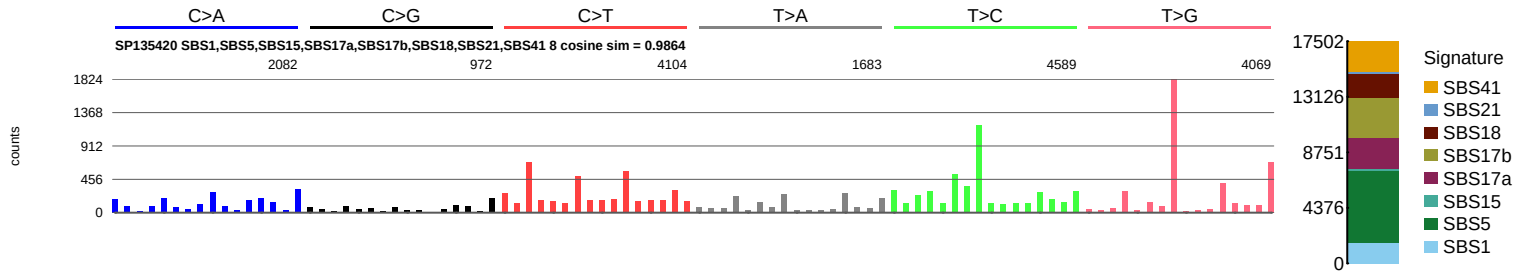
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98648
2078
1018
4096
1671
4577
4062
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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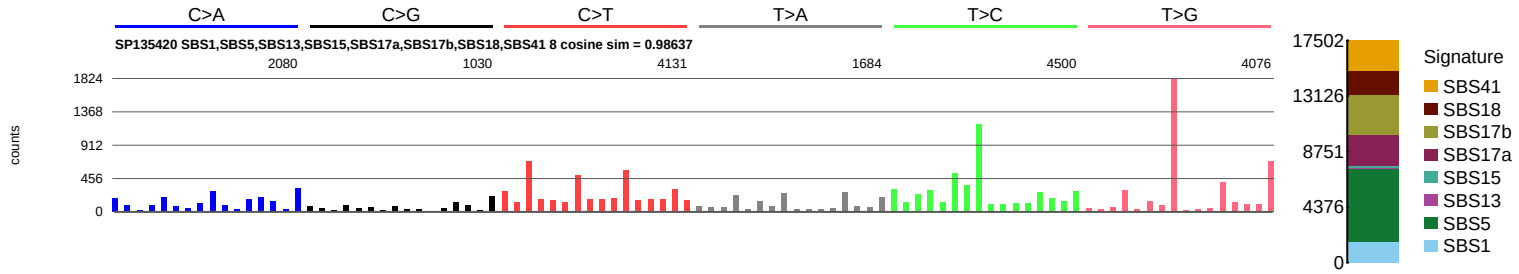
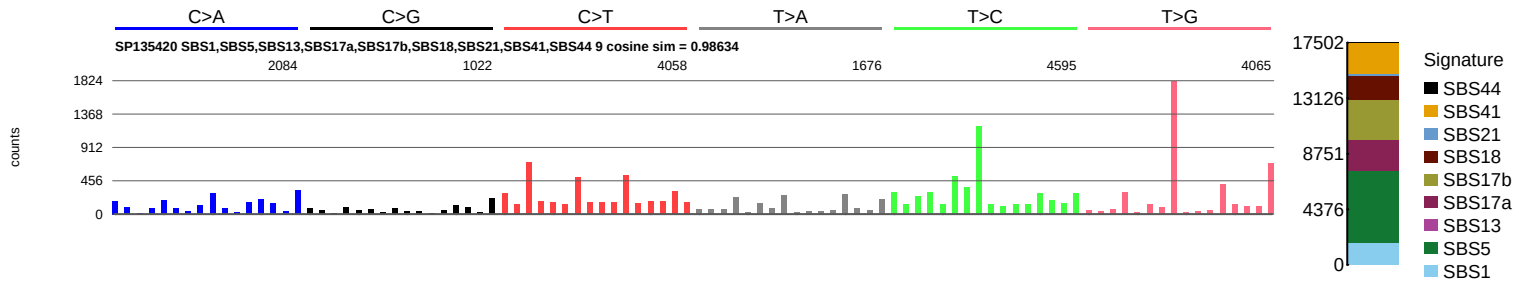
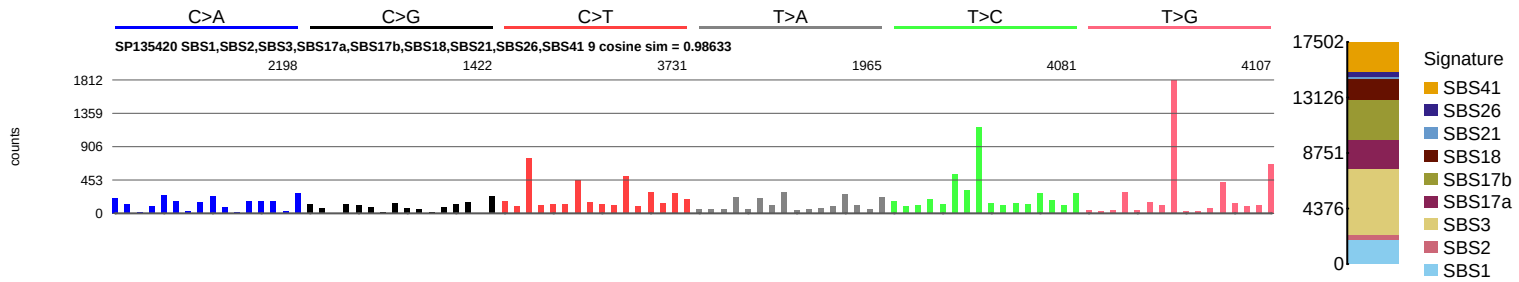
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9864
2082
972
4104
1683
4589
4069
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98637
2080
1030
4131
1684
4500
4076
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98634
2084
1022
4058
1676
4595
4065
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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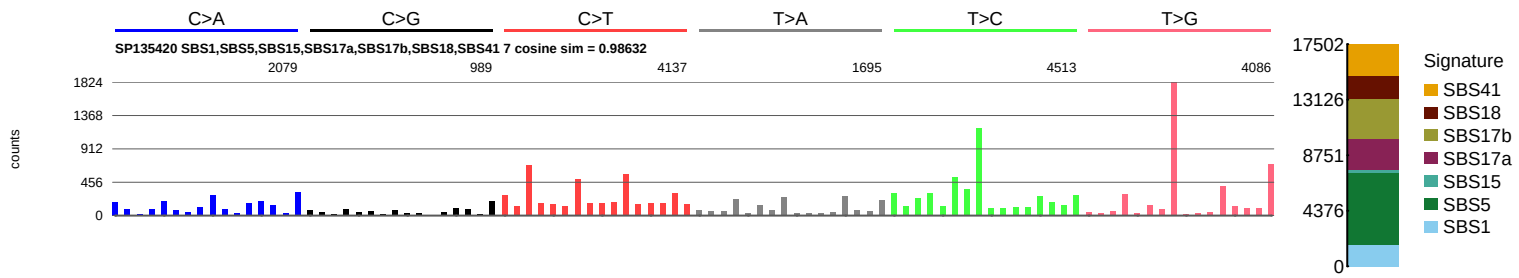
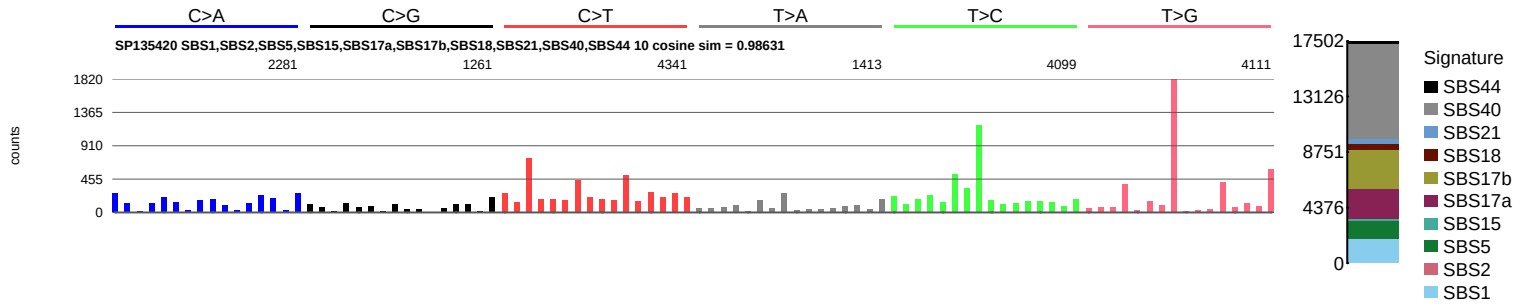
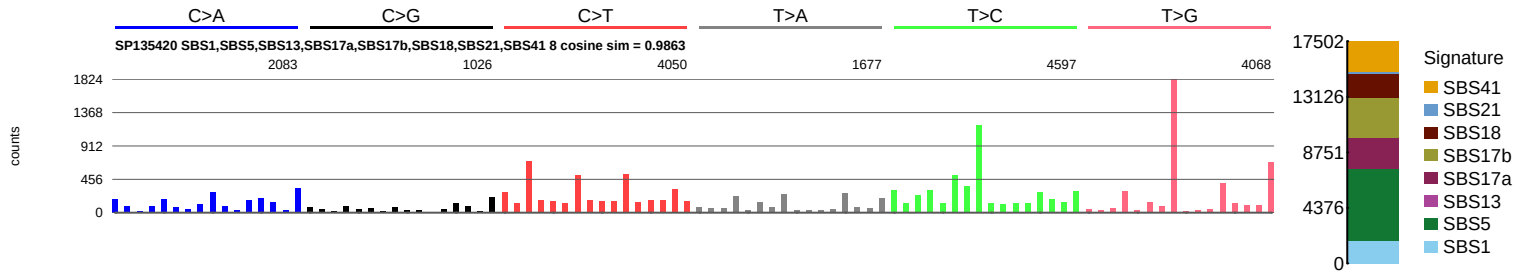
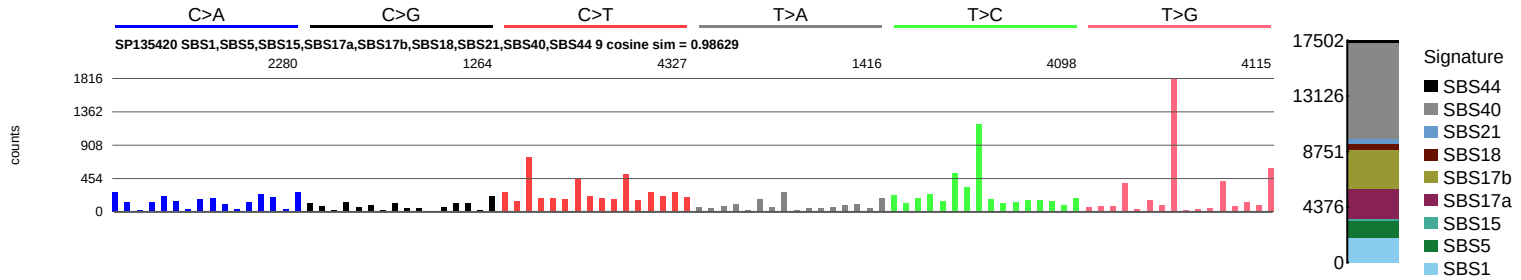
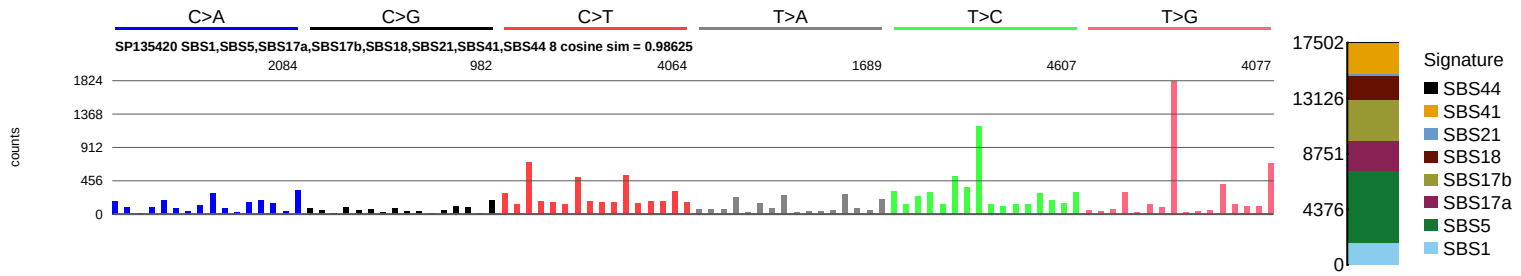
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98633
2198
1422
3731
1965
4081
4107
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

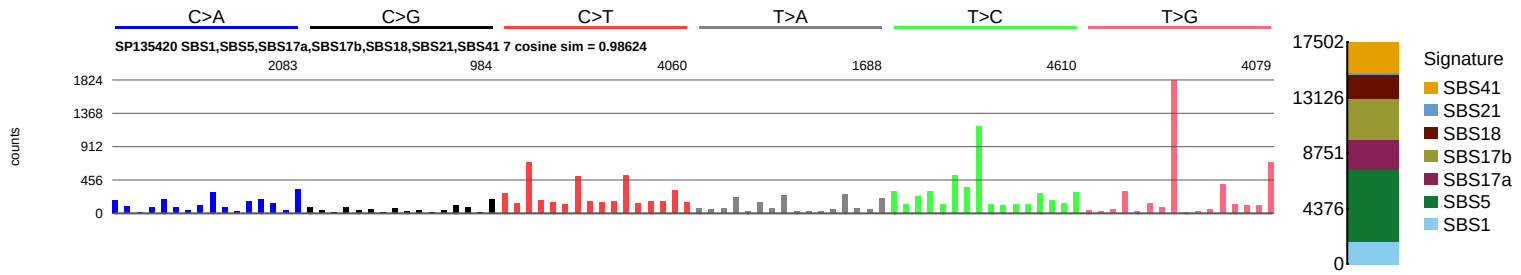

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98632
2079
989
4137
1695
4513
4086
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

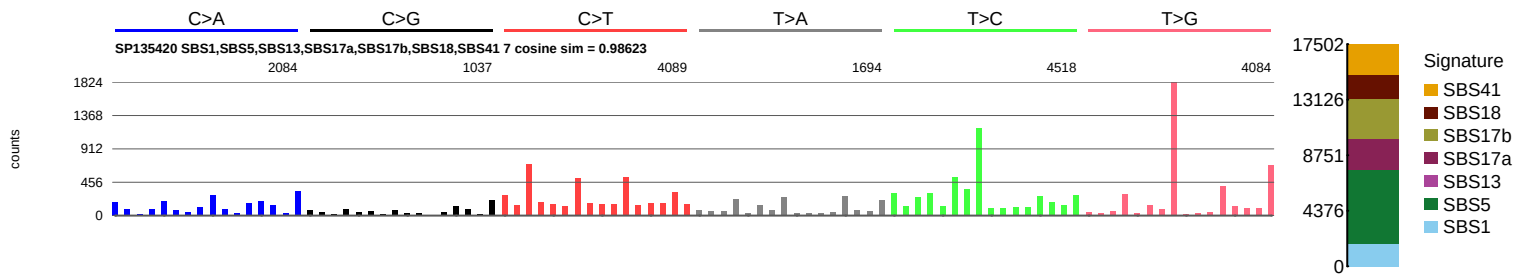

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98631
2281
1261
4341
1413
4099
4111
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

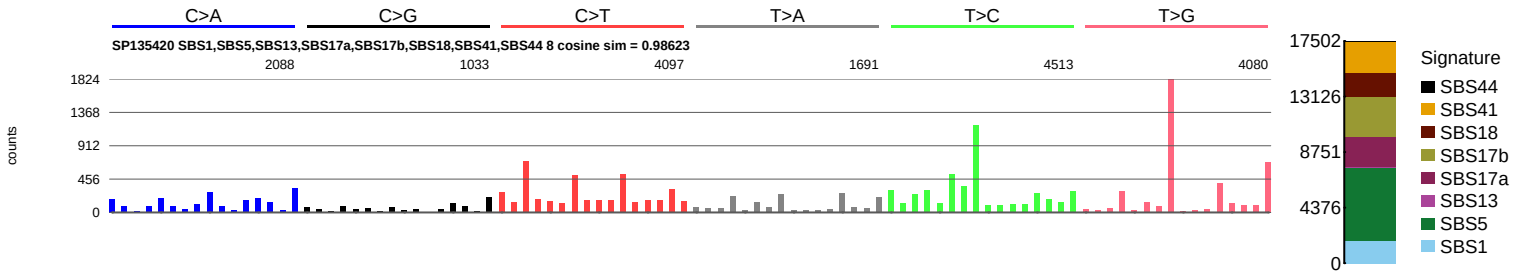

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9863
2083
1026
4050
1677
4597
4068
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98629
2280
1264
4327
1416
4098
4115
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98625
2084
982
4064
1689
4607
4077
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98624
2083
984
4060
1688
4610
4079
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98623
2084
1037
4089
1694
4518
4084
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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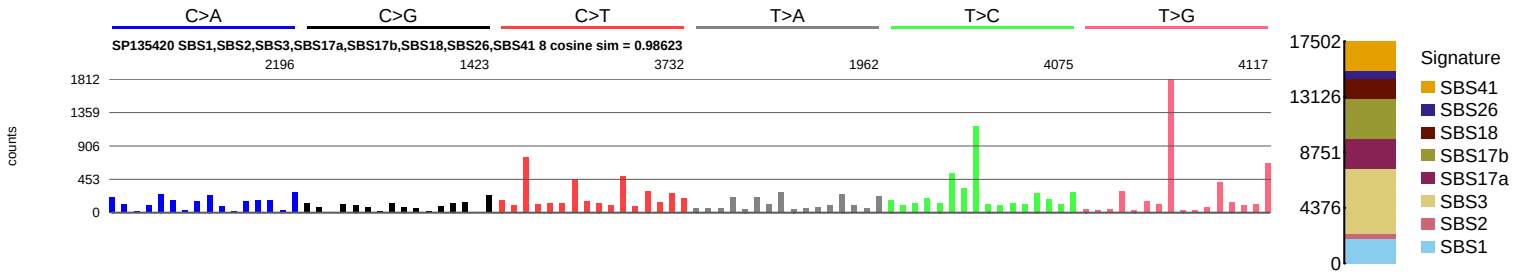
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98623
2088
1033
4097
1691
4513
4080
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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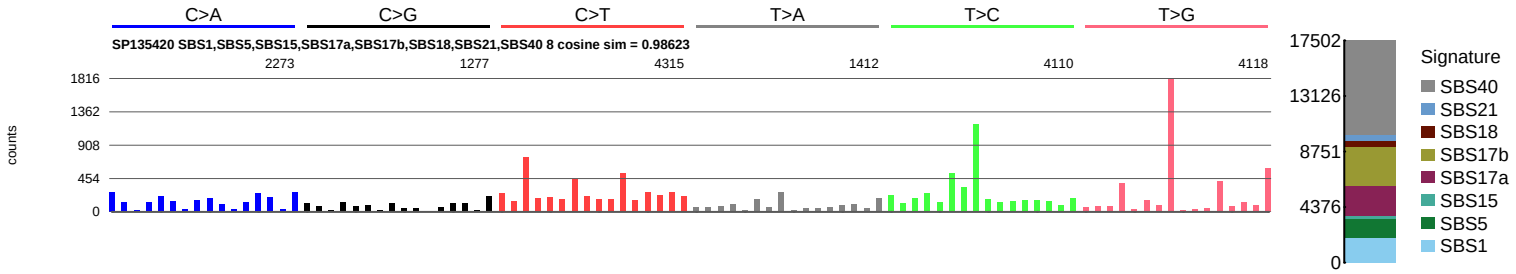
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98623
2196
1423
3732
1962
4075
4117
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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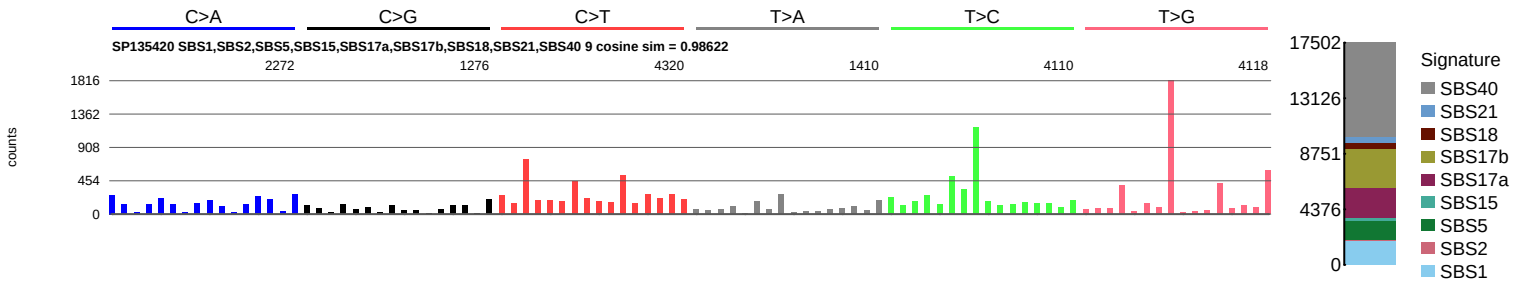
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98623
2273
1277
4315
1412
4110
4118
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98622
2272
1276
4320
1410
4110
4118
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
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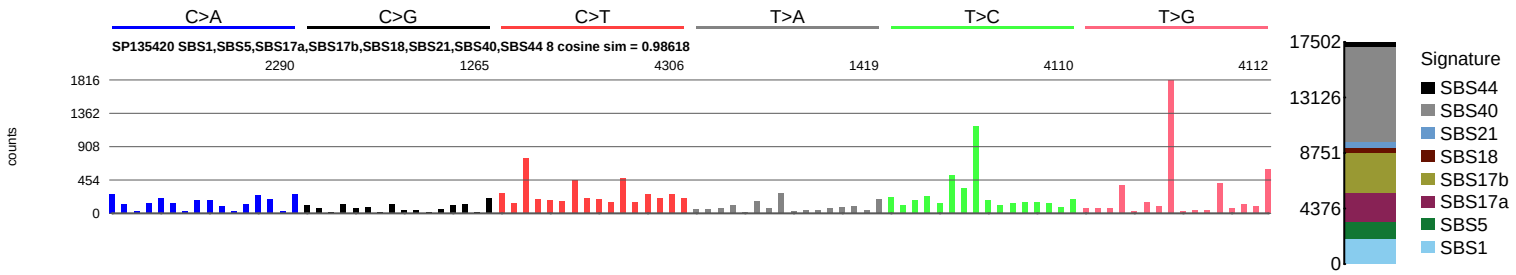
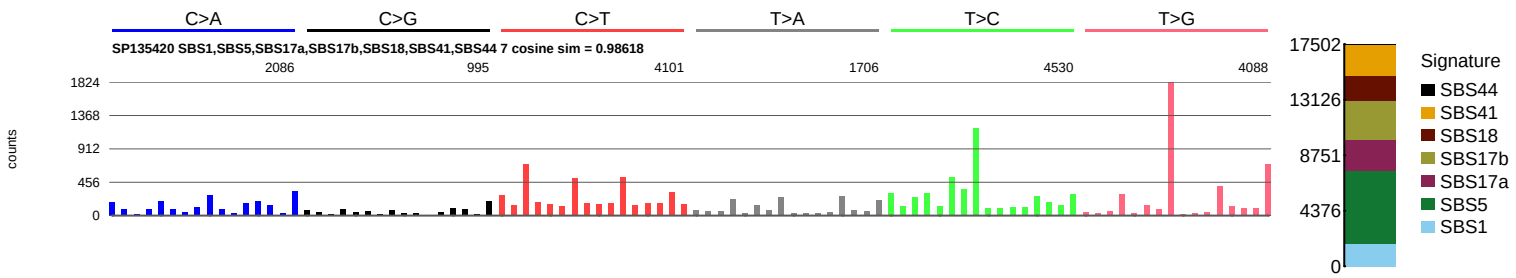
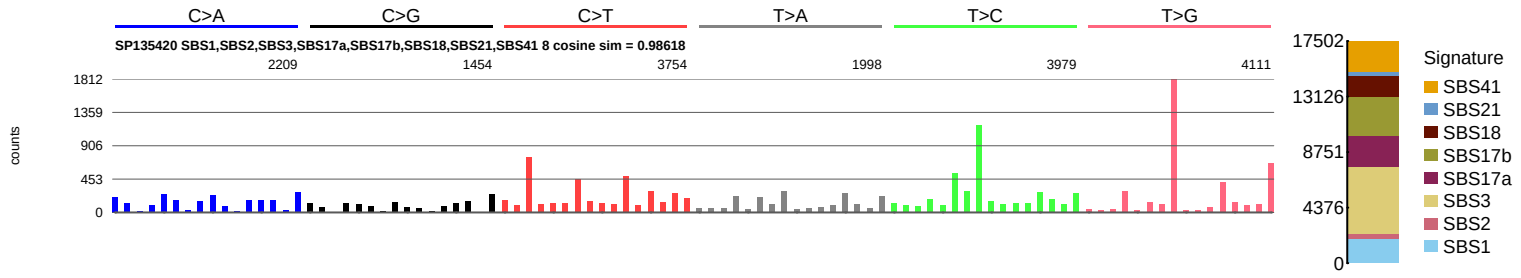
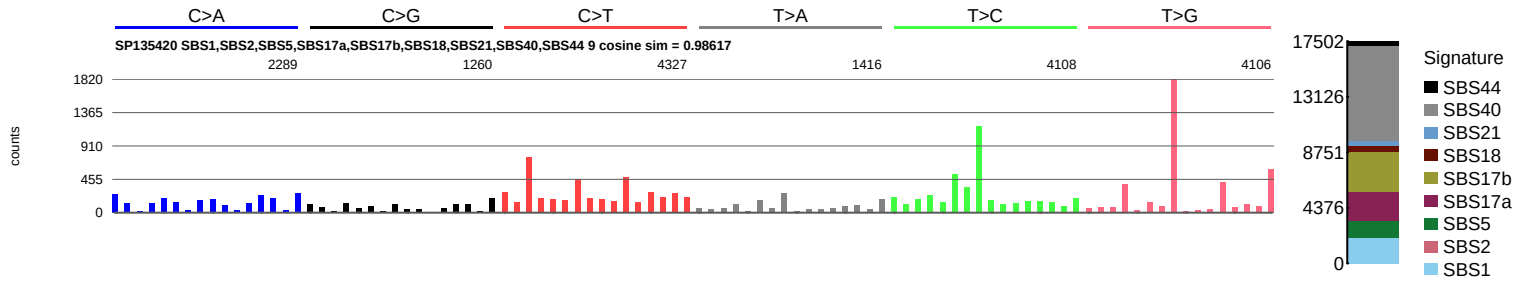
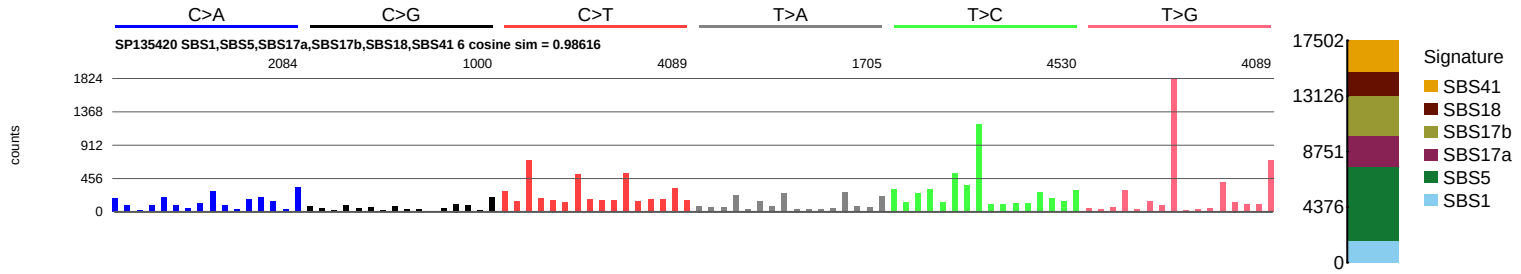
SBS2
SBS1

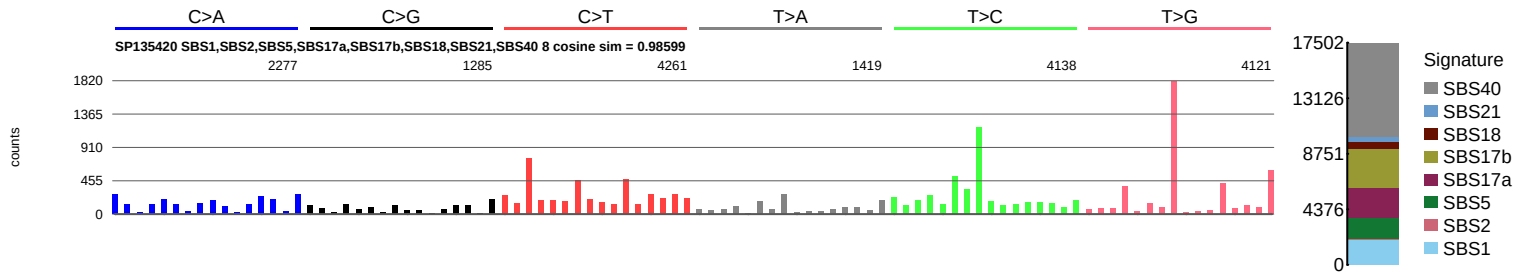

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98618
2290
1265
4306
1419
4110
4112
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98618
2086
995
4101
1706
4530
4088
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98618
2209
1454
3754
1998
3979
4111
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98617
2289
1260
4327
1416
4108
4106
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98616
2084
1000
4089
1705
4530
4089
counts
1824
1368
912
456
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98599
2277
1285
4261
1419
4138
4121
counts
1820
1365
910
455
0
17502
13126
8751
4376
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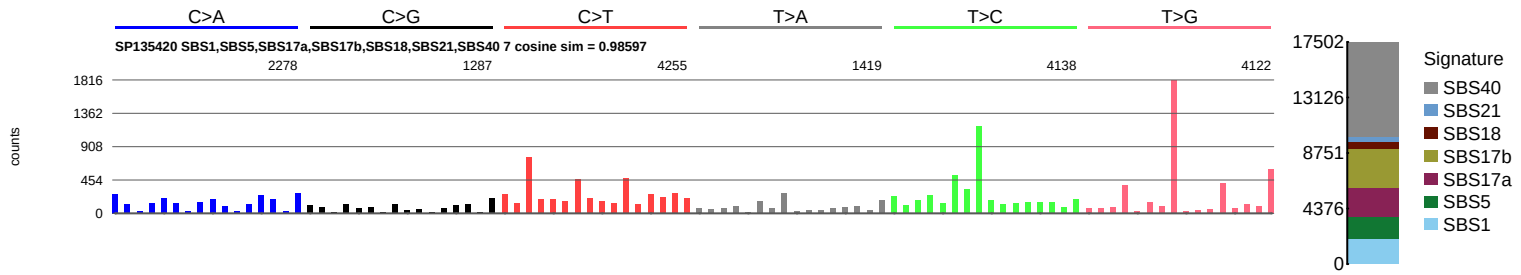
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98597
2278
1287
4255
1419
4138
4122
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
SBS40
SBS21
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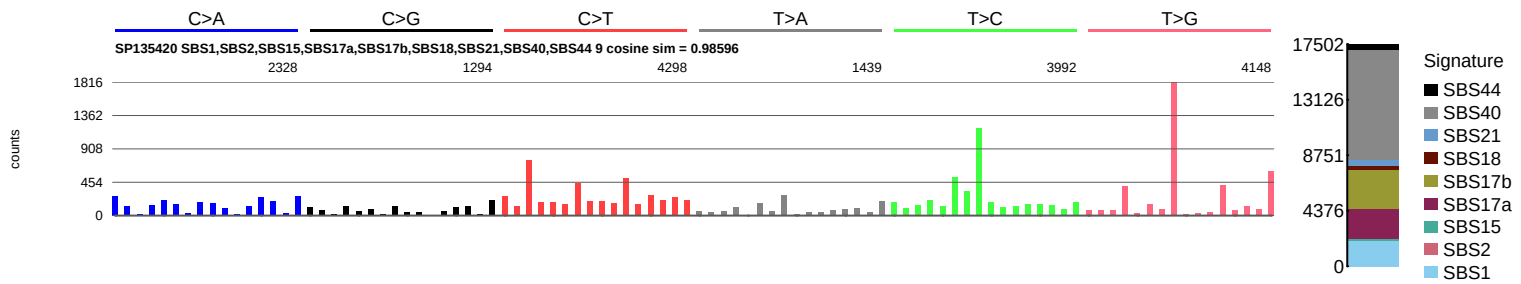
SBS17b
SBS17a
SBS5
SBS1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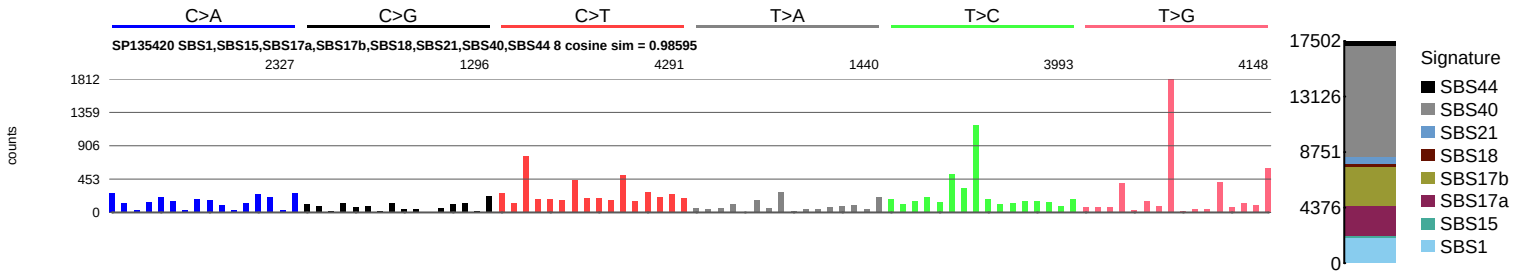
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98596
2328
1294
4298
1439
3992
4148
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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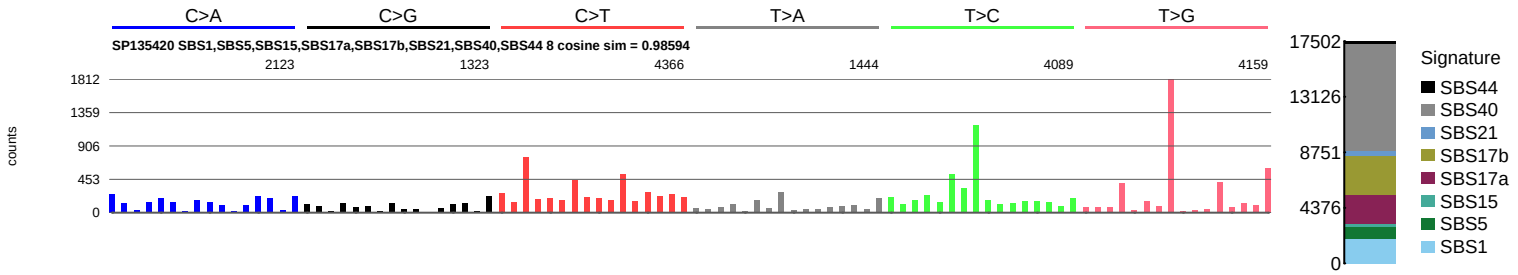
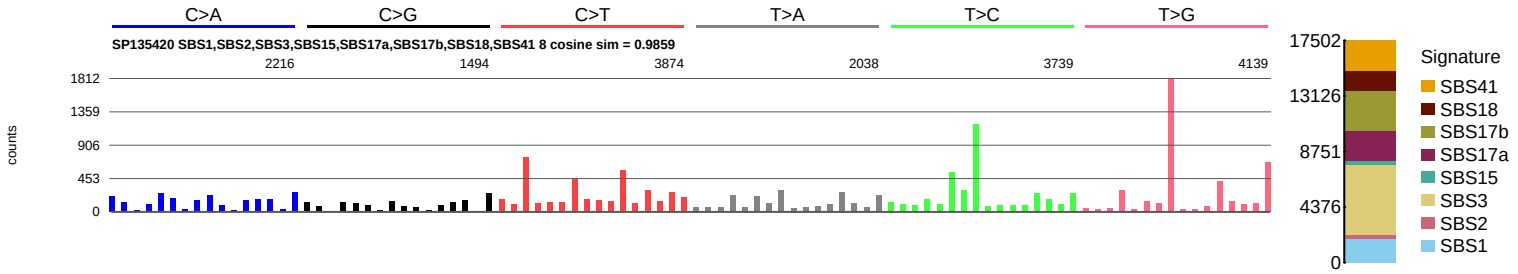
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98595
2327
1296
4291
1440
3993
4148
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98594
2123
1323
4366
1444
4089
4159
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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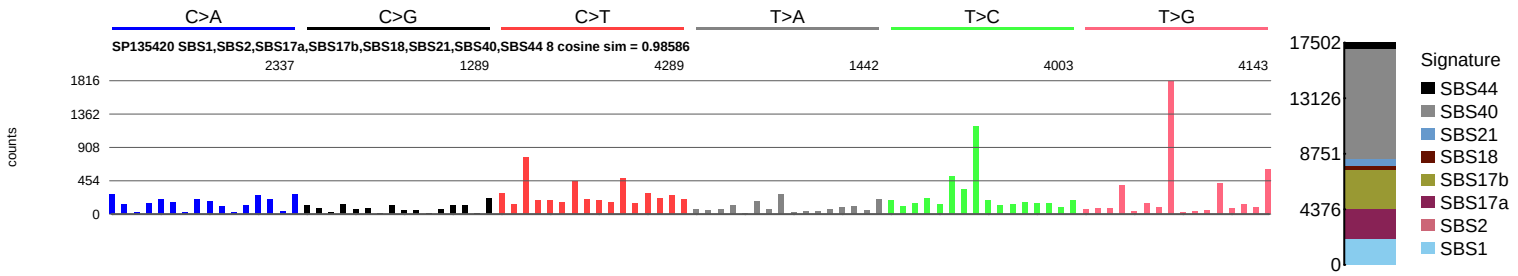
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9859
2216
1494
3874
2038
3739
4139
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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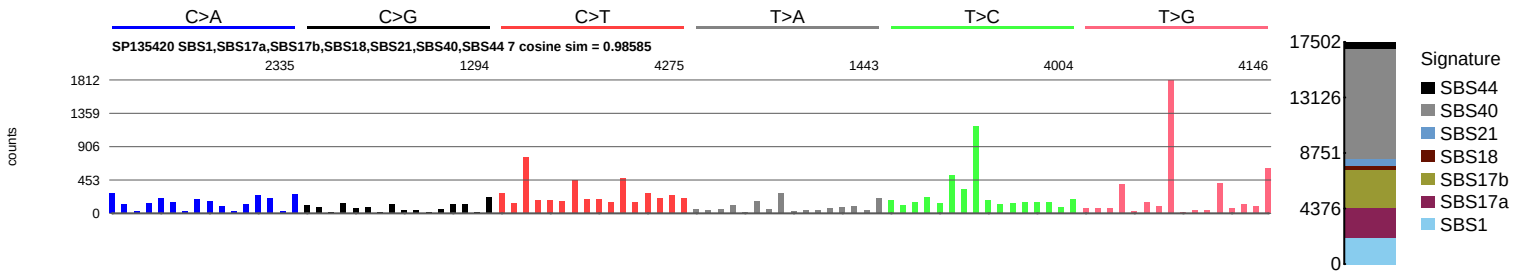
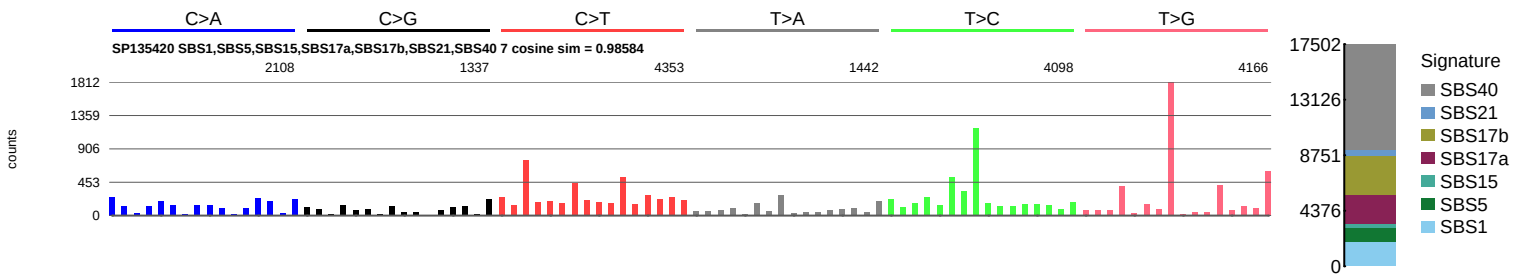
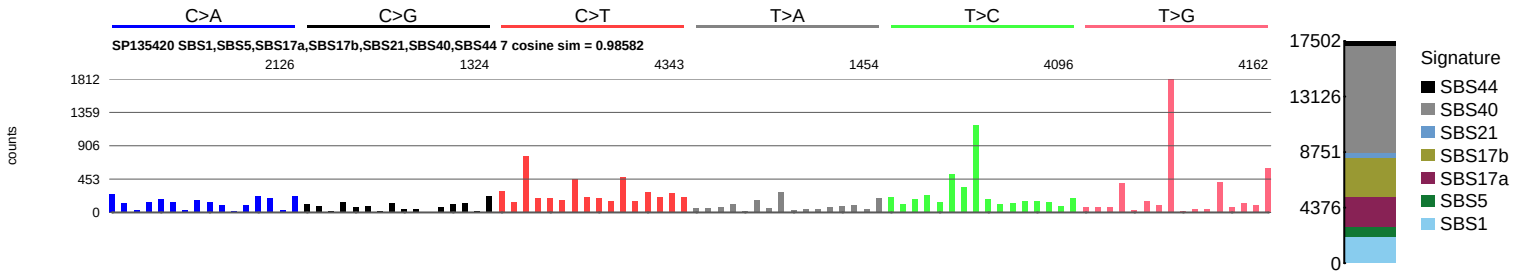
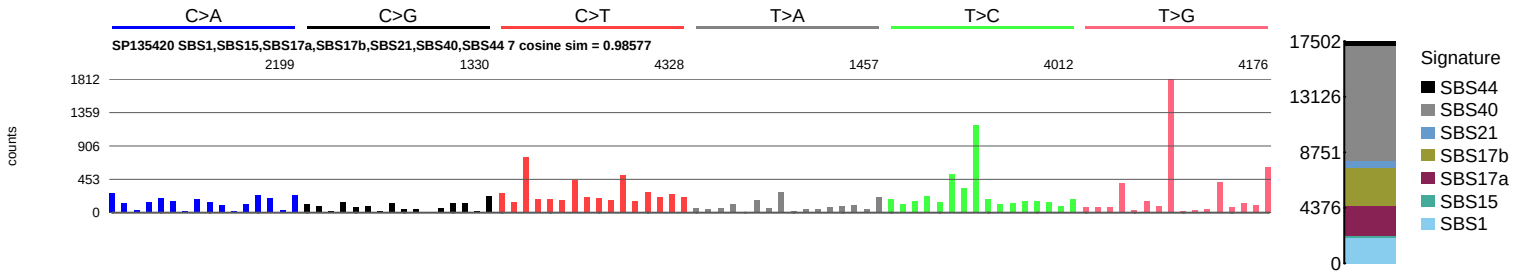
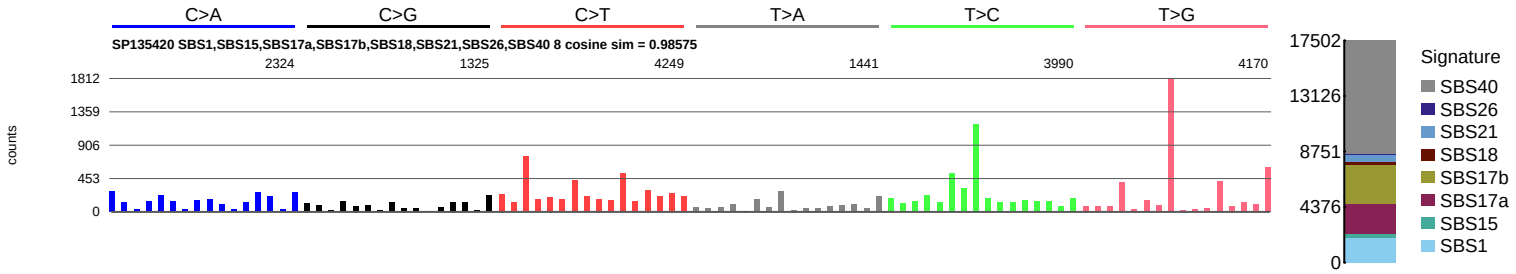
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98586
2337
1289
4289
1442
4003
4143
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

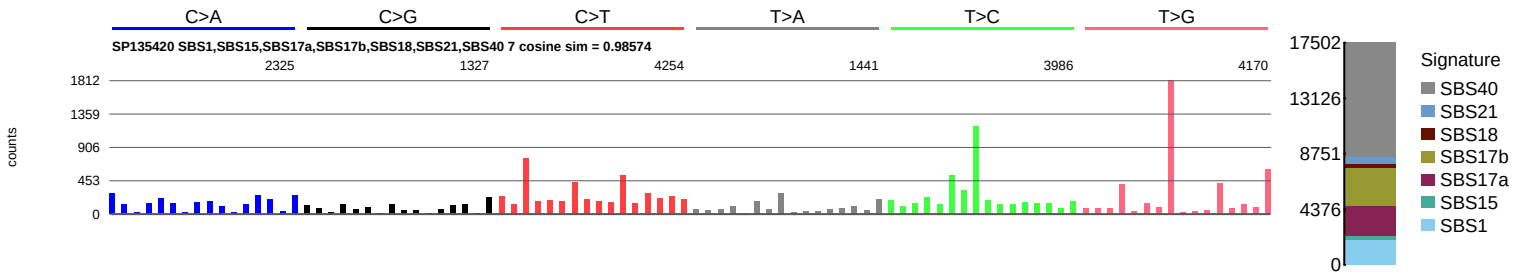

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98585
2335
1294
4275
1443
4004
4146
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

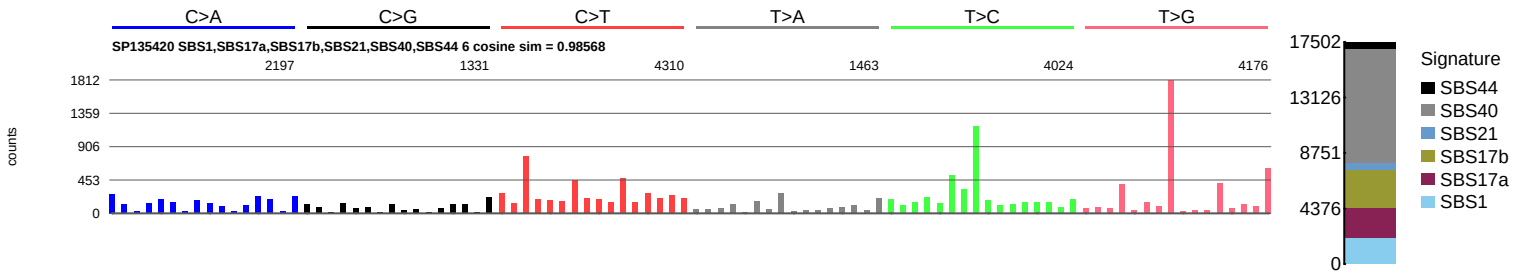

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98584
2108
1337
4353
1442
4098
4166
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

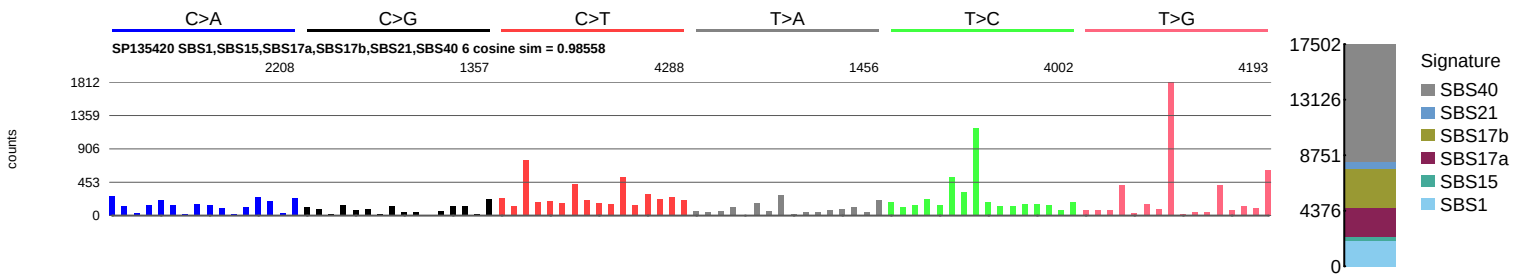

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98582
2126
1324
4343
1454
4096
4162
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

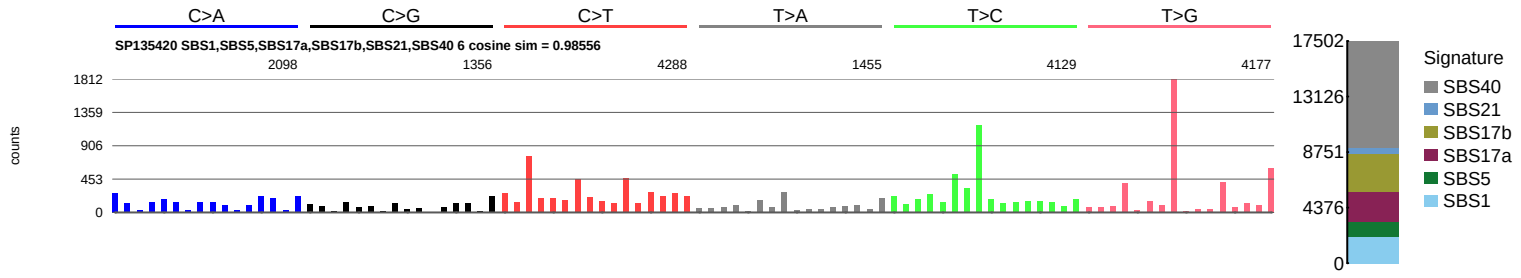

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98577
2199
1330
4328
1457
4012
4176
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

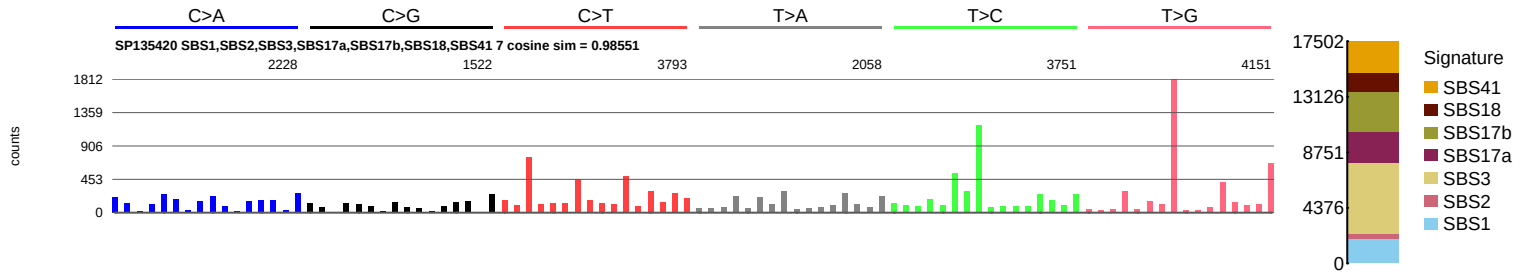

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98575
2324
1325
4249
1441
3990
4170
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

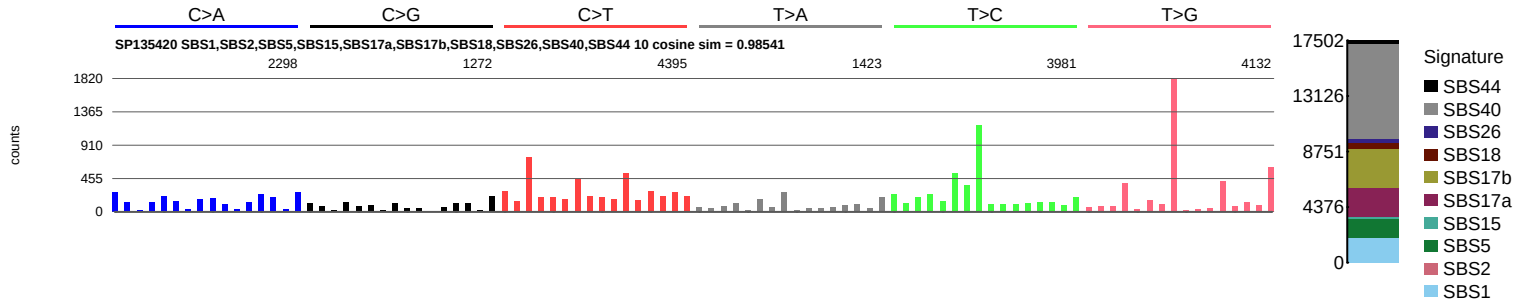

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98574
2325
1327
4254
1441
3986
4170
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98568
2197
1331
4310
1463
4024
4176
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98558
2208
1357
4288
1456
4002
4193
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98556
2098
1356
4288
1455
4129
4177
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98551
2228
1522
3793
2058
3751
4151
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98541
2298
1272
4395
1423
3981
4132
counts
1820
1365
910
455
0
17502
13126
8751
4376
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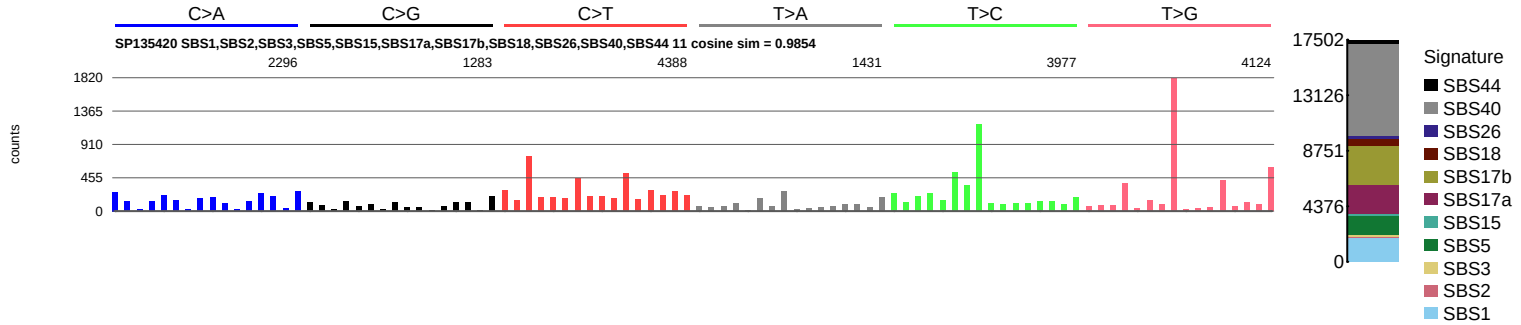
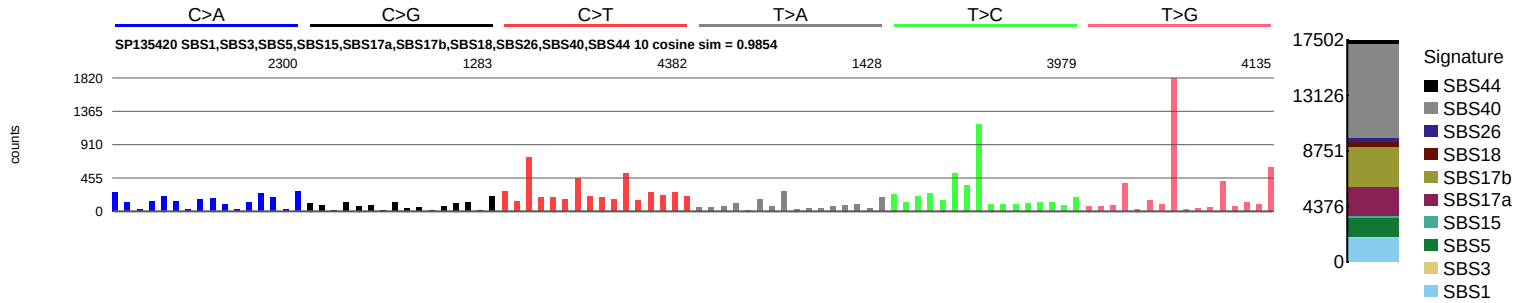
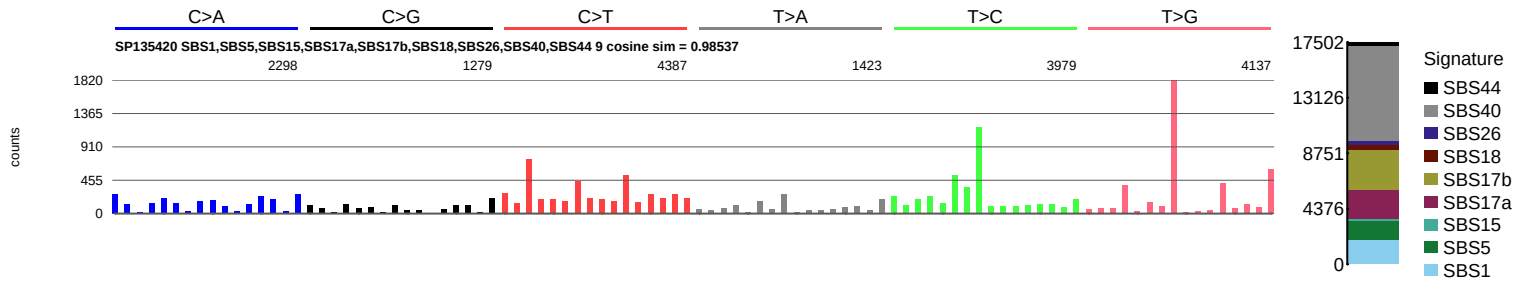
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.9854
2296
1283
4388
1431
3977
4124
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9854
2300
1283
4382
1428
3979
4135
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98537
2298
1279
4387
1423
3979
4137
counts
1820
1365
910
455
0
17502
13126
8751
4376
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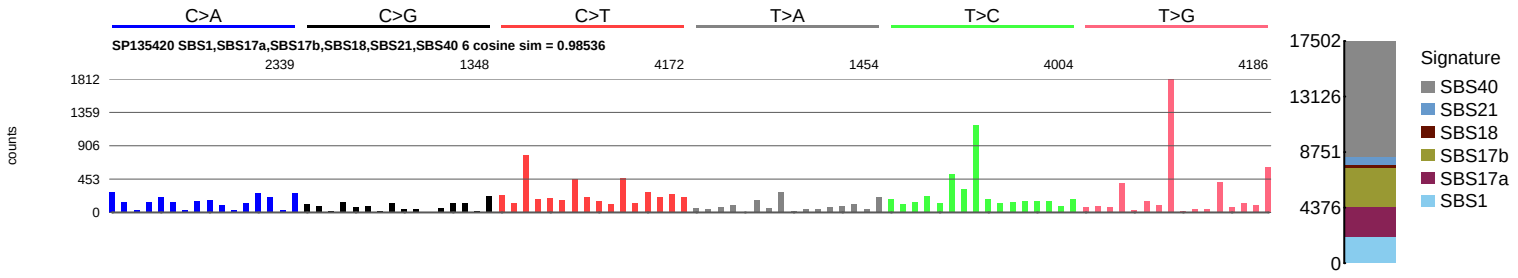
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98536
2339
1348
4172
1454
4004
4186
counts
1812
1359
906
453
0
17502
13126
8751
4376
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Signature
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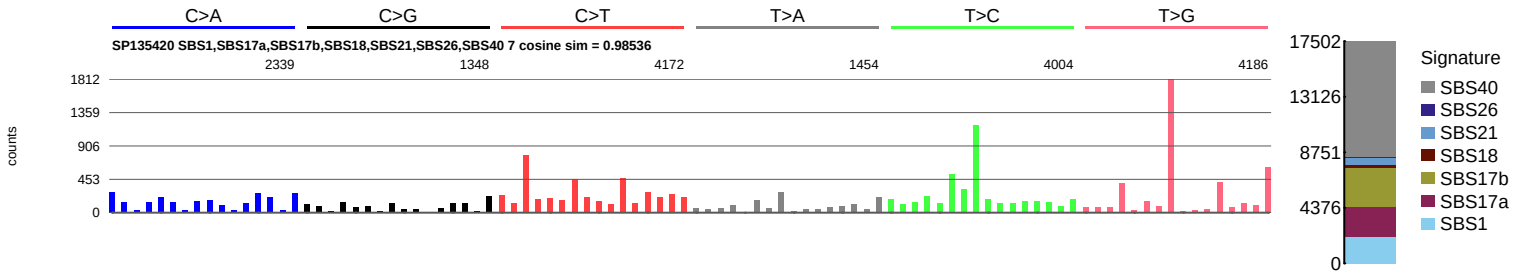
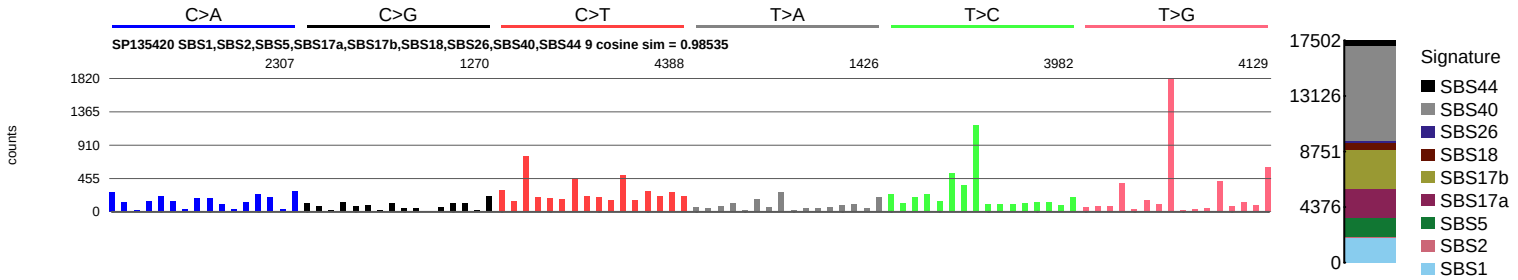
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98536
2339
1348
4172
1454
4004
4186
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98535
2307
1270
4388
1426
3982
4129
counts
1820
1365
910
455
0
17502
13126
8751
4376
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SBS40
SBS26
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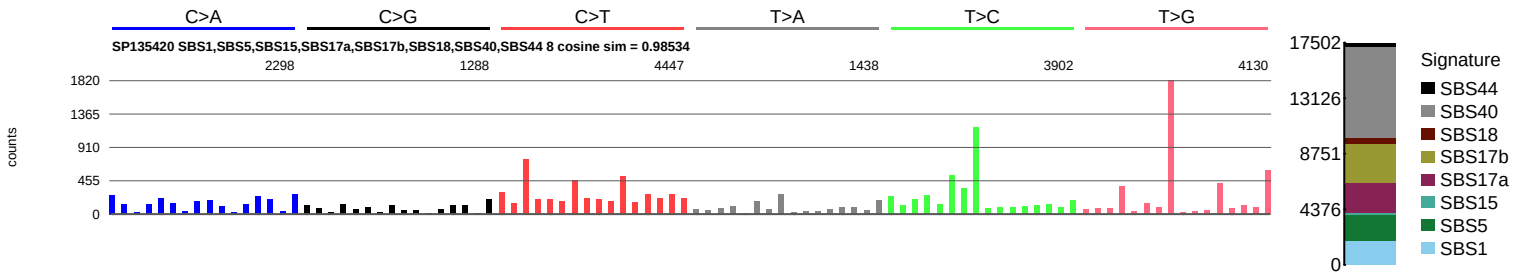
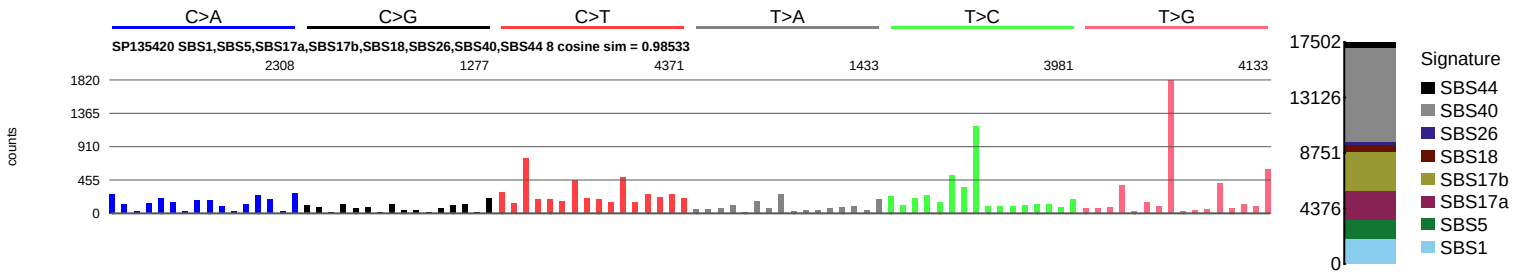
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98534
2298
1288
4447
1438
3902
4130
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98533
2308
1277
4371
1433
3981
4133
counts
1820
1365
910
455
0
17502
13126
8751
4376
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Signature
SBS44
SBS40
SBS26
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SBS17b
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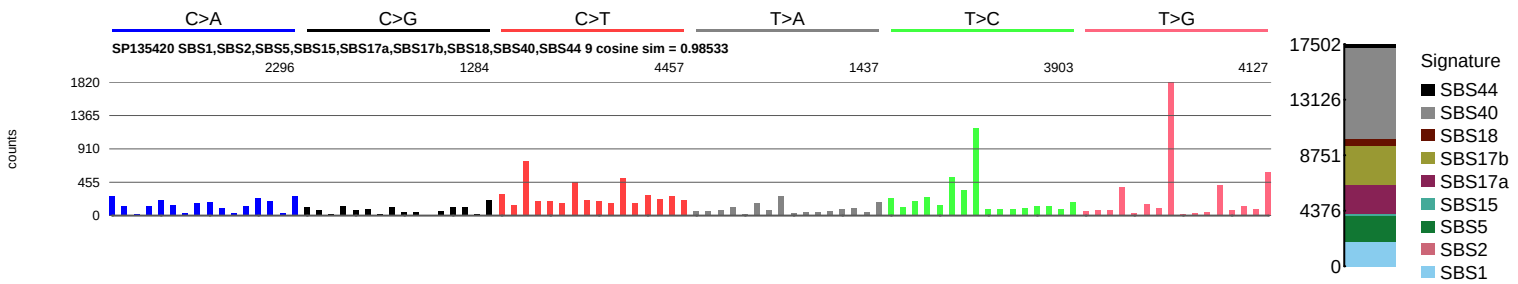
SBS5
SBS1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SP135420 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98533
2296
1284
4457
1437
3903
4127
counts
1820
1365
910
455
0
17502
13126
8751
4376
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SBS17a
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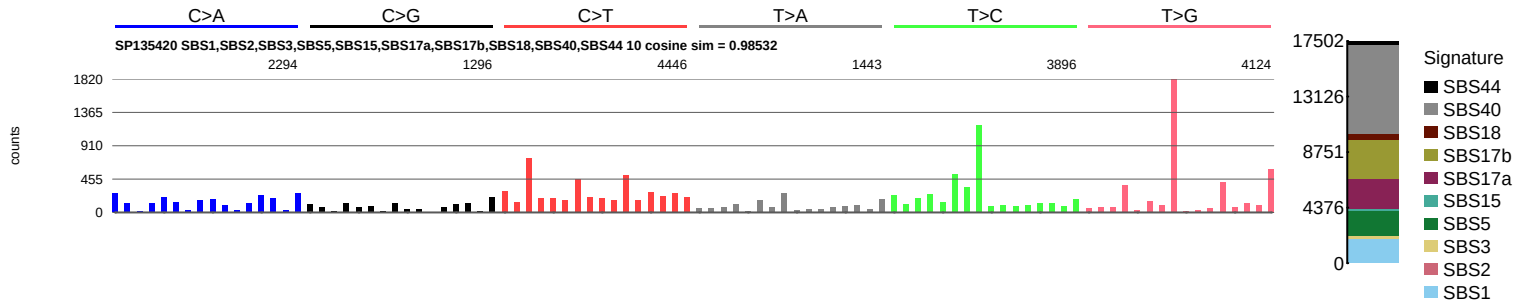
SBS1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2294
1296
4446
1443
3896
4124
counts
1820
1365
910
455
0
17502
13126
8751
4376
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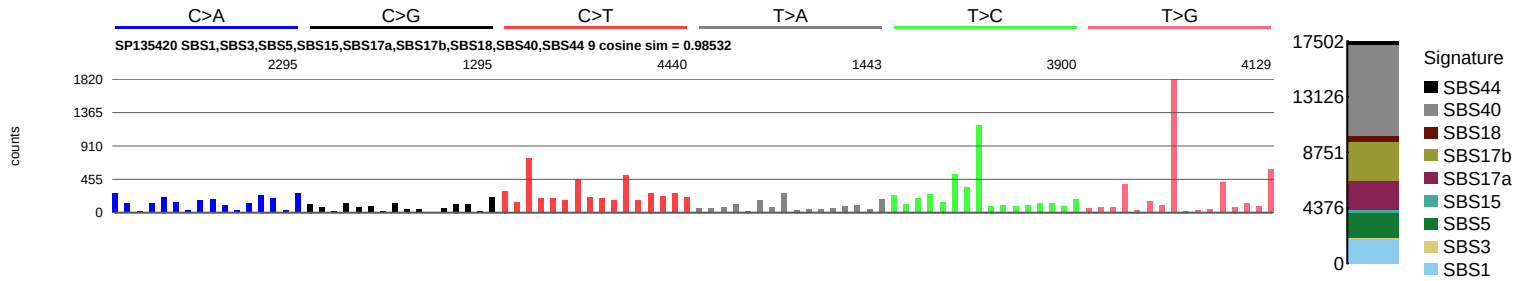
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98532
2295
1295
4440
1443
3900
4129
counts
1820
1365
910
455
0
17502
13126
8751
4376
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SBS17a
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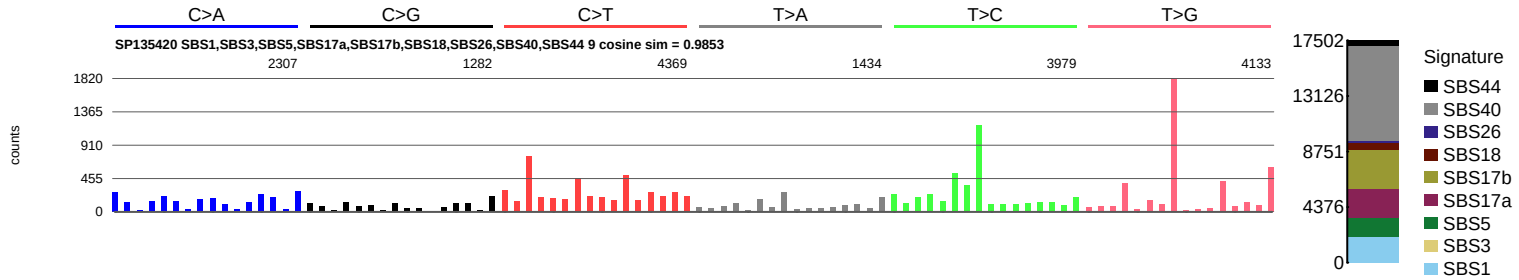
SBS5
SBS3
SBS1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SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9853
2307
1282
4369
1434
3979
4133
counts
1820
1365
910
455
0
17502
13126
8751
4376
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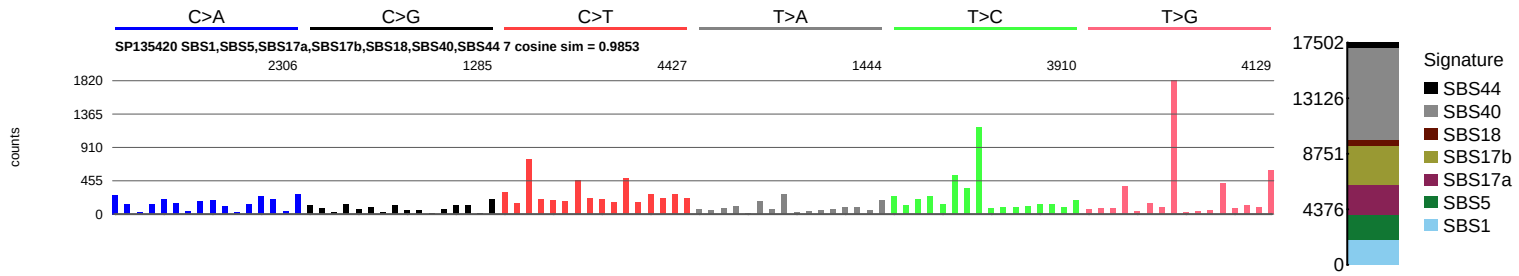
SBS3
SBS1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T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9853
2306
1285
4427
1444
3910
4129
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
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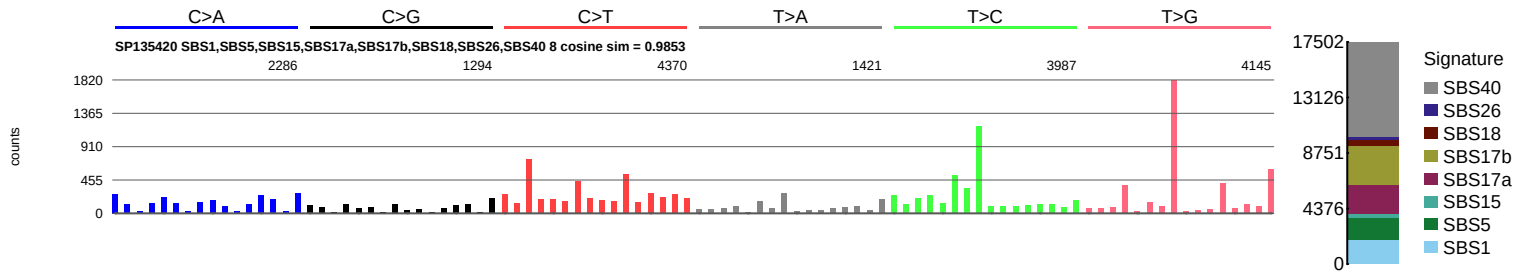
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>T
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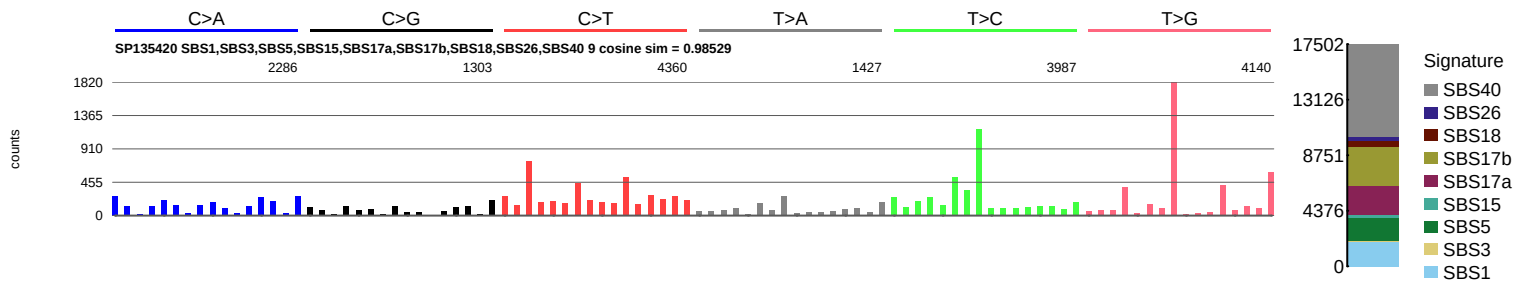
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9853
2286
1294
4370
1421
3987
4145
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98529
2286
1303
4360
1427
3987
4140
counts
1820
1365
910
455
0
17502
13126
8751
4376
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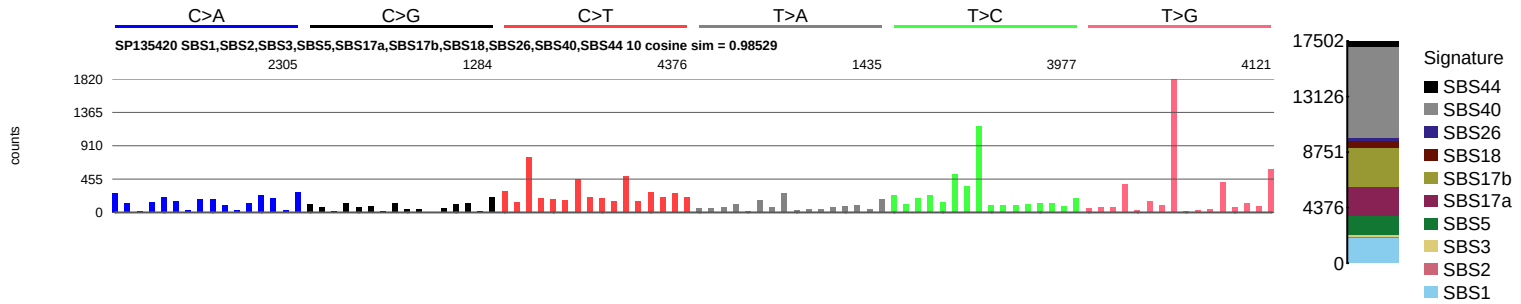
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98529
2305
1284
4376
1435
3977
4121
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
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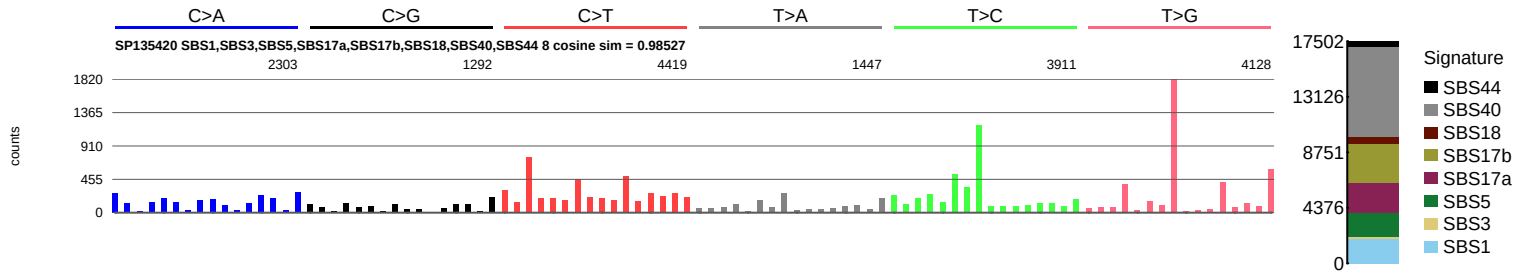
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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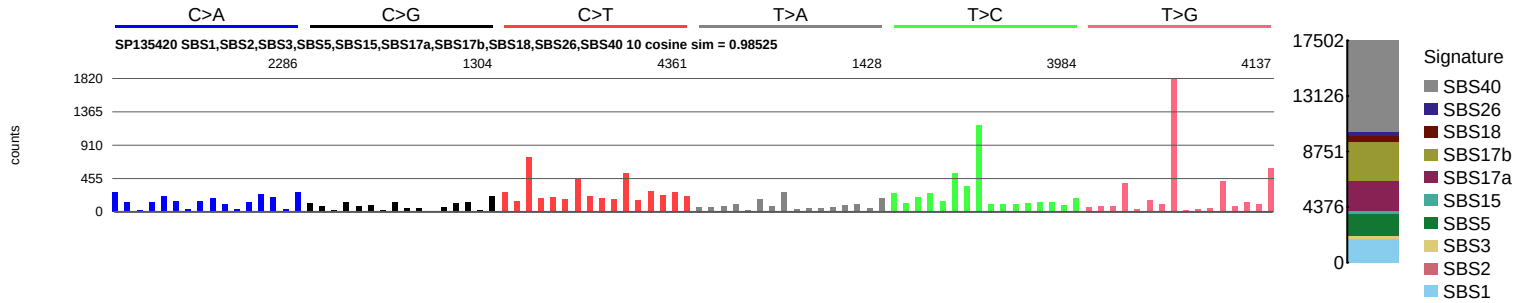
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98527
2303
1292
4419
1447
3911
4128
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
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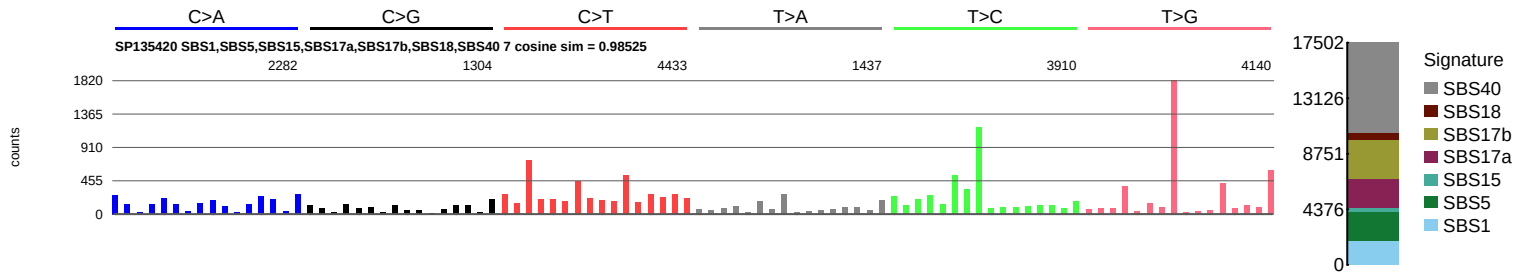
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98525
2286
1304
4361
1428
3984
4137
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98525
2282
1304
4433
1437
3910
4140
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
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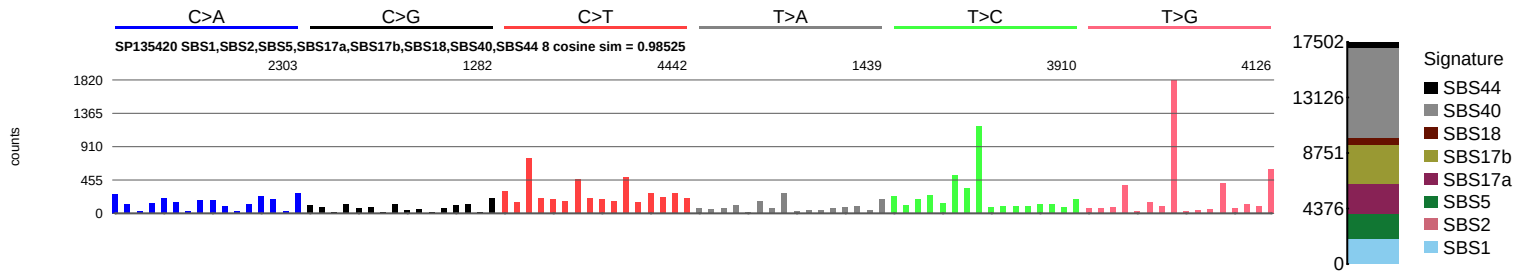
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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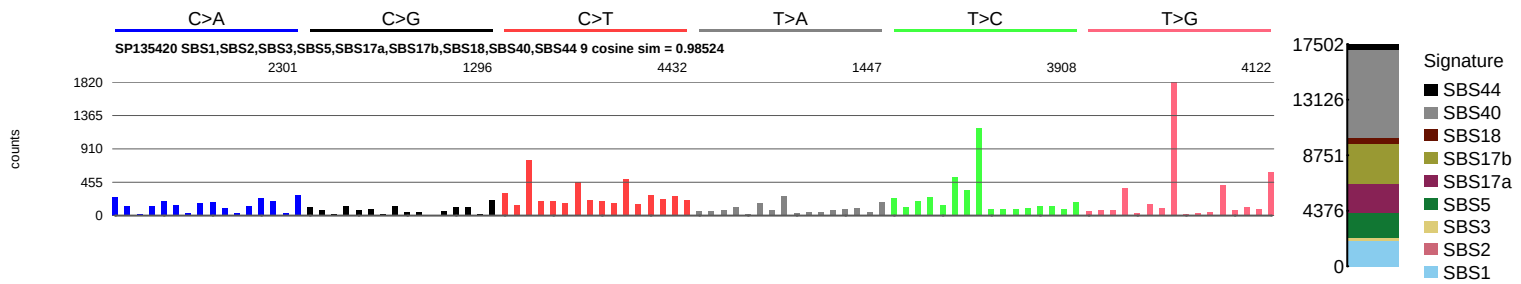
T>C
T>G
SP135420 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98525
2303
1282
4442
1439
3910
4126
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98524
2301
1296
4432
1447
3908
4122
counts
1820
1365
910
455
0
17502
13126
8751
4376
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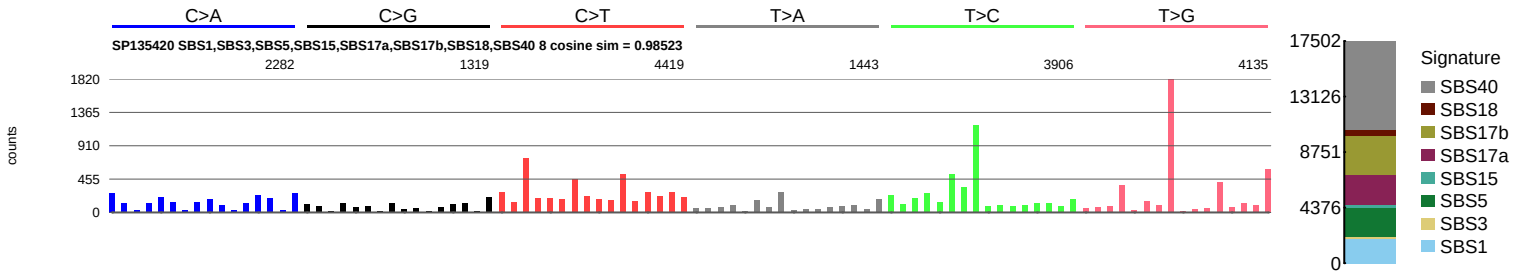
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98523
2282
1319
4419
1443
3906
4135
counts
1820
1365
910
455
0
17502
13126
8751
4376
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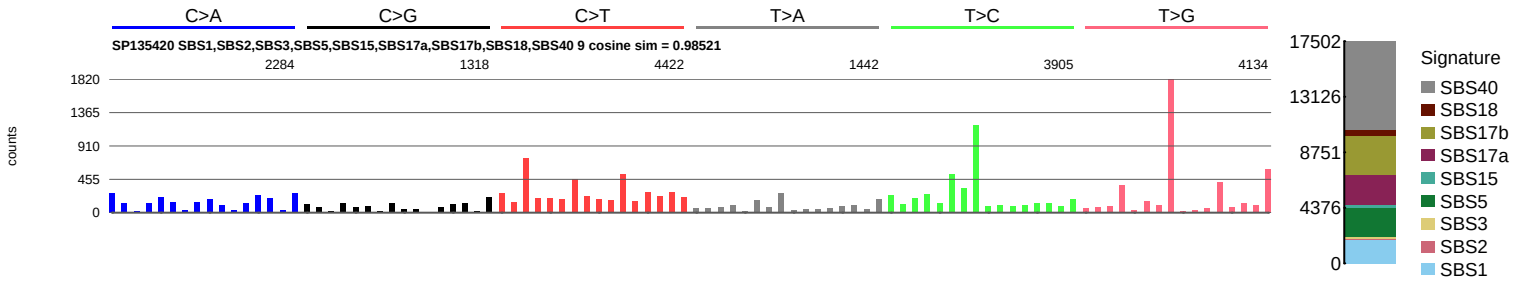
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98521
2284
1318
4422
1442
3905
4134
counts
1820
1365
910
455
0
17502
13126
8751
4376
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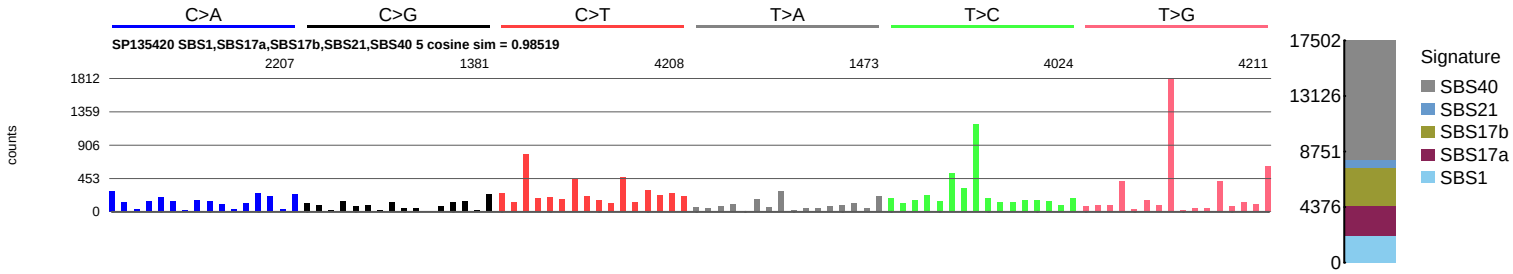
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98519
2207
1381
4208
1473
4024
4211
counts
1812
1359
906
453
0
17502
13126
8751
4376
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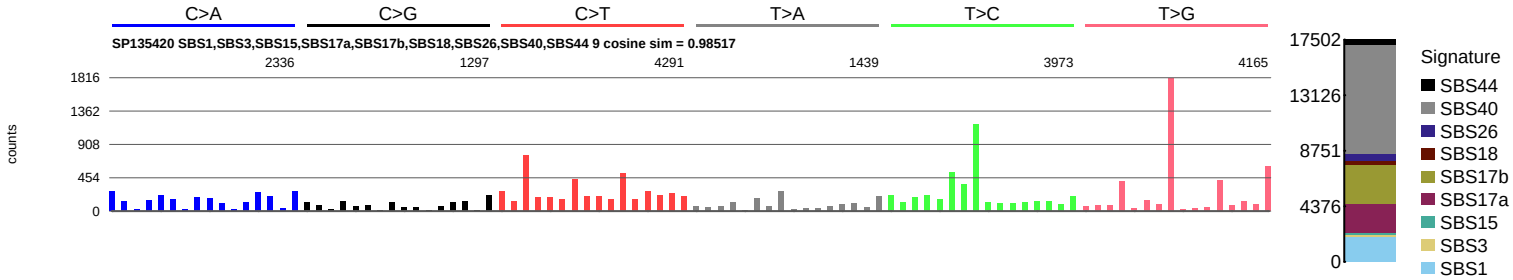
SBS21
SBS17b
SBS17a
SBS1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T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98517
2336
1297
4291
1439
3973
4165
counts
1816
1362
908
454
0
17502
13126
8751
4376
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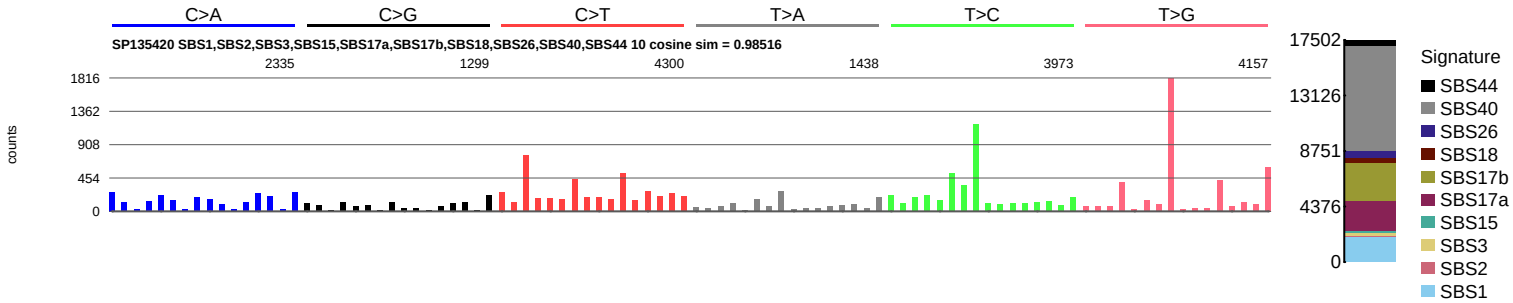
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98516
2335
1299
4300
1438
3973
4157
counts
1816
1362
908
454
0
17502
13126
8751
4376
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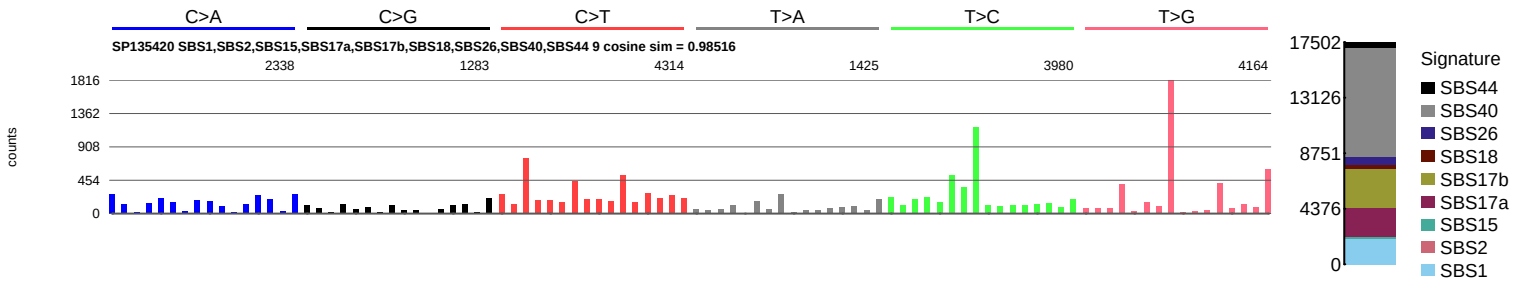
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98516
2338
1283
4314
1425
3980
4164
counts
1816
1362
908
454
0
17502
13126
8751
4376
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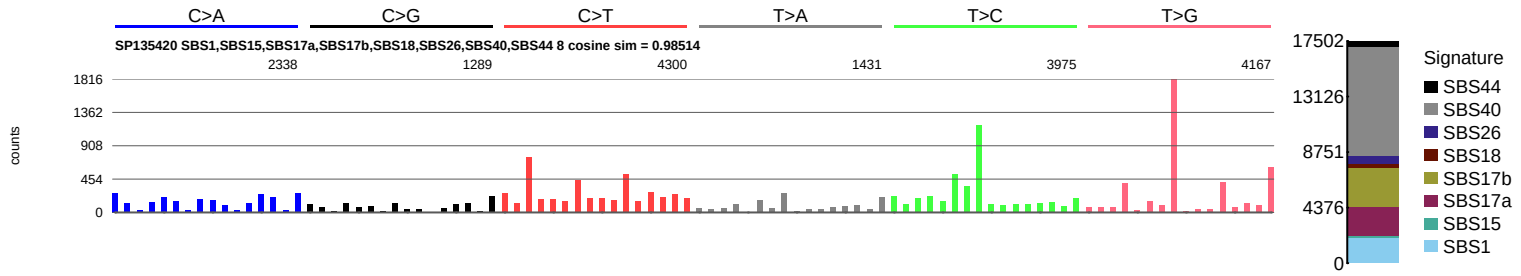
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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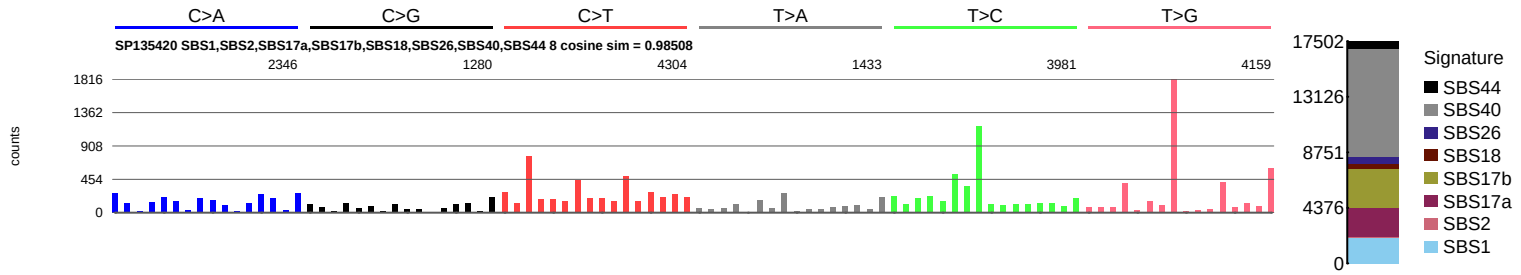
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98514
2338
1289
4300
1431
3975
4167
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98508
2346
1280
4304
1433
3981
4159
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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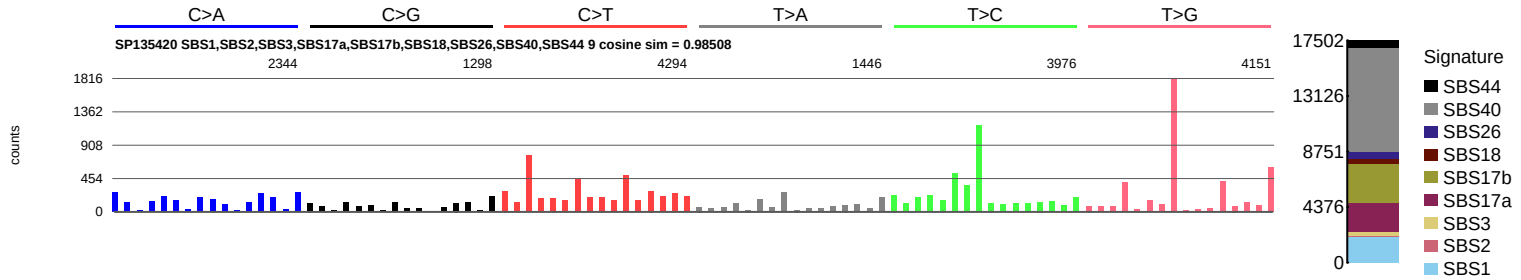
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98508
2344
1298
4294
1446
3976
4151
counts
1816
1362
908
454
0
17502
13126
8751
4376
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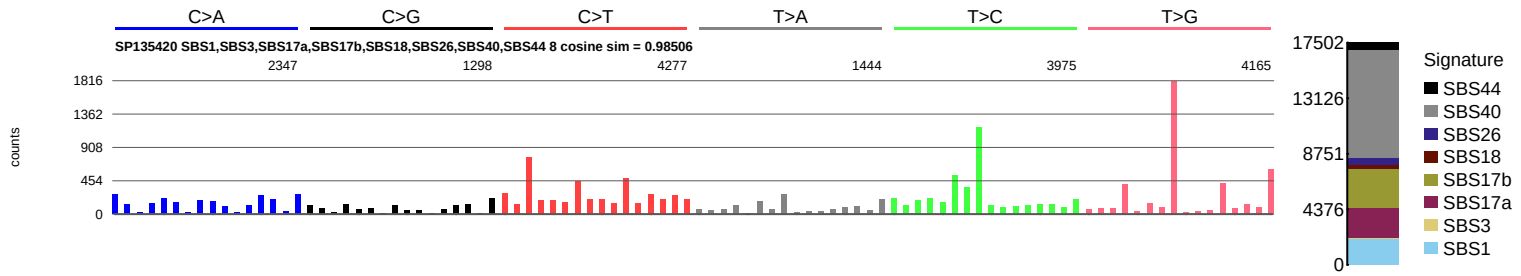
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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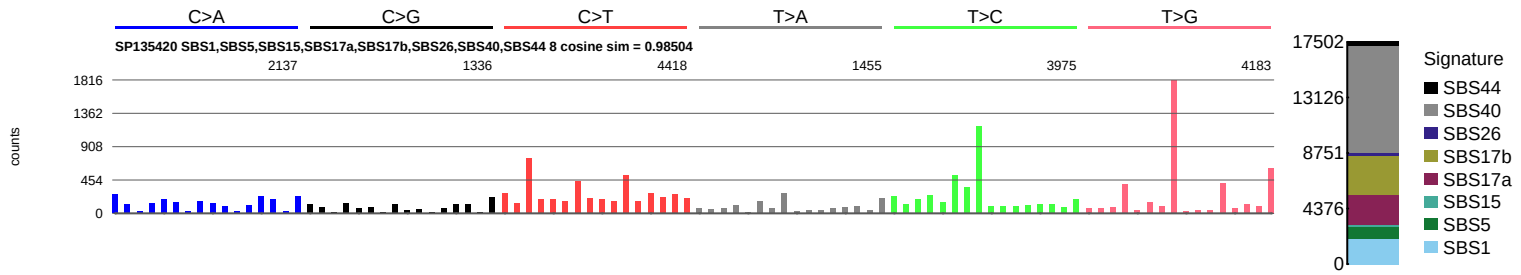
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98506
2347
1298
4277
1444
3975
4165
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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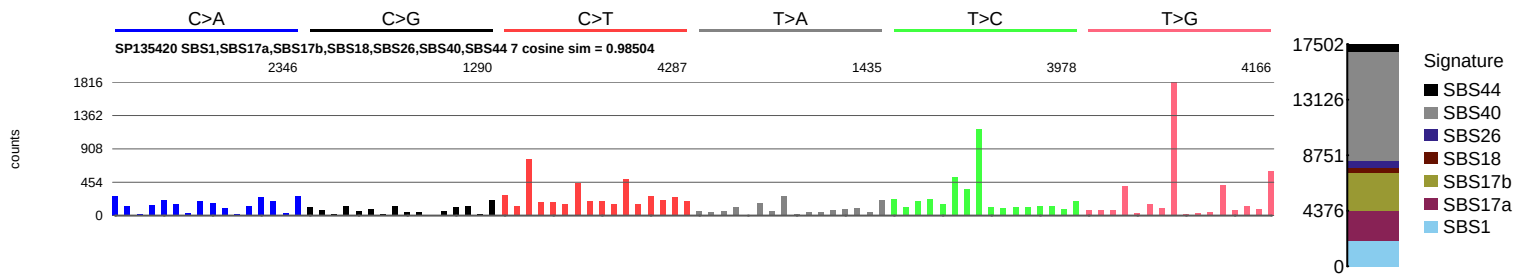
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98504
2137
1336
4418
1455
3975
4183
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1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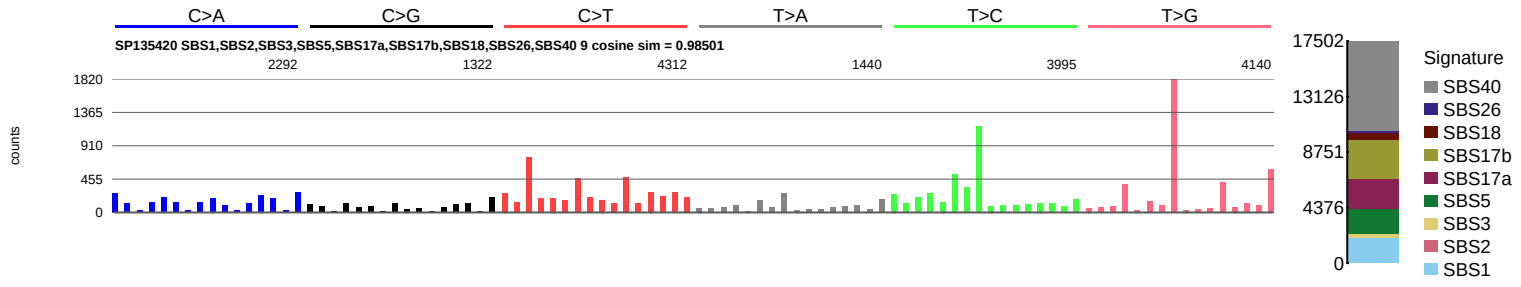
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98504
2346
1290
4287
1435
3978
4166
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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2292
1322
4312
1440
3995
4140
counts
1820
1365
910
455
0
17502
13126
8751
4376
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SBS40
SBS26
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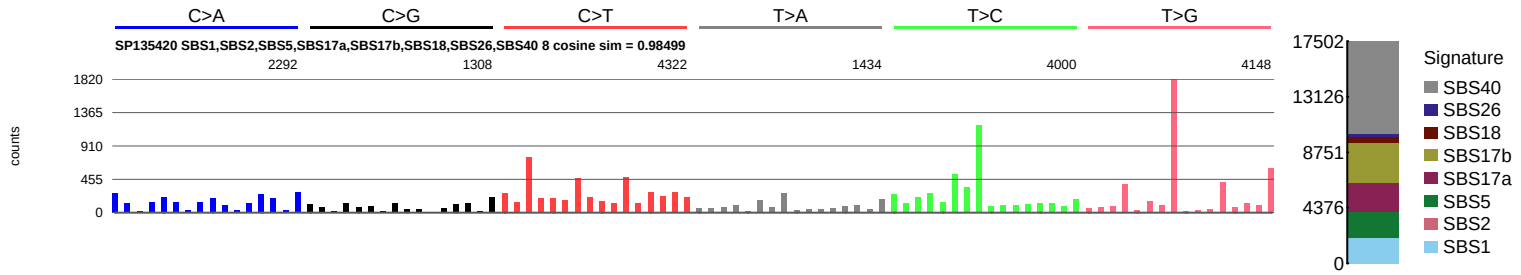
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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2292
1308
4322
1434
4000
4148
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
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SBS40
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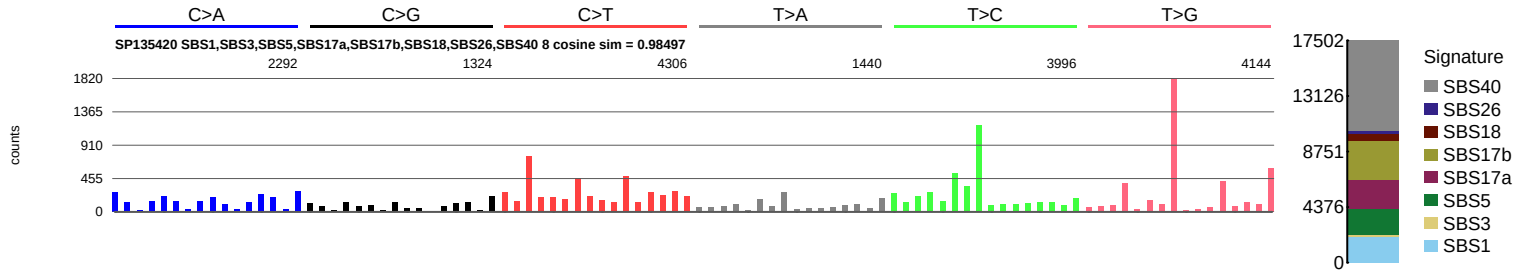
SBS17b
SBS17a
SBS5
SBS2
SBS1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SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98497
2292
1324
4306
1440
3996
4144
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
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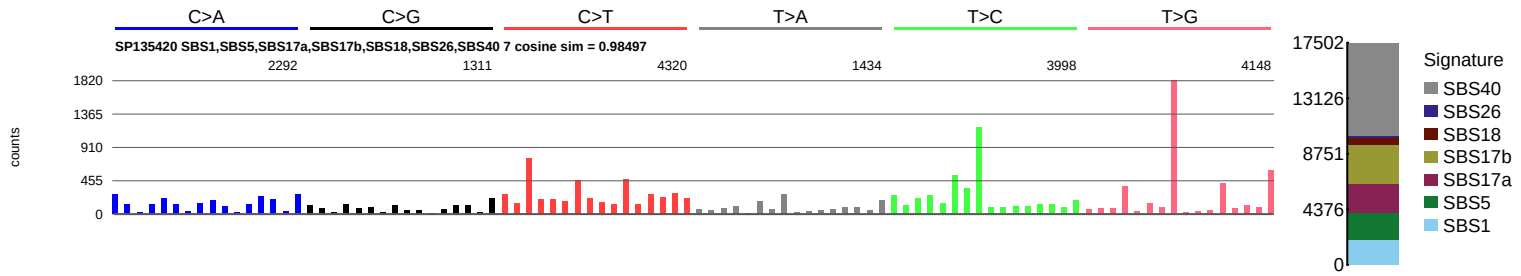
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
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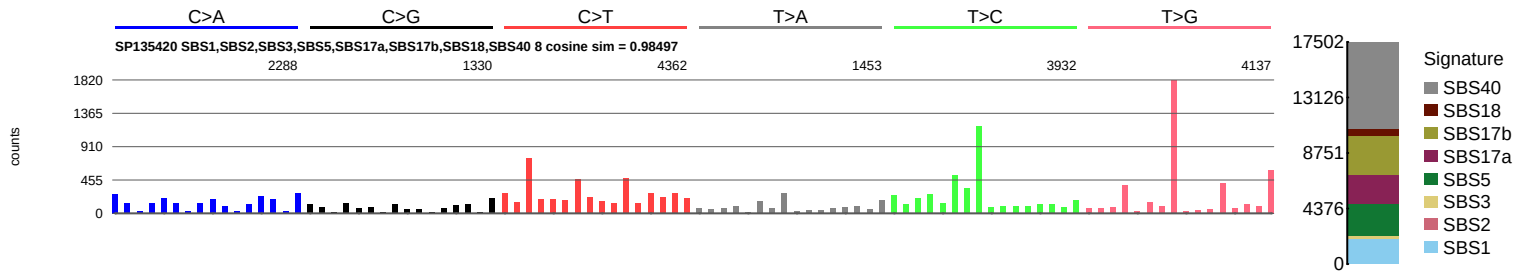
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98497
2292
1311
4320
1434
3998
4148
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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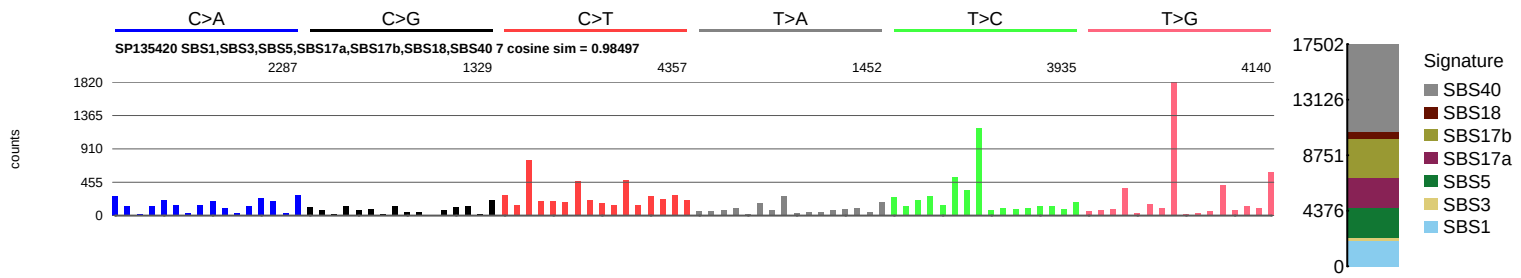
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98497
2288
1330
4362
1453
3932
4137
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>T
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T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98497
2287
1329
4357
1452
3935
4140
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
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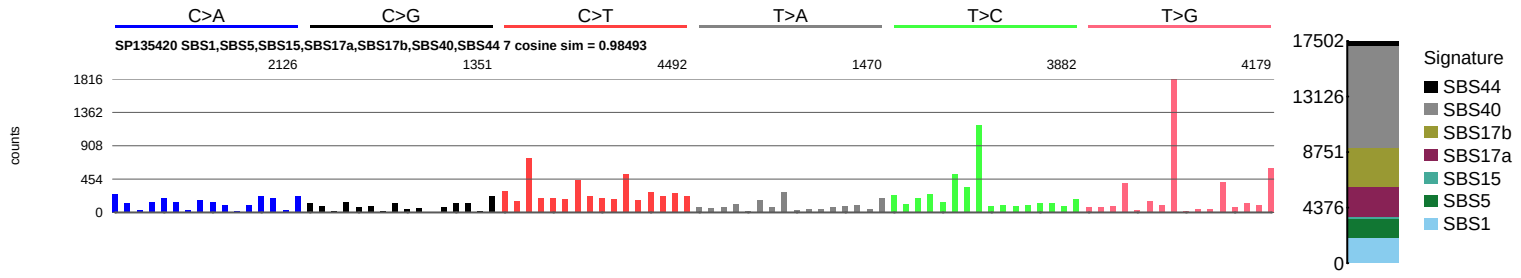
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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T>C
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SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98493
2126
1351
4492
1470
3882
4179
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
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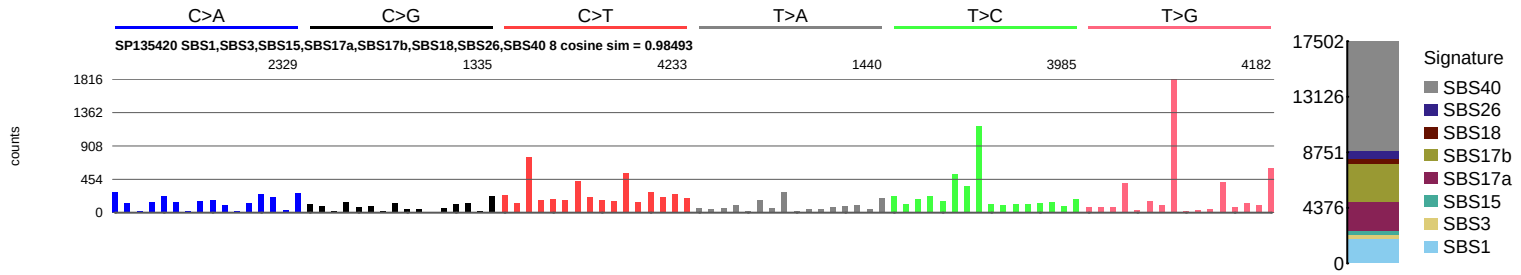
SBS17b
SBS17a
SBS15
SBS5
SBS1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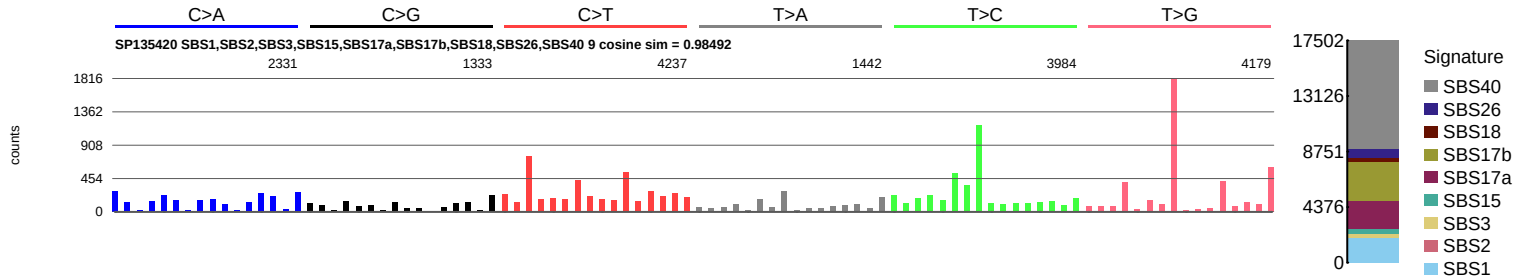
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98493
2329
1335
4233
1440
3985
4182
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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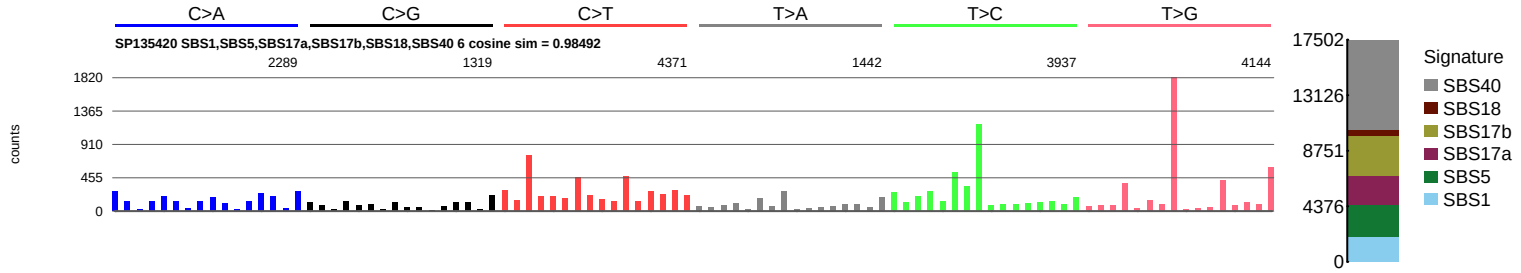
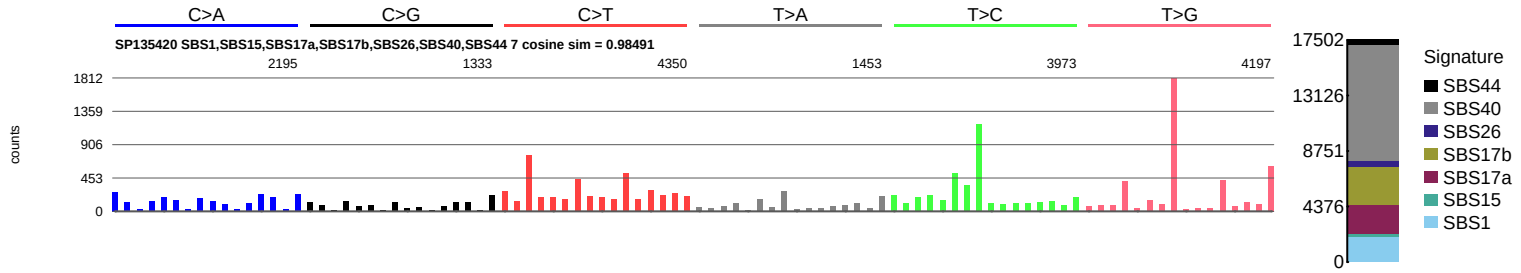
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98492
2331
1333
4237
1442
3984
4179
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98492
2289
1319
4371
1442
3937
4144
counts
1820
1365
910
455
0
17502
13126
8751
4376
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98491
2195
1333
4350
1453
3973
4197
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
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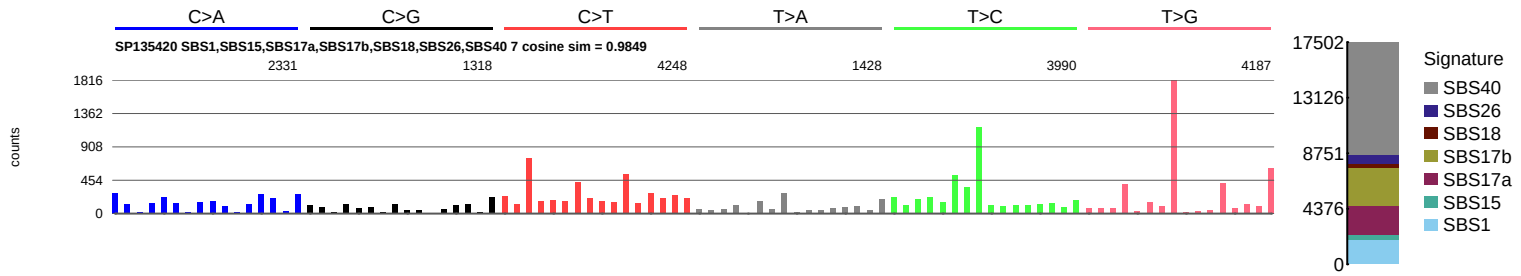
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.9849
2331
1318
4248
1428
3990
4187
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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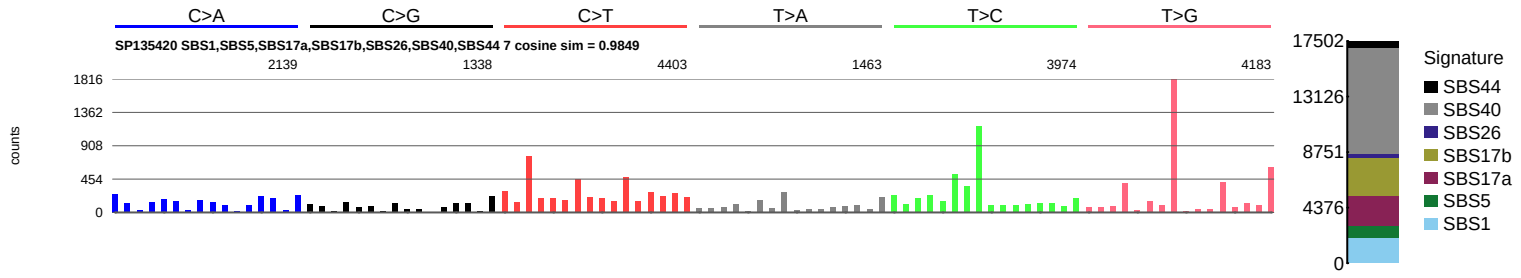
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.9849
2139
1338
4403
1463
3974
4183
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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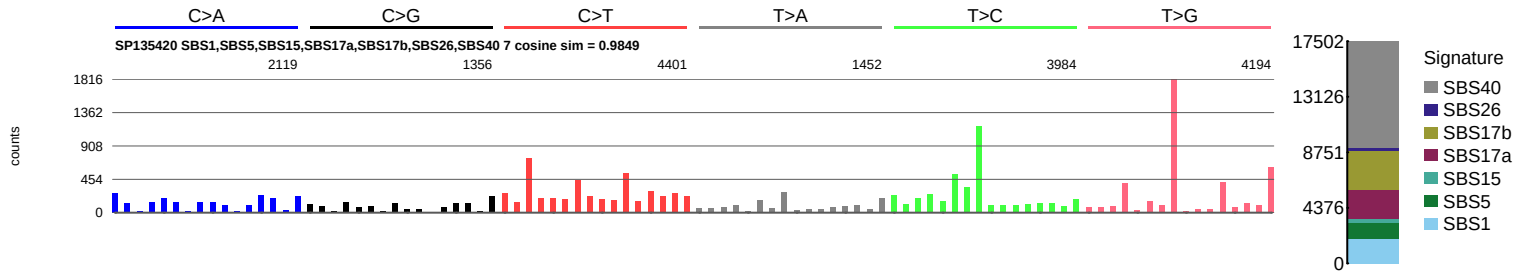
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.9849
2119
1356
4401
1452
3984
4194
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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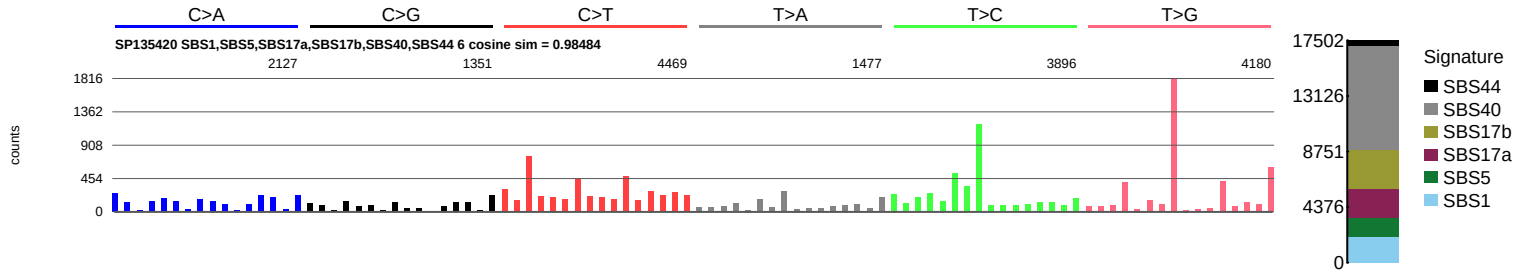
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
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T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98484
2127
1351
4469
1477
3896
4180
counts
1816
1362
908
454
0
17502
13126
8751
4376
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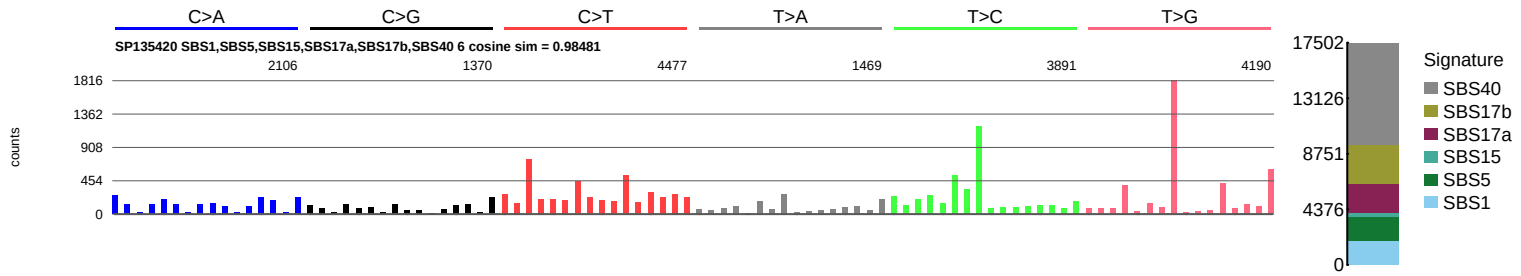
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS1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T>C
T>G
SP135420 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98481
2106
1370
4477
1469
3891
4190
counts
1816
1362
908
454
0
17502
13126
8751
4376
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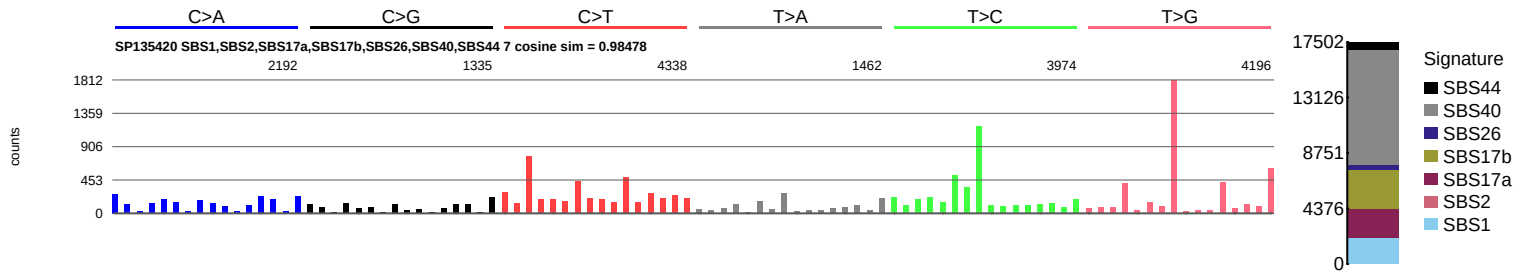
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98478
2192
1335
4338
1462
3974
4196
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
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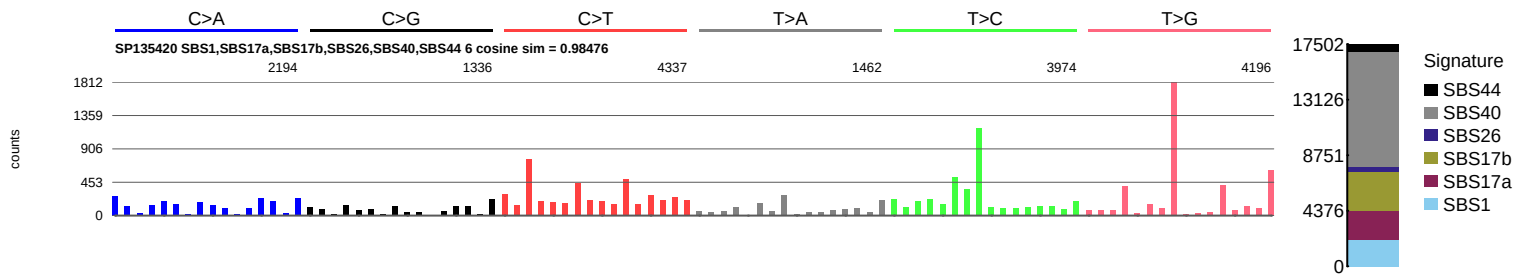
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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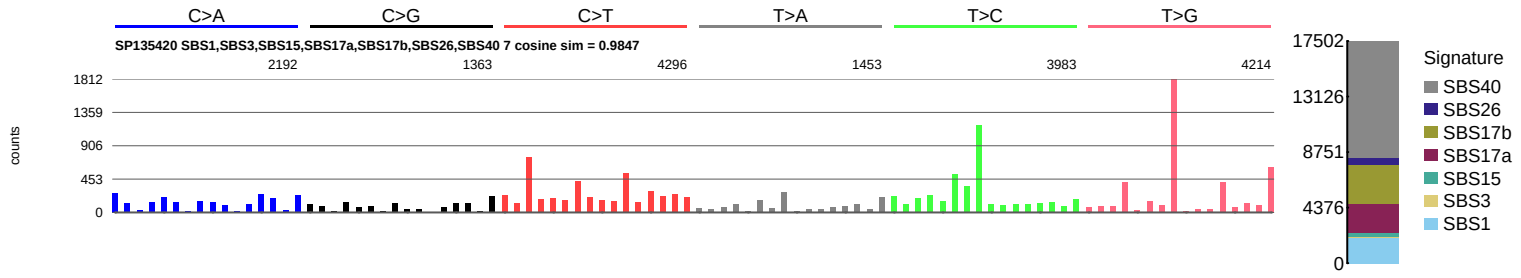
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98476
2194
1336
4337
1462
3974
4196
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.9847
2192
1363
4296
1453
3983
4214
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
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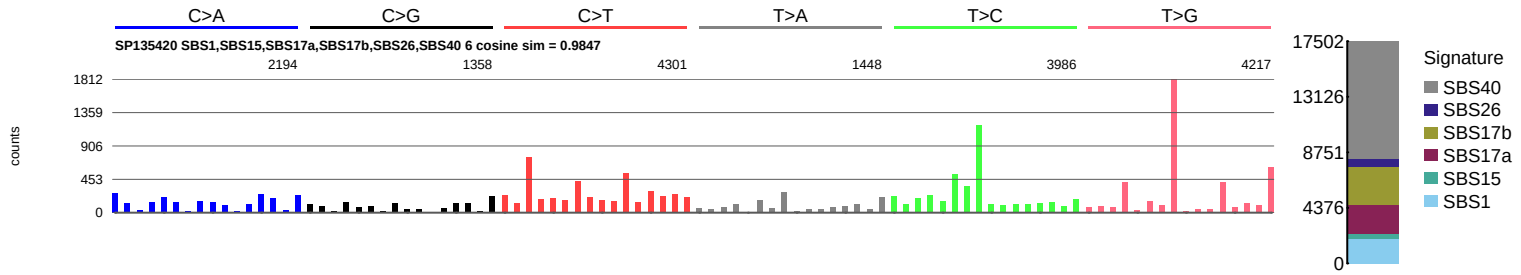
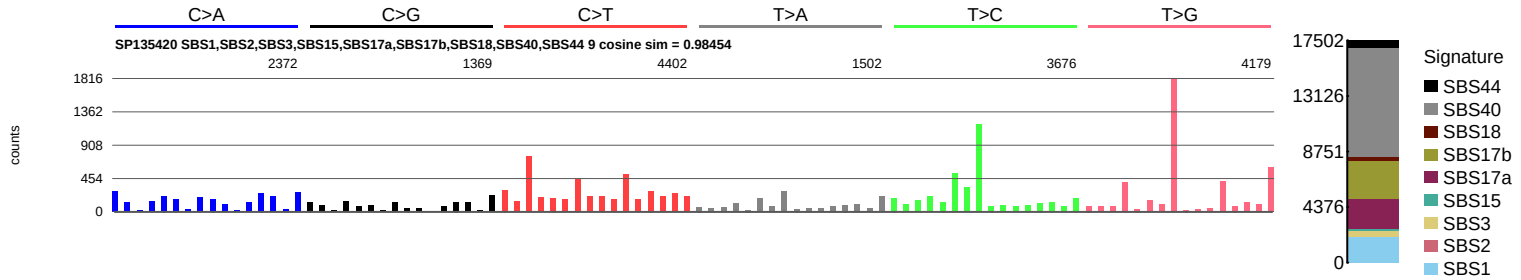
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.9847
2194
1358
4301
1448
3986
4217
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98454
2372
1369
4402
1502
3676
4179
counts
1816
1362
908
454
0
17502
13126
8751
4376
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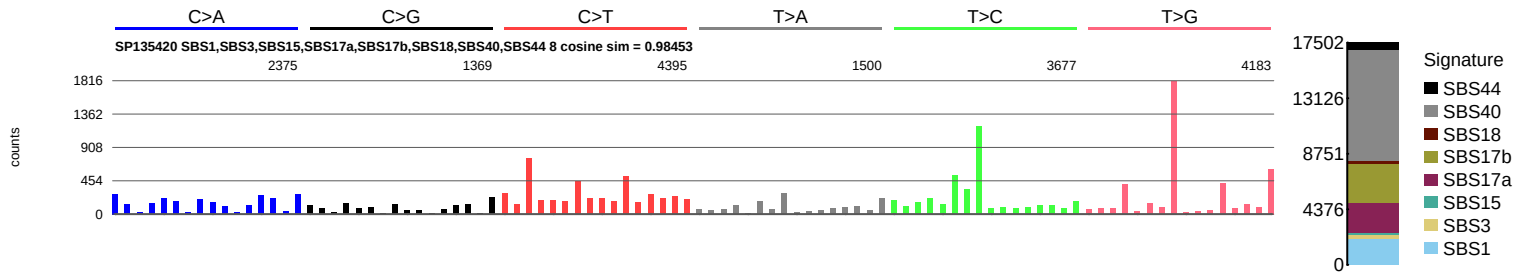
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98453
2375
1369
4395
1500
3677
4183
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
SBS44
SBS40
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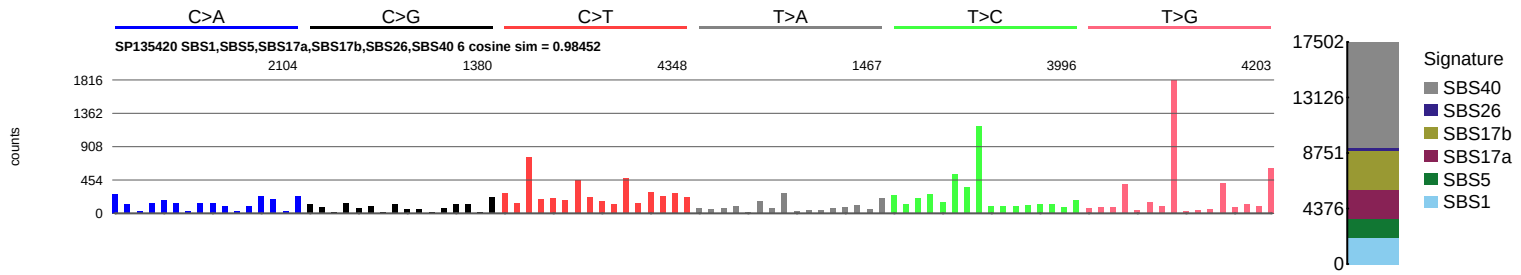
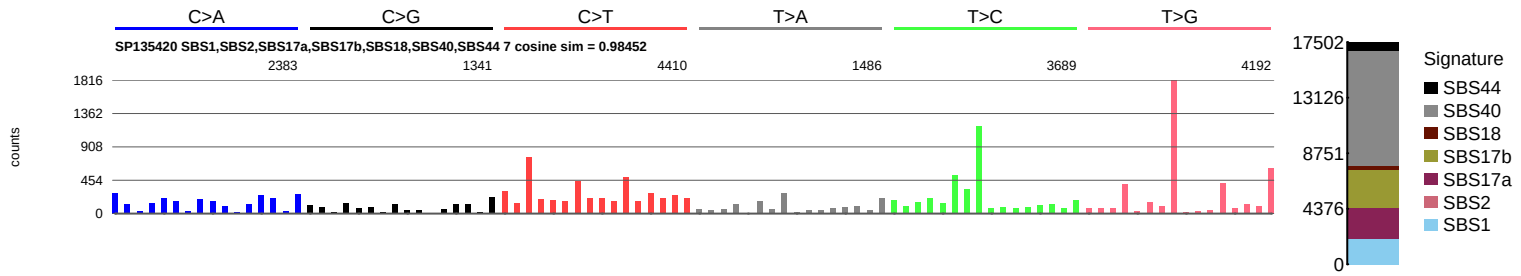
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98452
2104
1380
4348
1467
3996
4203
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1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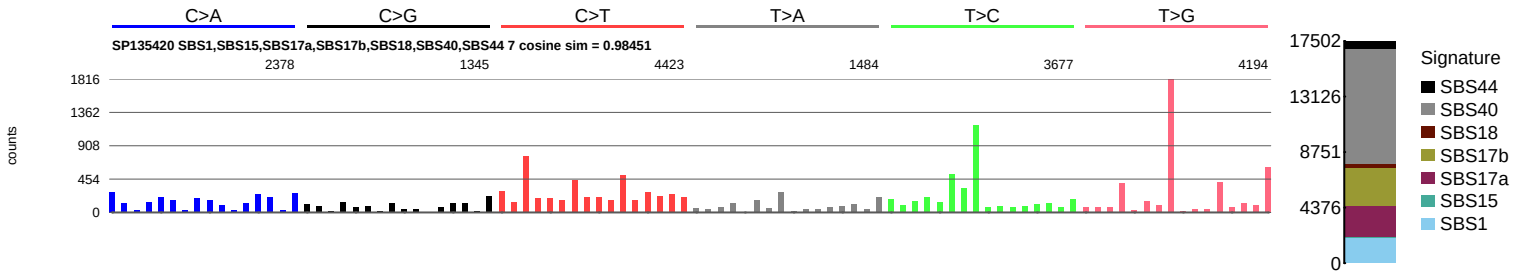
SP135420 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98452
2383
1341
4410
1486
3689
4192
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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T>G
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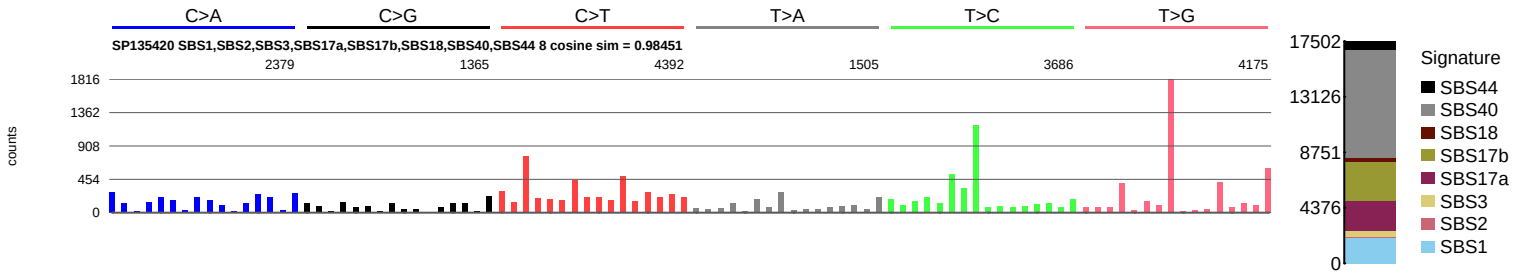
2378
1345
4423
1484
3677
4194
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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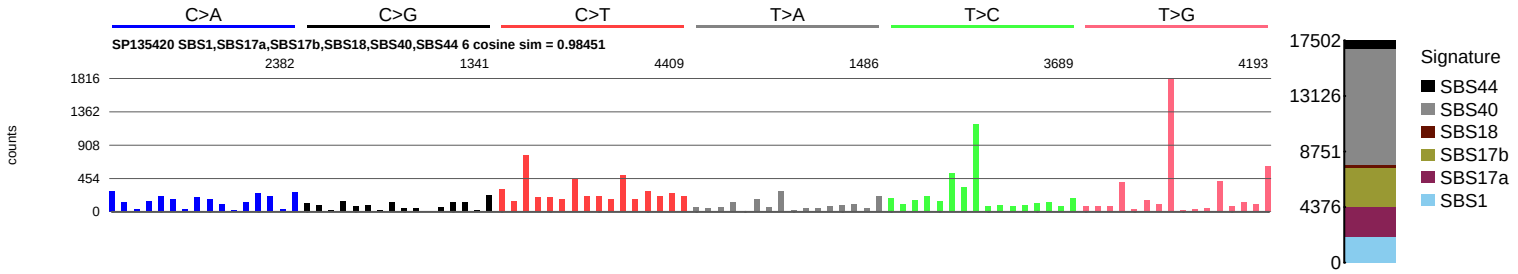
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98451
2379
1365
4392
1505
3686
4175
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98451
2382
1341
4409
1486
3689
4193
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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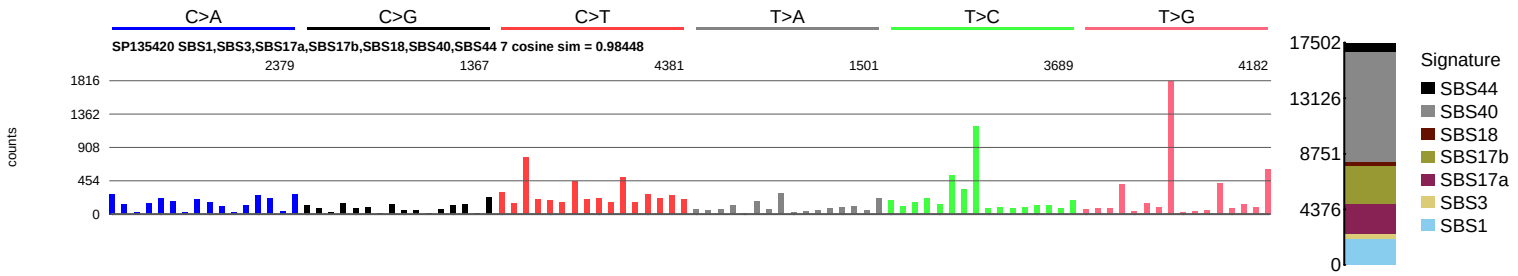
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1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SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98448
2379
1367
4381
1501
3689
4182
counts
1816
1362
908
454
0
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13126
8751
4376
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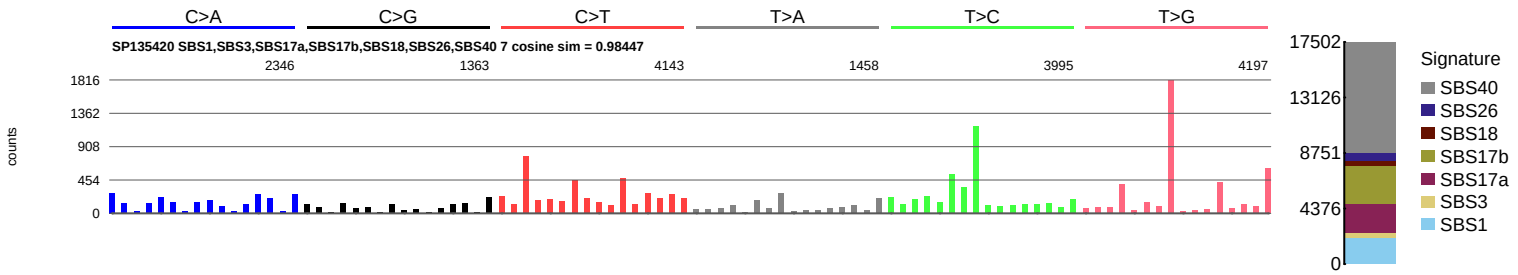
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1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SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98447
2346
1363
4143
1458
3995
4197
counts
1816
1362
908
454
0
17502
13126
8751
4376
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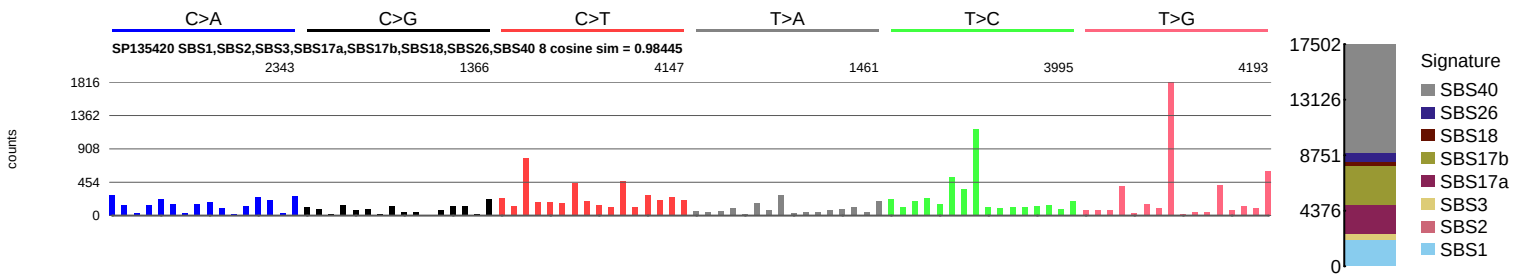
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>T
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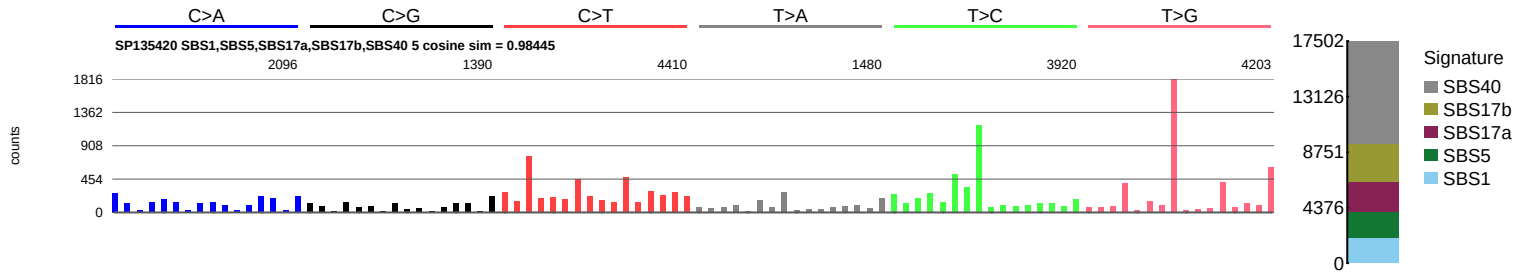
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98445
2343
1366
4147
1461
3995
4193
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
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T>A
T>C
T>G
SP135420 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98445
2096
1390
4410
1480
3920
4203
counts
1816
1362
908
454
0
17502
13126
8751
4376
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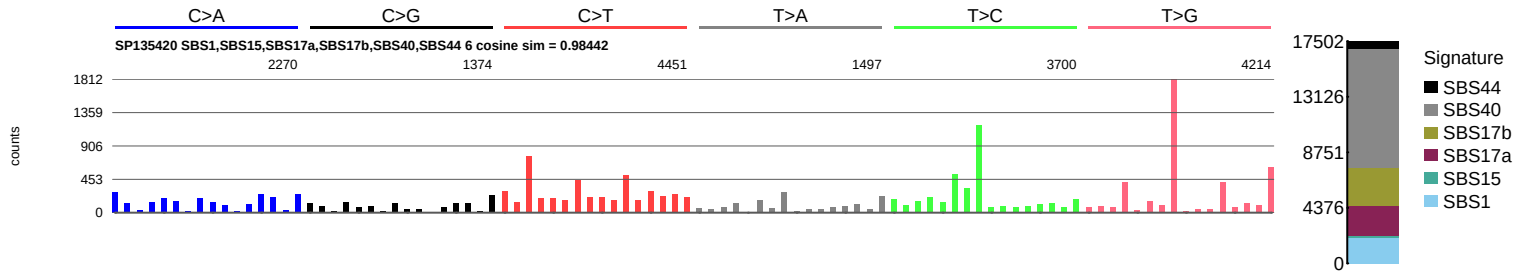
SBS40
SBS17b
SBS17a
SBS5
SBS1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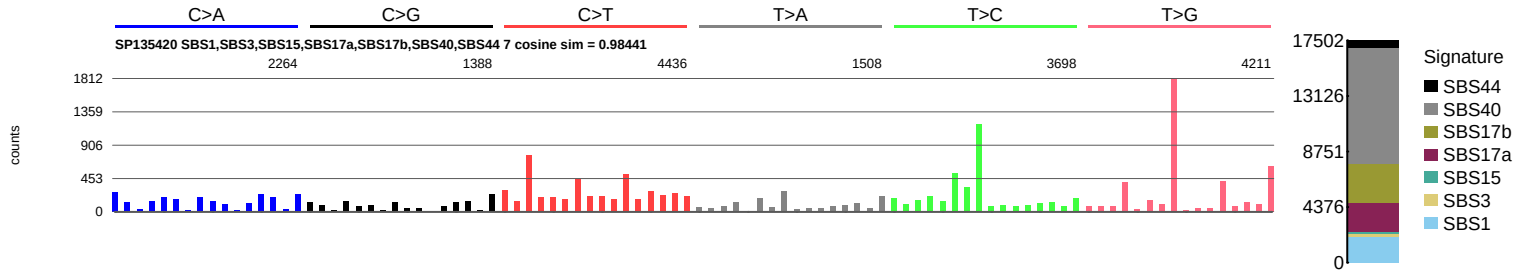
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98442
2270
1374
4451
1497
3700
4214
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1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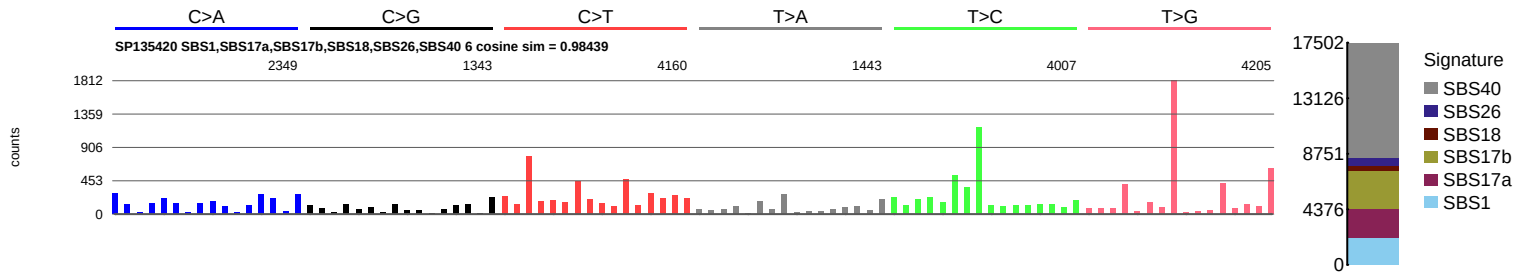
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98441
2264
1388
4436
1508
3698
4211
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1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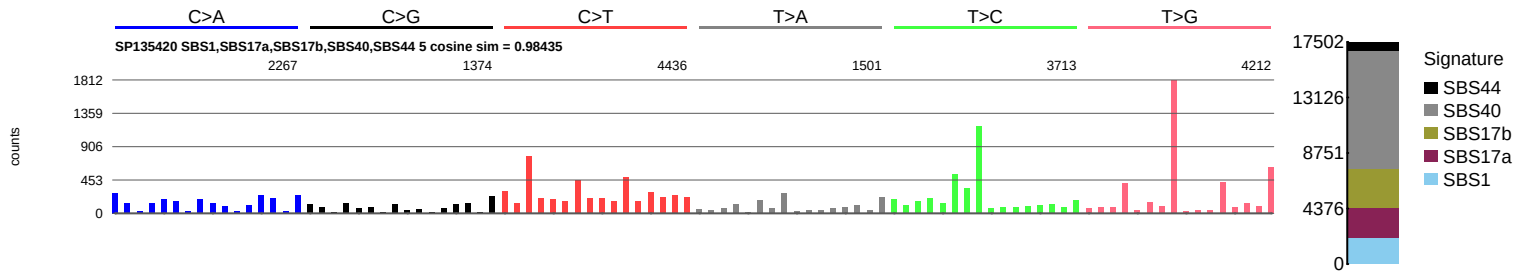
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98439
2349
1343
4160
1443
4007
4205
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98435
2267
1374
4436
1501
3713
4212
counts
1812
1359
906
453
0
17502
13126
8751
4376
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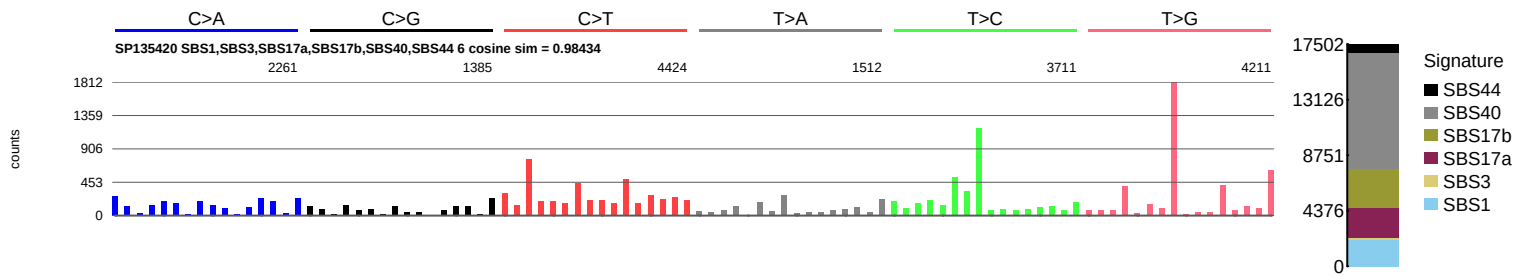
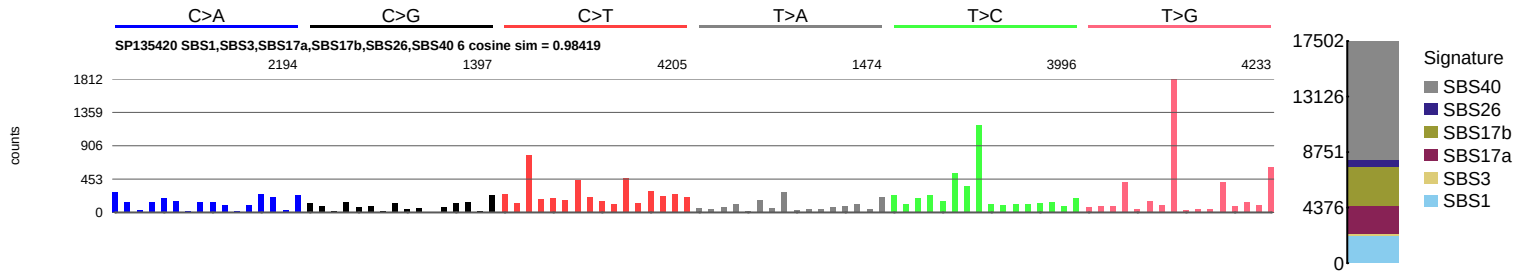
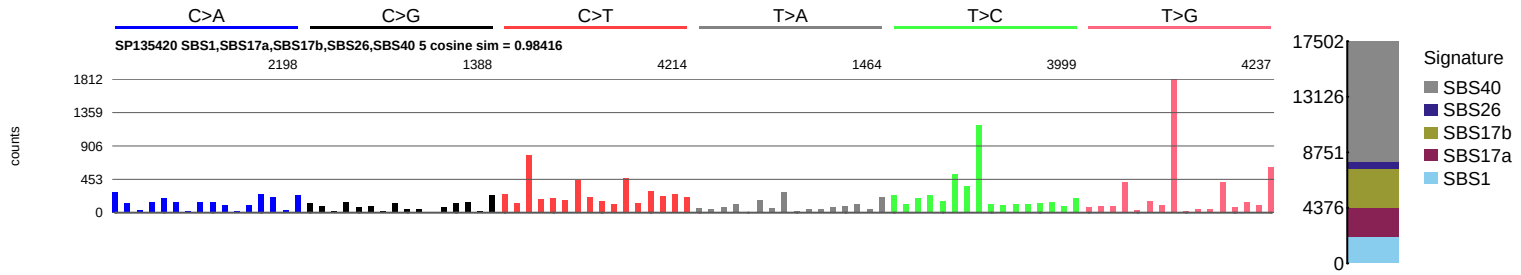
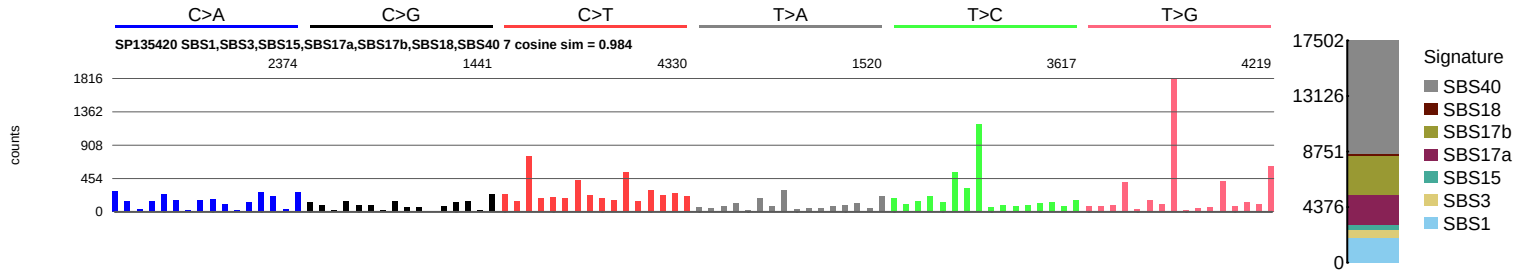
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98434
2261
1385
4424
1512
3711
4211
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98419
2194
1397
4205
1474
3996
4233
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98416
2198
1388
4214
1464
3999
4237
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.984
2374
1441
4330
1520
3617
4219
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
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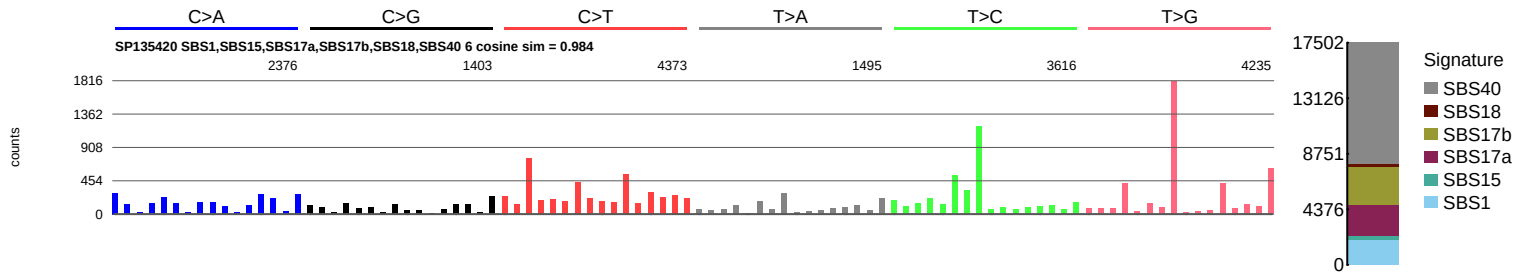
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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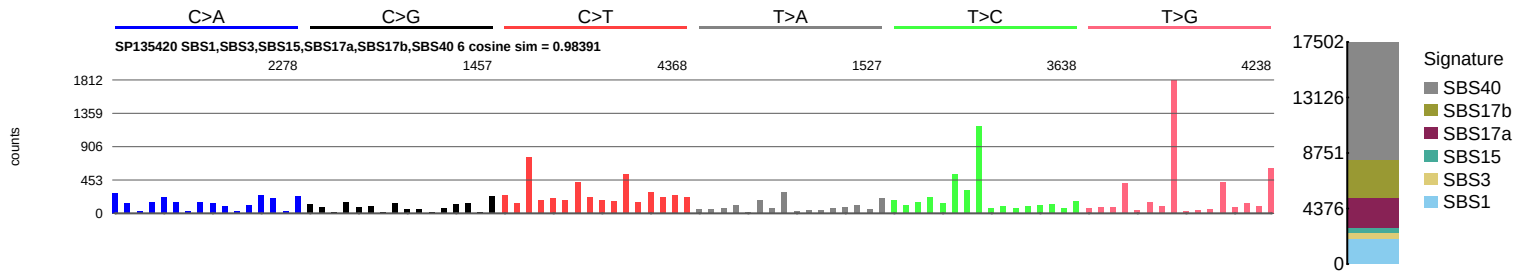
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.984
2376
1403
4373
1495
3616
4235
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98391
2278
1457
4368
1527
3638
4238
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
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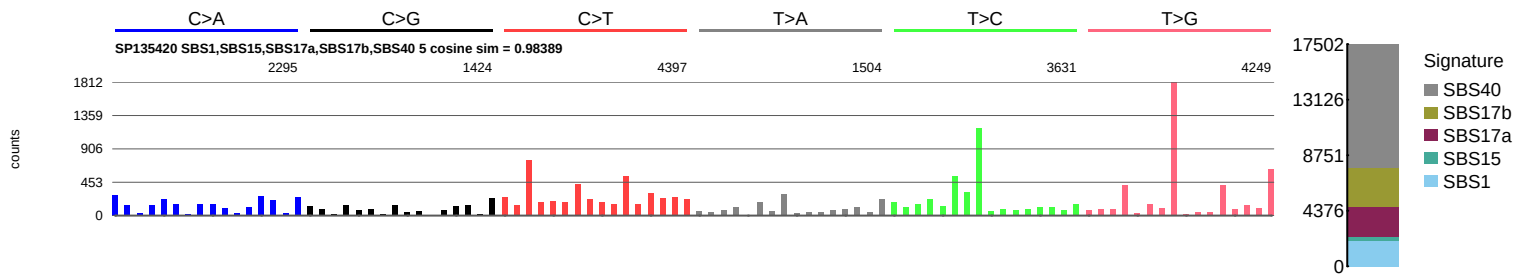
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98389
2295
1424
4397
1504
3631
4249
counts
1812
1359
906
453
0
17502
13126
8751
4376
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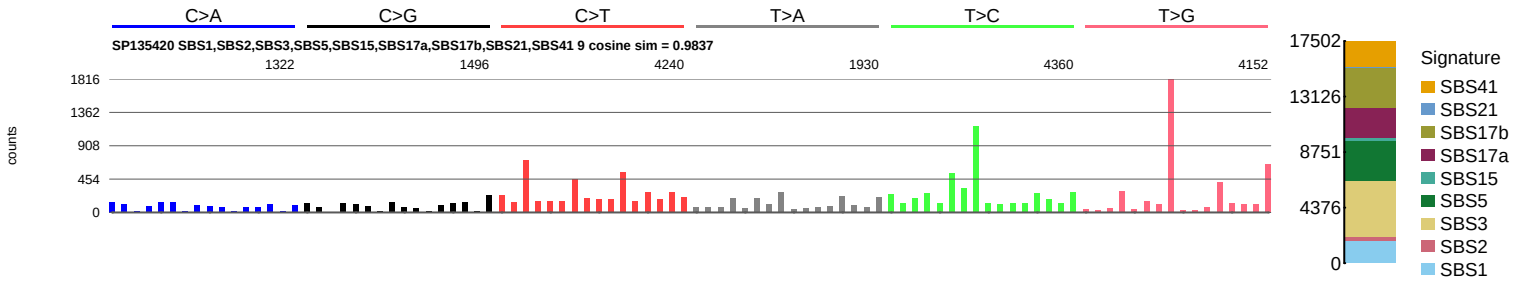
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.9837
1322
1496
4240
1930
4360
4152
counts
1816
1362
908
454
0
17502
13126
8751
4376
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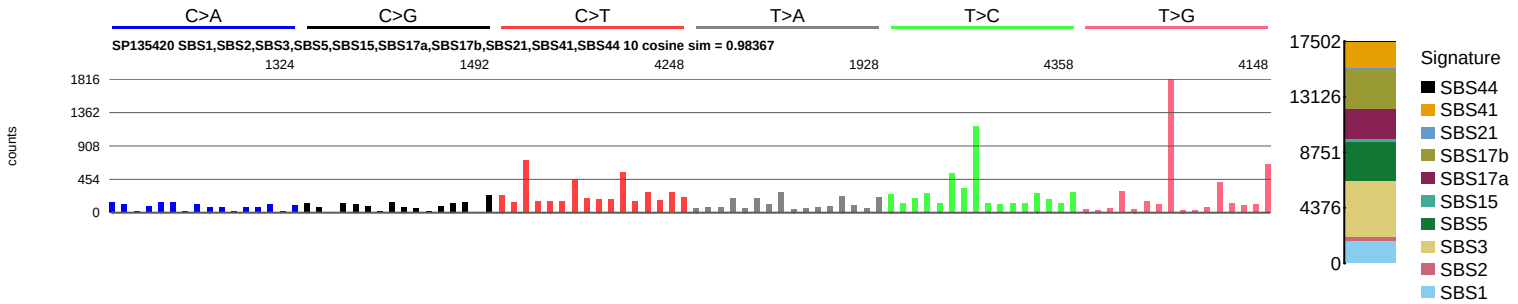
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.98367
1324
1492
4248
1928
4358
4148
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
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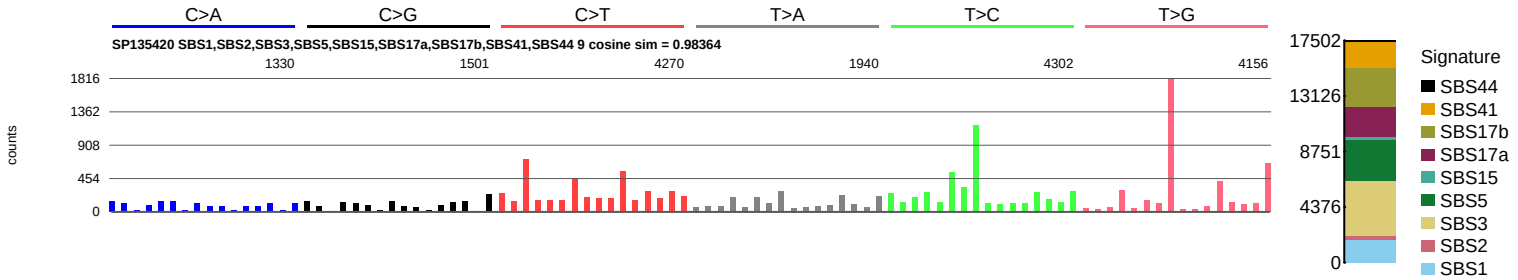
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98364
1330
1501
4270
1940
4302
4156
counts
1816
1362
908
454
0
17502
13126
8751
4376
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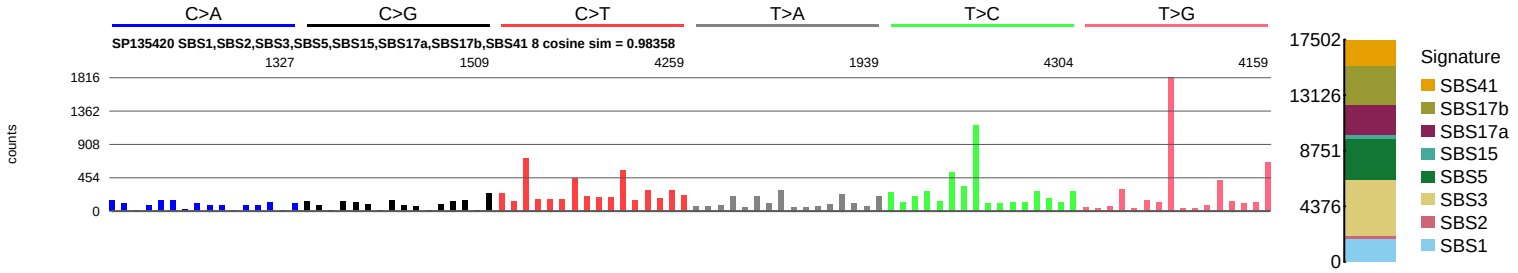
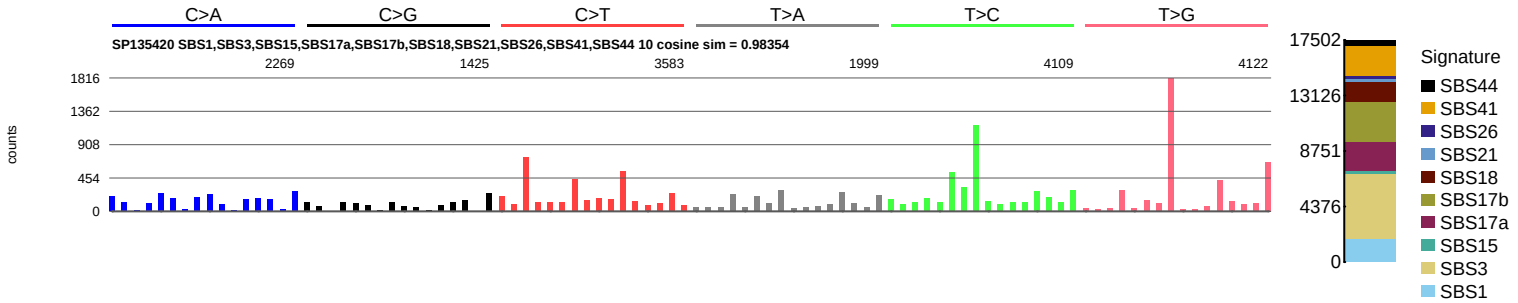
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98358
1327
1509
4259
1939
4304
4159
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98354
2269
1425
3583
1999
4109
4122
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
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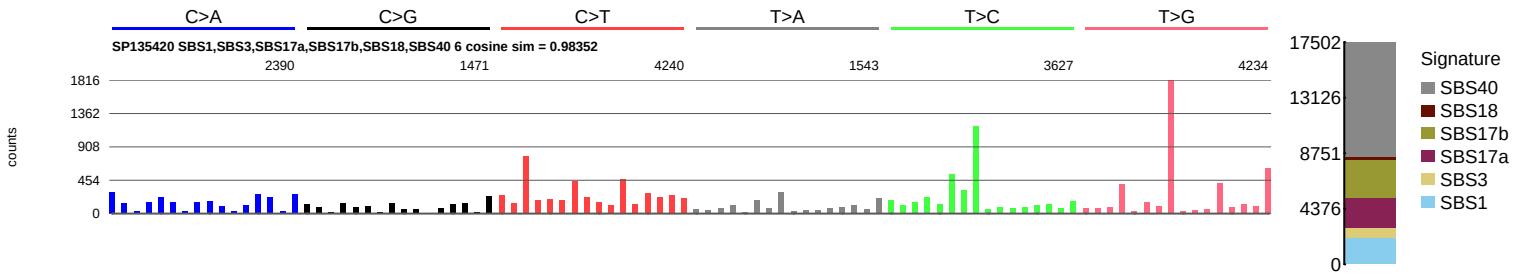
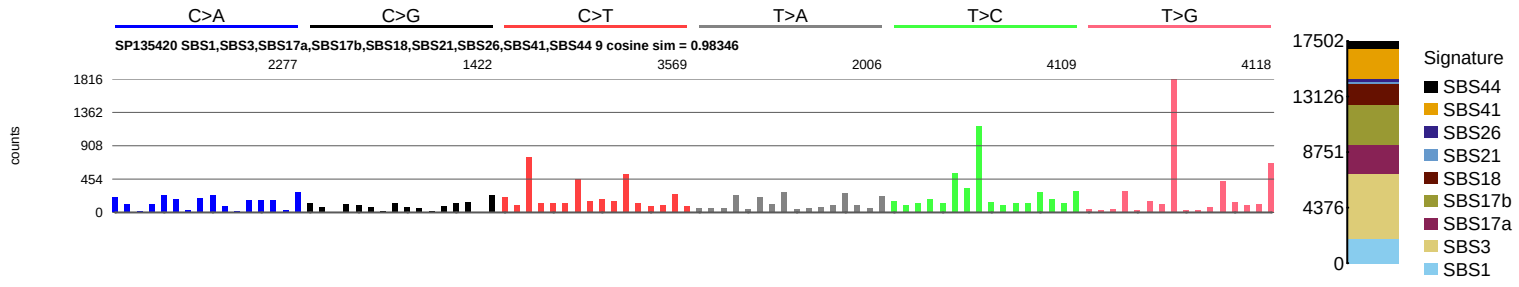
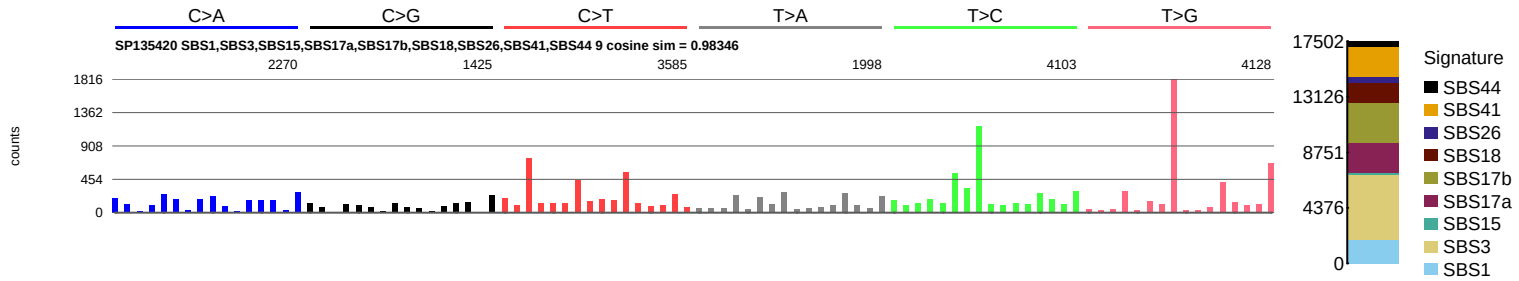
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98352
2390
1471
4240
1543
3627
4234
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98346
2277
1422
3569
2006
4109
4118
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98346
2270
1425
3585
1998
4103
4128
counts
1816
1362
908
454
0
17502
13126
8751
4376
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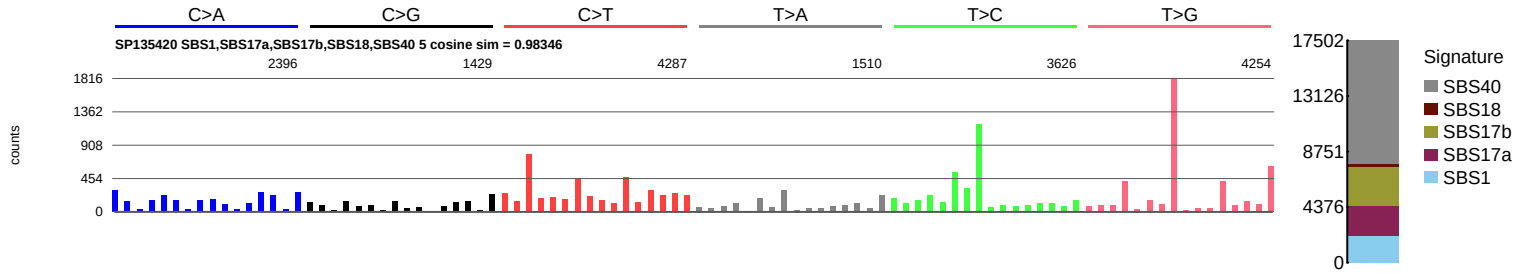
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP135420 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98346
2396
1429
4287
1510
3626
4254
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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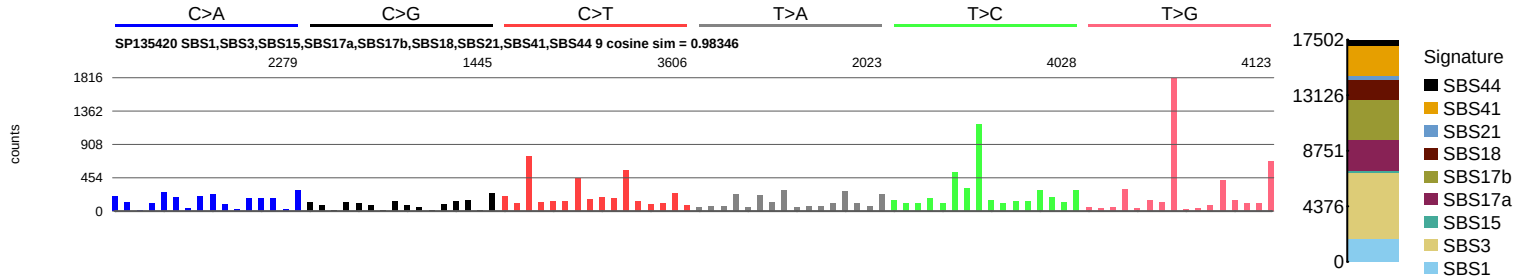
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
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T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98346
2279
1445
3606
2023
4028
4123
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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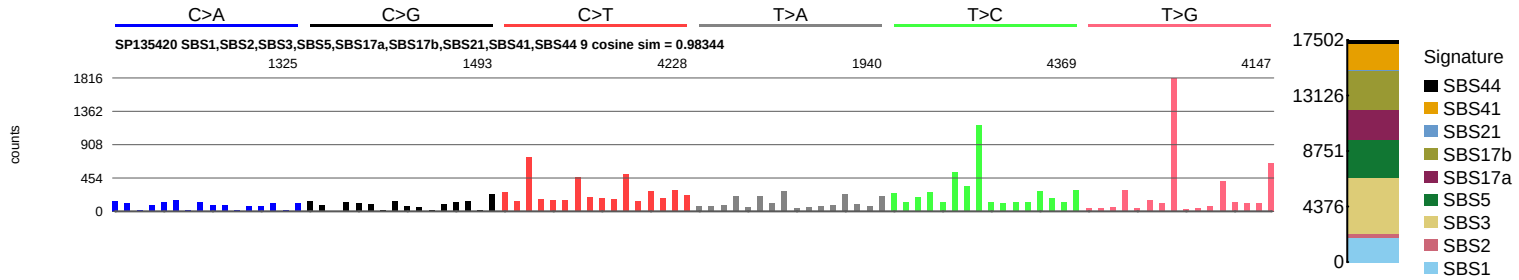
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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T>C
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SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98344
1325
1493
4228
1940
4369
4147
counts
1816
1362
908
454
0
17502
13126
8751
4376
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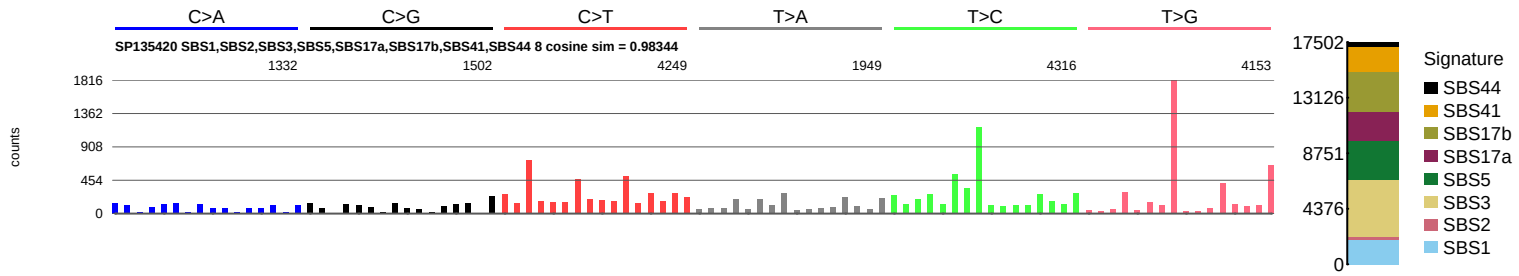
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98344
1332
1502
4249
1949
4316
4153
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
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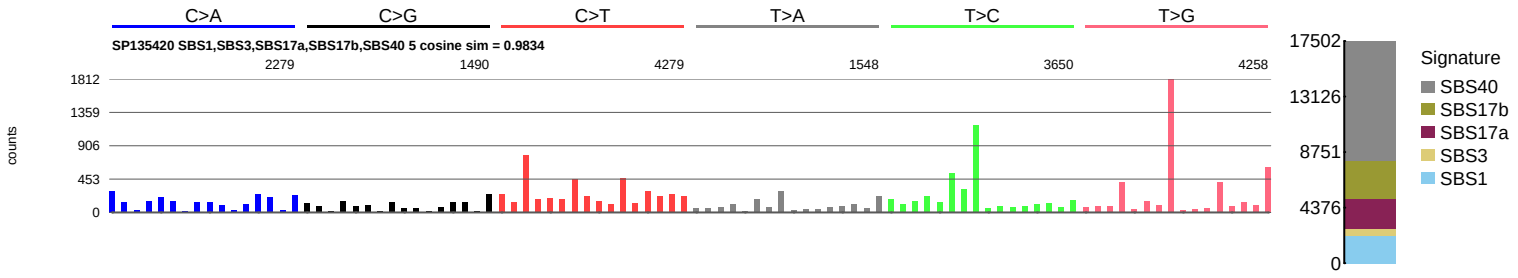
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS40 5 cosine sim = 0.9834
2279
1490
4279
1548
3650
4258
counts
1812
1359
906
453
0
17502
13126
8751
4376
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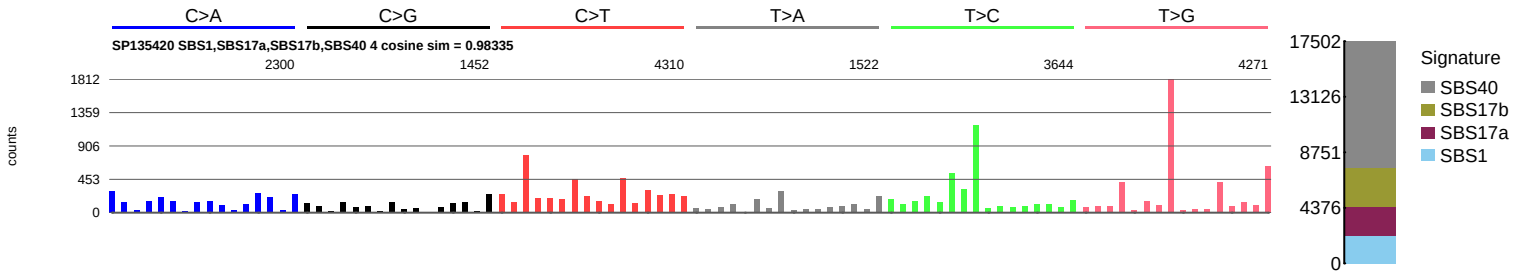
SBS40
SBS17b
SBS17a
SBS3
SBS1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C>T
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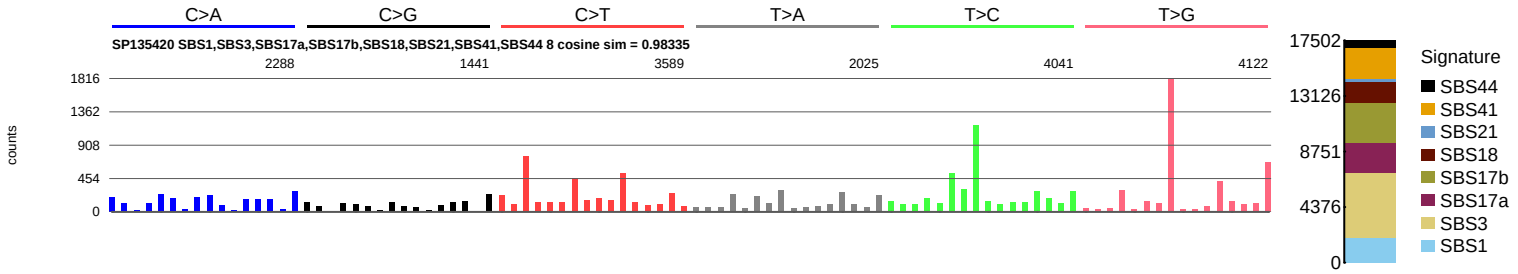
T>C
T>G
SP135420 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98335
2300
1452
4310
1522
3644
4271
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS40
SBS17b
SBS17a
SBS1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C>G
C>T
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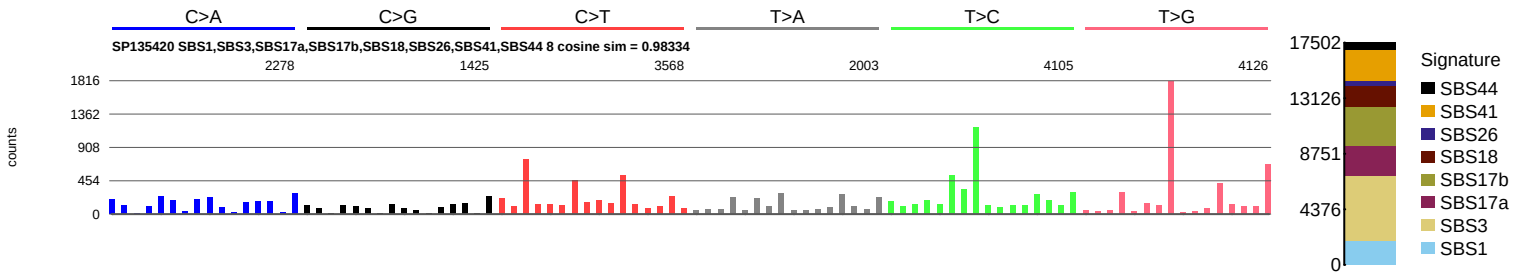
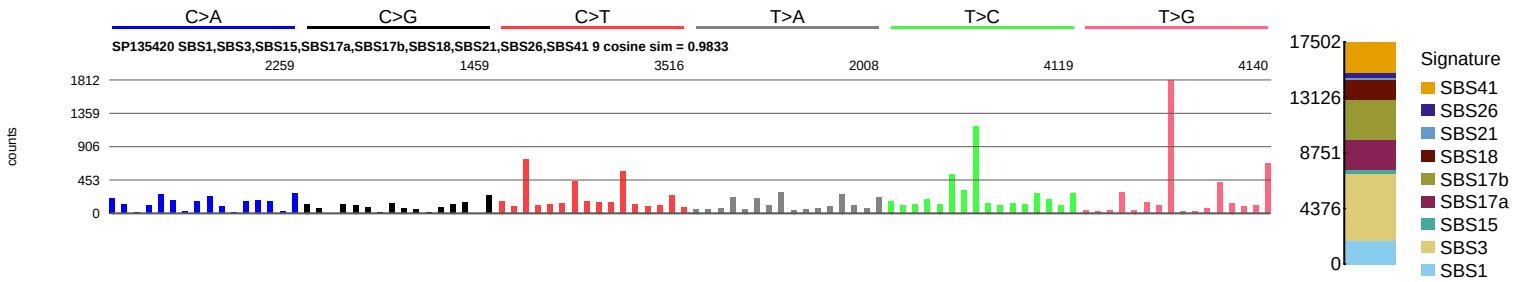
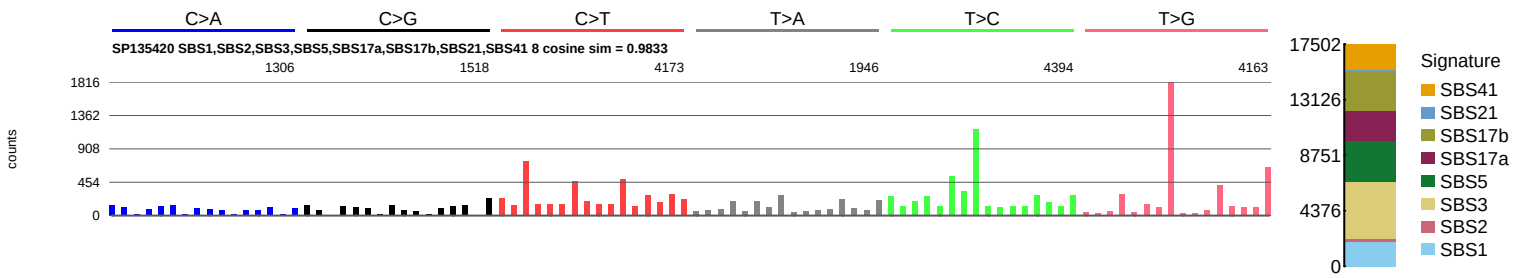
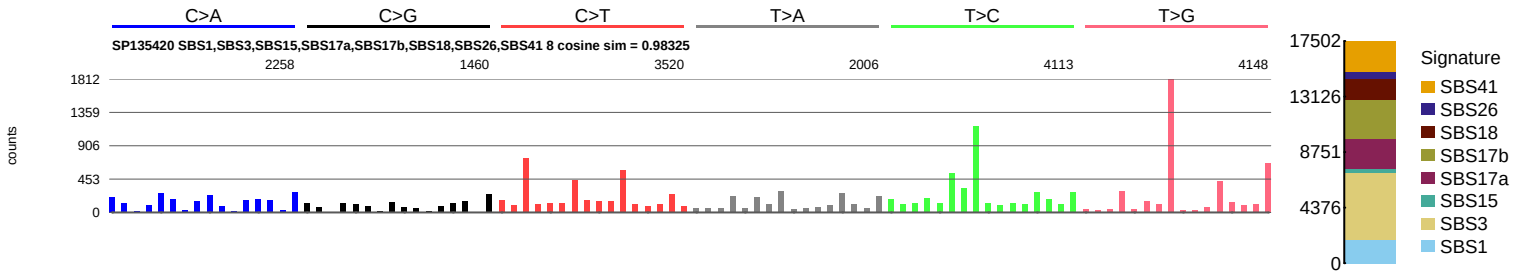
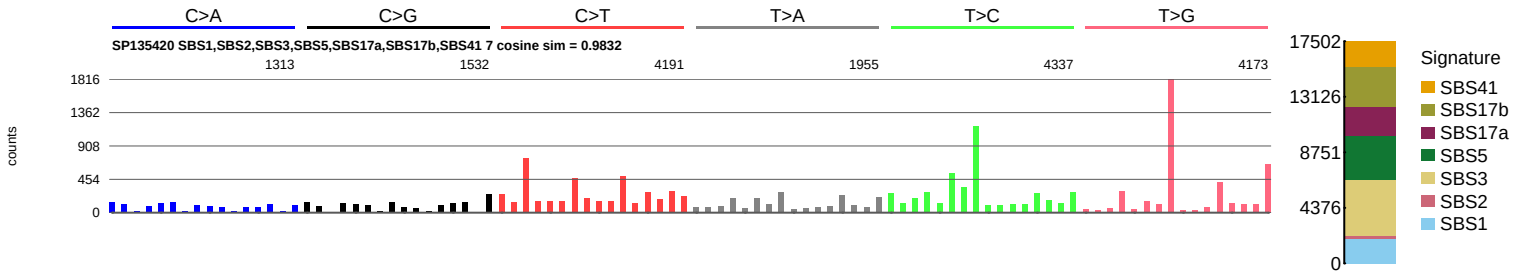
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98335
2288
1441
3589
2025
4041
4122
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

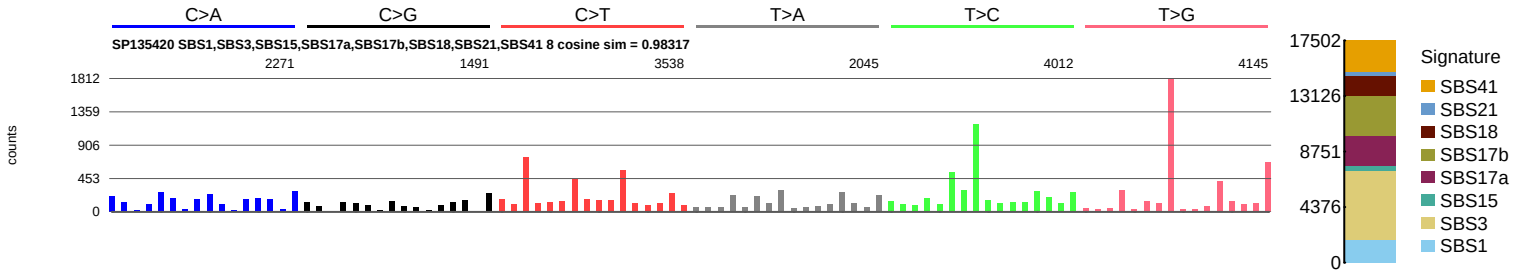

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98334
2278
1425
3568
2003
4105
4126
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

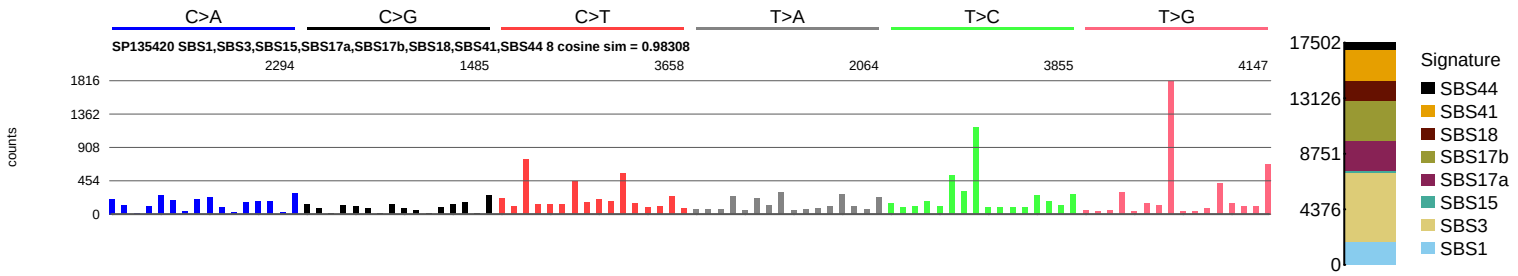

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.9833
2259
1459
3516
2008
4119
4140
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9833
1306
1518
4173
1946
4394
4163
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98325
2258
1460
3520
2006
4113
4148
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9832
1313
1532
4191
1955
4337
4173
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98317
2271
1491
3538
2045
4012
4145
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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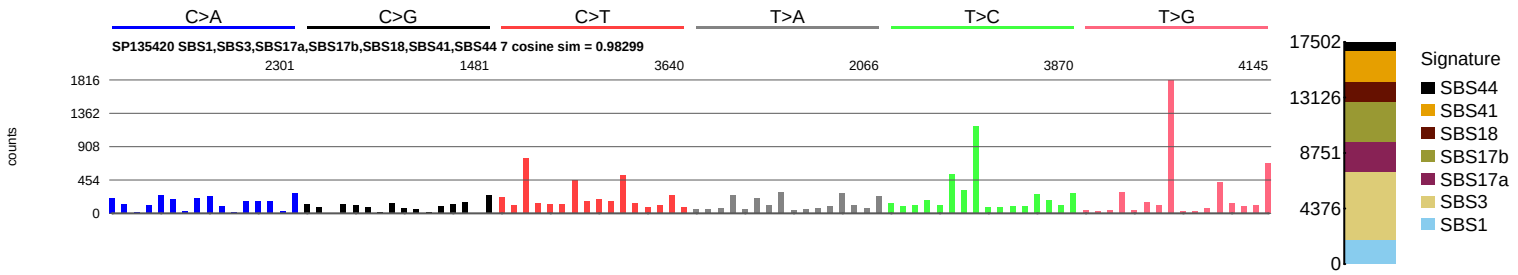
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98308
2294
1485
3658
2064
3855
4147
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98299
2301
1481
3640
2066
3870
4145
counts
1816
1362
908
454
0
17502
13126
8751
4376
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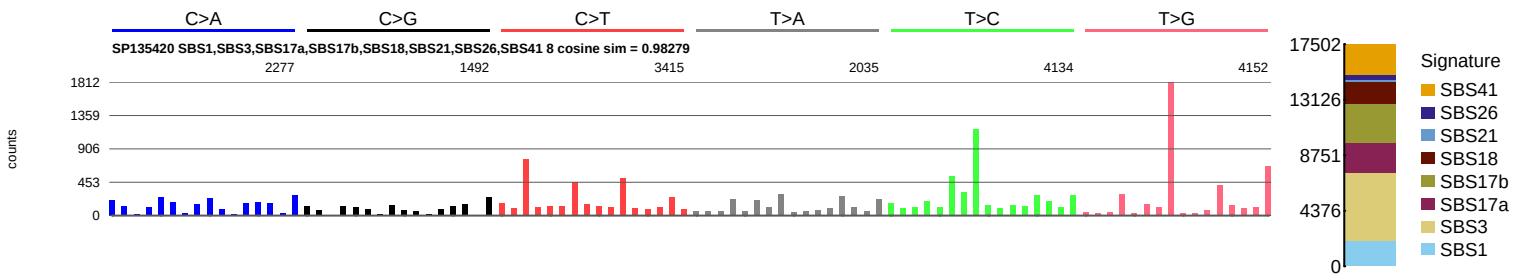
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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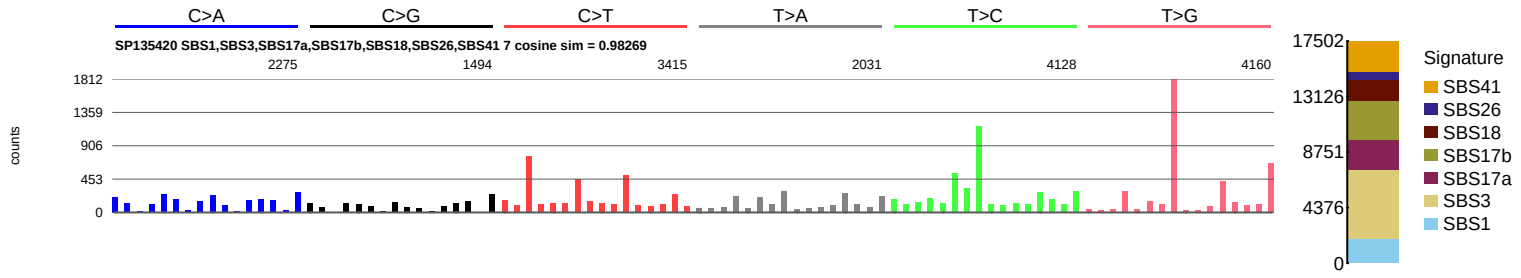
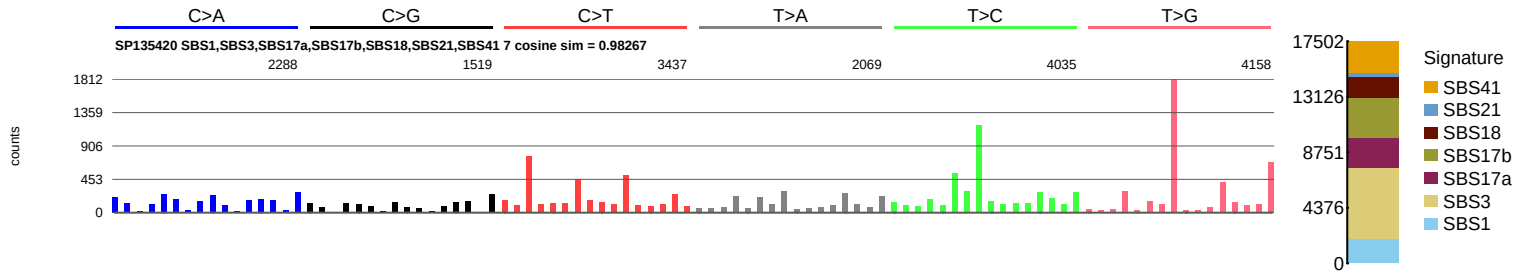
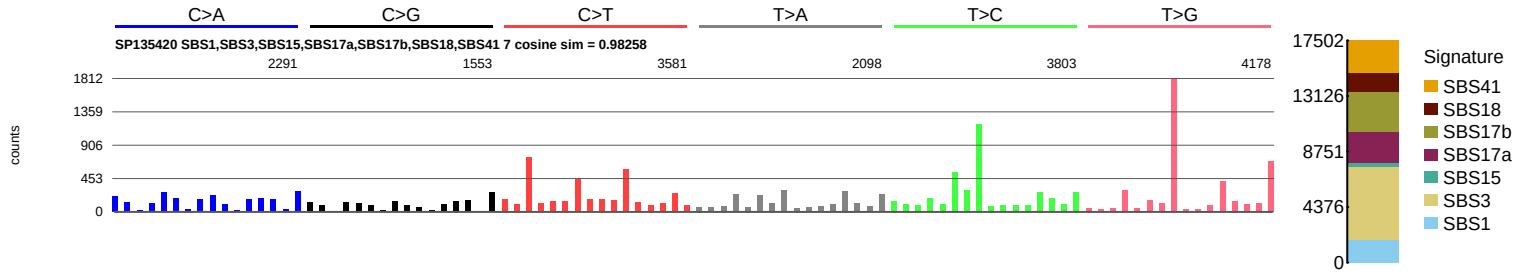
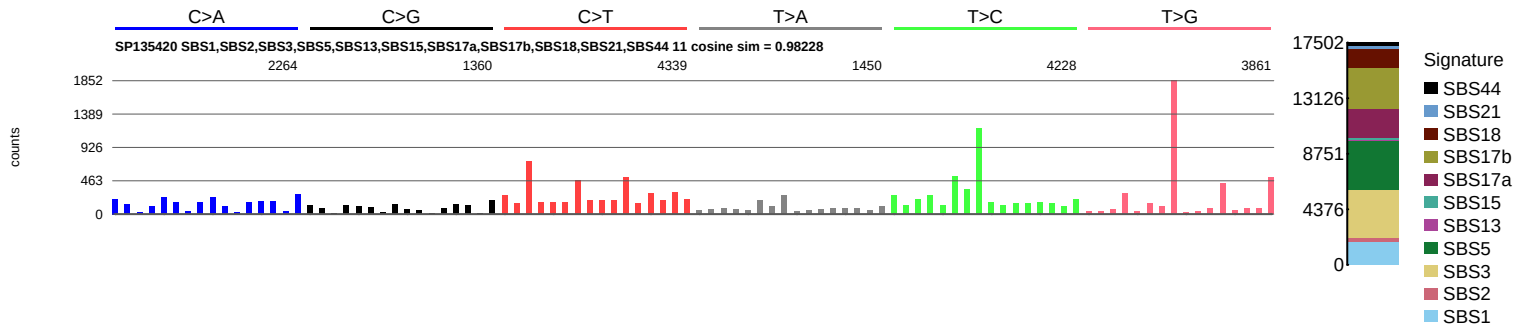
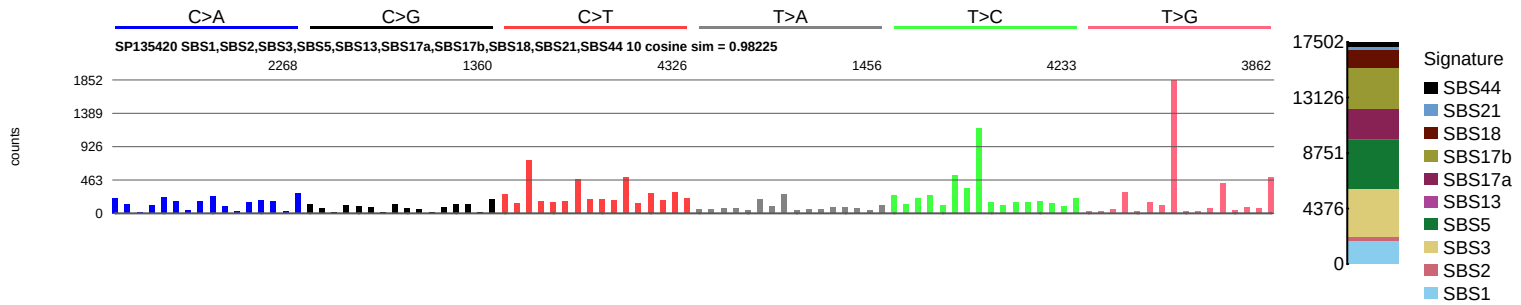
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98279
2277
1492
3415
2035
4134
4152
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

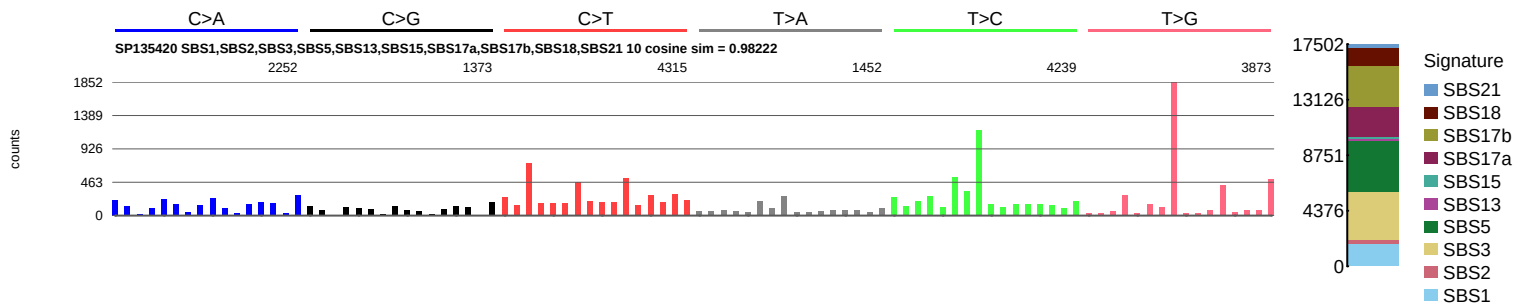

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98269
2275
1494
3415
2031
4128
4160
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98267
2288
1519
3437
2069
4035
4158
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98258
2291
1553
3581
2098
3803
4178
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98228
2264
1360
4339
1450
4228
3861
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98225
2268
1360
4326
1456
4233
3862
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98222
2252
1373
4315
1452
4239
3873
counts
1852
1389
926
463
0
17502
13126
8751
4376
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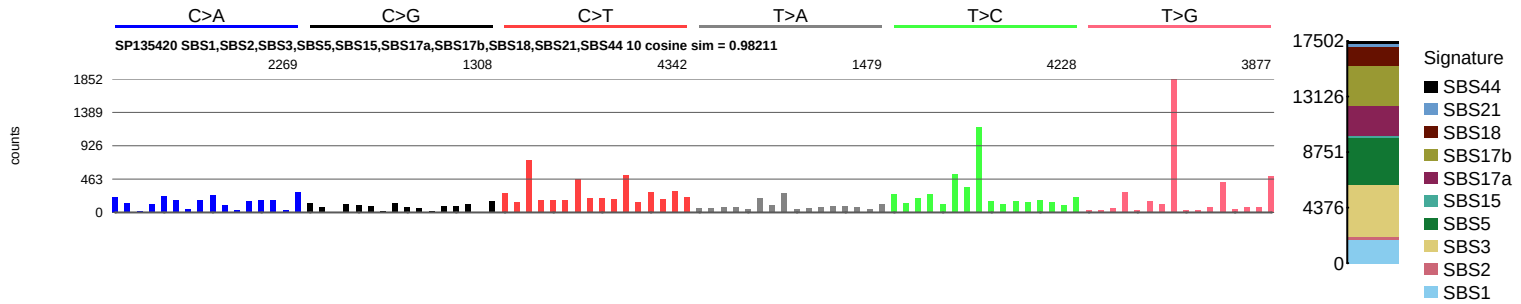
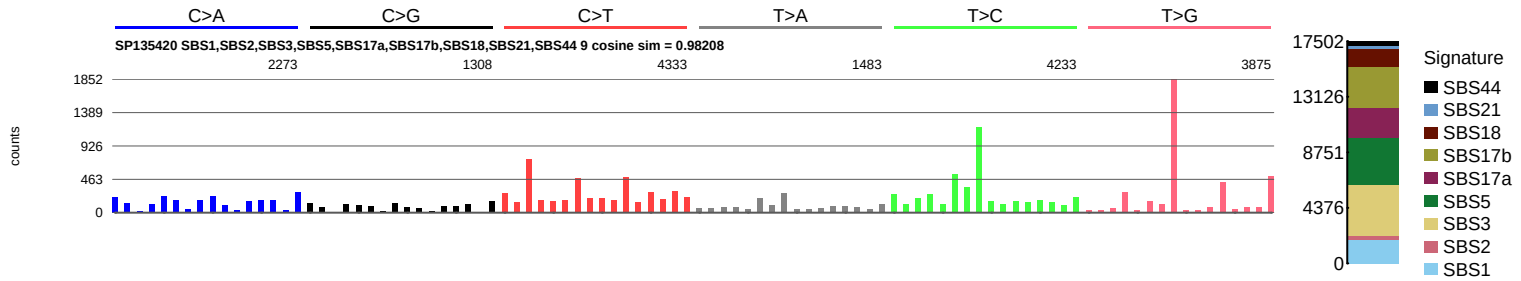
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98211
2269
1308
4342
1479
4228
3877
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98208
2273
1308
4333
1483
4233
3875
counts
1852
1389
926
463
0
17502
13126
8751
4376
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SBS21
SBS18
SBS17b
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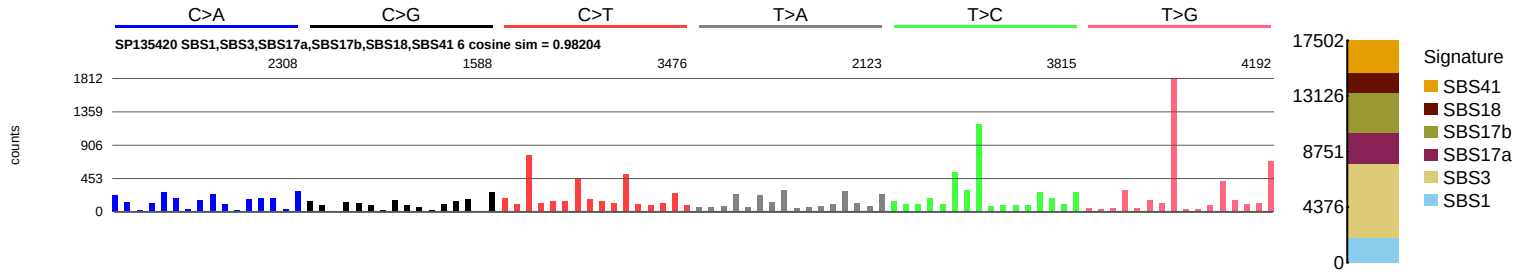
SBS1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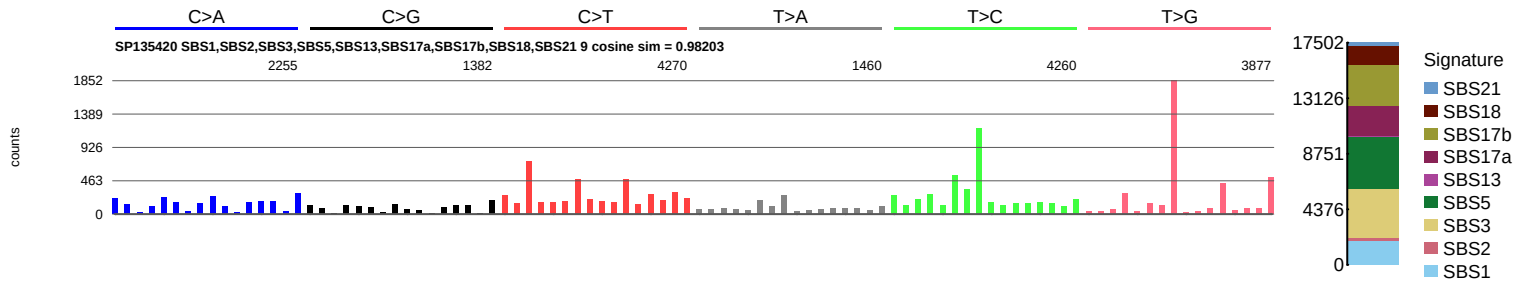
SP135420 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98204
2308
1588
3476
2123
3815
4192
counts
1812
1359
906
453
0
17502
13126
8751
4376
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98203
2255
1382
4270
1460
4260
3877
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
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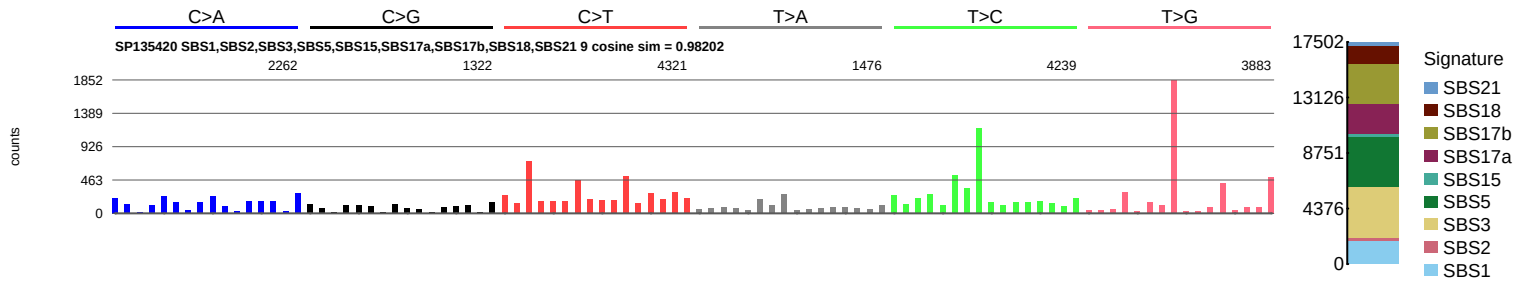
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98202
2262
1322
4321
1476
4239
3883
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
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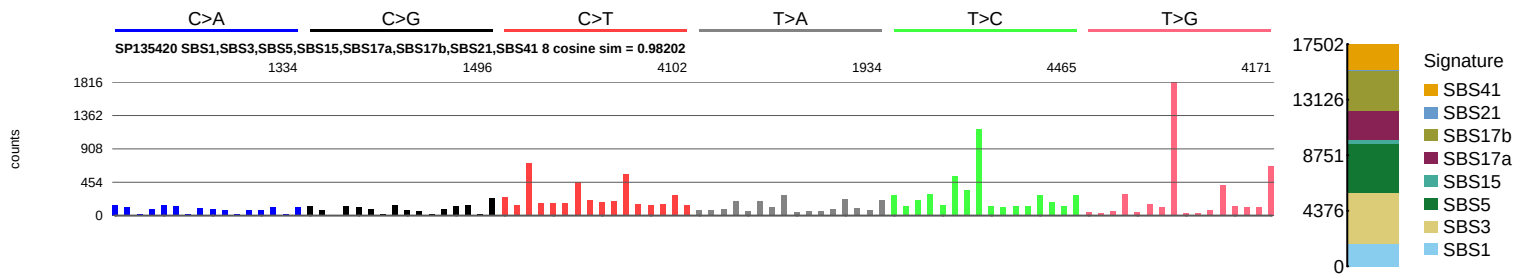
SBS2
SBS1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C>G
C>T
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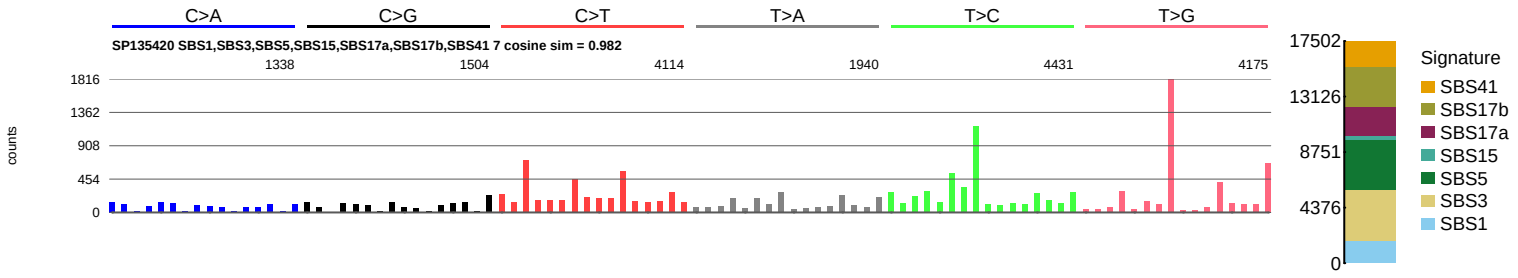
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98202
1334
1496
4102
1934
4465
4171
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.982
1338
1504
4114
1940
4431
4175
counts
1816
1362
908
454
0
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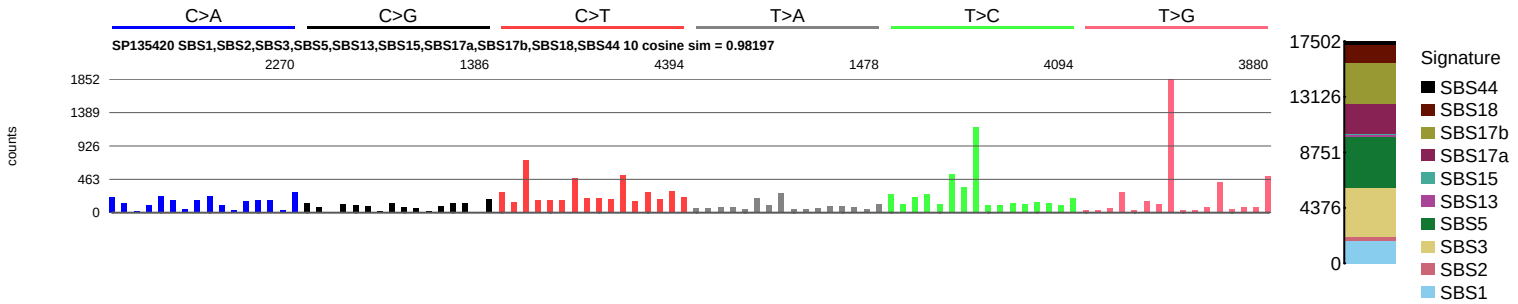
13126
8751
4376
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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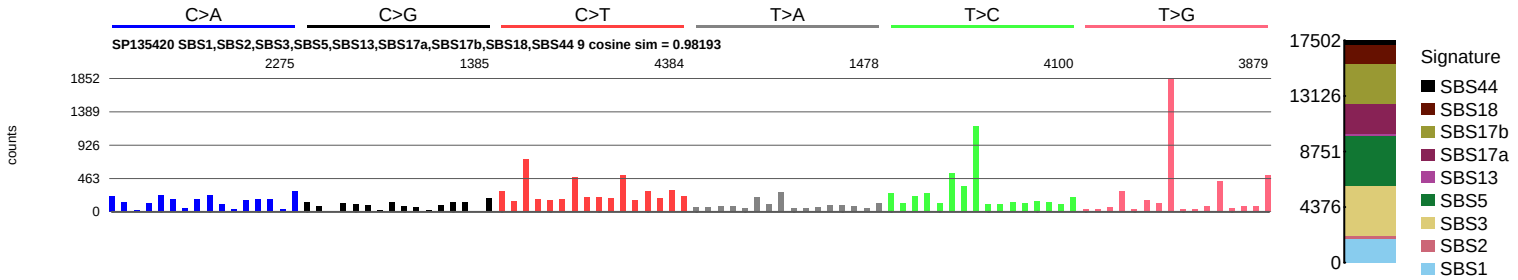
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98197
2270
1386
4394
1478
4094
3880
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98193
2275
1385
4384
1478
4100
3879
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
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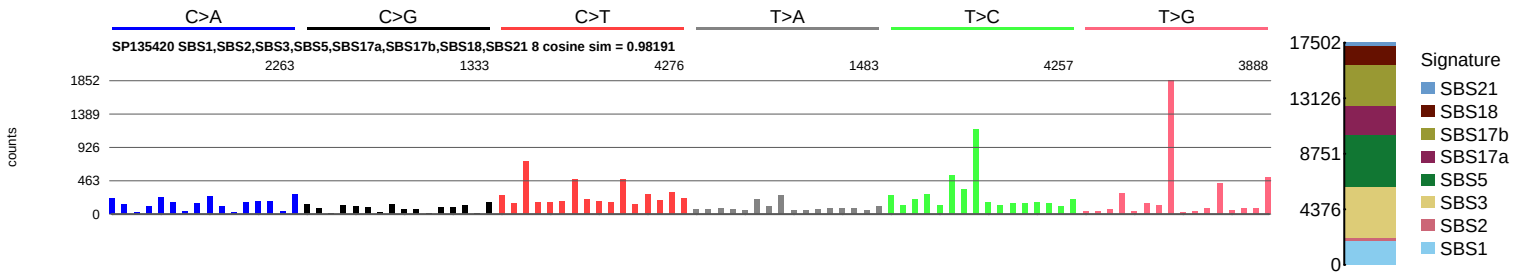
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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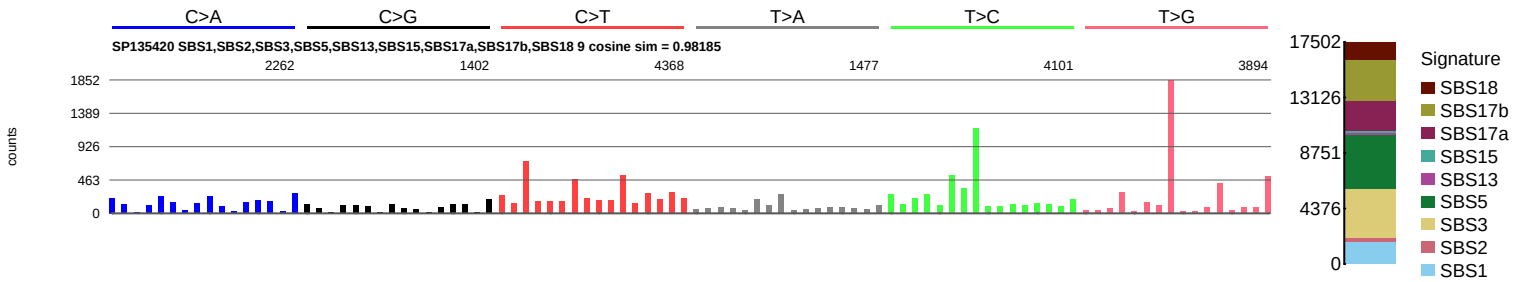
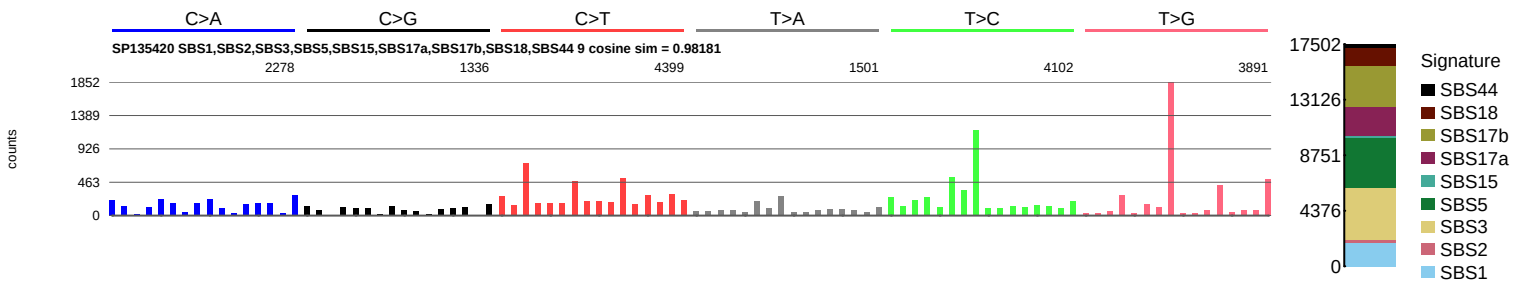
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98191
2263
1333
4276
1483
4257
3888
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98185
2262
1402
4368
1477
4101
3894
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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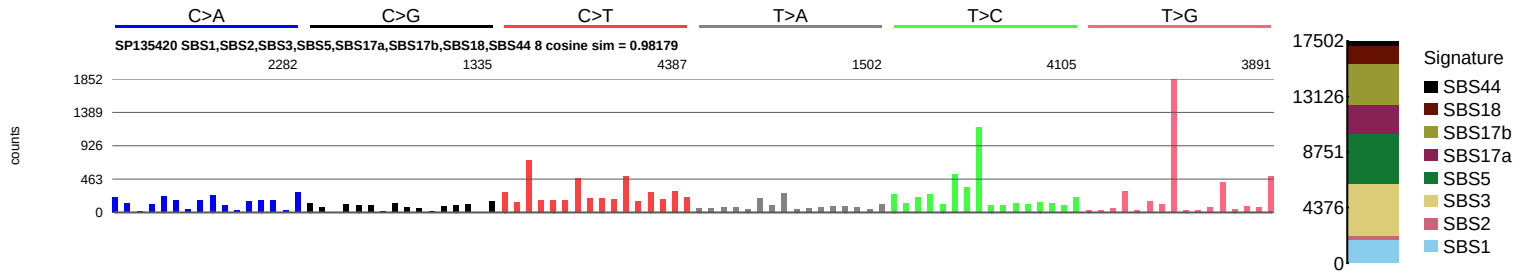
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98181
2278
1336
4399
1501
4102
3891
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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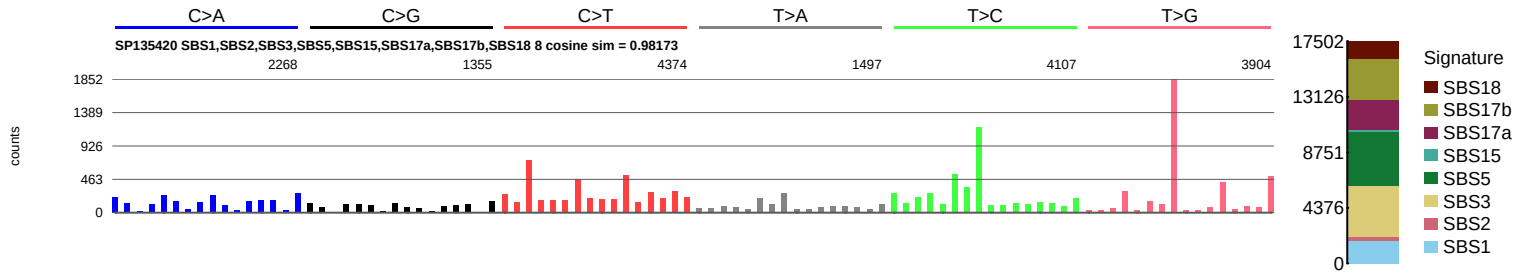
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98179
2282
1335
4387
1502
4105
3891
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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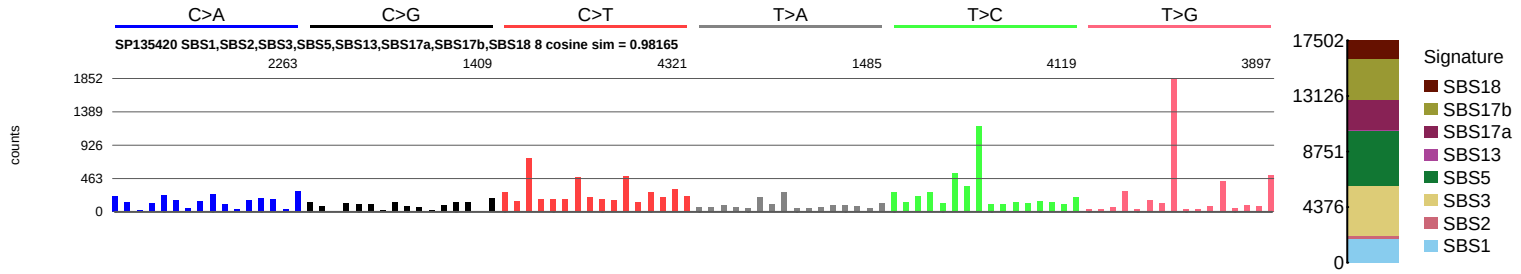
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98173
2268
1355
4374
1497
4107
3904
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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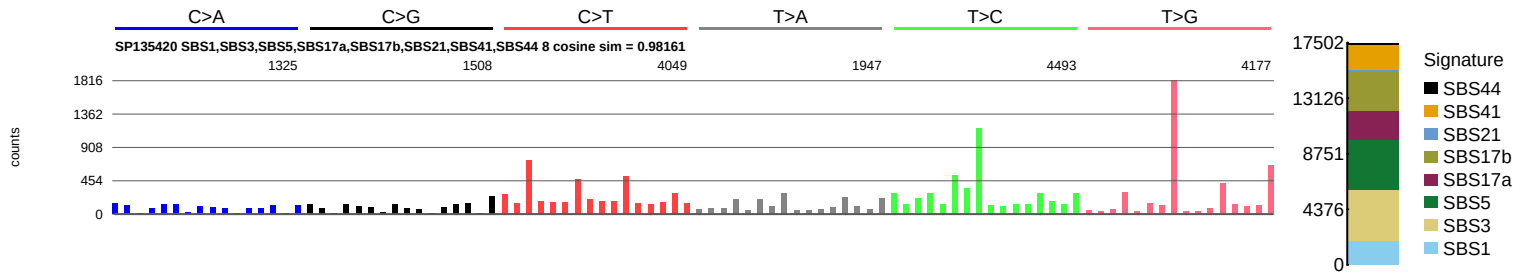
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98165
2263
1409
4321
1485
4119
3897
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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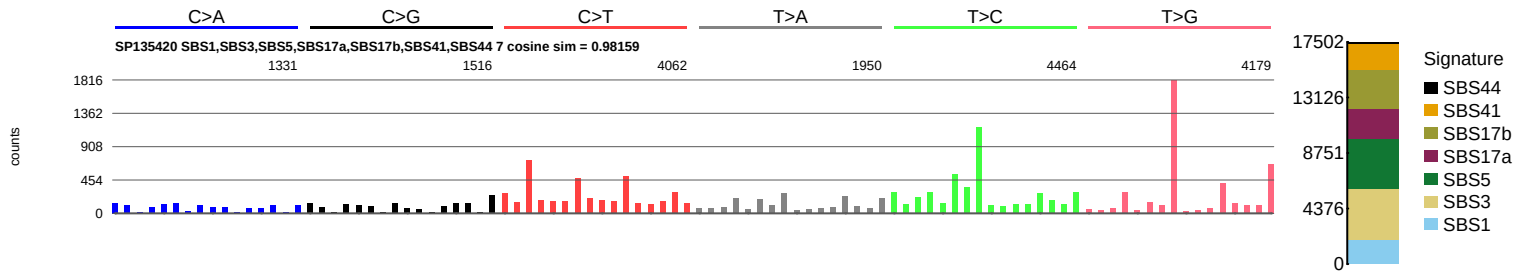
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98161
1325
1508
4049
1947
4493
4177
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1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T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98159
1331
1516
4062
1950
4464
4179
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
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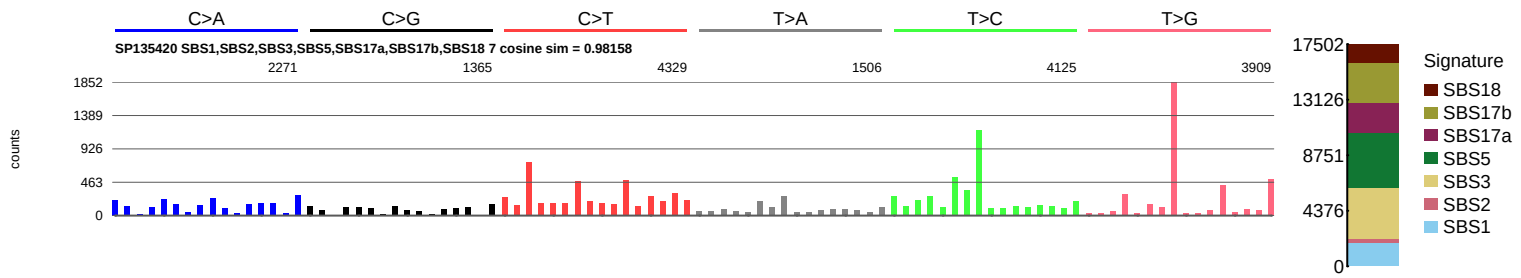
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1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T>C
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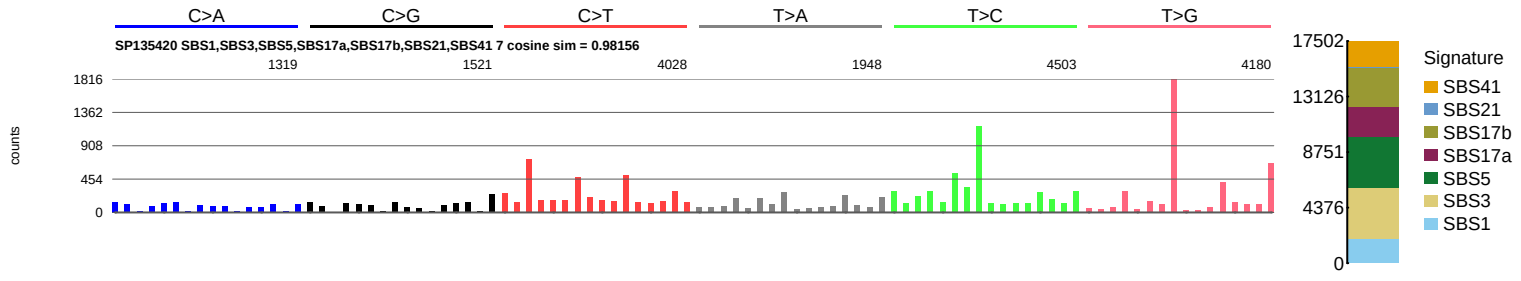
SP135420 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98158
2271
1365
4329
1506
4125
3909
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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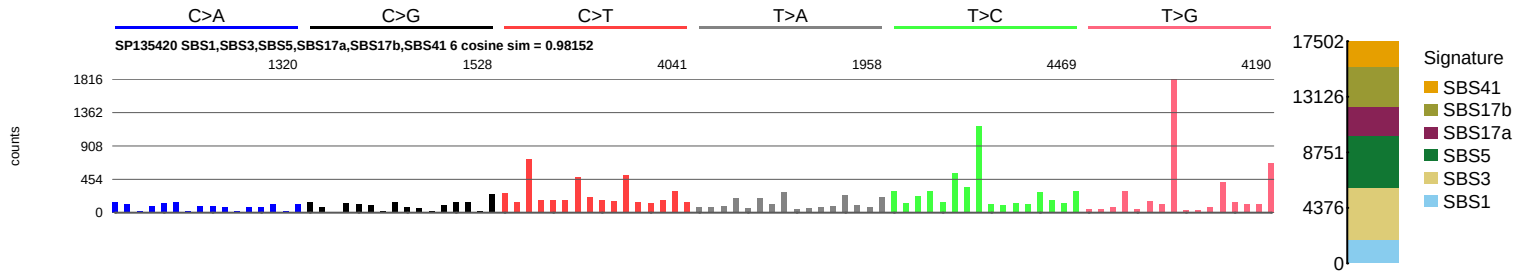
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98156
1319
1521
4028
1948
4503
4180
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
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T>C
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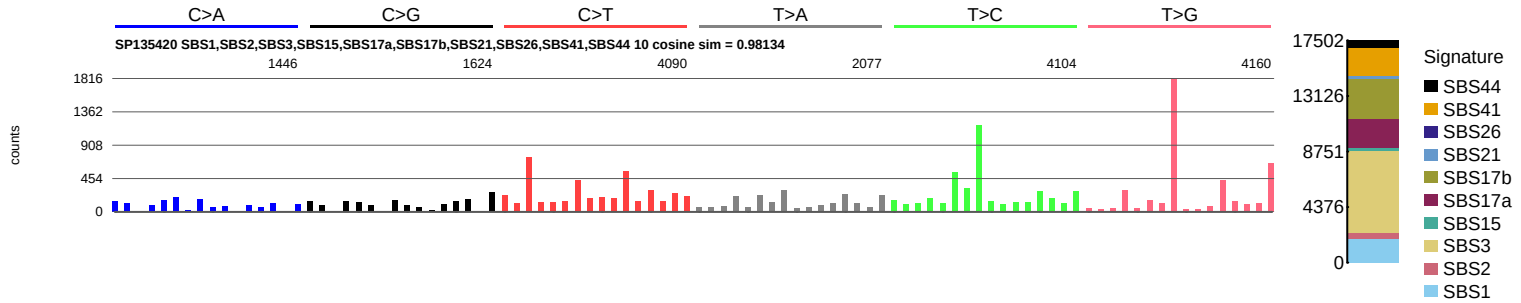
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98152
1320
1528
4041
1958
4469
4190
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98134
1446
1624
4090
2077
4104
4160
counts
1816
1362
908
454
0
17502
13126
8751
4376
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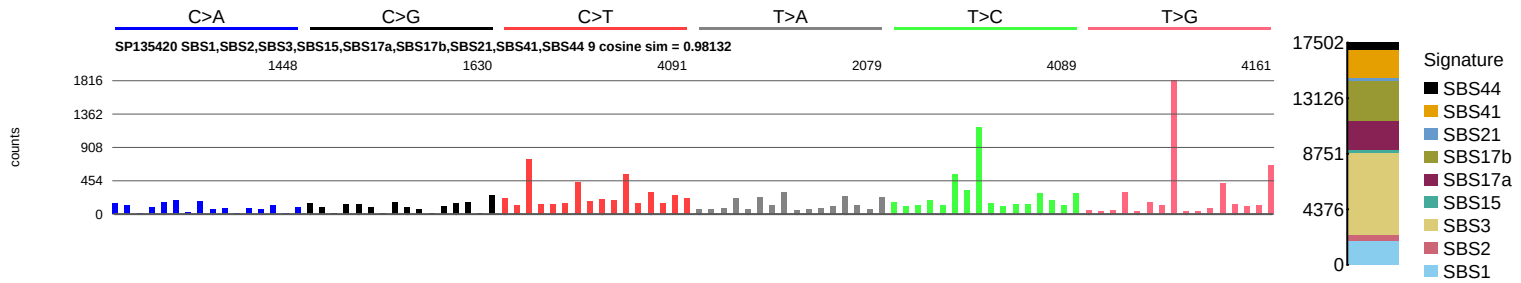
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98132
1448
1630
4091
2079
4089
4161
counts
1816
1362
908
454
0
17502
13126
8751
4376
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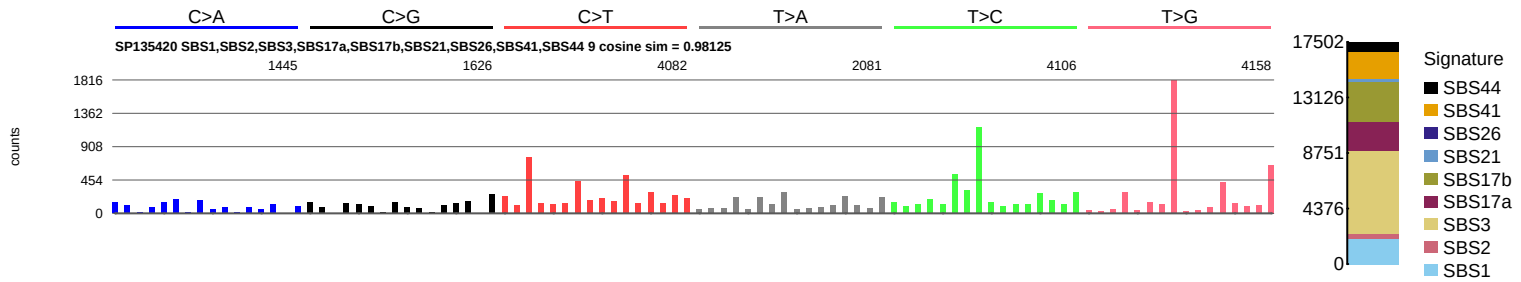
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98125
1445
1626
4082
2081
4106
4158
counts
1816
1362
908
454
0
17502
13126
8751
4376
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Signature
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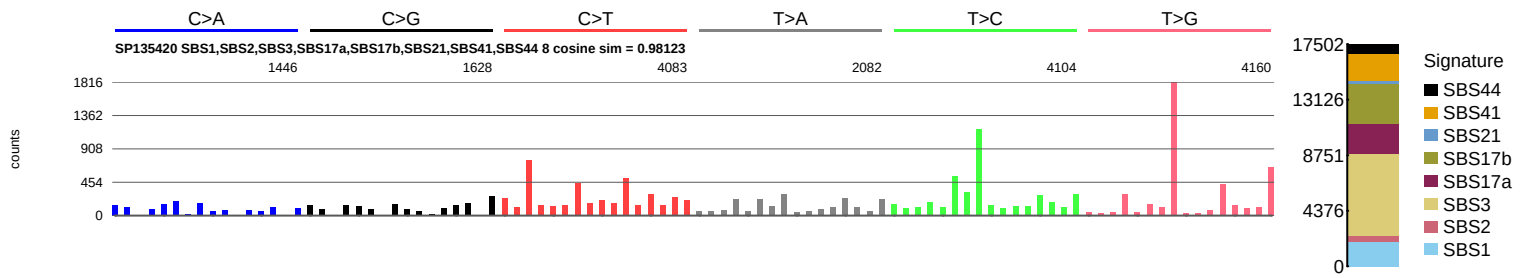
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98123
1446
1628
4083
2082
4104
4160
counts
1816
1362
908
454
0
17502
13126
8751
4376
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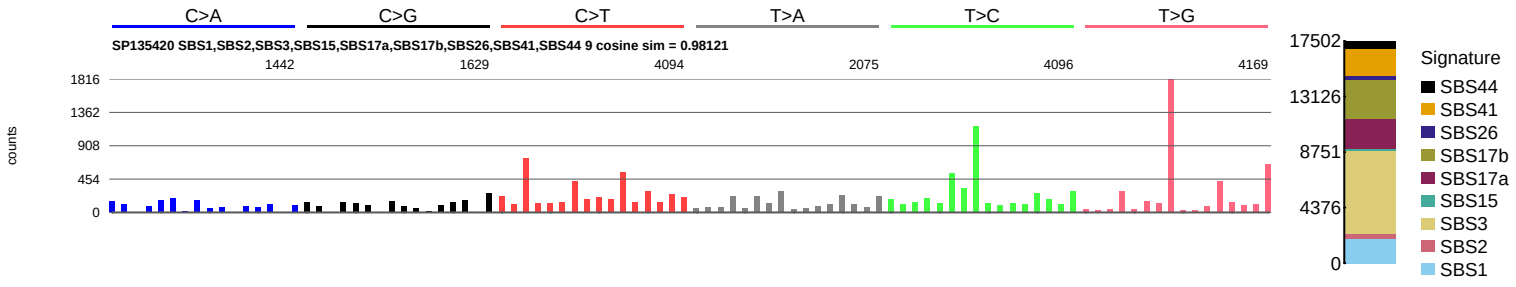
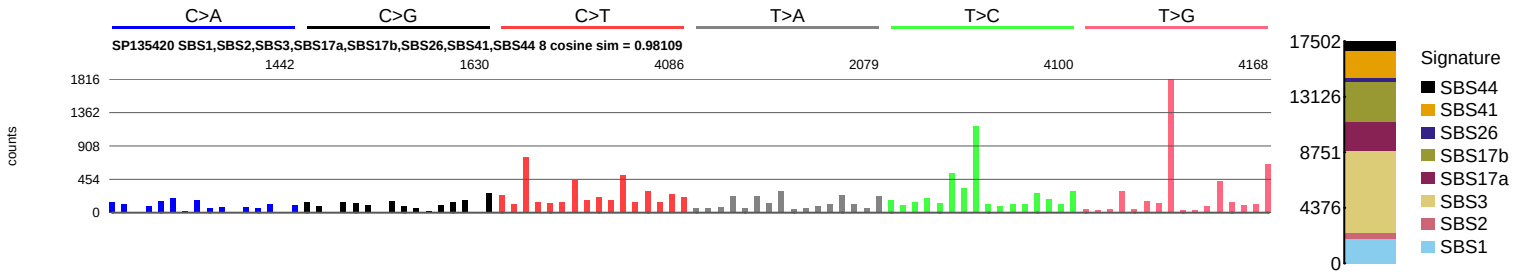
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98121
1442
1629
4094
2075
4096
4169
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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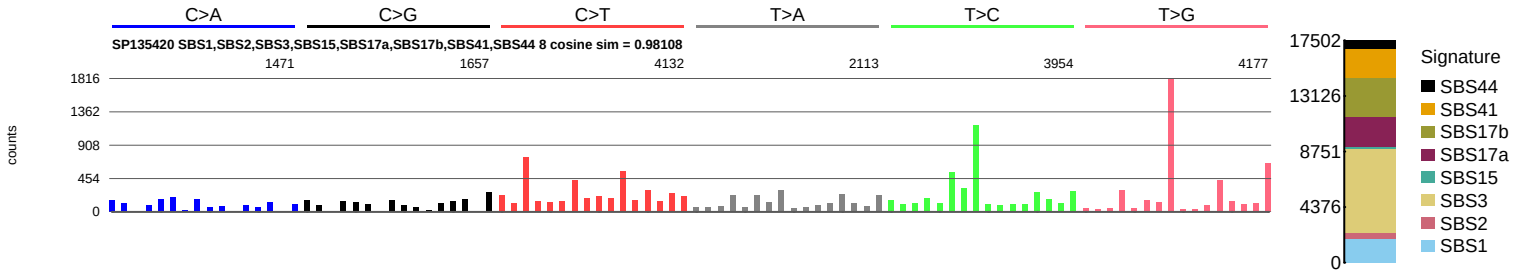
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98109
1442
1630
4086
2079
4100
4168
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98108
1471
1657
4132
2113
3954
4177
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
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SBS17b
SBS17a
SBS15
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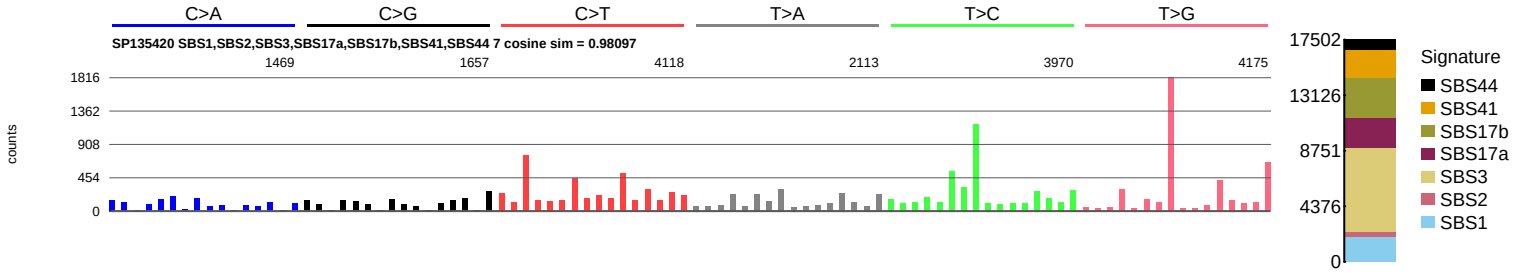
SBS2
SBS1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C>G
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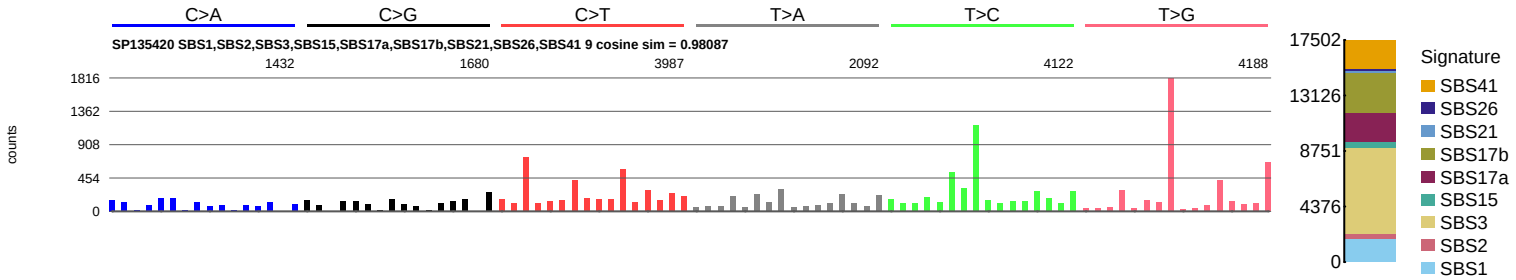
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98097
1469
1657
4118
2113
3970
4175
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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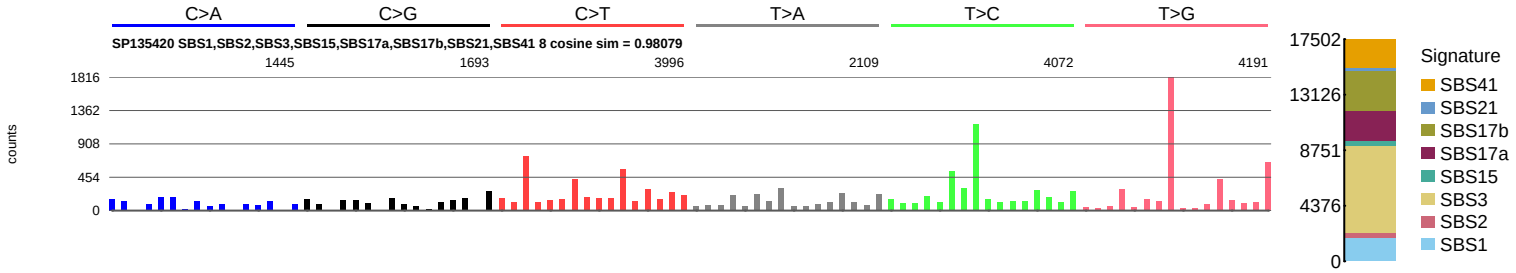
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98087
1432
1680
3987
2092
4122
4188
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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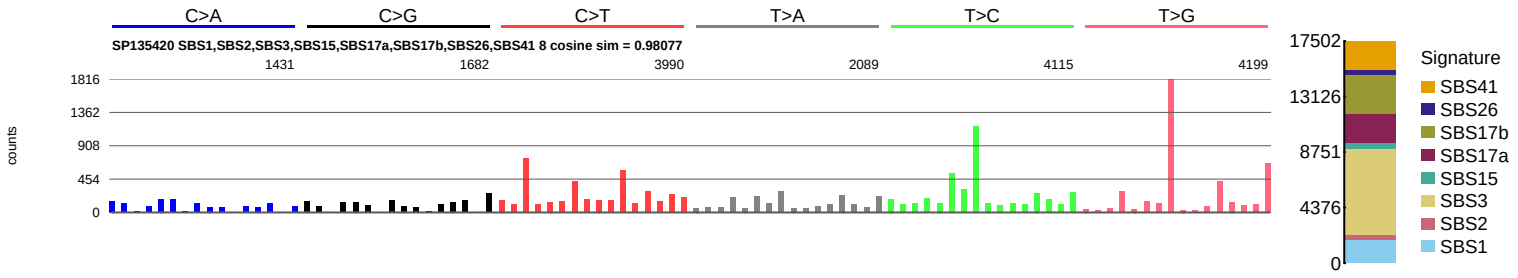
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98079
1445
1693
3996
2109
4072
4191
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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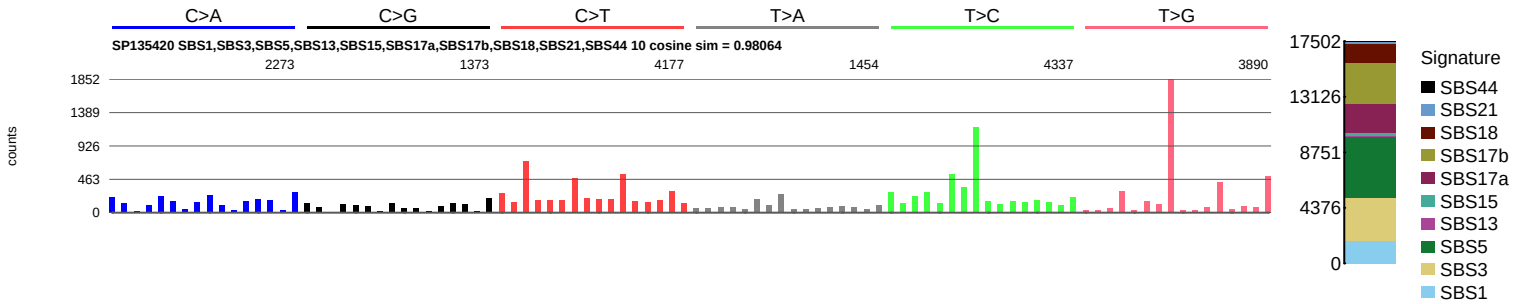
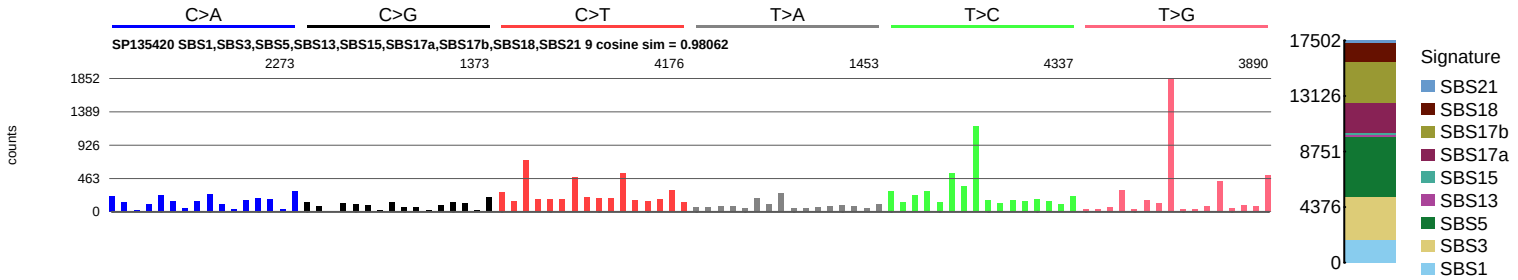
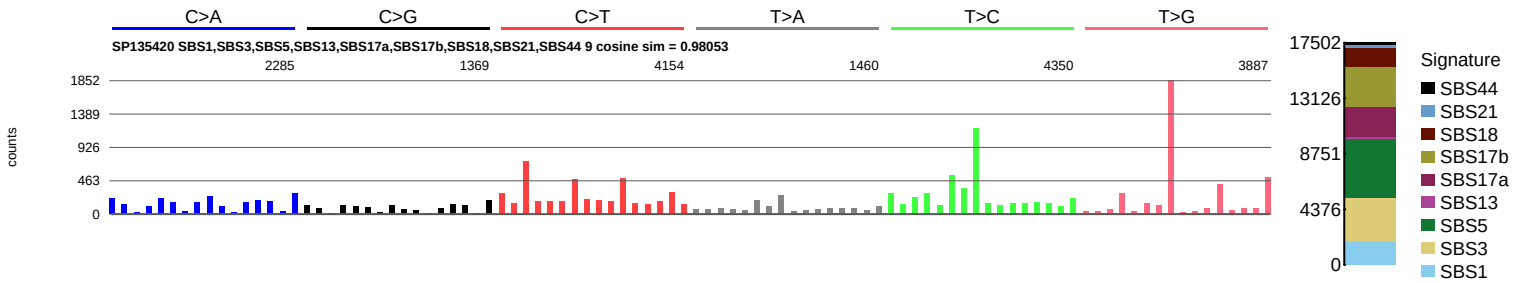
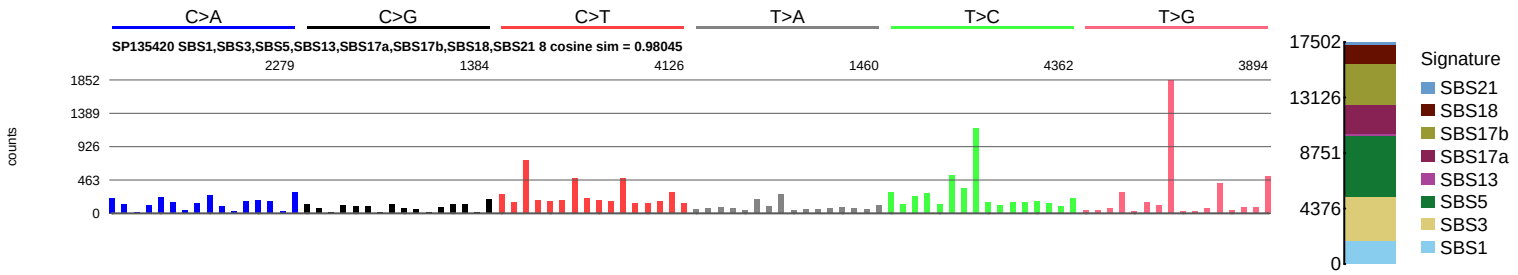
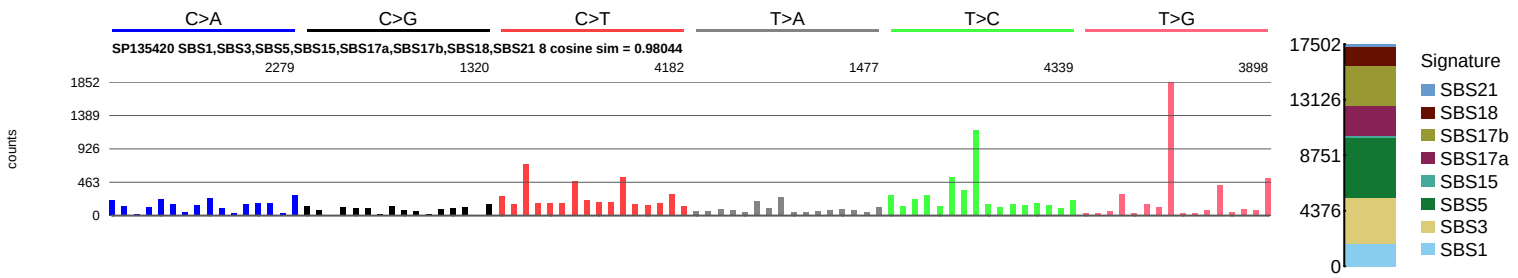
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98077
1431
1682
3990
2089
4115
4199
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

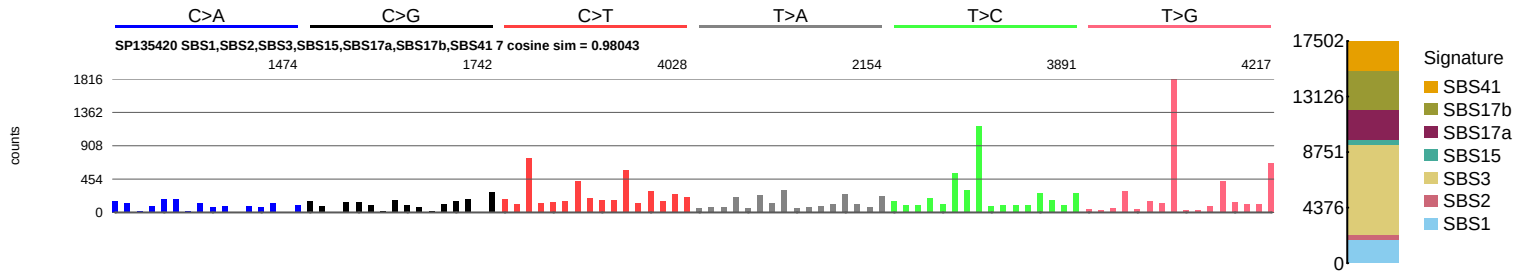

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98064
2273
1373
4177
1454
4337
3890
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

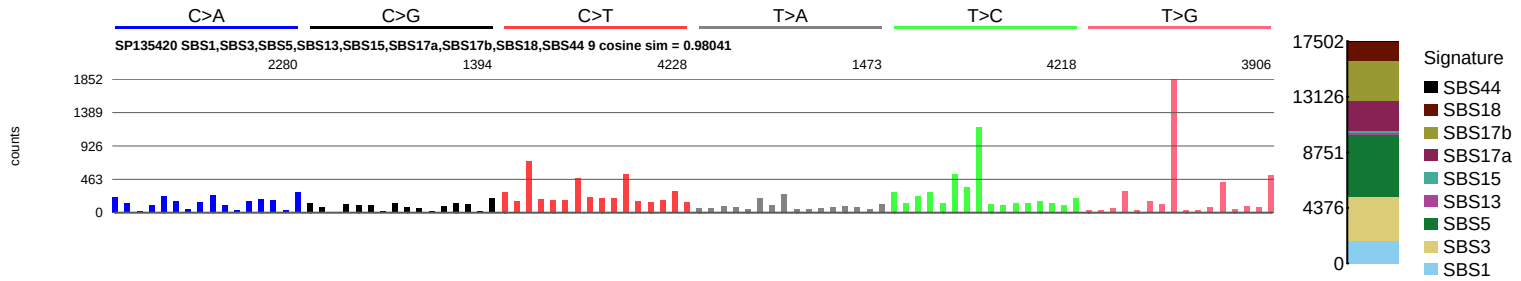

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98062
2273
1373
4176
1453
4337
3890
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

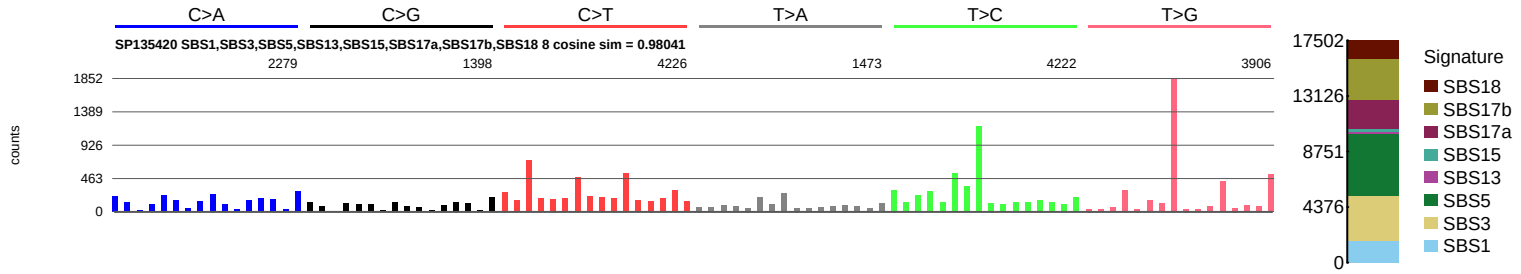

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98053
2285
1369
4154
1460
4350
3887
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

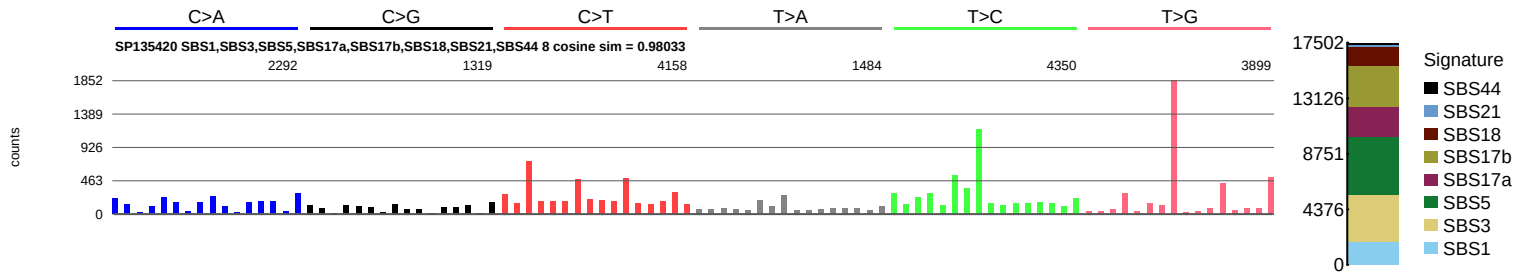

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98045
2279
1384
4126
1460
4362
3894
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

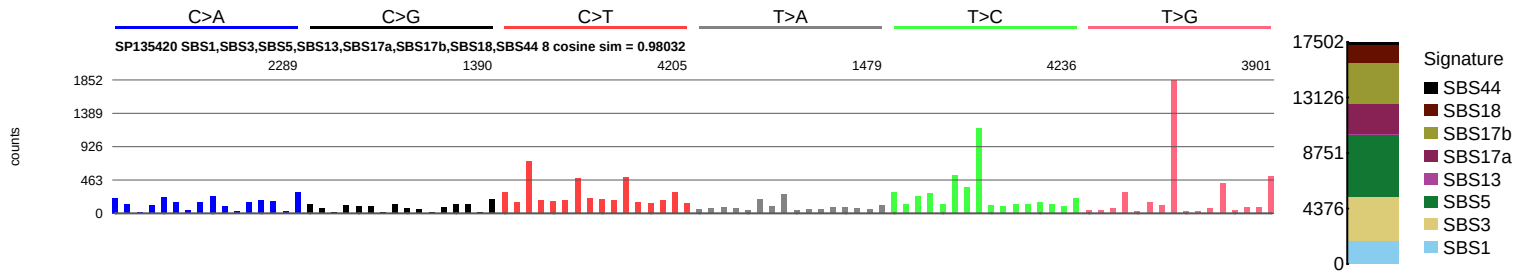

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98044
2279
1320
4182
1477
4339
3898
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

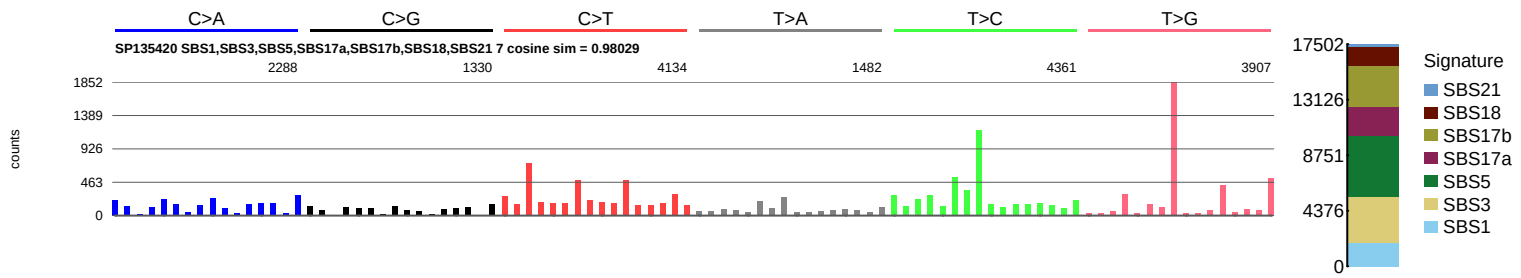

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98043
1474
1742
4028
2154
3891
4217
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

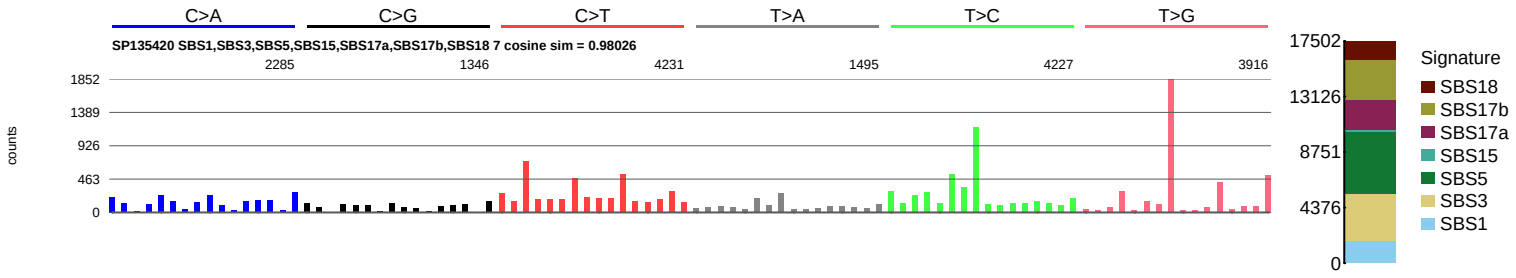

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98041
2280
1394
4228
1473
4218
3906
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

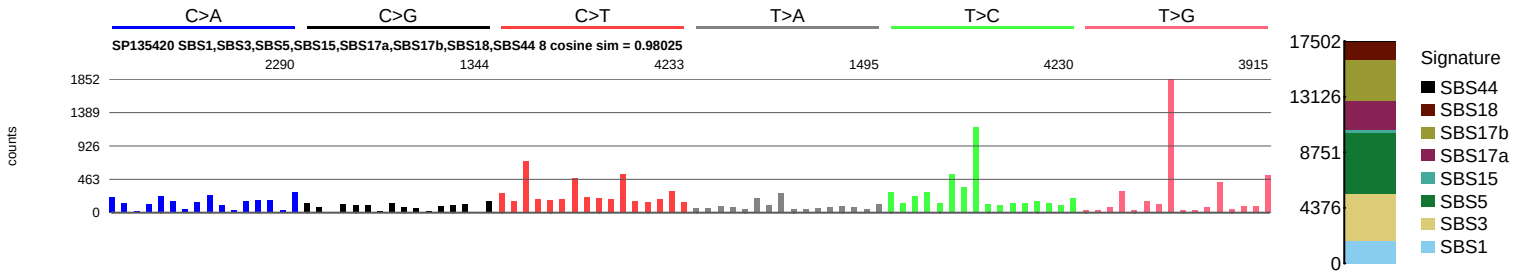

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98041
2279
1398
4226
1473
4222
3906
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

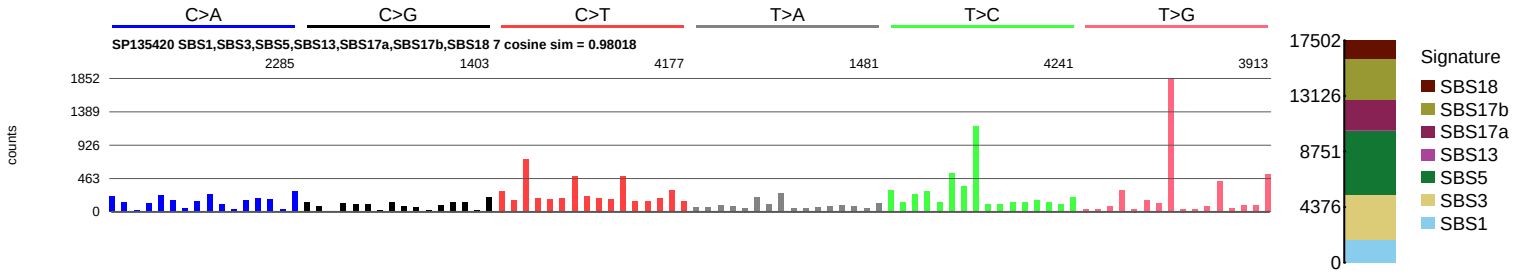

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98033
2292
1319
4158
1484
4350
3899
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

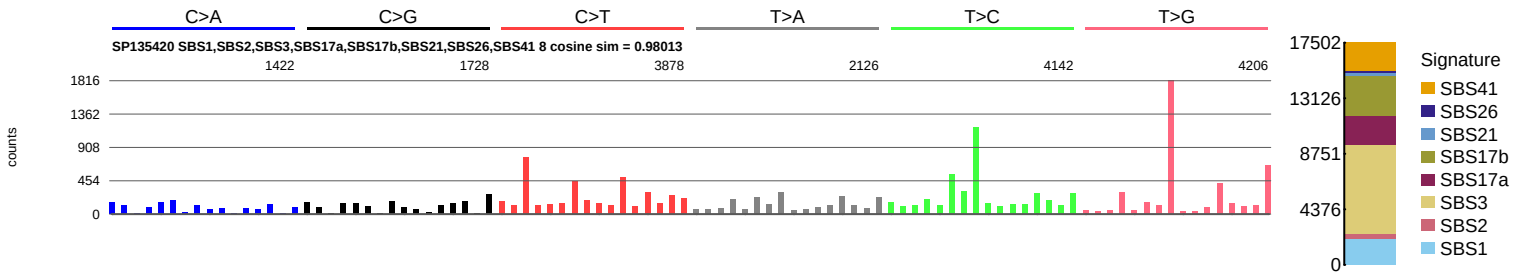

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98032
2289
1390
4205
1479
4236
3901
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

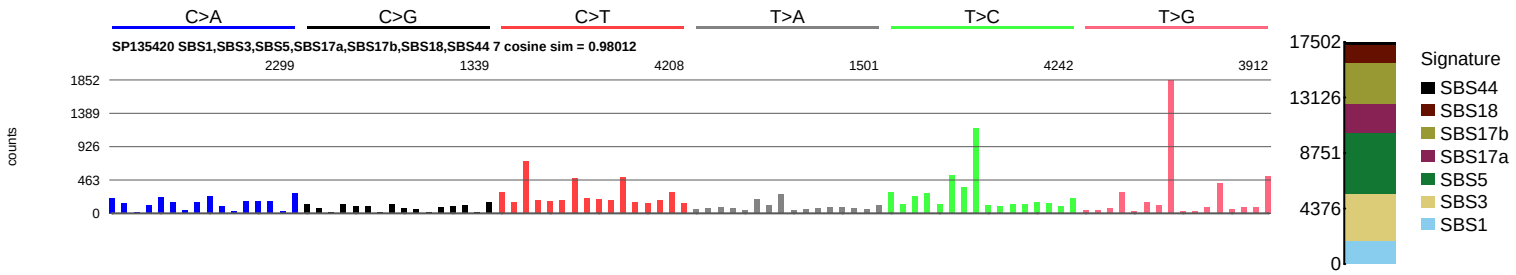

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98029
2288
1330
4134
1482
4361
3907
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

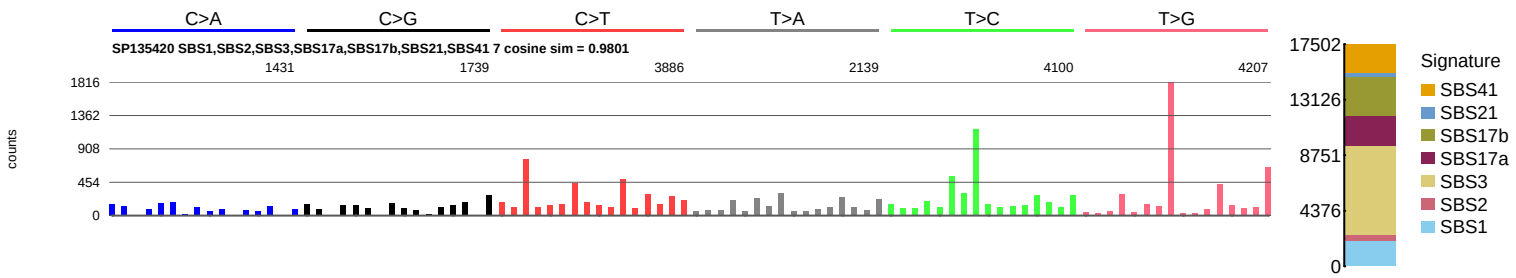

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98026
2285
1346
4231
1495
4227
3916
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

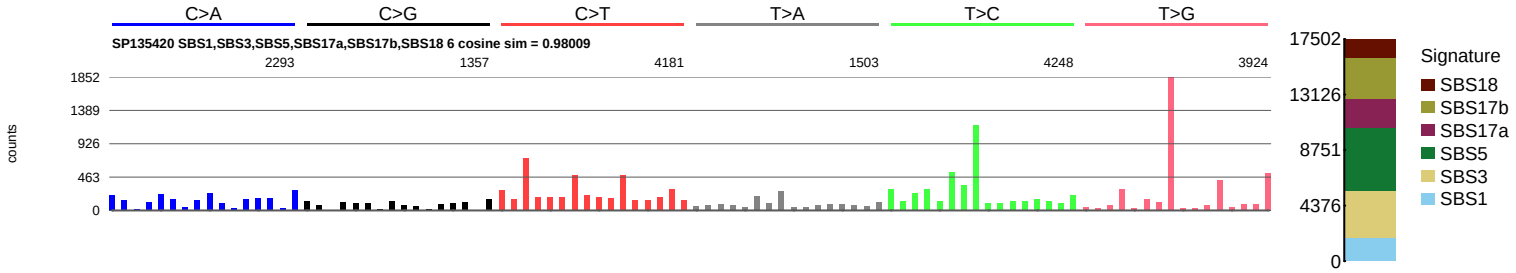

C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98025
2290
1344
4233
1495
4230
3915
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98018
2285
1403
4177
1481
4241
3913
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98013
1422
1728
3878
2126
4142
4206
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98012
2299
1339
4208
1501
4242
3912
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.9801
1431
1739
3886
2139
4100
4207
counts
1816
1362
908
454
0
17502
13126
8751
4376
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135420 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98009
2293
1357
4181
1503
4248
3924
counts
1852
1389
926
463
0
17502
13126
8751
4376
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1